

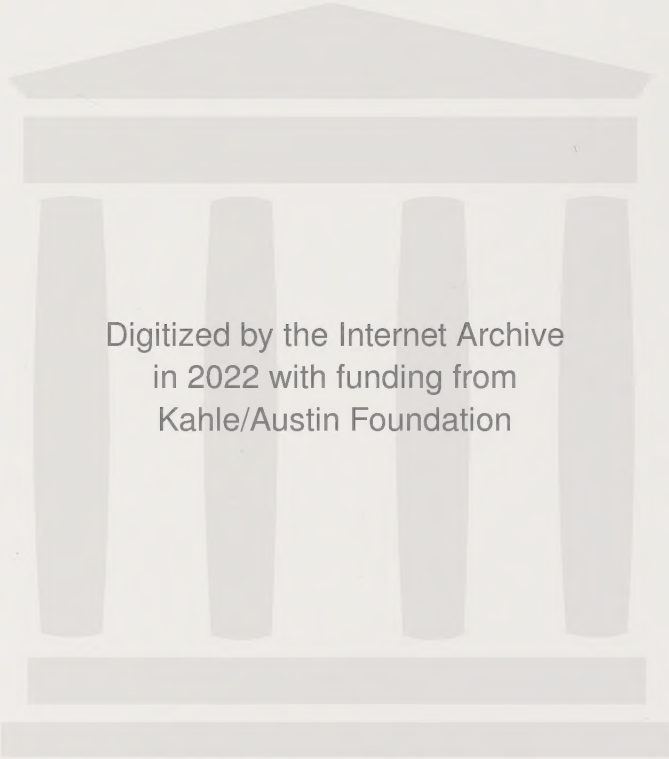
HELL'S KITCHEN

Cause, Prevention & Cure of
OBESITY, DIABETES & METABOLIC SYNDROME



By the creators of
DEAD DOCTORS DON'T LIE

J.D. Wallach, BS, DVM, ND
Ma Lan, MD, MS, LAc



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Dead Doctors Don't Lie!

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Dr. Joel D. Wallach, BS, DVM, ND
and
Dr. Ma Lan, MD, MS, LAc

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HELL'S KITCHEN

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Dedication

We would like to dedicate this book,
HELL'S KITCHEN
to our children Steve Wallach, Sarah Makis,
Jeremiah Wallach and Jennifer Wallach;
our grandchildren Jonathon, Zachary, Nikoloas,
Jacqueline, Natali, Shannon and Mollie;
and our first great grandchild Nashua.



About the Authors

Dr. Joel D. Wallach, BS, DVM, ND

Dr. Wallach has been involved in biomedical research and clinical medicine for more than 47 years. He received his B.S. degree (agriculture) from the University of Missouri with a major in animal husbandry (nutrition) and a minor in field crops and soils; a D.V.M. (veterinary medicine) from the University of Missouri; three year post doctoral fellowship (comparative pathology/medicine), Washington University, St. Louis, MO - The Center for the Biology of Natural Systems (The St. Louis Zoological Gardens, Shaw's Botanical Gardens, Botany Department, Washington University); Iowa State University, Ames, Iowa; Natal Parks Board, Umfolozi and Hluehluwe Game Parks, Republic of South Africa; St. Louis Zoological Gardens; The Chicago Zoological Gardens (Brookfield Zoo); Shedd Aquarium, Chicago, IL; Yerkes Regional Primate Research Center, Emory University, Atlanta, GA; a N.D. (naturopathic physician), the National College of Naturopathic Medicine, Portland, OR; Harbin Medical University, Harbin, Hei Long Jiang and the Shanghai Medical University, Peoples Republic of China.

He was a member of NIH site visit teams for facilities using exotic animal models for the study of human disease for four years and was a member of the 1968 National Science Foundation ad hoc committee that authored the 1968 Animal Welfare Act (humane housing, nutrition and care of laboratory and captive Exotic Species); a professor of nutrition at the National College of Naturopathic Medicine, Portland, OR; and a Consulting Professor of Medicine, Harbin Medical University, Harbin, Hei Long Jiang, People's Republic of China.

Dr. Wallach has appeared as an expert witness in State and Federal Courts as a comparative pathologist.

Dr. Wallach attained the rank of Lieutenant Colonel in the Missouri Air National Guard (131st Tactical Hospital), known as "Lindberg's own," and the Alaskan Air Reserves with responsibilities in public health and prevention and clean up of nuclear, biological and chemical attacks in North America and Europe.

Dr. Wallach was a recipient of the Wooster Beach Gold Medal Award (Association of Eclectic Physicians) for a significant breakthrough in the basic understanding of the cause and pathophysiology of Cystic Fibrosis. Dr. Wallach was nominated for the 1991 Nobel Prize in medicine (Association of Eclectic Physicians) for his work in the understanding of the genesis and pathophysiology of Cystic Fibrosis.

Dr. Wallach received the 2004 Guardian of the Constitution Award (Emord & Assoc.) for his proactive and successful litigation efforts in obtaining nutrition claims from the FDA for Selenium and Omega 3 essential fatty acids. He also received the 2004 James Lind Scientific Achievement Award (Emord & Assoc.) for "brilliant discoveries that have paved the way for greater health and longevity" for Americans.

Dr. Wallach has appeared frequently on local and national network and public radio and television (including a special on Cystic Fibrosis with ABC's 20/20), regional and national talk radio programs as an expert on trace mineral and rare earth deficiency diseases. Dr. Wallach is the host of his own syndicated talk radio show programs (Let's Play Doctor and Dead Doctors Don't Lie). Because of his free-wheeling style of humor and ability to zero in on the basic truth in health problems and medical politics he is widely known as the "Rush Limbaugh" of alternative health.

Dr. Wallach has given an average of 300 free health lectures per year for more than 15 years, literally reaching millions of enthusiastic fans. His tape, *Dead Doctors Don't Lie* (translated into eight languages) has sold more than 62 million copies world wide.

Dr. Wallach has more than 70 peer review publications to his credit including a major reference, *The Diseases of Exotic Animals* - published by W. B. Saunders. This 1,000 page illustrated text is listed by the Smithsonian Institute as, "A recommended reference for all professional zoological parks and aquariums."

Ma Lan, MD, MS, LAc

Dr. Ma Lan was educated in the People's Republic of China. Dr. Ma Lan received her MD from the Beijing Medical University, took her residency at the People's Hospital, Beijing and was a staff surgeon at the Canton Air Force Hospital; she received her MS (Master of Science) in transplantation immunology from Zhongshan Medical University, Canton, Peoples Republic of China; as with all Chinese doctors, Dr. Ma Lan was trained in Traditional Chinese Medicine (i.e. - acupuncture, herbs, manipulation, food medicine, massage and hydrotherapy) prior to entering the Western-style medical schools.

Dr. Ma Lan's research credits include being an exchange scholar at Harvard School of Medicine, Boston, MA; a research fellow in laser microsurgery at St. Joseph Hospital, Houston, TX; the Department of Orthopedic Surgery, the Medical College of Wisconsin, Milwaukee, WI; and the Department of Pharmacology, University of California, San Diego, CA.

Dr. Ma Lan has 10 peer review publications to her credit in the fields of transplant immunology and laser vascular surgery. Dr. Ma Lan attained the rank of Lieutenant in the Chinese Air Force with primary responsibilities as a general surgeon in the Canton Air Force Hospital, PRC.

Introduction

Despite extraordinary efforts by the federal government (U.S. Department of Agriculture, U.S. Surgeon General), various associations (American Heart Association, American Diabetes Association, American Cancer Society, American Medical Association, etc.) and private companies and authors (Atkins Diet, South Beach Diet, Zone Diet, Dr. Wallach's See Food Diet, etc.), the rates of diagnosis of overweight and obese Americans and diabetic Americans (in fact, the problem includes peoples of all industrialized nations) continues to go up at an ever increasing and an ever alarming rate - a metabolic pandemic!

The actual cause of obesity (we call obesity the Feedlot Syndrome), diabetes, the Metabolic Syndrome and other chronic degenerative diseases have eluded the "best" efforts of all - unfortunately there are no laws requiring doctors to cure people when there are cures available so they don't even look for the cure - result, 65 percent of Americans are overweight and 40 percent are obese; we Americans have an epidemic of 1 million new cases of type II diabetes each year; we rank 26th in the world when it comes to health and longevity (there are 25 other countries whose peoples are healthier and lived longer than we); we rank 29th in the world in live births and first year survivability of our babies - even though we spend more money annually for health care than all other countries combined (\$1.6 trillion for the U.S. alone out of \$2.7 trillion for the entire world).

Obesity, and the metabolic diseases associated with it, is a symptom of our doctor's, our health care system's and our government's abysmal failures or purpose. Obesity is not the disease, obesity is not the enemy, obesity is but a symptom of a deeper nutritional deficiency, just as a fever is not the cause of pneumonia, but rather it is a symptom of an infection - and we all know that just lowering the elevated body temperature (the fever) back to normal with aspirin does not kill the bacteria or virus - the cure for an infection requires a viable immune system and sometimes a wound disinfectant or an antibiotic.

Obesity is an anthropological problem, not a medical problem, therefore we can only solve the obesity problem by studying man, man's successes and failures throughout the years.

We looked back beyond written history and looked for clues that would show us what events, what technology, what people set us on the terrible path to the world pandemic of obesity - as a result we have found and identified the events millennia by millennia, century by century, decade by decade and year by year that have led up to our modern health crisis - we have identified the men, the technology and the time and date, Monday September 4, 1882 at 3:00 p.m. that set us on the "Road to Hell" and we have determined exactly how we got into "HELL'S KITCHEN."

We chose HELL'S KITCHEN as the title for this book because we wanted to illustrate all of the side effects of the technology that led to the epidemic of obesity, diabetes, Metabolic Syndrome and their complications. We chose HELL'S KITCHEN because the name conjures up the borough in New York that "cooked up" all of the known evils of gang warfare, drugs, prostitution, organized crime, petty crime, pickpockets and graft.

Knowing the basic, bottom line root cause of obesity and its associated diseases has allowed us to focus in on the cause, to eliminate the cause and approach it with all of our strength and resources. As a result, we have found the basic root cause of obesity - over a short 50 year period we (all modern people) willingly gave up a basic practice that humans from Europe, Asia, Africa, South America and North America have depended upon for tens of thousands of years for the advancement of mankind and for long healthy lives.

We as a people have literally discarded our health inheritance and heritage, as we would grandma's accumulations in the attic, the health and life supporting practices of throwing our wood, peat and kelp ashes (minerals left over when we burn carbon fuel) into our gardens, and cutting our valuable table salt with wood ash, and using ashes for food preparation "culinary" ashes - practices that we had used on a daily basis for thousands of years.

Mineral deficiencies result in cravings a.k.a., the munchies, that ultimately, produce obesity, diabetes and the Metabolic Syndrome. We can simply eliminate obesity, diabetes, the Metabolic Syndrome and other degenerative diseases in three to six months in America and in the world for that matter, if only we would all adhere to simple principals and supplement properly and cut back on most forms of

dietary sugars and carbohydrates.

We gave up the free byproduct of heating and cooking (wood ashes) that we humans had employed as a source of nutritional minerals before written history when we began to use electricity, natural gas and propane. Instead of replacing the lost source of nutritional minerals, we were convinced by the medical profession that we could replace nutrition with pharmaceuticals, surgery and health insurance - we chose poorly and we are paying the price in epidemics of nutritional deficiency diseases, higher medical costs and shorter lives.

Unfortunately, there is too much profit (\$750,000 per diabetic, and \$117 billion annually for treating obesity in America alone) in treating obesity and diabetes medically; there are no laws requiring doctors to cure people when there are cures available; therefore, medical doctors will never employ a cure even though the cure was available in 1957! If it is to be that we eliminate obesity in all of its metabolic forms and complications, it is up to each of us to employ the prevention and effect the cure - doctors are simply not going to do it.

Experts will attack our observations and conclusions as coming from the untrained, unworthy of an opinion, however, we would point out that Columbus, Franklin, Newcomen, Faraday and Edison were all simple men who made critical observations and in one way or another took action against the sea anchor of the dogma and common thinking of the day and as a result catapulted the capabilities of humanity forward beyond the dreams of the "learned" but ignorant scholars of the day.

We truly hope you will read HELL'S KITCHEN and employ the lifestyle and supplement suggestions in the book and pass on the information to those you love and care about. Only by taking responsibility ourselves can we solve the obesity (Feedlot Syndrome), diabetes and Metabolic Syndrome pandemics - just another diet will not solve the problem.

Dr. Joel Wallach and Dr. Ma Lan



Benjamin Franklin
1706 - 1790

American Longevity

To the PRINTER of the LONDON CHRONICLE.

SIR, I have often heard it remarked, that our Colonies in North America were unhealthy and unfavourable to long life; and more particularly so upon their first settlement. In opposition to this groundless notion, I here send you two paragraphs taken from the Pennsylvania Gazette of July 16, and the New-York Gazette of August 27, giving an account of the deaths of the first-born of the city of Philadelphia, and of the province of Pennsylvania:

PHILADELPHIA, July 16. "At Kennet, in Chester county, the 5th instant, died John Key, in the 85th year of his age, and the next day was interred in the burial place belonging to the people called Quakers, in that township, attended by a large number of reputable people, his neighbours and acquaintance. -- He was born in a cave, long

afterwards known by the name of Penny-Pot, near Race-street; and William Penn, our first proprietor, gave him a lot of ground, as a compliment on his being the first child born in this city. -- In the early part of his life his conversation was very engaging, and his company much sought after by those of his own age; in his decline he was much esteemed for his peaceable disposition. His constitution was very healthy till about 80, when he was seized with the palsy, and continued weakly till his death. -- About six years ago he walked on foot from Kennet to Philadelphia in one day, which is near 30 miles. -- It has been said that people in this province are short-lived; but when we consider how few children were born here 75 years ago, and observe how many old people there are still alive among us (who were born here) we shall rather think no quarter of the world can shew a greater number of aged persons then this province can, in proportion to the children born."

PHILADELPHIA, Aug. 24. On the 10th instant, at Brandywine Hundred, in New-Castle county, died Emanuel Grubb, in the 86th year of his age, and the next day was interred in St. Martin's church-yard, at Lower-Chichester, in Chester county, attended by a large number of his relations, neighbours, and acquaintance. -- He was born in a cave, by the side of Delaware river, not far distant from where he always lived, and died, and was the first child born of English parents in this province. His constitution was remarkably healthy during his whole life, having never been afflicted with any sickness till a few days before his death. His strength and activity were surprizing in a man of his age; he could mount and ride a horse with as much dexterity as a lad of 20. -- A few months ago, he rode from his own house to this city, and back again, which is upwards of 40 miles. - His memory was equally surprizing, and not in the least impaired, till his death; he could remember transactions of a late date equally well with those which happened in his younger days. He was exceeding temperate in his living, seldom making use of spirituous liquors; and for his peaceable and friendly disposition, was highly esteemed by all that knew him. This instance, among many, is a proof of the longevity of people born here.

It is worthy of observation also, that Governor Hutchinson, in his

valuable History of the Massachusetts's Bay, Vol. II. pp. 148 and 216, says,

"July 20, 1704, died at Marshfield, Peregrine White, aged 83 years and eight months, the first-born in Plymouth colony.

"Jan. 14, 1716, died at Salem, Elizabeth Patch, the first-born female in the old colony of Massachusetts's Bay, so that she must have lived 86 or 87 years; and April 14, following, died at Newport, in Rhode-Island, Mary Godfrey, aged about 87, being the first child born there. -- The longevity of the first-born in each of the three colonies, is worth noting."

As a well-wisher to mankind in general, I thought it worth while to send this to your useful paper, in order to allay the apprehensions of those, whose inclinations, business, or necessities may induce them to settle in that part of the British Empire. I must add at the same time, that I have myself lately travelled over the greater part of that extensive continent; and can with truth say, in spite of all that hath been boldly and positively asserted to the contrary, that I could not discover among that whole people, one grain of disaffection to their brethren in Britain, nor did I meet with a single person who had ever formed the most distant idea of throwing off their allegiance to the mother country. This it is doing them but bare justice to declare; and is all the return I can at present make them for the kindness and hospitality with which, purely on account of my being an Old England-Man, I was universally treated. I am, Sir, your constant reader,
F + S.

The London Chronicle, December 15, 1767

Benjamin Franklin

Chapter 1

“And the Lord God formed man of the dust of the ground, and breathed into his nostrils the breath of life; and man became a living soul.”

- Genesis 3:7

“I am a brother to dragons (dinosaurs), and a companion to owls”

- Job 30:29

“Praise The Lord from the earth, ye dragons, and all deeps.”

-Psalms 148:7

“And he laid hold on the dragon, that old serpent, which is the Devil, and Satan, and bound him (in fossils) a thousand years.”

-Revelation 20:2

Fire Power

It is often said that, “The road to hell is paved with good intentions” and that “A kingdom was lost for want of a nail!” Both of these statements are true when it comes to the world wide pandemic problem of obesity and all of its terrible consequences.

There have been so many political, medical and dietary approaches devised to solve the problem of obesity, yet they have all failed, because all of our historical approaches to obesity focused on the symptom - fat - fat in the food and fat in our bodies - none of the weight loss programs were focused on the cause. To date, without exception, all attempts at dealing with obesity, diabetes and the Metabolic Syndrome are focused on the most obvious symptom (the body fat) and the simple concepts of exercise and caloric restriction (CR) diets.

Believing that an obese person is fat because of a simple over

consumption of fat, carbohydrates and total calories would be the same as saying that wildfires are caused by ashes - because there are always ashes found after every fire!

This book will focus on the basic cause of obesity, thence on to the prevention and cure. As a result this book will feel and read differently than any other weight loss book that you have ever flipped through, read or used. To find out where we went wrong, we must begin at the beginning of the self constructed road to Hell. We will have to travel back in history - way, way back to before the beginning, and then methodically retrace our steps to find out exactly where we exited the highway to health that we had traveled relentlessly for millennia, where we suddenly veered from our safe, tried and true path and took the wrong fork in the road, reverse our course and get back on the right road - the right road to fitness, health and longevity!

Like an alcoholic, we must first admit we are lost and in trouble before we can begin to hope to extract ourselves from our downward spiral in health and our increase in weight. If we continue doing what we are presently doing in the world of weight loss and control, we will continue to get what we now have and we and our very cultures will be crushed by the health and financial cost of our own weight.

So to begin at the beginning, we will start by using the time line of anthropological and geological dating, not because we necessarily totally believe in this system, but because that's how the scientific clues to our obesity questions and problems are laid out.

For those wanting to use biblical dating, they can use the named geological periods rather than the geologist's chronological dates.

The Road to Eden

From the beginning, for "millions of years," every species of plants and animals on earth depended on a daily source of "solar income" or fresh supplies of energy from the sun or consuming another organism of some form that had temporarily stored the sun's energy. Plants are like solar panels, their light sensitive chlorophyll laden

leaves collecting the sun's radiant light energy as it arrives and converts it into chemical energy that humans and animals could eat, store and metabolically "burn."

Plants not only convert "solar income" into burnable chemical energy, they also store it for later use as cellulose, starch and sugar - storing it in their cells until they either use the energy themselves, decompose, burn or get eaten. Coal originated as plants that became buried deep within the earth, they were compressed under millions of tons of pressure and heated to thousands of degrees and ultimately became a fossil fuel.

Before man appeared on earth, before the dinosaurs (dragons), before the continents slipped into their 21st century positions, a living fossil covered the earth. They were members of a swampy plant culture, literally forests of strange trees and enormous ferns - "monsters of the vegetable world."

Before the Jurassic Period, before the Triassic Period, before the Permian Period, there was the Carboniferous Period, a long period of primitive earth's history named after the coal that was then being created.

According to geologists, for most of the earth's four and a half billion geological years, its land masses were starkly naked, while the cradle of plant and aquatic life unfolded unseen (except for the eyes of God) in the depths of the oceans.

In March of 2004, the International Union of Geological Scientists added the Ediacaran Period to the Pre-cambrian Eon, an enormous time span that covers ninety percent of the earth's geologic history. The new period takes its name from Ediacaran fossils, remains of the most complex animal life, that were found in enormous numbers in the Ediacaran Hills of South Australia.

The period begins around 620 million years ago with the rapid end of the global Marinoan glaciation - the catastrophic environmental calamity that entombed earth in ice for several million years. It ended 543 million years ago with the appearance of the Cambrian Period and the "explosion" of diverse forms of animal life.

The first toehold gained on the earth's beaches by plants occurred about 425 million years ago, and then ever so slowly, one could even

say in the beginning “shyly;” after a time, they “figured out” how to convert from an aqueous life of light filtered through water to an atmospheric life collecting and employing solar incomes directly from the sun, and they eventually flourished and covered the once bare land with a green tidal wave of plant life.

By the Carboniferous Period - about 360 to 290 million years ago, there was nothing shy about the growth and tangle of plants on earth - they dominated the earth!

Verdant jungles of a density unimagined today sprung up, at first they crawled and then they raced over the formerly barren earth, ravenously taking in the carbon dioxide from the atmosphere, using the new process of photosynthesis and the sun’s “solar income” to fuel their growth; the new “trick” was that plants “learned” how to store the excess energy produced in the form of cellulose (wood), plant starches and sugars.

Most coal beds were derived from the first waves of plants to leave the oceans and cover the land - animals soon followed. Coal is the concentrated remains of extinct plant life forms that once flourished and covered the planet.

Most prehistoric coal forests were dominated by the *Lepidodendron*, a huge tree six feet in diameter that could reach the dizzy height of 175 feet, the maximum growth potential of height that photosynthesis had ever attained to that date. Photosynthesis, a relatively new “discovery” had taken the single cell algae surrounded by the nutrient rich soup of the oceans, to an awesome dry land behemoth that converted the solar income of radiant energy into growth power and stored energy.

At its crown, the *Lepidodendron* branched into short stubby branches covered with narrow leaves three feet long. Unlike modern trees, the *Lepidodendron*’s heart wood was soft, pulpy and pithy; as long as water was in good supply the inner cells would remain turgid and kept the tree erect; during draught periods, these monster trees dehydrated, wilted, drooped and fell into the swamp.

The prehistoric jungles also boasted another botanical giant, the *Sigillaria*, a tree that had a long trunk that was characterized by a single fork at the crown; each fork was adorned with a large bundle

of long strap - like leaves.

The ancient jungles seethed with vegetarian and carnivorous insects. With little or no competition or predators, these insects reached gigantic sizes; some cockroaches attained lengths of 12 inches; dragonflies' wingspans reached almost three feet; and millipedes grew to six foot lengths.

Early in the Carboniferous Period, small newt-like vertebrates, not much different from fish, scuttled through the dense swampy undergrowth. By the end of the Carboniferous Period, fifteen foot long amphibian-like creatures were pulling their bellies through the primeval mud. During the same Carboniferous Period, quietly, and without much notice, some amphibians changed from laying gelatinous strings of aquatic eggs and began to lay hard shelled eggs incubated in nests built of decomposing plant material.

The shelled egg laying amphibians were followed by reptiles (some of which laid eggs and some gave birth to live young), dinosaurs (dragons - it is interesting to note that dragons are mentioned in the Bible thirty five times), birds and eventually mammals.

The earliest known and dated reptile (340 million years old) is called *Casineria*, it was six inches long and looked like a cross between a newt and a lizard; it had five toes, simple claws, a vertebral column and a foreleg anatomy that is characteristic of lizards.

At first, mammals were small shrew-like creatures that ate insects and worms. About 130 million years ago, a twelve pound fanged marsupial, *Repenomamus robustus*, devoured small asian dinosaurs, psitasaur, and the eggs of larger dinosaur species - now that mammals were beginning to eat dinosaurs, dinosaur populations began to decline.

Almost all of the plants of the Carboniferous Period wound up as coal because of low atmospheric oxygen levels (remember bacteria, algae and land plants had just begun to produce oxygen as a byproduct of photosynthesis) and they failed to decompose in the way that modern plants do in a twenty percent oxygen atmosphere.

Typically, when a plant dies today, oxygen penetrates it's metabolically dead cells allowing bacteria, fungi, amoeba and small insects to digest and break down the wood. By contrast, as the dense

mass of the plants of the Carboniferous Period died, they fell into oxygen-poor water or mud, or were covered by additional layers of dead plants or anaerobic sediment.

Sometimes this process of coal building was part of a great ebbing and flowing of the sea level along the coasts. During the Carboniferous Period, glaciers grew and alternatively shrank making the oceans rise and fall. As the glaciers melted, the rising oceans would flood the tropical forests with salt water and sediment. As the glaciers built up again, the sea levels would drop and drained the salt water that had been drowning and choking the forests.

Because significant levels of oxygen were not yet available to support the decomposition process in the flooded forests, the plants only partially decomposed, leaving an accumulation of black carbon that contained huge quantities of stored “solar income.” This spongy mass of carbon-rich “solar income” first became peat. After being squeezed and simmered by the tremendous pressure and heat of global geological forces, the peat ultimately hardened into coal.

In addition to the forest’s carbon that wound up trapped in the coal, the energy that had accumulated in the plant’s “solar savings account” over millions of years was also trapped.

The Taming of Fire

Fire is far older than man; it was set loose in the forests and tangles of brush by volcanic eruptions; lightening bolts; spontaneous combustion generated by the heat of decomposing plant material; and the friction of dry branches rubbing against each other at the pleasure of the wind. These early fires laid mineral rich ash onto the surface of the early “soil” which mixed a fine small particle sized mineral nutrient into the deep organic mulch that had built up over the years, decades, centuries and millennia.

Humans are one of the most efficient converters of “solar income” found in our food to mechanical energy. Humans need only about half the calories that a horse needs to exert the same level of physical energy. Our human metabolism is very efficient. Anthropologists believe that our large brains are the result of a constant stream of

high protein, high fat food supplies, "as we ate more meat our brain got larger and it allowed us to figure out how to get more meat"; our large brains allowed us to develop other methods to tap into the endless stream of "solar income" trapped by plants - fire!

By burning plants that we couldn't eat, such as trees and briar patches, humans and their cultures rapidly moved far beyond the creative limits of their digestive and metabolic systems. This revelation - fire, was the step that took humans farther than our earthly competitors - humans controlled fire, something that no other species would ever do! This new method of tapping into stored energy or "solar income" protected us from the harsh environment of the Ice Age and allowed us to control the temperature of our microenvironments. Humans were no longer flotsam on the sea! We had "fire power" in the broadest sense!

Early human remains from nine individuals were found in an Ethiopian fossil cache dated from 4.3 million years ago. The prehistoric humans, *Ardipithecus ramidus*, were found with a treasure trove of the oldest dated stone tools. Plant and animal fossil remains indicate that these early humans lived in low lying woodlands and grasslands teeming with monkeys, mole rats and wild cattle.

Anthropologists long believed that early humans "tamed" fire about 500,000 years ago, however, a recent analysis of burned antelope bones from caves in Swartkrans, South Africa, confirms that early man (*Australopithecus robustus* and *Homo erectus*) had built campfires and cooked meat roughly 1.5 million years ago - as many as a million years earlier than previously thought.

The cooked antelope bones were first described in 1988 by paleontologist, Bob Brain at the University of Cape Town. Brain suspected they had been burned by humans because the degree of charring indicated temperatures of more than 1,100 degrees F - heat levels that could be produced by campfires - heat produced by natural grass fires does not exceed 700 degrees F.

Anne Skinner, a chemist at the Williams College, Massachusetts, used Electron Spin Resonance, to examine the measurements of the free radicals left in the charred bones. Skinner said, "I'm very confident (the charred antelope bones) reached temperatures consistent

with a campfire.”

Fire, as a tool, separated man from other tool using animals - other animals metabolically used the sun's solar income, sticks, rocks, fiber, mud and water - only man had fire. Man tamed fire and used it for many thousands of years before he learned how to start and kindle it at will.

Many technologically ignorant peoples, including *Homo erectus* or *Java man* (before 250,000 years ago), found that starting a fire from cold fuel was a difficult and an exasperating task. In the beginning, when fire was “captured” it was fed and maintained for as long as possible, a primitive technique that is still practiced world wide yet today. Fire is easy to preserve or transport by the use of wet wood, lichens, dry fennel stalks, etc. In some cultures, hearth fires are maintained for generation after generation, oftentimes for the life of the house.

The first evidence we have for fire starting by humans is associated with the early Paleolithic Period of Europe (about 25,000 to 30,000 B.C.). Flints in the classic “strike-a-light” hand held shape and chunks of iron pyrite were discovered together in charcoal fire beds. However, iron pyrite is only found in very few places on earth and therefore, most primitive fire starters fall into the category of the fire drill (a bow used to spin an arrow shaft), fire saw and fire plow (pointed stick that is rapidly pushed back and forth in a groove created in a log or board). Humans eventually learned that they could start fire at will by generating sparks that were then showered into “tender” and with the simple generation of heat with some form of friction used to ignite tender.

With fire, humans were able to smelt metal, make tools, fire pottery, bake bricks, produce weapons, keep ourselves warm, eat food that was otherwise inedible (grain), keep predators at bay, destroy our enemies, develop specialties and unwittingly, we humans, with the ash byproduct that was generated by fire, produced the fertilizer and mineral supplement (“culinary” ashes) that would nurture and sustain ^{us} we humans for thousands of years to come!

Early humans who had not yet learned how to put their wood

ashes from their heating and cooking fires into primitive gardens lived a life that was short, hard, miserable, filthy, fearful, brutish and violent. They were often on the verge of starvation. They subsisted primarily in family units or family bands. They were driven by an internal, intrinsic and relentless salt and mineral hunger as were all living creatures. The hunger for salt and minerals didn't begin with the appearance of potato chips, they began with the appearance of the first human.

One isolated group of tiny people, dubbed the "Hobbit", *Homo floresiensis*, existed on the Indonesian island of Flores, 370 miles east of Java. They were the size of a *Homo sapiens* pygmy or midget only three foot tall with the body of a three year old, however, their brain was smaller than a grapefruit - a small human with a brain that was smaller than the brain of a chimpanzee.

There is some evidence that these tiny, tribal, and carnivorous humans accepted gourds filled with food that the modern human Floresians offered to them, until the 1500s, when Dutch traders arrived.

The "Hobbit" were clever enough to employ fire, make tools (points, blades, awls and tiny barbs) and hunt - cooking fire charred bones of the *Stegodon* (dwarf elephant), Komodo dragon (dinosaur), a wide variety of rodents and other animal species were collected from garbage dumps.

Early skeletons of the south Pacific "Hobbit" were dated as far back as 95,000 years ago, the more recent ones were dated at 13,000 - 18,000 years; however, the primitive "Hobbit" crafted stone tools dating back to 800,000 B.C.

The island of Flores was populated by hoards of an extinct dwarf elephant (*Stegodon*), monster rats and giant lizards (including the Komodo dragon), all of which were the primary food of the "Hobbit."

There is no evidence that the "Hobbit" learned to use the ashes from their heating and cooking fires to put in their food, salt or gardens - they disappeared leaving only a small fossil record. Only six skeletons have been found, suggesting the others were mineral poor and simply didn't survive the ravages of the millennia.

The small stature of the Hobbit demonstrates very clearly that

humans, like other life forms are not exempt from natural laws, in this case the "island rule." The island rule states that, "a combination of selective forces - geographic and temporal isolation, scarce resources and the absence of predators - will occasionally result in development of giant and dwarf species." The Hobbit simply followed the same historical course as the isolated desert Bushmen of Southern Africa.

Humans were territorial; they hunted and gathered in naturally productive and fruitful areas that had quantities small game, edible insects, their favorite plants and either clay or salt deposits for a source of minerals, a virtual Garden of Eden. There were endless fights between competing family bands and tribes over these flora and fauna rich Eden's.

Initially, the small family bands of humans were free of highly infectious diseases such as small pox, influenza, tuberculosis and the plague because they were living in small bands, isolated from travelers, strangers and new comers that could be carriers or hosts for disease causing viruses and bacteria.

They were, however, still susceptible to anthrax from the skins of contaminated animals, botulism from eating contaminated maggots that infested the decomposing carcasses of animals killed by large predators, and tetanus and gangrene acquired from soil, manure and feces contaminated puncture wounds.

Their offspring had a very low survival rate, and those that did survive were often deformed with congenital birth defects (Down's syndrome, heart defects, cleft palate, spina bifida, muscular dystrophy, clubbed hands and feet, etc.) caused by pervasive maternal nutritional deficiencies.

Injuries took their toll; fractures healed all askew or didn't heal at all, leaving the victim to remain deformed and dependant upon the good graces and surpluses of the rest of the band. If the disabilities were catastrophic, the victim was oftentimes eaten by his fellows. Revered relatives were often eaten when they died to perpetuate their spirit (mortuary cannibalism).

After 500,000 years ago, the human populations of western Eurasia and Africa began to diverge from each other and from east-

ern Asian populations in skeletal anatomy. Those that put wood, peat or kelp ashes from their cooking and heating fires into their food (remember the salt hunger - ashes are the minerals left over when the carbon of the fuel is burned away) or primitive gardens, were taller, their bones stronger, they were more fertile, delivered healthier babies and lived longer. Harder mineralized skeletons resulted in their bones surviving the ravages of rodents and scavengers, decomposition and the various destructive forces of the millennia - full skeletons of early man were now surviving into the 21st century.

Between 130,000 and 40,000 years ago the populations of Europe and western Asia produced many fossilized skeletons known as Neanderthals. They were the first humans known to bury their dead and care for their sick and infirm. Their stone tools, however, were still crude and lacked the standard shapes that would indicate specific functions.

African fossilized skeletons from 100,000 years ago had more modern skeletal features (humans continued to put the ash byproduct of fire into their food and gardens) than did their Neanderthal counterparts - they made the same crude stone tools. The bone evidence of their food prey indicated that their hunting skills were minimal and they only killed and ate small safely killed animals, including rats and rabbits. They had not yet reached the capability of killing pigs, buffalo and larger more dangerous species such as the elephant (mastodon, mammoth and the dwarf *Stegodon*). They could not yet catch fish - their living sites on the sea coasts and lake shores are devoid of fish hooks and fish bones.

Human developmental history (physical, emotional, spiritual and creative) appears to have found its footing and gained "traction" about 50,000 years ago - a time which Dr. Jared Diamond calls "the Great Leap Forward." The first real signals of the human leap include an increase in height and the appearance of standardized tools and ostrich shell jewelry.

Similar human developments appeared about 40,000 years ago in southwestern Europe, where abundant artifacts are associated with skeletons of Cro-Magnons. Soon after, the garbage dumps preserved at archaeological sites become more and more complex pointing to

biologically and behaviorally modern humans.

Cro-Magnon garbage heaps yielded stone tools, bone tools and bone fish hooks. Tools were produced in diverse and distinctive shapes so modern that their functions as needles, awls and engraving tools are straight forward and easily identified. Tools became more complex, moving beyond simple hand-held scrapers to multi-piece tools (stone axes) and weapons such as harpoons, spear throwers and bows and arrows.

These new weapons allowed the human to kill larger and more dangerous prey from a safe distance. The creation of ropes and threads for nets, twine and string permitted the invention of nets, lines and snares which allowed for the addition of small mammals, fish and birds to our human diet which promoted the expansion of our habitable territories.

Rubble from houses and scraps from sewn clothing point to an improved ability to survive frigid weather. Modern jewelry and tenderly buried skeletons show a new aesthetic and spiritual progression for humans - humans were now talking to God.

Of all of the Cro-Magnon's creations that have been preserved, the most remarkable is their artwork, including the famous cave paintings, statues and musical instruments. The impact of the life sized painted bulls and horses in the Lascaux Cave of southwestern France or the rhino, lion, horses, mammoth, panther, owl, bear and buffalo paintings of the limestone Chauvet grotto in southern France on a visitor is dramatic - you realize these ancient people were modern not only in their skeletons but also in their abstract minds!

The limestone caves, located in the Lascaux woods near Montignac, France, are in fact ancient museums of natural history: they include a main cavern sixty six feet wide and sixteen feet high, containing many steep galleries, magnificently decorated with engraved and painted figures. Archeologists date the site back to the last Ice Age, between 15,000 and 13,000 B.C. In total there are 600 painted animals and prehistoric symbols, and 1,500 engravings.

The Ice Age Chauvet grotto is known as the "Sistine Chapel" of prehistoric art. The grotto is found deep within the limestone hills of Auvergne, west of the Rhone Alps. The Chauvet grotto contains 447

paintings and 4,000 well preserved foot prints of men, children and dogs.

The last Chauvet artist produced a painting between 24,000 to 32,000 years ago. The artist's work was "signed" with a negative print of his hand, which was produced by blowing red ochre onto a non-arthritic hand that was held palm down against the cave wall. The artists also signed their work with a positive print of their hand by putting the ochre on their hand and then placing their print on their painting!

Art historian, Carole Fritz recorded, "I even cried the first time I saw it (Chauvet). It's just so intense, like sometimes when you are listening to (an) opera, it just moves something in your heart. And then you see these finger marks on the walls where they spread the pigment and dried their hands and you really have the feeling they just left the cave."

The Ice Age artists used mainly shades of red, yellow, black and brown, using charcoal and rare earth mineral pigments. The most dramatic of the paintings depict four sixteen foot long aurochs (large wild cattle); other animals include red deer, horses and stags, some of which appear to be crossing a river.

No human bones have been found in either of the caves, however, the Chauvet grotto has been a treasure trove of bones from fourteen different animal species including rodents, reptiles, birds and bears. The bear bones dominate - they have recovered 3,700 of them.

It is important to note that there were no grains or fruit seeds found in the caves, clearly indicating that the Cro-Magnon artists were primarily hunters and carnivores and that hunting dominated their lives.

At 40,000 years ago Cro-Magnons "moved into the neighborhood" of Europe with their modern skeletons (signifying the use of the ash byproduct of cooking and heating fires as a supplement mixed with salt to extend and cut the more valued salt and in their food as "culinary" ash), tools and more clever weapons.

Classical anthropologists believe that the Cro-Magnons, used their advanced technology, superior weapons to kill, infect or displace the Neanderthals - a theorized event that left little or nothing behind in

the form of evidence of hybridization between the Neanderthals and Cro-Magnons or a straight forward and ruthless genocide.

We don't believe the Cro-Magnons fought, captured, enslaved or slaughtered the Neanderthals, because they would have eaten them and left their charred, cracked and scraped bones in their garbage heaps with the rest of the debris of hunting. They would have created art work that depicted the great heroic battles, heroes, the slain enemy and victories.

We believe that the Neanderthals just failed to learn a basic survival technique, and allowed themselves to be "edged out" in a similar fashion as the prehistoric Bushmen allowed themselves to be "edged out" of the isolated and idyllic Cape of South Africa and into the Kalahari Desert, by the more advanced gun toting Dutch explorers from the Dutch East Indies Tea Company in 1462; and in a similar fashion as the island of Flores "Hobbit" allowed themselves to be edged out by the 16th century Dutch.

Neanderthals were primarily hunters and they developed the nutritional deficiencies of pure carnivores who had failed to develop the habit of using wood, peat or kelp ashes to mix with salt and season food and/or failed to put the ashes into their primitive or crude "gardens." One bad hunting year or a draught of several years could have wiped them out.

The number of humans surviving to "old age," (at least double the age of reproductive maturity) quadrupled at about 30,000 years ago - something equally as dramatic as the taming of fire happened here which we will talk about in due course.

Anthropologists looked at 750 fossilized dental and jaw fragments of early humans spanning over millions of years, from the Australopithecines to Neanderthals to early modern humans. Old age was defined as at least double the age which reproductive maturity was reached, which is also the point when the third molars usually erupt in the teen age years.

In calculating the ratio of old-to-young hominids, Rachel Caspari of the University of Michigan, at Ann Arbor and Sang-Hee Lee of the University of California, at Riverside found a trend of increased longevity across the human family. With "modern" humans, older adults

finally outnumbered younger ones for the first time at the 30,000 year mark.

The leap in longevity is thought to have been critical to the development of modern human culture, as elders passed down knowledge and helped to weld together complex societies. This “hand - me - down” knowledge, included, how to cook and preserve meat, how to use wood ashes to extend and cut what they perceived as the more valuable salt and to employ “culinary” ashes in the preparation of certain foods and the value of putting wood ash into their primitive gardens.

The results and benefits of incorporating wood ashes into their food either directly as salt extenders or “culinary” ashes or indirectly as a garden fertilizer produced stronger, taller, healthier, more fertile, more creative and longer-lived humans - we had not yet left the main road and humans were in fact accumulating good habits, becoming healthier, more fertile, living longer and honing better survival skills.



Chapter 2

“The aim of science is to seek the simplest explanation of complex facts. We are apt to fall into the error of thinking that the facts are simple because simplicity is the goal of our quest. The guiding motto in the life of every natural philosopher should be, seek simplicity and distrust it.”

- Alfred North Whitehead

Ash Power

It is clear and dramatic - something universally uplifting happened around 100,000 to 50,000 years ago. Dr. Jared Diamond, primary proponent of the “Great Leap” argues for an increase in the size and “perfection” of the human voice box, and thus he believes improved vocal and language skills, was the single event that vaulted humans forward in time, while others argued for a universal change in function and size of the human brain.

Our belief is that people learned, between 100,000 and 50,000 years ago, the value of the ash byproduct of their cooking and heating fires in a world wide parallel type of discovery similar to their discovery that they could tame fire. They learned the value of mixing wood, peat and kelp ashes with salt to extend and dilute the more valuable NaCl, employ them as “culinary” ashes and putting their wood heating and cooking byproduct (ashes) into a primitive type of garden for fertilizer.

After all, wood ashes were everywhere as a daily byproduct of cooking, heating and the enterprise of survival - it was natural to try and use ashes in many ways. Certainly with a salt hunger driving

them on a daily basis, humans would have put the ash in their mouth, just as a modern child puts everything in their mouth to see if they were tasty or usable.

The universal kitchen byproduct, ashes, were the minerals that were left when carbon fuels were burned. They were historically thrown out into the garbage piles, tasted and added to food as a spice to satisfy a craving as we salt our food today, to satisfy a salt hunger—a universal craving shared by all living humans and animals.

The ashes, initially thrown into the garbage piles, would have resulted in adjacent plants growing more verdant and being more productive - this seemingly simple observation led to the wood ash being routinely placed in the soil around favored food plants, resulting in greater production in terms of tons and bushels per acre and even more important, the fertilized gardens remained productive for longer periods of time. Modern agriculture was now in its infancy and the Great Leap Forward for mankind had begun in earnest!

While some human cultures flourished in the fertile river valleys and flood plains that were rich with mineral laden flood silt (Egypt, China, India, Europe) and mineral rich glacial waters (Hunza, Vilcabamba, Titicaca, etc.), the vast majority of early humans lived in harsh barren or heavily forested mountains, dry semi arid plains and seemingly sterile ocean and lake shores.

The majority of early humans obtained their minerals by eating the organs of their prey animals they killed, or directly or indirectly from wood, kelp, peat and manure (buffalo “chips”) ash that they directly consumed mixed with or without salt or indirectly by placing the ashes in their gardens.

There are primitive peoples yet today who add wood ashes, peat or kelp ashes directly into their food to satisfy their salt hunger and add the ashes to their gardens to extend the life of the garden and make it more productive - thus indirectly, yet regularly adding a constant stream of supplemental minerals into their food cycle.

Early man transitioning from hunter gatherer to farmer, “managed land” in a primitive, inefficient and heavy handed way by burning the mature forests and tangled brambles to allow the growth of a small number of “volunteer” edible shoots, tubers and seed plants

that always follow after the devastation of a wildfire. This process of humans transitioning from hunter gatherer to farmers took thousands of years and is still going on today in parts of the 21st century world - the process is expensive in terms of return and at best a gamble on the predictable results.

Investigators have long believed that the first organized gardens that we might describe as small "farms" appeared about 11,000 years ago - and then village life and civilization soon followed. Before "farming" was developed, humans were loose nomadic collections of family bands.

In June, 2004, archaeologists announced that they had discovered the remains of hoards of wild grains, grinding equipment, and a stone "oven" at Ohalo II, a 23,000 year old site on Israel's Sea of Galilee. Ehud Weiss, a paleoethnobotanist at Bar-Ilan University, Ramat Gan, Israel says, "his team found traces of 143 kinds of seeds including, ancestors of wheat and barley."

Weiss feels, "these people gathered volunteer and wild grains in the late summer and early fall, ground them into flour and baked them (in the oven)." "These findings are a breakthrough in (the understanding of) human diet and (food) technology - grinding and baking (grains) would have dramatically improved (the efficiency and usefulness of grains in) human nutrition," said Weiss.

Agriculture, as a science and commercial industry was born in the Fertile Crescent of Iraq 10,000 - 11,000 years ago with the early and clear domestication of eight "founder" crops - they were the start of large scale commercial agricultural systems not only in the local region, but also globally. Those eight founder crops were emmer wheat, einkorn wheat and barley; the lentil, pea and chickpea legumes; bitter vetch; and the fiber and oil crop - flax.

Time budget studies showed that early "farmers" spent more time and labor hours per day than hunter gatherers did, however, their food supply was eventually more dependable. Archaeologists have clearly shown that the first farmers were smaller and more poorly nourished, suffered from more diseases and died at a younger age than did the hunter gatherers - grain farmers didn't have access to the organ meat (liver, kidney, brains and hearts) that provided a de-

pendable dietary source of vitamins (vitamin B12, Vitamin C, vitamin A, vitamin D, etc.), essential fatty acids (omega 3) and minerals (selenium, copper, zinc) to the hunters every time they killed an animal.

Many early farmers simply didn't know the nutritional secret and value of their cooking and heating ash byproduct or maybe there were just fewer of the essential minerals that are required by humans in the soil from which their wood grew.

Early on, the simple availability of more calories and protein meant more people. Among the 250,000 species of plants and animals on earth only a relatively small number, several thousand perhaps, are edible for humans or are worth the effort to hunt, grow or gather.

A few hundred (about 800) have been "domesticated" and adapted for human use. Even among the several hundred food crops, most provide little food in the grand scheme of human diets and wouldn't themselves account for the calories required to support the rise in human numbers. Just a few dozen varieties of plants account for 80 percent of the modern agricultural production in the world today.

These dozen economic and dietary agricultural giants are cereal grains that include wheat, corn, rice, barley and sorghum; soybean legumes; the roots and tubers potato, manioc and sweet potato; sugar cane and sugar beets; and the banana. The four earliest domesticated fruits were the olive, grape, fig and date - these four were apparently the easiest to domesticate of all of the wild fruits.

Ancient Egyptians are usually credited with the domestication of the cat, however, in 2004 French archaeologists reported they had unearthed a large Neolithic village in Cyprus that contain the remains of a domesticated cat. The investigators reported the discovery of an intact skeleton of a cat that had been carefully buried beside the remains of a carefully buried human, along with a variety of tools and ceremonial objects.

The feline remains were dated back to 9,500 years ago - 5,000 years before the Egyptians began sculpting cats, or "miu" as the Egyptians called them. The two sets of remains were found less than 12 inches apart, positioned symmetrically with their heads facing west.

The investigators believe that the cats were used in Cyprus to

keep rodents out of their grain stores. The cat *Felis silvestris lybica*, somewhat larger than today's domestic cat, was only eight months old, was thought to be sacrificed to accompany its owner on the journey to the next world.

In September 2004, Silvia Gonzalez, a geoarchaeologist, reported on the radiocarbon-dating of a skull (Penon woman III) that had been found near Mexico City in 1959 - the analysis showed that the skull was 12,755 years old, considerably older than any previously known Native American.

Gonzalez believes that Penon Woman III was an ancestor of the Pericu, peripatetic hunter-gatherers, whose remains were dated back to 3,000 years ago and found throughout Baja California. Gonzalez suggests, "accounts left by 18th century Spanish missionaries of slim, shell-fish eating people in the Baja refer to the Pericu" - she also believes that, "the Spanish pushed the Pericu to extinction by forcing them into mission labor and simple corn diets."

In 4,000 B.C. pottery objects with surface glazes appeared and horses were domesticated in Eurasia.

In 2,500 B.C. the first free-standing glass objects of Egypt and Mesopotamia appeared.

In 1,500 B.C. the first glass vessels appeared.

The "founder" crops in the eastern zone of North America were limited to four plant species domesticated between 2500 and 1500 B.C. - a whopping 6,000 years after wheat and barley were domesticated in the Fertile Crescent. A species of North American squash was used as a container as well as a source of edible seeds; the balance of the three "founder" plants were produced only for their seeds - sunflowers, a small daisy called "sumpweed" and goosefoot (a leafy relative of spinach).

For 2,000 years these original North American founder plants were only available as less important foods in the total diet of Native Americans, while the indigenous Native Americans maintained their traditional hunter gatherer traditions and continued to rely on plentiful wild mammals, waterfowl, wild turkey, fish, shell fish and nuts and berries.

Farming did not contribute to any major part of the Native Ameri-

can diet until 500 to 200 B.C., when three additional seed crops appeared - knotweed, Maygrass and little barley.

All of these native North American founder crops were high in protein (17 - 32 %), compared with 8 - 14% in modern wheat, 9 % for corn and even lower percentages for barley and white rice. The sunflower and the sumpweed were also high in oil (45 - 47%). Sumpweed produced 32 percent protein and 45 percent oil. With the exception of the sunflower all the founder plants had serious downfalls including small yields at one tenth that of wheat, strong odors and one, the sumpweed produced hay fever-like allergies.

In contrast to the pre-founder crops, cereal crops alone account for one half of the calories consumed by the world's human population today, because they are technologically easy to grow and they are economically favorable to produce, convert into finished foods (bread, pasta, dry cereals, etc.) and store, they have become the "go to food."

Many potential foods were not useful for humans as they transitioned from hunter gather to farmer because they were indigestible such as twig bark that a deer might eat; poisonous such as certain roots, leaves and mushrooms; too much work was required to gather and prepare, such as seasonal and or small hard-shelled nuts and seeds; or too dangerous such as large game that were armed with horns or teeth or protected by aggressive herds.

Catching and domesticating wild herbivores by contrast was quick and easy compared with domesticating wild plants, which required clearing the land, sowing seed, irrigation, fertilization, weeding, harvesting, processing and storing.

To catch wild animals, a simple pen or fence only needed to be placed around water, a salt lick or ash dump, when the animals came into the pen to drink or lick the salt or ash minerals, a gate was pulled closed and the herd was started. (See Table)

The main reason why Eurasia was the place where most of the "founder" animal species were domesticated was that it was the continent that had the most potential species for domestication to begin with and lost the fewest to extinction over the last 40,000 years.

Of the world's 148 wild herbivorous animal species that were

Mammalian Candidates* for Domestication

	Eurasia	Sub-Saharan Africa	The Americas	Australia
Candidates	72	51	24	1
Domesticated	13	0	1	0
% domesticated	18	0	4	0

* A species of land dwelling wild herbivore or omnivorous mammal weighing more than 100 pounds.

Dates of First Evidence for Domestication of Large Mammals

Species	Date (B.C.)	Place
Dog	10,000	Southwest Asia, China, North America
Sheep	8,000	Southwest Asia
Goat	8,000	Southwest Asia
Pig	8,000	China, Southwest Asia
Cow	6,000	Southwest Asia, India, North Africa
Horse	4,000	Ukraine
Donkey	4,000	Egypt
Water buffalo	4,000	China
Llama/alpaca	3,500	Andes
Bactrian camel	2,500	Central Asia
Arabian camel	2,500	Arabia

candidates for domestication, only 14 made it to the production line. The other 134 species failed (they were doomed to “perpetual wildness” and to the risks of extinction) because they lacked one or more of the six characteristics necessary for successful domestication:

1. Diet: Every time an animal consumes a plant or another animal, the conversion of the food biomass into the eater’s biomass produces an efficiency of much less than 100 percent - usually about ten percent. It takes 10,000 pounds of corn to grow a 1,000 pound steer. If one wants to grow 1,000 pounds of carnivore, you have to feed them 10,000 pounds of herbivore fed on 100,000 pounds of corn.

Many species are too feed specific or too inefficient to lend themselves to domestication.

The exception is the dog (an omnivore), which was originally domesticated and raised as a guard animal and hunting partner, however, some breeds of dogs were domesticated and raised specifically for food in Aztec Mexico, Polynesia and ancient China (dog is still a popular “winter” dish in China).

2. Growth rate: Domesticated species must grow and mature quickly, which eliminates primates (baboons, chimps and gorillas) and large herbivorous species such as elephant, rhino and hippo - it is more efficient to catch and train wild individuals of large species than it is to raise them.

Rodents and lagomorphs (rabbits and hares), small herbivores (goats, sheep and cattle) and pigs grew and matured quickly.

3. Problems of captive breeding: many animals are secretive and will not readily reproduce in confined “factory” like environments.

4. Nasty Disposition: The grizzly bear, onager (wild Asian ass), zebra and African Cape buffalo are examples of potentially good candidates for domestication and sources of human food, however, their unchangeable and unmanageable dispositions relegates them to the “perpetual wildness” category.

5. Tendency to panic: Large herbivorous species react to danger from predators and humans in different ways. Some species are “flighty” and programmed for fast, instant escape by blind flight (deer and antelope) if they perceive a danger real or imagined. They will run into a fence or wall and literally kill themselves trying to escape in a walled enclosure.

Other species (goats, sheep or cattle) are calmer and tend to bunch up in the herd or flock when they feel threatened.

6. Social structure: Almost all species of large domesticated herbivores that have been domesticated from wild species have three social characteristics in common: 1) they live in herds or flocks; 2) they exhibit a highly developed dominance hierarchy; 3) herds would share overlapping and “movable” home ranges.

By growing and selecting and nurturing the species of plants and animals that humans could cultivate, domesticate and eat, we increased the density of our dependable food supply from 0.1 percent of an acre to more than 90 percent of an acre. An acre of land could then support significantly more people. Wandering and the need to follow and find herds, flocks and plants in productive and ripe states became unnecessary.

One acre could now feed 10 or 100 times more people than in a hunter gatherer society. The power and productive energy of the greater density of useable biomass and the sheer numbers of farmers and shepherds gave them a political strength, business and trade power and a military advantage over the less numerous and more environmentally vulnerable hunter gatherers.

In human populations, the possession of livestock instantly added to their health, fertility and longevity. The animals provided a steady stream of meat, milk, skins, fuel, fertilizer and power to carry loads, pull a plow and wagons. Humans no longer had to follow the migrating herds or wait for the expected annual seasonal gluts of meat, fish and birds as the herds, flocks, bands and schools returned to breeding or feeding grounds - humans could now control and depend on a constant year round supply of food - now we had what

seemed to be an unending food supply and fire.

Some animals such as cattle, sheep, goats, horse, reindeer, water buffalo, yak and camels produced significant quantities of milk allowing for the production of liquid milk, butter, cheese and yogurt - producing considerably more calories and protein over their lifetime, compared with simply being chased down, killed and eaten. The animal's organs, including the heart, livers, kidneys and brains were rich in stored minerals such as selenium, zinc and copper and vitamins including vitamin B12, vitamin C, vitamin D and vitamin A, the bones were rich in calcium and phosphorus and the brains were rich in essential fatty acids (Omega-3) extracted from plants and inorganic mineral sources that were not easily or metabolically accessible to humans.

Animals could eat and create usable food from grass and brush that were not directly usable by humans. Crop yields were increased by adding manure and manure/ wood ash mixes and composts to their gardens and fields. Even in the 21st century, despite the widespread availability of chemical fertilizer in industrialized nations, animal manure and wood ash mixes still make up the bulk of fertilizers for the gardens and fields of the third world's peasant and rural subsistence populations.

Dried manure was historically used as fuel for cooking and heating and is yet still used today (2005) as fuel for cooking and heating fires and the mineral rich ash then added directly or as part of a compost to the gardens and fields.

Animals, as a source of draft power were the natural result of domestication. They carried packs, loads and humans on their backs over long distances, up hills, over rough terrain and they pulled plows, carts and wagons.

At this point in time, human cultures were limited in size and complexity by the production limits of the hunting and gathering economies. The entire human hunter gatherer population of Europe is estimated to have been less than 100,000 people prior to the appearance of organized agricultural food production.

The upper limits of the numbers of individual humans in the earth's total population, was simply controlled by the availability and

quantity of food. Several good years in a row in terms of increases in animal populations and successful kills resulted in more babies surviving and greater numbers of adults living to older ages; and small migrations, poor kill efficiency, death of a primary hunter, famine and hostile weather would effect a cruel devastation on the human population.

The size of the individual hunter-gatherer bands was set by the amount of the available food. The group size rarely exceeded 200 individuals, while most bands were composed of only four or five families. Camps were moved quite frequently and as a result few personal belongings of any size or weight were accumulated.

At 11,000 to 15,000 B.C. village life began. Agricultural success beginning 10,000 years ago led to larger and more densely populated villages, towns and cities - perfect killing fields for measles, rubella, mumps, pertussis, smallpox, plague, influenza and tuberculosis.

Southwest Asia (Iraq) is credited with the earliest domestication of plants at 8,500 B.C. and animals about 8,000 B.C.; the eastern zones of the North American continent first domesticated plants and animals 6,000 years later.

The earliest known cultivated emmer wheat came from the "Fertile Crescent" of Iraq about 8,500 B.C. and only moved west to Greece at 6,500 B.C.; the first domesticated chick peas came from Turkey and northern Syria about 8,000 B.C. and did not appear in India until 5,000 years later.

The smelting of metal from ore started in the Near East about 5,000 B.C. Evidence points to copper as the first metal to be extracted from ore - copper carbonate, malachite and azurite were historically and commonly used as pigments, as a result, humans were quite familiar with various copper ores and other minerals.

In 4,500 B.C. the Chinese were the first to discover and record the medicinal powers of plants. The Egyptians must be recognized for fully developing the physical and chemical properties of aromatic essences or plant oils. The natural antiseptic and antibacterial properties of essential oils extracted from plants, and in particular cedar wood, myrrh and frankincense, made them ideal for preserving a

body for “it’s journey into the next world.”

Early drawings of Egyptian nobles and slaves depicted them as slim and fit - being overweight or obese was certainly not a common human problem during the time of the pharaohs.

About 3,400 B.C. the first wheel appeared near the Black Sea; it turns up over much of Europe and Asia over the next few centuries. All of these Old World wheels are of the same simple design - a solid wooden disc constructed from three planks fastened together by two or more cross ties.

As civilizations grew and prospered, and nations fought amongst them selves for political power, trees disappeared, they were simply gobbled up by hoards of men demanding fuel, timber for building and burning and clearing forested land for pasture and growing crops. The combination of these ever growing human populations and relentless demands for wood depleted the same forest stores of plant “solar income,” resulting in an energy crisis.

The numbers and sizes of our fires, our meals, our towns, and our economies and ultimately human numbers, were all controlled by the limited ability of the woody plants available to us to keep up with demand and convert the sun’s energy into a type of energy that we could use as food, fuel and building materials.

Many early craftsmen turned to a “hybrid” fuel that would be a significant bridge between wood and coal - charcoal. Charcoal is wood that has already been partially burned. For thousands of years charcoal has been made by heaping wood into piles that were set on fire and then buried causing a slow oxygen depleted smolder that produced almost pure carbon. The charcoal burned cleaner and hotter than wood, however, the process of making charcoal was energy wasteful as it takes four pounds of wood to make one pound of charcoal.

In these early energy crises, coal presented some lucky cultures with the stored energy of millions of years of solar income. Solar income, that had been deposited in a “solar savings account” of a size that no one at the time could fathom. Coal would give these early energy seekers the power to change almost every relationship that humans had with nature, the earth and the sun itself!

At 3,300 B.C., a thirty five year old Neolithic man ("Otzi") was murdered by being shot in the back with an arrow. His body was later found in 1991 in a melting glacier in the Tirolean Otztal Alps on the Italian-Austrian border. He had well trimmed hair and was clothed in a grass cape, the skins of ibex and deer, and leather shoes stuffed with grass for insulation.

Otzi carried a bow made from the wood of the yew tree, a quiver filled with arrows that were tipped with flint points, and a copper axe.

Radiographs of Otzi's well preserved body showed significant levels of degenerative arthritis of the neck, lumbar region of the lower back, and one hip; Otzi had calcifications of his arteries including carotid arteries, coronary arteries, aorta and iliac arteries - diseases thought to be the results of the dietary sins of modern western man.

Otzi had the cardiovascular disease (arteriosclerosis) and the degenerative bone disease (osteoporosis and osteoarthritis) of modern humans, however, he was "fit" in terms of the BMI (Body Mass Index) which measures body fat versus lean mass.

"Cardiovascular disease" and osteoporosis and osteoarthritis are all multiple vitamin, antioxidant and mineral deficiency disease complexes, that indicate that the Paleolithic diet was not in of itself perfect. Whatever Otzi was eating, his diet was chemically less than perfect - it certainly was deficient in vitamin C, calcium, magnesium and antioxidants - however, Otzi wasn't obese.

On December 2001, a Peruvian archaeological site was uncovered that was documented through carbon dating. to be the oldest city in the Americas, at 3,100 B.C. Construction of Caral, one of twenty cities in the coastal Peruvian zone of Norte Chico, started nearly 400 years before the first pyramid was built in Egypt and when most people in the world were still hunters and gatherers. The discovery was reported in the December 23, 2004 issue of the journal *Nature*.

The "Andean" society remained basically the same for 1,200 years before being overrun by their more warlike neighbors - "that is the longest that any civilization is known to have survived," said Jonathan Hass of Chicago's Field Museum, who led the expedition.

“We are seeing the emergence of centralized decision making, government and religion out of pristine conditions,” Hass said. “They were not following a pattern established by someone else. They were developing it on their own. An Andean culture was being invented in this area.”

It had long been thought that the Americas were behind Europe, Africa and Asia in developing civilizations. The new date shows that the Americas, separated from the other centers of civilization by then, impassable oceans, developed in a parallel way to Europe and Asia. The Andean society was built mainly on cotton farming and trade with fishing villages along the coast of what is now modern day Peru. “Here is a unique system,” Hass said. “There wasn’t anything like this in the world.”

The radiocarbon dating showed that the Supe Valley and the nearby Pativilca and Fortaleza valleys, construction began there even earlier - about 3,000 B. C. and spread over 700 square miles.

The attraction for the Andean society for this location was the cold Humboldt Current that runs north from Antarctica along the Peruvian coast. The current is filled with marine life, which served as an “endless source” of high protein food. They also “figured out that if you take water out of the rivers and put it on desert land, the desert blooms and becomes very productive,” said Hass.

The Andeans dug canals by hand to bring water to their fields and gardens. They grew guava, bean, peppers and fruit trees; however, their most important crop was cotton, which was traded to the coastal villagers, who used the cotton to make fishing nets - “you can’t catch anchovies without nets.” Anchovies were the Andeans primary protein supply.

The Andean diet dominated by small fish, would provide high quality protein, minerals (i.e. - calcium, zinc, selenium, magnesium, iodine, etc.), vitamins (A, D, E, etc.) and the essential fatty acids.

There is evidence that the Andeans had communal hearths and had feasts that would coincide with successful harvests and fishing seasons. The Andeans developed an idyllic society where crime rarely occurred and few tried to attack them. Eventually, however, warfare was brought to them and these peaceful people were defeated and

Norte Chico was abandoned - only scattered subsistence farmers reside there today.



Chapter 3

"Nothing is too wonderful to be true."

- Michael Faraday

"Beware of doctors, as they would bring death by medicine!"

*- Cato, the Roman
(234 - 149 B.C.)*

Codes, Oaths and Medicine

By 3,000 B.C., the warm and fertile zones in the Middle East, irrigated by the Tigris and Euphrates Rivers (calculated by some to be the location of the Garden of Eden) had generated many of humanity's first steps into civilization - Ur on the banks of the Euphrates (Sumerians), 100 miles up river from the Persian Gulf; Babylon, farther up river; Assyria, centered on Assur; and later Nineveh on the Tigris, near what is now Mosul in 21st century Iraq.

All of the Mesopotamian kingdoms ("lands between the rivers") have produced extraordinary archeological artifacts and written legacies, which have allowed us to understand their civilizations, health, infrastructures and governments.

The Mesopotamian legacy included 30,000 clay tablets, rich with cuneiform writings; approximately 1,000 of the tablets found in the Library of Assurbanipal (600 B.C.) were devoted to medical diagnosis, therapy and prognosis.

The main works, "The Treatise of Medical Diagnosis and Prognosis," are made up of 3,000 reports on forty tablets. It is comprised of a list of diseases:

1) "The patient coughs continuously. What he coughs up is thick and frequently bloody. His breathing sounds like a flute. His hand is cold, his feet are warm. He sweats easily, and his heart activity is disturbed." -Tuberculosis.

2) "Stinking disease and distended bellies." - nutrient deficiency diseases (beriberi, pellagra, hypoproteinemia), symptomatic of the birth of grain based agricultural societies.

What was clearly missing in all of these early health and disease descriptions was obesity.

The Xia Dynasty (2205 - 1818 B.C.) fell after the Yie and Lo Rivers dried up; the Shang Dynasty fell after the Yellow River went dry. Primitive "slash and burn" agricultural techniques were thought to be the reason. After the fall of the Shang and Zhou (1122 -255 B.C.) Dynasties, farming methods changed - wood ashes from heat-ing and cooking fires were added to subsistence gardens for fertil-izer. Another garden enrichment technique devised by the Chinese was to put plant debris and hand pulled weeds into holes and then burned to produce ash - the vegetables where then planted on top of the ash.

The sixth king of the first dynasty of Babylon, Hammurabi (1728 - 1686 B.C.) was a powerful king who brought Babylon to a feared and respected position amongst nations. Hammurabi's greatest legacy included mathematical works, dictionaries, astrology and mystical writings, however, his finest works were gathered together in "The Code of Hammurabi."

The Code included 282 laws that deal with the control and regu-lation of society in general, family life, trades and professions - in effect a legal code. The laws included medical instructions for physi-cians. It's guidelines include fee schedules for therapy, with a sliding scale based on the patient's social or royal rank.

There were ruthless fines and punishments "for incompetence or failure." "If a physician has performed a major operation on a lord with a bronze lancet and has saved the lord's life....he shall receive ten shekels of silver (a greater amount than the annul pay of a trades-

man); but if he caused the death of such a notable, his hand would be chopped off. A doctor causing the death of a slave would have to replace him."

The earliest Egyptian medical writings were produced in 2,000 B.C.

In 1,600 B.C., a "Book of Wounds" was produced near Luxor, Egypt. The work listed a head to toe inventory of forty eight case reports. The surgical cases treated included wounds, fractures, abscesses, circumcision, etc. Broken bones were set with ox-bone splints that were held in place with resin soaked bandages (casts). The papyrus lists large numbers of "dressings, adhesive plasters, braces, plugs, cleansers and cauteries."

The Ebers papyrus (1,550 B.C.), found in Thebes is over sixty feet long, it is the oldest surviving medical "book." The Ebers papyrus contained information on fifteen diseases of the abdomen, twenty nine of the eyes, eighteen of the skin, twenty cough treatments. Seven hundred drugs and 800 formulae are discussed (herbs, minerals and animal parts and extracts).

Notably missing in all of these Mesopotamian health writings are descriptions and treatments of obesity.

Egyptian medicine credited vegetables and fruit with great healing powers; they used tree resins (myrrh, frankincense and manna). Egyptians, like Mesopotamians, used plant extracts such as senna, colocynth and castor oil. Medicinal recipes included ox spleen, hog brains, tortoise bile sweetened with honey, and animal fat. Antimony, copper and other minerals were used as astringents and disinfectants.

"Multiple" ingredient pills, ointments, poultices, fumigations, inhalations, gargles and suppositories contained ingredients ranging from leeks, lapis lazuli, garlic, onion, tamarisk, cereals, spices, condiments, resins, gums, dates, hellebore, opium to cannabis were on the list of common "medications."

By 1,000 B.C. the populations together referred to as "Greeks," were flourishing on the shores of the Aegean Sea, in Ionia (the western shores of Turkey), the Greek mainland (the Peloponnese) and the related islands.

The accumulations of early Greek medical information were recorded in the Homeric epics, dating from before 600 B.C., but also certainly offering older data. Greek scholars have documented 147 battle wounds in the *Iliad* (106 spear thrust, 17 sword slashes, 12 arrow hits, and 12 sling shot wounds).

One of the most famous survivors of arrow wounds was King Menelaus of Sparta, whose physician “pulled the arrow from the king’s flesh, sucked out the blood” and applied a topical salve.

According to Greek mythology, Asclepius was a heroic warrior and “blameless” physician, he was the son of the god Apollo and a mortal woman. Homer described Asclepius as a tribal chief and a highly regarded wound healer; his sons became physicians who were called Asclepiads. As the “father-god” of medicine, Asclepius, was usually described as a bearded sage with a staff and snake (the origin of the caduceus sign of the modern physician, with its two snakes intertwined, double-helix like, on a winged staff; the shedding of the snake’s skin symbolized the renewal of life). Asclepius was frequently pictured with his daughters Hygeia (health or hygiene) and Panacea (cure-all).

The Greeks are legendary for employing religious healing methods including exorcisms, diviners, shamans and priests.

By contrast, Hippocratic medicine, the foundation of Greek written medicine, quite clearly bases the medical healing art on natural philosophy rather than the supernatural. The Hippocratic writings never attributed disease or cures to the gods.

Most of what we know about Hippocrates (460 to 377 B.C.) is a collection of myth and legend. The earliest of information claims that he was born on the island of Cos and that, “he lived a long and virtuous life.” The sixty writings that make up Hippocrates’ Corpus were in reality derived from many scholars (like the Bible), they were dispersed, mixed up, fragmented and then put together again like a puzzle.

Hippocrates’ Corpus, as we know it, was put together about 250 B.C. in the library at Alexandria; some texts were added at a later date - medical historians of modern times still argue over which parts of the Corpus are genuine and which have been composed and added

by students or admirers.

In May, 2004, a team of Egyptian and Polish archaeologists, unearthed the ruins of the University of Alexandria - the alma mater of Archimedes, that was in it's peak of glory 2,300 years ago when Alexandria was the center of the scholarly world.

It was at the University of Alexandria that Archimedes invented a water pump design that is still used today; Euclid organized and developed the rules of geometry; Hypsicles divided the zodiac into 360 equal arcs or degrees; and Eratosthenes calculated the diameter of the earth. Other scholars edited the works of Homer and produced the Septuagint, the ancient Greek translation of the Old Testament.

Hippocrates observed that man is "governed by the same physical laws as the cosmos, hence medicine must be an understanding, empirical and rational, of the workings of the body in it's natural environment."

The profound concept in the Hippocrates Corpus was that "health was equilibrium and illness an upset."

Followers of Hippocrates specialized in bedside medicine, and even at this early date blamed their treatment's failures on the patient's failure to follow directions:

"Make frequent visits; be especially careful in your examinations, counteracting the things wherein you have been deceived at the changes. Thus you will know the case more easily, and at the same time you will also be more at your ease. For instability is characteristic of the humours (body fluids including blood, bile, urine, pus and mucus), and so they may also be easily altered by nature and by chance.....Keep a watch also on the faults of the patients, which often make them lie about the taking of things prescribed. For through not taking disagreeable drinks, purgative or other, they sometimes die. What they have done never results in a confession, but the blame is thrown upon the physician."

The therapeutic methods of the followers of Hippocrates was "expectant" - they waited and watched their patients, talking, winning trust and giving a helping hand to the "healing power of nature" (vis

medicatrix naturae) highlighted in Hippocratic text *On Ancient Medicine*.

The Hippocratic Oath forbade the use of a knife, even for the common kidney and bladder stones (the results of raging osteoporosis). Additional works referred surgery to those competent in handling war wounds - surgery was considered an "inferior trade" of the hands, while the work of a true physician was a work of the head and mind.

The preferred Hippocratic methods of treatment were steeped in dietary regulation. Humans can't eat forage like herbivores, so a good diet was crucial for good health - "The first cook was the first physician."

Followers of Hippocrates made no claims for "miracle cures," however, they did promise "that they would first and foremost do no harm (*primum non nocere*)" and looked at themselves as friends, advocates and representatives of the sick.

Even though the Hippocratic Oath is taken by modern graduates of medical schools, the Oath's origins are unknown other than it originated somewhere around the fifth and third centuries B.C. The Oath did not reflect the general medical standards of conduct of the day. The Hippocratic reverence for human life (Hippocrates was against abortion and infanticide) was not shared by Plato and Aristotle. The Hippocratic Oath was the early form of the western style concept of a "profession" as a morally self-regulating discipline among those sharing a trade and committed to helping others.

The shortcomings of the Hippocratic method of healing included a lack of deep knowledge of the internal physiology of the human body, however, its departure from contemporary thinking of disease was that "sickness was a disturbance in the health of the patient" was a landmark shift in thought.

The preferred Hippocratic therapies were founded in the regulation of diet. He reasoned that humans, unlike sheep and goats, could not eat plant roughage diets and therefore a good and proper diet was essential to good health. But the Hippocratic teachings on diet (*diatetica*), meant more than a simple meal of food and drink, his view was that "diet" included an entire lifestyle including exercise,

sex, bathing, sleep and spirituality - in effect Hippocrates was the first physician to consider "fitness" in all of its forms essential to health and longevity.

"Life is short, the art long, opportunity fleeting, experience fallacious, judgment difficult," offers the first of the Hippocratic aphorisms, outlining the arduous, but honorable labor of the physician of the time.

Hippocratic medicine taught two principals that are no longer regarded as sacrosanct by the modern physician:

- 1) The role of a physician was a lofty and selfless role.
- 2) The understanding of sickness requires the understanding of nature.

The Hippocratic Oath

I swear by Apollo the healer, by Aesculapius, by Health and all the powers of healing, and call to witness all the gods and goddesses that I may keep this Oath and Promise to the best of my ability and judgment.

I will pay the same respect to my master in the Science as to my parents and share my life with him and pay all my debts to him. I will regard his sons as my brothers and teach them the Science, if they desire to learn it, without fee or contract. I will hand on precepts, lectures and all other learning to my sons, to those of my master and to those pupils duly apprenticed and sworn, and to none other.

I will use my power to help the sick to the best of my ability and judgment; I will abstain from harming or wronging any man by it.

I will not give a fatal draught to anyone if I am asked, nor will I suggest any such thing. Neither will I give a woman means to procure an abortion.

I will be chaste and religious in my life and in my practice.

I will not cut, even for the stone, but I will leave such procedures to the practitioners of that craft.

Whenever I go into a house, I will go to help the sick and never with the intention of doing harm or injury. I will not abuse my position to indulge in sexual contacts with the bodies of women or of men, whether they be freemen or slaves.

Whatever I see or hear, professionally or privately, which ought not be divulged, I will keep secret and tell no one.

If, therefore, I observe this Oath and do not violate it, may I prosper both in my life and in my profession, earning good repute among all men for all time. If I transgress and forswear this Oath, may my lot be otherwise.



Greek philosophers looked at health and fitness as two of the greatest assets of life that a human could have. The earliest philosopher to write about Hippocrates' thoughts and methods was Plato (427 - 437 B.C.); he divided human nature into three functions - reason, spirit and appetites. The three functions, according to Plato, were located in the brain, the heart and the liver respectively - and all three were constantly in competition.

Through reason and spirit, the philosopher could conquer a human's sordid animal desires. The Republic's recognition and distinctions between reason, appetite, mind and body, was of the greatest philosophical and psychological value.

Plato's pupil, Aristotle, additionally theorized significant new views on the constitution of human life. A doctor's son from Stagira in Thrace, Aristotle (384 - 322 B.C.) moved to Athens to become a pupil of Plato; he later became the tutor and mentor of Alexander the Great (356 - 323 B.C.).

Aristotle, Alexander's personal physician and tutor, made and recorded many profound physiological observations.

Aristotle was a promoter for the systematic description of nature and he established a method of empirical study into the natural world: zoology, botany, and meteorology. He believed that the natural world was organized and that nothing occurred by accident - following that line of reasoning, anatomical reality was described in terms of their function (teleology). Aristotle's methods of scientific investiga-

tion guided medical research for almost 2,000 years.

Aristotle was the first scientific investigator to advance medical and biomedical science through the art of animal dissection. His description of a three chambered heart perplexed many of his followers, however, reptiles (i.e. - snakes, lizards, turtles, etc.) and amphibians (i.e. - frogs, toads, newts, etc.) classically have three chambered hearts, suggesting that Aristotle did his early dissections on reptiles and amphibians.

Aristotle observed chicken embryos developing in eggs and believed that the observable beating heart was the first sign of life and therefore was the center of life. He concluded that the heart and blood vessels were vessels that contained blood in an organized fashion as opposed to blood simply being a diffuse pool in the body. He believed that blood itself was the "nutrient" that vessels absorbed from the intestines concluding, "this explains why the blood diminishes in quantity when no food is given, and increases when much is consumed." Thus Aristotle gave the first recognition of anemia's association with nutritional deficiencies.

Failing to appreciate the central nervous system and the peripheral nerves as a system separate from the cardiovascular system, Aristotle credited the heart as the seat of the soul, and the source of movement and sensation further adding to his view the central importance of the heart. He believed the brain was simply a regulator, a cooler that "prevented the blood from overheating and boiling."

Aristotle's presentations of physiology and psychology in, *On the Soul*, *On Sleep and Waking*, *On Sensation* and *On Memory* fascinated physicians and scientific philosophers; 2,000 years later his works and teleological observations guided physiologists such as William Harvey (who finished the puzzle on how blood circulated through a closed vascular system, that connected arteries and veins through capillaries which Aristotle couldn't see for lack of a microscope).

Aristotle's princely student, Alexander the Great destroyed his enemies even as a boy and complained that, "There were no more worlds to conquer." Alexander, mortally wounded, died in 323 B.C. - on his death bed he exclaimed, "I die by the help of too many physicians." Following Alexander's death, basic science gained an

important status in the government of King Ptolemy (323 - 282 B.C.).

King Ptolemy's most important cultural contribution, the Alexandrian Library and the Museum (sanctuary of the Muses) produced a hybrid vigor in scientific thinking as a result of a blending of the Greek and Egyptian texts of the day. The library was the "Mecca" of the known scholarly world, with an accumulation of more than 700,000 writings and manuscripts, an observatory, zoological gardens, 13 massive lecture halls and research laboratories.

A section of the library was destroyed in 48 B.C. during riots that followed Julius Caesar's entrance into Alexandria; sometime later Christian zealots called for the destruction of the Temple of the Muses and the pagan idols within. According to legends, in A.D. 395, the last remaining resident scholar (a female mathematician), Hypatia, was dragged from the museum and stoned to death by rioting crowds. The Muslim sacking of Alexandria in the 7th century completely destroyed the last vestiges of the museum and library.

Following Alexander's conquest of the "known world," Hellenistic thought reached from the Persian Gulf to the Island of Sicily; Greek became the status language of royalty. A positive result of the "Greeking" of the world was a substantial increase in the accumulation of scientific information on animals, plants, minerals and "drugs."

Alexandria was a magnet for advanced medical and scientific thinkers, including Herophilus of Chalcedon (330 - 235 B.C.) and his fellow academic, Erasistratus of Chios (330 - 255 B.C.). Herophilus practiced medicine in Alexandria and achieved fame by doing human dissections in public. He wrote eleven scholarly texts, including three on anatomy - it was he that discovered and gave names for the prostate gland and the duodenum (Greek for twelve fingers - length of the upper small intestine). He wrote extensively on the diagnostic value of the pulse, therapeutics, ophthalmology, nutrition and diets, and midwifery.

The personal renderings of Herophilus and Erasistratus were destroyed by rioters; however, through their later follower's writings, including Cornelius Celsus (A.D. 60), we learned that they experimented on and dissected living humans. These reports were probably correct since the Greeks used their conqueror and occupier sta-

tus to subjugate their "inferiors," especially condemned criminals.

Herophilus differentiated between arteries and veins for the first time. His observations pointed out that the walls of arteries were much thicker than those of veins. His most notable discovery was a description of the nerves and their origin in the brain; he correctly concluded for the first time that the nerves, not the blood vessels, were the carriers of motor impulses from the central nervous system to the peripheral tissues including muscle.

Greek medicine metastasized throughout the Mediterranean, including Italy. The southern Italian cities of Elea, Tarentum and Metapontum claimed more of a Greek cultural following than Roman. Rome was different - the "no-nonsense" Roman thinking stated that "one was better off without doctors." "Romans had no need of doctors," insisted writers such as Cato (234 - 149 B.C.), because they, "the Romans were healthier unlike the weaker Greeks." "Beware of doctors," he wrote, "as they would bring death by medicine!"

"It is our duty, my young friends," declared Cicero (106 - 43 B.C.), "to resist old age; to compensate for its defects by a watchful care; to fight against it as we would fight against disease; to adopt a regimen of health; to practice moderate exercise; and to take just enough food and drink to restore our strength and not overburden it." Thus Cicero was one of the first individuals of royal status to call for a national program of individual responsibility for health and fitness - and to consciously avoid overeating and obesity.

The medical giant of the Roman era was Galen (129 - 216 A.D.). Galen's importance to medicine over the millennium was due in great part to his voluminous writings. More of his writings survived than any of those of any other author - he produced more than 350 original works on such wide ranging subjects as the soul to blood letting - more than all other Greek writings together.

Galen was the son of a wealthy architect; as a result, he enjoyed an education that was unequalled for his time - when he was but sixteen, his father steered him toward medicine. He studied with Alexandrian medical scholars, and traveled and studied medicine and "drugs" throughout Egypt, India and Africa.

Galen returned home to Pergamon (now known as Bergama, Turkey) in 157 A.D., where he was appointed physician to the gladiators, a post which allowed him to expand his anatomical knowledge and surgical expertise, "as the wounds of battle allowed windows into the living body."

Pergamon was a rustic town, so in 162 A.D., Galen left for Rome, where his public debates against Methodists and well attended public anatomical displays raised his fame to almost royal levels. Galen was famous for parlor tricks, which included the severing of nerves in the neck of a pig. As the nerves were cut, one by one, the pig would continue to squeal until Galen cut the laryngeal nerves - the squealing noise stopped even though the pig continued an attempt to squeal.

Galen became the physician to senators and Emperors, and as a result he developed the first modern medical ego so typical of M.D.s:

"I have done as much for medicine as Trajan did for the Roman Empire when he built bridges and roads through Italy. It is I, and I alone, who have revealed the true path of medicine. It must be admitted that Hippocrates already staked out this path...he prepared the way, but I have made it passable."

In the absence of scholarly universities, colleges and degrees and the artful nature of medicine, students followed the teachings of individual physicians, often sitting at their feet out of respect and following him on his rounds. Medical authors often publicly debated philosophies resulting in a flourishing of rival schools of medical thought.

Ancient scholarly pursuits gained a systematic approach to the *materia medica*, through the writings of two unique philosophers. Theophrastus (371 - 287 B.C.), a student of Aristotle, became the director of the Peripatetic school of Athens. His two works on plants deal with their descriptions and Investigations of Plants (*De historia plantarum*) and their origins (the *Causis plantarum*) and Explanations of Plants - through these writings he laid the foundation for the field of modern botany.

The *Investigations* classifies plants into the groups of trees, shrubs and herbs. All of 550 species and varieties are identified, through habitats ranging from the Atlantic to India (his information and material on Indian plants were collected by scholars of Alexander's expedition in the 320s B.C.) - his work led to the systematic review of medicinal botany and the widespread appearance of botanical gardens.

The second most important post Galen scholar was Dioscorides (40 - 90 A.D.), a Greek surgeon attached to Nero's army. His *De material medica*, published in Greek, however, better known by its Latin name appeared as five books:

Book I - Aromatics and essential oils (i.e. - saffron, oils, salves, shrubs and trees).

Book II - Animals, cereals and herbs.

Book III - Roots, juices, herbs and seeds.

Book IV - Roots and herbs.

Book V - Vines, minerals (including salts of lead and copper).

His books were designed to list the pharmaceutical values of the plants and other "drugs."

In the days of the Greeks and Romans, health, fitness, athletics and "diet" (philosophy) were everything. Obesity was not an option for the Greek or Roman citizen. These cultures were consciously putting their wood ashes, dutifully collected each morning from their cooking and heating fires, into their gardens, they cut and extended their dietary salt (Greek and Roman soldiers, tradesmen and artisans were paid "salaries" in the form of salt causing salt to be a substance of great value and it was often traded ounce for ounce for gold) with wood ashes, and wood ashes were used as "culinary" ashes for the preparation of many grain dishes.



Chapter 4

"All men dream, but not equally. Those who dream by night in the dusty recesses of their minds wake in the day to find that it was vanity; but the dreamers of the day are dangerous men, for they may act their dreams with open eyes, to make it possible."

- T. E. Lawrence

The Seven Pillars of Wisdom

Hospitals – The Road to Hell

The appearance of medieval hospitals, were in effect logical extensions of religious altruism. Those established in the west were initially small and originally erected for the benefit of religious pilgrims. Later versions of hospitals were more expansive and provided a wider service for more people - St. Leonard's in York had 225 beds for the sick and poor in 1287; even larger were the secular hospitals of Milan, Siena, and Paris.

In Florence, a city with a population of 30,000, there were more than thirty hospital foundations by the 15th century. Many had only ten beds, while some others boasted hundreds of beds. In England hospitals and almshouses numbered about 500 by 1400, however, only a few were of any significant size. London's St. Bartholomew's dates back to 1123; and St. Thomas from 1215. At Bury St. Edmunds, six hospitals were endowed between 1150 and 1260 to "care for lepers, pilgrims, the infirm and the aged."

The Crusades played a large role in hospital expansion; crusad-

ing orders, including the Knights of St. John (later to be called the Knights of Malta), the Knights of Templar, and the Teutonic Knights - all built hospitals in the Mediterranean and Germanic territories. By 1300 non-military brotherhoods, including the Order of the Holy Spirit, were supporting clinics from Alsace to Poland; the order of St. John of God built insane asylums in 1500 in Spain and about 200 hospitals in the New World.

Originally, small hospitals were hostels or hospices that lacked medical support. By 1215, physicians were on full time staff at the Paris Hotel Dieu, near Notre Dame, and Sta Maria in Florence, from 12 beds in 1288 "for the sick and the poor," to a medical staff of ten doctors, a pharmacist and several assistants including female surgeons by 1500.

The Golden Age of hospital building during the 1200s occurred at the same time as the founding of universities in Italy, Spain, France and England. Paris University was founded in 1110; Bologna in 1158; Oxford in 1167; Montpellier in 1181; Cambridge in 1209; Padua in 1222; and Naples in 1224.

By the 1230s Montpellier was attracting medical students from distant places. Initially medical schools developed in an eclectic fashion like Topsy. However, the staff eventually formed formal faculties for mutual protection and job security.

There were significant philosophical differences between the clerically administered Hospitals of Paris, Cologne, and Oxford, where the religious beliefs of the faculty dominated all things and the non-denominational universities such as Montpellier and Italy, where the arts and law faculties dominated. At that time a Bachelor of Medicine (MB) required seven years of study and Medical Doctorate (MD) required ten years of study.

As a result of the stringent requirements for medical degrees, few students applied and fewer graduated: Bologna issued sixty five degrees in medicine and only one in surgery between 1419 and 1434; Turin only 13 between 1426 and 1462. The one large medical school, which was considered a model of superior service and training was in Padua, here medical students made up ten percent of the student body; its medical faculty was relatively large, boasting 16 teachers in

1436, by comparison Oxford had only a single MD on the faculty.

Degreed physicians were at the top of the "pecking order," however, they were not the only health practitioners. Larger towns and cities were magnets for a wide variety of "healers." In 1400, Florence was invaded by medical graduates from Padua and Bologna, bone setters from Rome, and families specializing in eye diseases, hernia and the stone. Herbalists, midwives, hawkers of home remedies, and priests sold religious cures.

The numbers of health practitioners continued to increase eventually resulting in the appearances of guilds, apprenticeships, examinations of applicants, locations of pharmacies, supervision of drugs, food and herbs. By 1236, Florentine physicians and pharmacists joined forces and formed a single guild that was officially recognized as one of the cities seven major crafts.

Medical organizations formed in many different ways. In southern Europe surgeons and physicians were equals and it was encouraged for physicians to also become surgeons. In Frederick II's regulations for the Kingdom of Sicily (1231), one could only get a license to practice after five years of schooling which included courses and qualifications in surgery.

In Italy, the opportunity to take courses in surgery encouraged physicians and surgeons to be cooperative and "happy bed fellows." Outside of Italy, surgery was excluded from the university curricula. In northern Europe surgical training and practice were organized as guilds with required apprenticeships, and were therefore regarded by licensed physicians as "quacks."

In Paris, the surgeon's organization was established in 1210 after the College of St. Cosme was chartered. Its members were differentiated by "long robes" and "short robes." Only the long robes were allowed to perform surgery. Training was primarily practical and the college granted three possible degrees: a bachelors degree, a simple license, and a masters. A three sided faculty "war" developed between the various physicians on the faculty - the physicians, surgeons and barbers.

The introduction of anatomical dissection to the curriculum further added to the animosity because the dissection was under the

direction of physicians and the knives were wielded by the surgeons.

The earliest civic contracts for “public” physicians were recorded in Reggio, Italy in 1211 and Bologna in 1214, where the contractors were responsible for treating soldiers as well as the common citizen. Contracts commonly imposed a residence requirement, outlined the doctor’s private and public duties, and set scales of fees.

The civic contract system flourished and in 1300 showed up in all the large towns and cities in northern Italy. By 1500, civic doctors were being hired in northern France, Flanders, several German cities - only England dragged their toes and failed to hire civic doctors.

In 1516, the conflict was resolved by the surgeons agreeing to be supervised by the physicians - both quickly became allies and attacked the barbers. In the Germanys and England, barber-surgeons were quite common; in Italy, Spain, and southern France barber-surgeons never were accepted.

In 1518, the College of Physicians was chartered to regulate and license the practice of medicine in London.

Hospitals were the home of surgical teachings and surgeries for the poor. In 1780, John Howard had inspected Europe’s hospitals and found them dripping with filth, pus, gangrene and infections - surgery was better performed at the patient’s home. Hospital surgical mortality was 300 to 500 percent higher than for surgeries performed at home. “It is like a plague,” groaned John Bell, “no operation dare be performed, every cure stands still, every wound becomes a sore, and every sore is apt to run into gangrene.” At the Paris Hotel Dieu, “at the hearing of the word (surgery), the patients gave themselves up for lost.”

Simpson coined the term “hospitalism” defined as “a general morbid condition of the building, or of it’s atmosphere, productive of disease.” Simpson believed hospitals should be burnt down every once in awhile,

In England, the Crimean War (1854 - 56) brought to the public view the need for sweeping changes in hospital management. Florence Nightingale (1820 - 1910) became a heroine; she was from a wealthy well educated family, born in Florence during a family vacation, she pursued nursing as a mechanism to see the world. After a

short stint with the Daughters of Charity in Paris in 1853, she was appointed superintendent in London of the Establishment for Gentlewomen during illness, eventually becoming superintendent of nurses at King's College Hospital.

The start of the Crimean War in 1854 was her opportunity to put her theories on cleanliness to work. At the English base at Scutari, across the Bosphorus from Istanbul, hospital conditions were dreadful. Rumors of the neglect of the sick and wounded were confirmed by newspaper stories by William Russell, a *London Times* reporter: "Not only are there not sufficient surgeons.....Not only are there no dressers and nurses....there is not even linen to make bandages."

The *London Times* demanded action, England responded; Nightingale offered her services to her friend Sydney Herbert, the Secretary of War. Herbert confessed, "There is but one person in England that I know of who would be capable of organizing and superintending such a scheme." A contingent of thirty eight nurses - ten Roman Catholic sisters, eight Anglican sisters, six St John's House nurses, and fourteen from various hospitals were gathered and sent over the vigorous objections of the top military leaders.

On November 4, 1854 Nightingale and her nurses pulled into Scutari, where almost 2,000 wounded and sick soldiers lay in filthy rat infested wards. Three hundred scrubbing brushes were promptly ordered. As the cleanliness crusade was in full flower, the Battle of Inkerman was exploding, and the hospital was flooded with armies of the wounded.

Nightingale organized the care of the wounded and ultimately called in an additional eighty nurses to bolster her war on filth. She prepared and served meals, supplied bedding, and dealt with the laundry, "I am a kind of General Dealer in socks, shirts, knives and forks, wooden spoons, tin baths, tables and forms, cabbages and carrots, operating tables, towels and soap," she exclaimed. In six months she cut the hospital death rate from forty percent down to two percent. When she returned to England in 1856, she was surprised to find herself heralded as a heroine.

Nightingale possessed a drill instructor's fortitude and firm hand. Her success in promoting the profession of nursing was owed to her

demand that in her *Notes on Nursing* (1859) on the necessity for nursing applicants to receive an intensive training in nursing theory and practice.

Nightingale was a proponent of facts, referring to statistics "as the most important science in the world."

Not much has changed in "modern" 20th and 21st century hospitals since the old days of medicine; sure, there is more chromium, more diagnostic machinery (CT scans, MRI, PET scans, ultrasound, etc.), yet the mortality statistics are still abysmal.

In 1990 a survey by the World Health Organization (WHO) and the Center for Disease Control showed that America ranked 17th in longevity when compared with the top thirty two industrialized nations. The entire world spent \$2.7 trillion for health care; the United States alone spent \$1.6 trillion for health care, yet ranked 17th in longevity. Despite enormous health care spending that eclipsed the gross national product of most nations, the United States dropped to 24th in longevity ten years later in June of 2000 and 26th in longevity by March 2005!

In 1996, a Harvard School of Public Health survey showed that 180,000 Americans were killed each year in hospitals alone as a result of medical negligence; 1.3 million Americans were physically injured each year in hospitals alone as a result of medical negligence.

In March 11, 1998, the Center for Disease Control reported that 2,000,000 Americans are infected each year in hospitals alone as a result of medical negligence; and despite around the clock nursing care and intravenous antibiotics 90,000 died.

In November 1999, according to the Institute of Medicine, medical mistakes and errors ranked as 8th top killer of Americans in hospitals each year. In December 2002, a Johns Hopkins study elevated medical mistakes to the number three cause of death in Americans each year killing 250,000.

In 2004, the Center for Disease Control, listed poly-pharmacy (multiple drugs interacting in a negative way) as the cause of untoward or bad drug reactions in 2.2 million Americans; despite heroic efforts, 140,000 to 200,000 died as a result of medical negligence.

Each year 5.8 million American casualties, killed, wounded and infected, are perpetrated against hospital patients by medi-

cal doctors and yet no one seems to care - an unbelievable score that dwarfs the horror of all terrorism and wars!

How can this happen? Many factors are involved:

- 1) Americans have a low functioning immune system for the lack of raw materials (minerals) necessary to maintain and repair itself.
- 2) Doctors simply don't pay attention to details of diet, supplementation, treatments, surgeries and prescriptions - this environment encourages obesity, diabetes and the Metabolic Syndrome.
- 3) Doctors don't wash their hands, change their clothes or change shoe covers between patients.
- 4) Doctors lie to insurance companies (private and government) to get paid for procedures that patients don't need or get.
- 5) According to the insurance industry American doctors are responsible for seventy five percent of all health insurance and Medicare fraud.
- 6) Doctors get kickbacks for prescribing drugs that patients didn't need.

December 8, 2004, the *Los Angeles Times* featured a front page story that revealed the current state of the "modern" American hospital system. The story described the terrible conditions at the King/Drew Hospital and "how whole departments fail a hospital's patients. A culture of mismanagement pervades nursing, orthopedic surgery, the residency program and the pharmacy. Individual employees' shortcomings often made matters worse."

"Patients languished unattended by nurses."

"There was so much stuff at that place (King/Drew)... You couldn't focus on patient care."

According to the *Los Angeles Times*, "Mistakes and lax supervision have debilitated King/Drew's pharmacy and doctor-training programs, which affect nearly every patient. The Department of ortho-

pedic surgery was crippled by employee misbehavior, absenteeism, profiteering, even the commission of felonies.”

At King/Drew Hospital it was every doctor, every nurse and every pharmacist for themselves - the patients are relegated to the status of a vehicle that the hospital staff rode to obtain money, drugs, sex and power - it was and still is a modern Sodom and Gomorrah.

To get off of the Road to Hell, to eliminate the weight and obesity problems of urban man, we must begin to take control of our own health programs (including weight control and anti-obesity programs), support our body's raw material needs and boycott the existing medical system of hospitals and surgery - doctors, hospitals and the medical system are so busy trying to “get their share” they no longer have time for the serious treatment and prevention of the overweight and the obese.



Chapter 5

"The formulation of a problem is often more essential than its solution, which may be merely a matter of mathematical or experimental skill."

- Albert Einstein

The Energy Crisis

In 1268 Norman castles and great manors had chimneys to draw the coal smoke outside, it would be another 200 years before this luxury was found in the humble houses of common people. The commoner's fire sat on a raised stone hearth in the middle of the main room to avoid setting the wooden walls on fire. The smoke would fill the room until it seeped through gaps in the walls or roof - the dusty ashes left over from the wood's combustion were taken out every morning, mixed with salt (three to one or ten to one to cut and conserve the salt) for their diets, used as "culinary" ashes to prepare grain dishes and then they placed the balance of the household ash byproduct in the vegetable garden as fertilizer.

From an indoor air quality standpoint, the majority of Englishmen had not advanced much beyond the conditions suffered by *Homo erectus*, who, over hundreds of thousands of years earlier, had huddled over wood fires inside large huts made of sticks and branches.

The 13th century Englishmen were still getting their daily supply of minerals, either directly (mixed with salt and placed on their food) or indirectly (placing the ashes in the garden) from their wood ashes.

Beginning in 1285, during the reign of Edward I, various commissions were established in London to address the problem of coal

smoke, which early environmentalists complained, “had infected and corrupted” the air.

During the summer of 1306, bishops, barons and knights from all over England ventured from their country estates and manors and traveled to London town. They arrived to take part in a new political experiment called Parliament. When they arrived in London they were met with an awful odor - fumes and smoke from burning coal. Blacksmiths and other tradesmen had begun burning coal in place of wood, filling London’s air with sooty, smelly smoke.

By the summer of 1306, enough coal was being burned by blacksmiths, brewers (20,000 wagon loads of wood were used each year to distill and ferment beer and whisky), and others who required large quantities of fuel, that there was an organized resistance against it leading to a royal ban of coal use in London and other cities.

Despite new laws threatening steep fines and the destruction of furnaces, coal use continued to increase in volume. Some sources say that one transgressor of the coal ban law was, “hanged, tortured, or decapitated.”

The reason for London’s rising coal use was obvious. The population of London and of Britain generally, had been increasing, and as a result the forests were in a severe state of decline. Forests near London were consumed first because of easy access; they were burned for firewood, cut for building timber, or simply burned to clear land for crop production or pasture for livestock.

Wood ash supplies for garden fertilizer, cutting salt and culinary uses declined. People did not like to use coal ashes or “clinkers” for their food, salt or gardens because they imparted a burned petroleum smell and taste. The reduced supply of wood ashes caused a general decline in the health and immune defenses in the populations of Europe - “for lack of a nail a kingdom was lost.”

As supplies of energy from living forests declined and became scarce, and more costly, the demand for energy from coal rose significantly. The coal merchants, watching the burgeoning cities and the shrinking forests surely thought coal was nearing a boom in sales. As luck would have it the energy crisis was solved, not by coal, but

rather by a plague that devastated Europe's populations.

In October 1347, the crew of a trading ship that docked in Sicily brought the Bubonic Plague to Europe. The "Black Death" was already wiping out the populations of China, India, and the Middle East; in its first European rage, continuing until 1351, it would kill almost one third of Europe's population - 25 million souls were lost.

Some modern scholars speculated that the plague was particularly deadly because the general population had already been weakened by years of bad weather, bad harvests, reduced supplies of wood (and therefore ashes) and famine. At the time, most residents of Europe viewed the plague as evidence of the wrath of God! After careful consideration, the medical staff and faculty of the prestigious University of Paris determined that the plague was caused by "an unusual alignment of the planets that had occurred on March 20, 1345."

In England, the plague would break out again and again and again, at least three additional times before 1400, and Europe's population would continue to be ravaged by disease until 1500. By 1500 only 3 million people lived in England, about half the number that had lived there before the appearance of the plague. As deserted and abandoned farms and lands were reclaimed by the forests and trees became large enough to provide the needs of the decimated population, coal sales went into the tank - the coal market literally dried up, and wood ashes were back!

In the late 1500s the English suffered an energy crisis when their population grew and again because of an increased demand for and consumption of wood, the forests shrank. The English learned to tolerate what had been thought of as intolerable - they became the first western culture to mine and burn coal on a massive preferential national scale. The coal burning public and businesses filled the air of London and other cities with the most polluted and foul air in the history of humanity.

The English went on to power an industrial revolution with coal and transform the world of industry, agriculture, human health and the fate of the planet itself - the energy from coal was a double edged sword.

The nobles led popular demonstrations by the populous denounc-

ing the use of coal in the city - King Edward I banned the use of coal for political expediency. The original bans were ignored, so more desperate laws were passed to punish first offenders with "great fines and ransoms." Second offenders had their furnaces destroyed.

In 1346 through 1352 the plague (Black Death) killed twenty five percent of Europe's population with some cities recording a loss of seventy percent of their populations.

In 1455 Gutenberg produced a printing press from a wine press, and printed his first Gutenberg Bible.

During the Edo period (1603 - 1867) of Japan, rice straw was used to make a variety of products including sandals, hats, mats and rope. The manufacture of products from rice straw was a common cottage industry used to supplement the meager Japanese cash income.

The old and used straw items were burned as fuel in the kitchen ovens and heating stoves - farmers and "ash traders" purchased or traded for the ash which was then resold for fertilizer. In this way, almost all of the 10 million tons of the annual production of rice straw was returned to the soil.

The carbon dioxide produced by burning the straw was used by the growing rice plant, along with the sun's energy (solar income) and the photosynthesis process to manufacture the carbon chains required by all living things.

In 1661, John Graunt, a little known London draper theorized that by making tables out of London mortality data and analyzing it, he could create more useful information. His resulting observations, published in 1662, resulted in Charles II, recommending Graunt's election to the prestigious and newly formed Royal Society - adding that, "if they found any more such tradesmen, they should be sure to admit them all without further ado." Graunt's analysis confirmed that infant and childhood mortality was higher than earlier believed, and that thirty percent of London's deaths were children under the age of four or five years old. The records also showed that lung related deaths reached between twenty percent and twenty five percent of the total deaths. He recorded that, "Seasoned residents lived nearly as long in London as elsewhere, yet new-comers, and children

do not, for the Smoaks, Stinks, and close Air are less healthful than that of the Country; otherwise why do sickly Persons remove to the Country Air?"

In 1677 a smallpox outbreak wiped out 700 people in Massachusetts - twelve percent of the population. During the 1702 Massachusetts outbreak of smallpox, three children of Cotton Mather (the Puritan leader who had supported the burning of "witches" during the Salem witch hunts) were stricken with the disease - a few years latter he was introduced to the practice of smallpox "inoculation" by his black slave who had been vaccinated as a child in Africa. When he checked with other black slaves who had been brought to Boston from Africa, Mather learned that vaccinations were in fact a standard practice in Africa.

Benjamin Franklin

Benjamin Franklin was born on Sunday, January 17, 1706. Boston was by then an incorporated town for 76 years. Boston was no longer a simple, isolated religious outpost - it was a bustling, growing port city. Boston had more than 1,000 permanent houses, a 1,000 seagoing ships registered at it's port and 7,000 permanent residents - the population would double every twenty years.

Even as a child, Franklin was a creative, yet practical inventor. One of his first inventions was a set of flippers for his hands and feet to improve his swimming speed. He made two oval palettes, with holes for his thumbs, and "I also fitted to the soles of my feet a kind of sandals."

Early in life Franklin became infatuated with kites. While flying his kites, Franklin would strip, wade into a pond or lake, floated on his back, and used the kite and wind power to pull him across the body of water. "Having engaged another boy to carry my clothes round the pond," he recalled, "I began to cross the pond with my kite, which carried me quite over without the least fatigue and with the greatest pleasure imaginable."

Franklin was originally slated by his father to be a minister, and to that end he was sent to the Boston Latin School at the age of eight

to prepare for entry into Harvard. Franklin did well at Harvard, moving up from the middle of his class to the number one slot, which resulted in his being advanced to a higher grade level. Because Franklin's father couldn't sustain the tuition costs, Franklin was forced to drop out of school.

Young Franklin was described as skeptical, puckish, curious and irreverent, "Dr. Franklin, when a child, found the long graces used by his father before and after meals very tedious," his grandson stated. "One day after the winter's provisions had been salted 'I think, Father,' said Benjamin, 'if you were to say Grace over the whole cask - once for all - it would be a vast saving of time.' "

At age ten, with only two years of schooling, Franklin went to work full time in his father's candle and soap shop. It was not the most fun job that a ten year old kid could have; he spent his working hours skimming rendered tallow from boiling tubs of "stinking" fat; trimming wicks, adding wood ashes to the soap and filling candle molds was considered a mindless repetitive activity by young Franklin. His father took him on long walks to visit various tradesmen to find a trade or art that might better interest him.

As it turned out, his knowledge of many crafts and skills allowed him to become a first rate tinker and inventor. Franklin, at age twelve, went on to sign on for nine years of indentured service as an apprentice and to acquire the training and the experiences of a publisher, printer and newspaperman.

Franklin was directed to produce poetry based on news stories, the stories were well received, to which Franklin wrote, "flattered my vanity." Herman Melville would write that, "Franklin was everything but a poet."

Benjamin Franklin's four year old son, Francis died of smallpox in 1736, causing him to become an advocate of vaccination for smallpox.

The print industry was a natural course of life for Franklin, "From a child I was fond of reading," he recalled, "and all the little money that came into my hands was ever laid out in books." Books became the most important influence in his life. The love and respect of books was one of the common traits shared by the Puritanism of Mather.

and the Enlightenment of Locke, worlds that would blend in the character and personality of Benjamin Franklin. "Often I sat up in my room reading the greatest part of the night, when the book was borrowed in the evening and to be returned early in the morning, lest it should be missed or wanted," admitted Franklin.

Franklin ultimately became the most popular writer in Colonial America. His original pen name was Silence Dogwood - "a slightly prudish widowed woman from a rural area (written by an energetic unmarried Boston teenager who had never spent a night outside of Boston)." Franklin disguised his handwriting to fool his older brother, the publisher of the *Courant*, who published the first essay of Silence Dogwood on Monday, April 2, 1722.

Franklin wrote and published fourteen Silence Dogwood essays between April and October 1722 - all were designed to promote social improvements for the misfortunate, especially women. Further, in the summer of 1722, Franklin's older brother James went to jail for criticizing the government on their failure to capture pirates, allowing him at age 16 to print and publish the paper himself for three issues - he boasted, "I had the management of the paper, and I made bold to give our rulers some rubs in it, which my brother took very kindly, while others began to consider me in an unfavorable light as a genius that had a turn for libeling and satire."

Mrs. Silence Dogwood quoted in full an essay from an English newspaper that defended free speech. "Without freedom of thought there can be no such thing as wisdom," it declared, "and no such thing as public liberty without freedom of speech."

In January of 1723, when Franklin's older brother James wrote, "Of all knaves, the religious knave is the worst" he added, "too much of it (religion) is worse than none at all," James was immediately forbidden by the General Court to publish an issue of the *Courant*, without him first submitting each issue to the authorities for approval before publication.

James continued his attacks, which resulted in the General Court forbidding James Franklin from further publishing the *Courant*. On Monday, February 11, 1723, there appeared on the *Courant* masthead - "Printed and sold by Benjamin Franklin."

James returned to run the newspaper, and in a jealous rage beat Benjamin as he would a truculent apprentice, rather than as a brother and fellow author. So, at the age of seventeen years of age, Franklin, on the afternoon of Wednesday, September 25, 1723, took a sailing ship heading for New York. Franklin, at seventeen years of age was almost six foot tall, slim, muscular and barrel-chested.

While on his journey, Franklin met the only printer in New York, one William Bradford. Bradford suggested that Franklin go to Philadelphia instead of New York and apply for a job with his son Andrew Bradford who ran the family print shop and weekly newspaper there.

Philadelphia was the second largest village in America with a population of 2,000 people - only Boston was larger. Through a variety of adventures Franklin befriended Sir William Keith, the governor of Pennsylvania. On a false promise of a letter of credit, Franklin went to London in November 1724 to buy equipment to set up a new print shop back in Philadelphia.

Franklin found a job with Samuel Palmer's well respected London print shop. It was an exciting time in London with famous writers such as Swift, Defoe, Pope, Richardson, Fielding and Chesterfield. On July 1726 Franklin set sail on his return trip to Philadelphia.

Franklin became the first printer in America to manufacture type. One of the most frequently used contemporary typefaces, a sans-serif font dubbed FRANKLIN GOTHIC that is often used in newspaper headlines, was named after Franklin in 1902. Franklin created a copperplate press to print paper money for the New Jersey assembly so beautifully ornate that the bills could not be counterfeited.

In 1732, in Philadelphia, Franklin printed the first issue of his *Poor Richard's Almanac*. Franklin used the pseudonym Richard Saunders - in his new almanac he used "pithy maxims" to enliven his meteorological forecasts. "Poor Richard" held that, "thrift and hard work led to wealth and virtue."

Franklin's publishing business flourished, and his ability to compete with Philadelphia's other printer, Andrew Bradford grew more aggressive. Through the early 1730s, they exposed each others errors and bickered over local and national political events. The high

born Bradford, and his *American Weekly Mercury* newspaper, were followers and supporters of Pennsylvania's "Proprietary faction," which supported the Penn family and their appointed governors. The common born Franklin and his *Pennsylvania Gazette* were vigorously anti-establishment and supported the rights of the elected Assembly.

Their politics crossed purposes during the 1733 reelection campaign of the Assembly's speaker, Andrew Hamilton, an anti-Proprietary leader who had aided Franklin's bid to take a government printing job from Bradford. Franklin was fascinated by Hamilton's anti-aristocratic populism. "He was no friend to power," Franklin wrote. "He was the poor man's friend."

Hamilton won the election and in 1736 he was able to get Franklin appointed to the post of clerk of the Assembly. Private profit and public service were combined. "The clerkship, Franklin agreed, "gave me a better opportunity of keeping up an interest among the members, which secured to me the business of printing the votes, laws, paper money, and other occasional jobs for the public, that, were on the whole, very profitable." This experience also taught him the useful skill of winning over enemies. After one high-bred member spoke out against him, Franklin decided to hold him close (hold your friends close, and your enemies closer):

"I did not, however, aim at gaining his favor by paying for any servile respect to him, but, after some time, took this other method. Having heard that he had in his library a certain very scarce and curious book, I wrote a note to him, expressing my desire of perusing that book, and requesting he would do me the favor of lending it to me for a few days. He sent it immediately, and I returned it in about a week with another note, expressing strongly my sense of the favor. When we next met in the House, he spoke to me (which he had never done before), and with great civility; and he ever after manifested a readiness to serve me on all occasions, so that we became great friends, and our friendship continued to his death. This is another instance of the truth of an old maxim I had learned which says, - He that has once done you a kindness will be more ready to do you another, than he whom you yourself have obliged."

Franklin was able to take away the Philadelphia postmaster job

from Bradford when Bradford was exposed for problems with his book keeping. The commission was given to Franklin in 1737. "I accepted it readily," Franklin noted, "and found it of great advantage, for though the salary was small, it facilitated the correspondence that improved my newspaper, increased the number demanded, as well as the advertisements to be inserted, so that it came to afford me a very considerable income."

Between 1740 and 1744, Lord Anson's (1697 - 1762) round-the-world expedition launched with great hope and 1955 men. Before the expedition ended, however, 320 died from fever and dysentery and 997 had died from scurvy by 1744. Scurvy, characterized by swollen, spongy, bleeding gums, large bruises, swollen joints, lassitude, heart failure and death was a significant danger for sea faring crews and passengers.

Dr. James Lind (1716 - 1791), a Scottish naval surgeon, steadfastly protested against poor shipboard accommodations, spoiled food and polluted water. In his written works, *Treatise on the Scurvy* (1753), Lind relied on his personal experience to reason that citrus fruits were effective antiscorbutics, supporting his hypothesis by the world's first controlled nutritional clinical trial on board the *HMS Salisbury* in 1754:

"Twelve scurvy patients were chosen, Lind gave two of them a quart of cider; two received oil of vitriol; two were given vinegar; two sea water; two were given oranges and lemons; and two were given an electuary of garlic, radish, Peru balsam, and myrrh. The two given oranges and lemons were fit for duty in six days, and put to the task of nursing the others, who in fact remained sick."

Initially Lind did not look at scurvy as a deficiency disease, but rather he felt that scurvy was successfully treated with citrus as a plant medicine (an herbal medicine) the same way that malaria was treated with Jesuit's bark.

On February 16, 1741, after several business skirmishes with competitors Franklin published *General Magazine*, his first general interest magazine. In the 1700s the term magazine, referred to an

eclectic collection of material abstracted from newspapers among other sources. Franklin's version of a magazine was taken from the ten year old, *Gentleman's Magazine* originating in London: official proclamations, reports on government proceedings, discussions of paper money issues, a few poems and a report on Whitefield's orphanage.

After courting several women and fathering an illegitimate child, Franklin and Deborah Read moved into a common law relationship.

In 1743, Franklin proposed the formation of the American Philosophical Society, which was to be an American version of the Royal Society think tank of England. It would be housed in Philadelphia, and its members would include illustrious scientists and thinkers from other cities.

The members would exchange their studies by post, and abstracts would be distributed to the members quarterly. The subjects to be investigated were to be practical rather than theoretical and included: "newly discovered plants, herbs, trees, roots, their virtues, uses, etc...improvements of vegetable juices, such as ciders, wines, etc.; new methods of curing or preventing diseases, ...improvements in any branch of mathematics... new arts, trades, and manufactures... surveys, maps and charts ...methods of improving the breeds of animals ...and all philosophical experiments that let light into the nature of things."

By the spring of 1744, the society began meeting regularly; early members included Thomas Godfrey, a pedantic mathematician and Cadwallader Colden, a scholar and an official from New York, who became a life long friend and Franklin supporter.

In 1747 Franklin proposed a volunteer military force that would be independent of Pennsylvania's colonial government - the Pennsylvania Militia was born.

In 1747 after the King George's War (1744 - 1748), Europe's War of Austrian Succession and the War of Jenkin's Ear and the French and Indian War in Canada (Franklin's 16 year old son William marched off to fight in this war), Franklin wrote a motivating pamphlet, "Plain Truth," signed simply "a Tradesman of Philadelphia." Franklin's depiction of what might happen if Philadelphia was invaded and sacked by French or Spanish privateers:

"On the first alarm, terror will spread over all...The man that has a wife and children will find them hanging on his neck, beseeching him with tears to quit the city....Sacking the city will be the first, and burning it, in all probability, the last act of the enemy.....Confined to your houses, you will have nothing to trust but the enemy's mercy....Who can, without the utmost horror, conceive the miseries of the latter when your persons, fortunes, wives and daughters shall be subject to the wanton and unbridled rage, rapine and lust."

So, who would save the colony? Franklin called on the new "middle class!" "We, the middling people," he crowed, "The tradesmen, shopkeepers and farmers of this province and city." He went on to say, "At present we are like separate filaments of flax before the thread is formed, without strength because without connection, But Union would make us strong."

"The pamphlet had a sudden and surprising effect," Franklin wrote. Soon 10,000 men from all over Pennsylvania signed the charter and formed more than 100 militia companies. Franklin further organized a lottery to raise 3,000 British pounds sterling to furnish the Militia Association with cannons and equipment. Franklin soon led a delegation to New York to urge Governor George Clinton to approve the sale:

"He at first refused us peremptorily; but at dinner with his council, when there was great drinking of Madeira wine, as the custom of that place then was, he softened by degrees, and said he would lend us six. After a few more bumpers he advanced to ten; and at length he very good-naturedly conceded eighteen. They were fine cannon, eighteen-pounders, with their carriages, which we soon transported and mounted on our battery."

Franklin, didn't realize it, but his charter of the Militia Association, in its spirit as well as wording, conceived and preceded the Declaration of Independence three decades later:

"Being thus unprotected by the government under which we live, we do

hereby, for our mutual defense and security, and for the security of our wives, children and estates....form ourselves into an Association."

Franklin's print shop grew into a "highly successful, vertically integrated media conglomerate." He boasted a printing press, publishing house, newspaper, an almanac series and partial control of the postal system. The highly successful books he published included Bibles and psalters to Samuel Richardson's novel *Pamela* (Franklin's 1744 reprint of *Pamela* was the first novel published in America).

Franklin created a series of profitable partnerships and franchises from Newport and New York to Charleston and Antigua - the network of enterprises provided Franklin a passive stream of income - and he invested wisely.

In 1748, at age forty two, Franklin retired and sold his printing business to David Hall, his trusted foreman; the agreement provided him with half the businesses profits for eighteen years, which amounted to 650 British pounds per year (the average income at the time was 25 pounds per year). Now he would have "leisure to read, study, make experiments, and converse at large with such ingenious and worthy men as are pleased to honor me with their friendship."

Franklin gave up his print shop to focus and engage his energies on science, then politics, then diplomacy and statecraft. As Poor Richard declared in his almanac that year, "Lost time is never found again." His intellectual curiosity and infatuation with the orderliness of the greater universe drew him to science. His scientific interests would continue through his whole life, including research into the Gulf Stream, meteorology, the earth's magnetism, electricity, heating and refrigeration systems.

Franklin didn't have any formal academic training in science or math to be considered a great theorist of the likes of Einstein; his interests of what he fondly called his "scientific amusements" caused some contemporaries to dismiss him as a mere tinker. However, even during his life he was held up as the most famous scientist alive - as Harvard professor, Dudley Herschbach stated, "His work on electricity was recognized as ushering in a scientific revolution comparable to those written by Newton in the previous century or Watson and

Crick in ours.”

An example of Franklin’s practical application of science was his 1740s invention of an iron wood-burning stove that could be built into fireplaces to maximize heat output into the house while at the same time minimizing, fuel requirements, smoke and drafts - using his understanding of convection currents and heat transfer, he produced an original and unique stove design. The resulting wood ash byproduct of minerals were still being mixed with food as “culinary” minerals, with table salt and mixed into the soil and/or compost of the garden.

The stove was designed to allow the heat and smoke from the wood fire to rise up and warm an iron plate on the top. In 1744, he contracted a contemporary to construct the new stove and two of his brothers and several friends to market the stove throughout the north-east.

The new “Pennsylvania Fireplaces,” as Franklin called them, were initially popular at a price of 5 British pounds. One testimony letter in the Boston Evening Post stated, “They (the stove) ought to be called, both in justice and gratitude, Mr. Franklin’s stoves.” The letter writer went on to say, “I believe all who have experienced the comfort and benefit of them will join with me that the author of this happy invention merits a statue.”

During a visit to Boston in the summer of 1743, Franklin and others were entertained one evening by a traveling scientific showman from Scotland called Dr. Archibald Spencer. Spencer featured amazing demonstrations of science that were designed to amuse the host’s guests. He revealed Newton’s theories of light and exhibited a device that measured blood flow - both were interests of Franklin’s.

Dr. Spencer performed electricity tricks, such as creating static electricity by rubbing a glass tube and drawing sparks from the feet of a boy hanging by silk cords from the ceiling. “Being on a subject quite new to me,” Franklin recalled, “They equally surprised and pleased me.”

A 100 years earlier, Galileo and Newton had explained and demystified gravity, however, another awesome force of the universe electricity, was no more understood than it had been by the prehis-

toric people who had tamed fire. There were individuals, like Dr. Spencer, who used the wonders of electricity as a parlor trick to entertain admiring audiences; however, no one had as yet put the power of electricity to the task of real work.

The Abbe' Nollet, court scientist to France's King Louis XV, connected 180 soldiers and then 700 monks and "made them jump as if one for the court's entertainment by sending a pulse of static electricity through them." Franklin, however, was the ideal person to convert electricity from the status of a simple parlor trick into a functional, practical and working science.

Franklin's initial experiments involved collecting an electric charge and then studying its various properties. He employed his friends to draw charges from a spinning glass tube and touch each other to see if sparks flew. These early experiments resulted in the discovery that electricity was "not created by the friction of rubbing the tube, but collected only."

Many of Franklin's originally coined terms such as battery, charged, neutral, condense, and conductor are still used today in the fields of electricity and the greater subject of energy. "He has written equally for the uninitiated as well as the philosopher," the 19th century English chemist Sir Humphrey Davy noted, "and he has rendered his details as amusing as well as perspicuous."

Franklin's discovery that the creation of a positive charge was accompanied by the generation of an equal negative charge became known as the "conservation of charge" and the "single-fluid theory" of electricity. The concepts were easily formulated in Franklin's "book keeper" mind. His observations were a scientific breakthrough and discovery of historic greatness - "As a broad generalization that has withstood the test of 200 years of fruitful application," Harvard professor I. Bernard Cohen declared, "Franklin's law of conservation of charge must be considered to be of the same fundamental importance to physical science as Newton's law of conservation of momentum."

Franklin also discovered a basic feature of electrical charges, "the wonderful effects of points" - that shortly would produce his most famous practical application. He electrified a small iron ball and

dangled a cork next to it, which was repelled to a degree based on the size of the ball's charge. When he brought the tip of a pointed piece of metal near the ball, it drew away the charge. By contrast, a blunt piece of metal did not draw a charge or spark as easily, and if it was insulated instead of grounded, it failed to draw a charge.

Franklin continued his electricity experiments by collecting and storing electricity in an early capacitor called a Leyden jar (named after the Dutch town it was discovered in). These jars had a metal foil sleeve on the outside; on the inside of the glass jar, separated from the foil by the glass insulation, was lead or water or metal that could be charged up through a wire. Franklin demonstrated that when the inside of the jar was charged, the outside foil had an equal but opposite charge. He also correctly determined that the electrical charge in the Leyden jar was stored in the glass itself.

In a practical progression of his electrical storage experiments, Franklin lined up a series of parallel glass plates separated by metal - charged them up, wired them together, and discovered a new electrical device - "what we called an electrical battery."

Franklin was able to use the electrical battery to turn the iron fence surrounding his house into the first electrified security fence; he created an electrical toy spider that would hop like a live one; he electrocuted turkeys with his electrical battery for the entertainment of his guests before the birds were roasted for dinner with electricity, thus inventing the first electrical execution and cooking devices.

Franklin commented by letter to a friend that at one of his parties, "A turkey is to be killed for our dinners by the electrical shock; and roasted by the electrical jack, before a fire kindled by the electrified bottle; while the health's of all the famous electricians in England, France and Germany are to be drank in electrified bumpers, under the discharge of guns from the electrical battery." "The birds killed in this manner eat uncommonly tender," he wrote, thus he became a cooking pioneer of electrically roasted turkeys!

This early Franklin experiment, in which he killed and cooked turkeys with electrical devices, was perhaps the very first close call humanity had with a possible epidemic of obesity, type II diabetes and the Metabolic Syndrome, for there were no wood

ash byproducts resulting from this form of cooking! No wood ashes to consume with salt, no culinary ashes - no minerals to put in the garden!

Franklin kept a log of his electrical experiments - he noted in November 1749, that there were twelve similar properties in electrical sparks and lightening, including:

1. Producing light.
2. Color of the light.
3. Crooked discharge directions.
4. Swift motion.
5. Being conducted by metals.
6. Crack or noise in exploding.
9. Destroying animals.
10. Sulphurous smell.

He made a mental connection between this surmise about lightening and his earlier experiments on the efficiency of pointed metal objects to draw off electrical charges - "Electrical fluid is attracted by points. We do not know whether this property is in lightening. But since they agree in all particulars wherein we can already compare them, is it not probable they agree likewise in this?" He was so enthused by this prospect, he added - "Let the experiment be made."

For millennia, lightening was universally considered to be a supernatural phenomenon or an expression of God's wrath. During one thirty five year time line in Germany alone (1700s), "386 churches were struck by lightening and an excess of 100 bell ringers were killed by heavenly bolts. In Venice, 3,000 people were killed when tons of gunpowder stored in a church was struck."

Franklin noted that, "The lightening seems to strike steeples of choice and at the very time the bells are ringing; yet they continue to bless the new bells and jangle the old ones whenever it thunders. One would think it was now time to try some other trick." He concluded that the "water vapors in a cloud can be electrically charged and when these electrically charged clouds pass over high tree, lofty towers, spires, masts of ships ... draw the electrical fire and the whole

cloud discharges.” His proposal led to some practical advice - “Dangerous therefore it is to take shelter under a tree during a thunder gust.”

Before he even tried his lightening experiments himself he detailed his thoughts in two famous letters to Collinson in 1750, which were presented to the Royal Society in London and then widely published. The basic concept was to use a tall metal rod to draw some of the electrical charge from a cloud, just as he had used a needle to draw off the charge of an iron ball in his workshop. In his letter he listed specific details:

“On the top of some high tower or steeple, place a kind of sentry box big enough to contain a man and an electrical stand. From the middle of the stand, let an iron rod rise....upright twenty to thirty feet, pointed very sharp at the end. If the electrical stand be kept clean and dry, a man standing on it when such clouds are passing low might be electrified and afford sparks, the rod drawing fire to him from the cloud. If any danger to the man be apprehended (though I think there would be none) let him stand on the floor of his box, and now and then bring near to the rod the loop of a wire that had one fastened to the leads; he holding it by a wax handle (i.e. - insulating him from it). So the sparks, if the rod is electrified, will strike from the rod to the wire and not affect him.”

Franklin's letters were abstracted in London by *The Gentleman's Magazine* in 1750 and then published as an eighty six page booklet the following year; his letters were translated into French in early 1752 and quickly became a local public and international sensation.

Earlier, King Louis XV of France commissioned Comte de Buffon and Thomas-Francois D'Alibard to perform Franklin's proposed experiment and construct a sentry box with a forty foot iron rod. On May 10, 1752, just after 2:00 p.m. a storm cloud passed over the village of Marly on the northern suburbs of Paris, a retired soldier, drafted to perform the experiment drew sparks as Franklin had predicted. An enthusiastic prior grabbed the insulated wire and repeated the experiment six times!

Franklin correctly surmised that “lightening rods” could render

safe one of the greatest natural dangers faced by humans. "Houses, ships and even towns and churches may be effectually secured from the stroke of lightening by their means," he predicted.

Franklin had been waiting for the steeple of Philadelphia's Christ Church to be finished; impatient, he developed the idea of using instead a kite. To that end he enlisted the efforts of his son William to help fly the silk kite. A sharp wire was attached to the top of the kite, and a key was attached near the base of the wet string, so that a wire could be brought near it in an effort to draw sparks.

Initially Clouds passed over to no effect. Franklin almost fell into despair until he saw several of the strands of the string straighten out and stiffen. He put his knuckle to the key and drew an electrical spark.

Franklin was literally moments away from discovering a commercially viable source of clean household energy - electrical appliances and the end of the ash byproduct of cooking and heating fires! However, the onset of the American Revolution and the birth of a nation siphoned off his creative juices and energies.

It would take an additional 150 years before Edison's work produced the practical generation of electrical power and the flurry of invention of patented electrical appliances that would replace wood fueled heating and cooking fires and eliminate once and for all the health and life supporting kitchen and heating byproduct of wood ash - minerals.



Chapter 6

“With all things being equal, the simplest explanation is normally the right one.”

- Hakim’s Razor

The Fuel Revolution

After having tamed fire 1,500,000 years earlier, humans had created many unique ways to use the power of fire to change natural materials into everything from metals, pottery, bricks, and glass, to salt, soap, and ale. However, in all those hectic and creative millennia of baking and boiling, melting and smelting, it didn’t quickly come to them that fire held another power - the power of movement and motion.

For the average human, the back-breaking work of transporting loads from one farm to another, or from one market to another was performed only with the power of muscle, water, or wind. This fact of work powered only by humans, animals, water or wind limited the production and growth that a culture could expect to accomplish in each century. There was only so much mechanical lifting, turning and pulling that humans and animals could do and only so many hours in the day.

In the 1700s, a machine was invented that allowed humans to grow beyond the historical constraints and limits by turning “solar income” fuels into enormous amounts of mechanical torque, lift and forward motion! The invention was the steam engine and the fuel was coal. The steam engine would soon become the center of the industrial revolution, causing it to be called, “The most important

invention in the creation of the modern world.”

Coal, “motivated” the steam engine and its inventors, who were searching for a solution to the problem of coal mine flooding as the mine shafts were driven below the water table - because wood had become scarce, coal had become a necessity of life and the problem of flooding had to be solved.

In addition to the threat of mine gasses, miners had to worry about water. Rainwater poured into the tunnels, when the depth of the mines fell below the water tables, ground water seeped into the shafts and more frequently than not unearthed underground streams and springs quickly filled the shafts.

Sudden mine floods occurred when diggers broke through their shaft into an adjacent abandoned mine that had filled with water. The results were catastrophic, drowning the miners immediately, or worse trapping them in the mine until they slowly died of oxygen deprivation or starved to death.

To keep the mines dry, operators employed gravity by digging tunnels or “watergates” from the valleys up into the mine; others used various means of power including wind mills, windlass, chain of buckets and simple buckets carried on the backs of diggers. The greatest amounts of water in the largest mines was raised by horse power, however, the cost of maintaining fifty to sixty horses and their teamsters was too burdensome for even the largest of mines.

By 1700, coal production was ten times greater than it had been in 1550. England was now mining five times more coal than all other nations combined, she was now totally dependent on coal as a fuel for home use and powering industry. Even if England optimized her wood fuel production from her forests, ceased the use of wood for building and manufacturing, the forests could no longer replace the dependency on coal - wood fuels were once and for all gone in England.

The flooding of the mines remained a threat to England’s very existence as a world power - as a result, in 1660 Charles II created the Royal Society; an independent group commissioned to study and promote the theory and application of science.

The original members were the great philosophers, thinkers and

scientists of the day, including Sir Issac Newton, Gottfried Leibnitz, Christopher Wren, Samuel Pepys, John Evelyn, and Robert Boyle - "Religion, nationality, profession - all were subordinated to this ultimate aim of technical mastery, subordinated to a degree perhaps never before found in history."

One of the more important concerns of the illustrious group was coal, yet coal was still a mystery to them. In the 17th century, people still believed that coal was a living organism and still alive. The Royal Society members, although initially confused about the origin of coal, they agreed for certain to one thing - England's coal supply was threatened by mine flooding. They focused all of their attention on pumping technology, and on atmospheric pressure.

Scientists had only recently begun to appreciate that the atmosphere had weight and volume and it fills vacuums with a powerful force. They learned and appreciated the fact that the force of air opposing a vacuum could be put to work, however, producing and controlling the vacuum was problematic.

Denis Papin, a French Huguenot living in England, found the solution - steam! Papin had previously proven his knowledge of steam by demonstrating to the members of the Royal Society his steam pressure cooker which he named the "Digester" or Engine for softening bone so they could be completely eaten.

Between 1690 and 1695, Papin invented a small contraption consisting of a moveable piston within a brass cylinder. When he heated water that filled one end of the cylinder, the resulting steam vigorously drove the piston upward. When he removed the flame, the steam condensed and produced a vacuum - the piston was then vigorously pushed back toward the vacuum by the outside atmospheric pressure.

Papin didn't believe his first steam piston could be built large enough to be of practical use. The inventor who actually put the steam piston to work, was not a member of the Royal Society or London's scientific elite, rather it was Thomas Newcomen, a simple ironmonger or "tinker" from a small rural town.

The members of the Royal Society tried to downplay the importance of Newcomen's steam engine, leading many historians to be-

lieve that Newcomen's lack of credentials was the problem. One historian went on to say, "to the small scientific world of London, whose center was the Royal Society, it seemed inconceivable that such a man could excel the brightest intellects of the day in the power of invention."

Newcomen's steam driven device was similar to the Papin piston, however, it was much larger and sported a separate boiler in which to generate the steam. After the steam was let into the bottom of the cylinder, cold water was added to condense the steam; the resulting vacuum forced the piston in a downward stroke. As the piston moved up and down, it rocked an overhead beam. The opposite end of the beam moved the rod of a vacuum pump in a similar fashion to those driven by horse power.

The first practical Newcomen steam engine was "quietly and without ceremony" installed at a coal mine in 1712. The engine was called a "fire engine," and the men who fed coal into the engine's boiler and keep the engine's fire alive were called "firemen."

Newcomen's steam engine was an immediate success and much revered by mine operators - one engine did the work of pumping more efficiently and more economically than fifty horses! Those historians who know Newcomen's story, write of him and his work with an almost religious reverence, calling his steam engine, "the most wonderful invention which human ingenuity had yet produced," and credited him with giving "to the whole world.... The art of converting fuel into useful power for the benefit and convenience of humanity."

In 1761 an English publisher printed an indispensable book for the American homemaker - *The Colonial Housewife; or, Accomplished Gentlewomen's Companion*, which features "every article which can add to the knowledge of the housewife." The *Companion* offered the housewife more than 700 recipes, including such New World dishes as West Indian turtle and roast turkey, along with selected recipes from France, the Netherlands, Spain and Italy. The author also included 300 home remedies for ailments that range from cancer to dropsy.

By the 1760s, hundreds of Newcomen engines (because they used

atmospheric pressure rather than steam pressure were often called atmospheric engines) were pumping water from mine shafts throughout England and Scotland; some of Newcomen's engines were built to gigantic proportions for the day, reaching cylinder diameters of more than six feet and lengths of ten feet.

Initially the engine was used only by coal mine operators because it was a fuel hog, and coal was for all practical purposes free at the mouth of the shaft.

In 1736, James Watt, the son of a Scottish carpenter was born. He was the only one of five children born to his parents to survive to manhood; however, he was very frail and suffered from continuous migraines that kept him in bed for weeks at a time. According to a 19th century biographer, Watt's fragile health left him with "an almost feminine delicacy and sensitiveness, which made him shrink from the rough play of robust children"; as a result, he was home schooled during his early years, he spent his time playing with his father's carpentry tools and intently watching steam pouring from tea kettles.

Watt excelled in math and studied the art of constructing mathematical instruments. It was as an instrument maker, only twenty years old, that he was asked to fix a small model of a Newcomen engine that had been reverently kept at Glasgow University.

The Newcomen model, like the full sized engine, was extremely wasteful, requiring enormous amounts of coal to keep it going. Watt observed that as steam was injected into the system and then cooled with water, heat was wasted in the cyclical process of reheating and cooling the cylinder. Watt revolutionized the Newcomen steam engine by installing a separate condenser; the cylinder would be attached to another container that was surrounded with cool water, after the cylinder filled with steam, the steam passed into the cooler container where it condensed. In this way, the cylinder itself would always remain hot and ready for the next injection of steam!

According to Dr. Joseph Black, a good friend of Watt's, "this capital improvement flashed upon his mind at once, and filled him with rapture."

Watt, after suffering many challenges ("of all things in life, there

is nothing more foolish than inventing”), eventually found the perfect partner, the Birmingham industrialist Mathew Boulton, and one of the most successful partnerships in history was started.

Boulton was a manufacturer and an industrial visionary. Together, Boulton and Watt formed the Lunar Society - members included Joseph Priestly, the scientist who discovered oxygen, sulfur dioxide, and carbon monoxide; Joseph Wedgwood, the pottery giant; and Erasmus Darwin, inventor, poet and grandfather of Charles Darwin.

Boulton built his financial empire with buckles, rather than coal, iron or textiles. He employed 1,000 workers in his Soho manufactory, where they made precision crafted goods, with a range from buckles, buttons and filigree to classic scientific instruments including clocks, thermometers and telescopes. Boulton’s manufactory became so famous that it drew tourists, including Catherine the Great.

Boulton had a serious interest in steam engines because the water wheels that powered the Soho facility could not be depended upon for a continuous source of power. He had unsuccessfully tried his hand at inventing and designing steam engines, he even exchanged letters on the subject of engine design with his good friend Benjamin Franklin.

In 1773 Watt moved to Birmingham and began working on improvements on his steam engine at Soho. Boulton provided skilled artisans, financial backing and best of all emotional encouragement to Watt.

The all important cylinder was produced by John Wilkinson, the pioneering iron founder. In 1776, while the American Revolution was in full swing, James Watt installed his first two engines, one to pump water out of a mine shaft and the other to inflate the bellows in Wilkinson’s iron foundry.

While Watt improved the steam engine in many ways, his finest moment was in getting the Newcomen engine to extract four times the work force or horsepower out of a lump of coal. The steam engine’s new fuel efficiency moved it out from the darkness of the coal mine and into the business world to find a home in the world’s factories.

The first civilized vaccination was performed in 1796 by Dr. Edward Jenner, an English physician. He noted that milkmaids who

had contracted the mild disease cowpox never contracted smallpox: Jenner removed some material from a milkmaid's cowpox pustule and scratched it into the skin of an eight year old boy. The boy soon developed cowpox. Sometime later, Jenner injected the boy with the smallpox virus and he failed to contract the disease. During the late 19th century, research by the French chemist, Louis Pasteur and the German bacteriologist Robert Koch produced many new vaccines including those designed for anthrax, rabies and tetanus.

England's coal industry had become relatively large by the 1700s just to meet the domestic and non iron industrial requirements of the nation; it would expand again tenfold by 1830; and double again by 1854. Between 1842 and 1856 the number of English coal mines quadrupled again.

The steam driven industrial revolution's impact on human endeavor has been compared to the discovery of fire, the invention of the wheel, and to the tasting of the apple from the Tree of Knowledge.

Not until the 1800s would it be scientifically established, and commonly accepted, that coal itself, not only the plant fossils held within, was plant material and prehistoric.

By 1790, "potash fever" hit the young American country, trees were burned by the thousands for the ash which was exported to Great Britain for fertilizer. The first U.S. Patent Number 1, was for "the production and use of wood ash as fertilizer."

On April 17, 1790, Benjamin Franklin died at the age of 84. In the largest public gathering ever seen in the United States, 20,000 people attended Franklin's funeral. By August a national census revealed there were 3,929,625 Americans living in the United States.

The census tally showed a vast population growth since the early colonial days, when a 1726 census showed a population of only 32,442 people in New Jersey compared with 200,000 in 1790. New Hampshire had a population of 52,700 in 1767 compared with 200,000 in 1790.

The census revealed that the United States was made up of many ethnic groups. The largest was British, followed by 400,000 Irish. There were 260,000 Scottish and 2,000 Jews. Some ethnic groups

were concentrated in particular areas; a third of Pennsylvanians were of German decent and New York had 80,000 people of Dutch decent; 19,000 Swedes and 9,000 French Canadians are scattered throughout several states.

In the mid-1800s Ralph Waldo Emerson wrote this about coal:

“Every basket is power and civilization. For coal is a portable climate. It carries the heat of the tropics to Labrador and the polar circle; and it is the means of transporting itself whithersoever it is wanted. Watt and Stephenson whispered in the ear of mankind their secret, that a half-ounce of coal will draw two tons a mile, and coal carries coal, by rail and by boat, to make Canada as warm as Calcutta; and with its comfort brings its industrial power.”

In 1822, Dr. William Beaumont, a U.S. Army surgeon stationed at Fort Mackinac on Lake Erie, was called to a remote trading post on the Canadian border to treat Alexis St. Martin, a nineteen year old trapper who had been the victim of a shotgun wound to the abdomen. He recovered, however he was left with a permanent abdominal fistula or tunnel connected into his stomach from the outside (it had to be plugged during meals to keep the food from extruding to the outside):

“On August 1, 1825, at 12 o’clock, I introduced through the perforation, into the stomach, the following articles of diet, suspended with a silk string, so as to pass in without pain: viz. a piece of high seasoned a’ la mode beef; a piece of raw, salted fat pork; a piece of raw, salted, lean beef; a piece of boiled salted beef; a piece of stale bread; and a bunch of raw, sliced cabbage.”

The cabbage and bread were digested first, then the pork and boiled beef; the boeuf a’ la mode took the longest. Beaumont’s studies revealed that the digestive processes worked, not by fermentation as it was commonly believed, but rather by stomach juices (acid) dissolving the food.

Coal was not just another simple fuel, and “no mere article of

commerce." Coal represented man's victory over the elements of nature - it became the foundation of civilization itself:

"With Coal, we have light, strength, power, wealth and civilization; without Coal we have darkness, weakness, poverty, and barbarism."

For centuries, Britain led the world in coal production, and as a result, the industrial revolution started there; Britain became the most powerful country on the planet, it created an industrial society, one that no one could have imagined - the genie was out of the bottle!

In the United States, coal transformed an empty wilderness into an industrial superpower at warp speed. Without coal we would have wallowed longer in the poverty, backbreaking hand labor and the ignorance of the pre-industrial world; however, we might have discovered a more graceful, a more gentle, and a more friendly path to modernization, than the one we eventually took.

Michael Faraday, the scientist acclaimed by all the world, had little formal education and had taught himself everything. Since the 1820s his had been a career of "marvelous discoveries in many different fields, that of induced electricity appearing as the most remarkable of all."

"With his horseshoe magnet and disc rotating between it's poles, Faraday, had demonstrated before the Royal Society in 1831 (some 41 years after Franklin's death), how mechanical energy could be converted into electrical energy - thus presenting to the world, the first miniature model of the dynamo (generator.)"

Faraday published a two volume collection of his experiments and theories called, *Experimental Researches in Electricity* - Thomas Edison, as a young boy devoured Faraday's written works on electricity and formed his views on electricity based on the work of Franklin and Faraday.

Faraday's "dynamo" seemed to provide the answer to the problem of a cheap supply of electric current. This early dynamo, powered by steam engines, was gradually improved over time by many inventers and tinkers - by the 1860s lighthouses along England's coast were lit by giant arc lights.

In 1876, at the Philadelphia Exposition, Moses G. Farmer exhibited three arc lights powered by his privately built dynamo. In 1877 a newspaper reported that Paul Jablochhoff, a former Russian officer of engineers had invented "electric candles" which were introduced as street lamps in Paris.

Coal, steam and electricity now had a toe hold in the daily life of human culture - it was clear that wood as a fuel for the hearth and industry was no longer viable because the maximum rate of re-growth of trees could no longer keep up with the growing demand for fuel and power.

Unfortunately at this point, territorial expansion, financial empires, industrial growth and mass transportation (i.e. - roads, canals, railroad, transoceanic ships, etc.) were the driving force in all industrialized nations - the race was on! As a result of the excitement generated by new sources of power and business growth, no one paid any attention to the human price that would be paid over the horizon of time.

The source of the seemingly endless flow of wood ashes (mineral nutrition) that had supported human development in the fields of intellect, art, industry, health, longevity and science had begun to be pinched off - we had moved from wood to coal as the common fuel, we had moved from wind, water and muscle power to steam and electricity to run our machines, farms and industries in almost a single breath.



Chapter 7

"Genius is ninety nine-percent perspiration and one percent inspiration."

-Thomas Edison

Hell's Fork

The chemical fertilizer industry began with Justus von Liebig, the German chemist, who in 1840 established the beginnings of the farm chemical mega-industry. Liebig gained the support of scientists of the day and the "practical" farmer looking for quick yields. Liebig burned various food crops in the laboratory and analyzed the resulting ash, and he found that the minerals phosphorus and potassium were common to all. Liebig created the first NPK or nitrogen, phosphorus and potassium fertilizer - this was the beginning of the NPK school of thought in agriculture.

The new commercial chemical fertilizers were initially used by the larger commercial farms of Europe, however, in the 1800s, eighty five to ninety percent of all food was still produced by family operated subsistence farms and wood ashes from cooking and heating fires and animal manure were still the principle source of garden and field fertilizers - the general population of the world had not been divested of their historical source of minerals just yet.

Thomas Edison

The 1830s and 1840s ushered in the "Canal Craze" and the Erie Canal carried pioneer families and great quantities of goods into the middle of America. Villages, towns and cities bankrupted themselves

to attract the builders of canals that gave the fledgling nation a mass transit and heavy freight system.

In 1847, Thomas Alva Edison was born in Milan, Ohio. Driven by the industries supported by the Erie Canal, Milan was a boom town. Six hundred wagons arrived each day from a 150 mile radius, filling twenty barges with 35,000 bushels of wheat! Fourteen grain elevators lined the basin - almost all roofed with shingles manufactured by Sam Edison, "Little Al's" father.

By the age of eight, "Little Al" was described by one of his teachers as a little "addled." Edison recalled, "I was always at the foot of the class, I used to feel that the teachers did not sympathize with me."

Edison missed years of school as a result of chronic upper respiratory infections, one of which left him with significantly damaged hearing (there are many stories on how Edison lost his hearing including being boxed on the ears and having been pulled on a moving train by his ears). He was the kind of person who learned by actual hands on experience rather than rote learning - perhaps he was a victim of ADD.

One of Al's daily household chores as a small boy was carrying the wood ashes produced by the cooking and heating fires to the outside and he then put them into the garden - the common kitchen chore of the day.

America, by 1851, had progressed, in the words of Walt Whitman, to "a nation of which the steam engine is no bad symbol." To the American people, still in the pioneering stage of the "New World," - "every new method which leads by a shorter road to material comfort, every machine which spares labor and diminishes the cost of production....seems the grandest effort of the human intellect."

In 1851, the typical American pioneering characteristic included ingenuity and the manual skills necessary to produce food in the rocky farm land of New England. Subsistence farming in Massachusetts and Connecticut turned to a variety of cottage industries to produce additional income streams.

Americans produced good glass products, pottery and kitchen utensils of copper and iron. Shipyards flourished, the art of naviga-

tion was advanced, the textile industry grew, gun makers and clock makers where amongst the respected trades and tradesmen.

The American farm boy (the "whittling boy") such as John Fitch and Eli Whitney, constantly dreamed thoughts of tools, machines and practical appliances, were free to leave behind the intellectual and creative shackles of European restrictions and old ways and try new methods of farming, tool making and business in general. This pioneering environment engendered the appearance of the "Yankee inventor."

"Even though their means and (scientific) knowledge were limited, and troubles beset them at every step....(they) founded most of our important industries." According to E. L. Bogart, an economic historian, "invention became a national habit."

By 1851, the revolutionary invention of the cotton gin, sewing machine, steamboat and the telegraph which were featured at the London Industrial exposition, drew world wide attention.

According to Tocqueville, "the young American grew up ignoring the general laws of mechanics, but rather, directed their thoughts to the purely practical part of science."

The Erie Canal was a route by which the skills and products of the Yankee inventors of the American northeast moved toward the Midwestern frontiers. European tourists were awed by the unique skills of the Americans, including Cyrus McCormick, inventor of the mechanical reaper, who "revolutionized" agricultural implements, while in Europe, agricultural tools and methods continued to remain the same since the time of the Roman Empire.

Large numbers of American women "liberated themselves from the bondage of needle and thread by acquiring Mr. Howe's sewing machine; many also utilized new-fangled egg-beaters, apple-peelers and clothes-wringers flabbergasting European visitors."

Even in far-flung outposts, "Americans had a passion for rapid communications and intelligence, reading many newspapers that arrived by post, carrying on commerce by means of the newly invented electromagnetic telegraph."

In 1853, Edison at age six, started a fire in his fathers barn, "just to see what it would do," - Little Al received a severe public beating

at the hands of his father as a punishment and as an example for other children who might have contemplated mischief.

Little Al's father was a jack of all trades, however, he initially made his fortune in the shingle manufacturing, lumber and feed business. The collapse of the family fortune - "the lake shore railroad came in 1853, bypassed Milan and drove through Norwalk to Toledo with the result that commercial traffic followed the line of the railroad and was diverted from Milan, which eventually lost 80 percent of it's population."

Sam Edison's financial collapse resulted in the whole family moving north where Sam became a light house keeper in Port Huron, Michigan; young Edison's mother, Nancy, took Little Al out of public school and took over his education. She constantly read to him, going over the classics such as Gibbon's, the *Rise and Fall of the Roman Empire* and Sear's, *History of the World*.

At age nine, when Little Al asked questions that she could not answer she gave him R. G. Parker's, *A School Compendium of Natural and Experimental Philosophy*. The book illustrated basic home experiments in chemistry and electricity - Little Al carefully ran each experiment.

At age ten, Edison began reading Benjamin Franklin's notes on electricity, including the details of generating static electricity, and the turkey electrocutions. Edison tried the trick of vigorously rubbing the fur of two huge tomcats, whose tails he had attached to wires - the shocked cats severely scratched him.

Edison, at the age of eleven, partnered up with another boy to start his first commercial enterprise. The two friends prepared a large garden to grow and sell vegetables. They tilled the soil, laid in wood ashes and animal manure for fertilizer, seeded the rows, watered and hoed weeds. It didn't take long - Edison was soon driving a horse cart around town selling onions, lettuce, cabbage and peas. The first summer's income was \$300.

Edison did not get excited about farming and "hoeing corn in the hot sun was unattractive." He abandoned the fledgling farm.

At age thirteen, Little Al's first paying job was to sell newspapers on the three hour train trip from Port Huron to Detroit selling copies

of the *Detroit Free Press*. He also multitasked by selling fruit, peanuts and sandwiches to the passengers - he learned the value of delegating by paying other boys to sell his snacks in other rail cars he couldn't get to during the trip.

He developed a habit of hanging around the editorial offices and composition rooms of the *Free Press* to glean early news to accelerate his newspaper sales. At the end of the first year in the Civil War, on April 6, 1862, his diligence paid off, he hit a home run - the proof of the next days headlines featured a story that said as many as 60,000 soldiers had died in the battle of Shiloh (actual deaths were 24,000). He had only enough money to buy 300 papers so he boldly walked into the offices of the *Free Press*' famous editor, Wilbur B. Storey and procured 1,000 copies on credit.

Edison had already bribed the railroad officials to wire ahead to all the station managers that a major battle had taken place. He was mobbed at every station and the papers sold out even though he raised prices - his net profit was an outrageous \$150, six times the annual income of most Americans at the time.

During the following summer, Edison snatched a three year old boy from in front of a run away boxcar, and the overjoyed father offered him telegraph lessons as a reward. Five months latter, Tom (as he grew older Little Al preferred to be called Tom) traveled throughout the Midwestern United States as a "tramp" telegraph operator with hundreds of other boys.

The young "tramp" telegraph operators were known for gambling, cursing, drinking, smoking, playing jokes and hanging out with women - they had time, money and all the adult men were off fighting the war. Edison was particularly fond of chewing tobacco (pica), gambling and played jokes on his fellows; however, he spent most of his time reading in deserted boarding houses and tinkering with telegraphic devices in railway stations on the night shifts.

Between 1857 and 1858, young Edison rejected reading and writing, when an opportunity arose to play with telegraphs, batteries and chemicals - he developed an obsession for learning everything that was known about the "new science" of electricity. As he worked with his early telegraph constructed of scrap metal, "Edison the boy

approached, unwittingly, the mainstream of electrical experimentation since the days of Benjamin Franklin.”

In 1863, Charles Banting, an obese undertaker, noticed that obese people oftentimes died younger than slim people. He gave up carbohydrates and promptly lost 100 pounds - he was so excited about his rapid weight loss that he published a small pamphlet - *How to Eliminate Corpulence* (obesity).

This pamphlet was the first public instruction on how to lose excess weight - this sentinel case was the harbinger of our future obesity problems and yet physicians failed to take note.

In 1868, a friend wrote Edison from Boston - he wrote back asking if there were any jobs. The response said Edison should come to Boston with all haste and appear in the Western Union office in that city.

Edison arrived in Boston in 1868, after having held jobs with the Western Union Company and the military. He was driven continuously to insomnia by his ponderings on electrical and telegraphic equipment - magnets, springs, cylinders, rotating gears, armatures, batteries and rheostats filled his thoughts. He rented Charles Williams Jr.'s instrument workshop that had previously been occupied by Alexander Graham Bell. Edison, who had become an electrical “addict,” modernized the stock telegraph tape printer and collaborated with other telegraphers to sell his machines and a stock-and-gold quotation service.

About this time, Edison invented an electric rat killer (using the same techniques that Franklin had used to electrocute turkeys) that consisted of two plates placed on the floor and connected to a storage battery - when the rats stepped on the two plates they completed a circuit and were immediately killed. He later produced a smaller version to kill cockroaches - the predecessor of the electrical blue light mosquito and fly zapper.

In October of 1868, Edison filed for a patent for the first electric vote recorder - the patent, No. 90,646, was granted in June 1, 1869. As a telegraph press reporter, Edison had noticed a great waste of time in recording the voice votes during legislative sessions of Congress, he conceived of a time saving device which he whittled out of

wood with his own hands. Edison's crude vote counter would be the fore runner of the 20th century voting machines - in Edison's words:

"In front of each member of the House (there were) two buttons, one for aye and one for no. By the side of the Speaker's desk, was erected a square frame, in the upper part of which there were two dials, corresponding to the two classes of votes. Below the dials were spaces in which numbers appeared. When the vote was called for, each member pressed one or another of the buttons before him and....the number of votes appeared automatically on the record. All the Speaker had to do was to glance at the dial and announce the results."

Edison immediately took his voting machine to the Massachusetts Legislature, where he presented his idea to a committee chairman - he was sure he would reap the handsome sum of \$50,000 dollars. To his total dismay, the chairman immediately rejected his machine saying, "it would infringe on the sacred right of the minority to filibuster!"

Boston failed to meet the financial demographics necessary to grow his business so Edison moved to New York. In 1869, Edison arrived in Manhattan (now aged 22) and found a job with Dr. Sam Law's Gold Indicator's wire service as a piecework assistant.

Almost as soon as he started, the main office device failed and broke down - Wall Street came to a screeching halt and hundreds of brokers' runners screamed at the office's door for information. All of the office's experts failed in their efforts to repair the machine - Edison pulled off his second coup by promptly fixing the device which had failed because a small spring coil had broken off and jammed the drive gears. Edison, at age 22, was now elevated to hero status in the financial world and the rapidly expanding growth industry of electronics.

A short time later he went to the directors of a Western Union subsidiary to offer a machine that connected stock tickers from outside offices to the main machine in the home office - they paid him a blinding fee of \$30,000!

He stretched himself and contracted to provide private telegraph

machines, electrical equipment and 1,200 fast model stock tickers for Western Union. He formed a partnership with a Newark machinist, and by laboring for sixteen hour days and surviving on coffee, apple pie and cigars he fulfilled his contract and delivered all his machines on time.

Edison bought out his partner. To keep the cash flow going he worked as a foreman, supervising fifty or more pieceworkers - at night he built a laboratory equipped with the most up to date scientific instruments. He slept in his chair for six weeks while he consumed huge stacks of current scientific literature and he conducted experiments day and night.

In the early 1870s, Edison hired three individuals who proved to be crucial to his future successes - Charles Batchelor, a British textile machinist; John Kruesi, a Swiss clockmaker; and Edward Johnson, a railroad and telegraph engineer. Batchelor could translate Edison's crude drawings into a detailed drawing; Kruesi built working models that were presented with patent applications; and Johnson would complete the detailed patent applications, production contracts and payroll.

In 1871, J.B.H. Dumas, proved through a terrible human experiment that an artificial diet with only the known nutrients of the day was missing something and could not by itself support life.

Responsible for the care of a nursery of starving children and having no food, Dumas made an artificial liquid diet. Even though, Dumas' diet contained the known proteins, carbohydrates and fats the diet failed to support and nourish the children. Laboratory investigations confirmed his concerns, mice could thrive on a milk diet; however, the mice could not survive on milk substitutes composed of fats, proteins and carbohydrates.

Feeding mice on a mix of compounds found in milk (proteins, fats, carbohydrates and salts), Nicolai Ivanovitch Lunin (1853 - 1937) demonstrated that the mice died; he concluded that there must be some small quantities of unknown factors in milk.

Edison's 1872 notebook contained more than 100 sketches of revolutionary electronic and mechanical devices and that year he won 34 patents. In 1875 Edison gave his 71 year father a task - find a

piece of property on which he could build a laboratory.

In 1873, the fifty four year old Lydia E. Pinkham, of Lynn, Massachusetts became America's first millionaires. She manufactured and marketed, Lydia Pinkham's Vegetable Compound as a cure for "female weakness" - the product was soon being touted as "the greatest remedy in the world." It was favored for dealing with dysmenorrhea, PMS and the hot flashes and night sweats characteristic of the "female change of life."

In March 1876, Edison built his thirty foot by one hundred foot laboratory building in Menlo Park. Every room on the first level displayed a banner sporting a quote from an English painter, Joshua Reynolds, "There is no expedient to which a man will not resort to avoid the real labor of thinking." Each clock had it's spring removed, the length of the work day would be determined by Edison, who frequently would work for 24 hours stealing only cat naps on the floor and then sleeping for eighteen hours.

In 1876, Alexander Graham Bell invented the first telephone. Born in Scotland, Bell started is career as an elocution teacher, working with deaf children.

In 1877, Thomas Edison built his first version of the phonograph; he published an article proposing ten uses for his new invention. They included preserving the last words of dying people; recording books for blind people to listen to; announcing clock time; and teaching spelling.

Reproduction of music was not seriously considered as a use of the phonograph by Edison. A few years later, Edison complained to his assistant that his invention had no commercial value. Within a few years he had a change of heart and finally entered into the office machine business with his phonograph.

Other entrepreneurs created "jukeboxes" by arranging for a phonograph to play popular music for the price of a coin. Twenty years later, Edison finally agreed that the main use of his phonograph was to record and replay music.

In August 1878, Edison returned to Menlo Park from a vacation (his employees described him as highly animated and excited). He had left in July sick and exhausted to the Rockies to view a total

eclipse of the sun with a group of scientists and academics. One academic, George Baker of the University of Pennsylvania, had talked excitably about a system of lights an inventor, Moses Farmer, had set up at an Ansonia, Connecticut foundry. They were known as arc lights, because the light was produced by an arch of elongated sparks reaching between two carbon rods the diameter of pencils.

Bright as searchlights, arc lights had been employed since the 1860s in British and American Lighthouses and certain public meeting places. The arc lights were too bright, too dangerous and too expensive to operate to be practical for private home or small business use.

It wasn't the eight large arc lamps and the foundry that generated Edison's high level of excitement, it was the electrical generating system itself - the electric light was produced by a small crude dynamo or generator and the current was miraculously carried by wires one quarter mile to the foundry!

Edison observed for the first time that electricity could be generated in one place, sent over a distance through insulated wires and then efficiently divided and directed to multiple lamps. Edison immediately saw a potential profit.

A reporter for the *New York Sun* who had been invited on the trip, recorded Edison's moment of epiphany, "Edison was enraptured....He ran from the instruments to the lights and from the lights back to the instrument. He sprawled over a table with the simplicity of a child, and made all kinds of calculations. He estimated the power of the instrument and of the lights, the probable loss of power in transmission, the amount of coal the instrument would save in a day, a week, a month, a year, and the result of such saving on manufacturing."

Edison's habit was to reduce the size of machines, to miniaturize them, in order to save energy and reduce heat production. Rather than sending generated electricity through carbon rods to produce an arc light, which was useless for home and small business lighting, Edison transmitted it along a wire to a tiny filament in a small incandescent light fixture. He worked for forty eight hours straight at Menlo Park, "I discovered the necessary secret, so simple that a bootblack might understand it," he crowed! In his usual bold fashion he went

public with his ideas only one week after his visit to Ansonia.

Edison's "wild" claims were widely published in newspapers, "he had not only found the way to create an incandescent bulb, but he also would be able to light the entire lower part of New York with one engine and 15 or 20 dynamos: I have it now! With a process I have just discovered, I can produce a thousand - aye, ten thousand (lamps) from one machine. Indeed, the number may be said to be infinite....with the same power you can run an elevator, a sewing machine, or any other mechanical contrivance, and by means of the (electrical) heat you may cook your food."

More than 100 years earlier, Benjamin Franklin had killed and cooked turkeys with electricity provided by storage batteries. Edison saw the future - he envisioned the world run by electrical devices.

By envisioning a process by which we could cook our food in the near future with electricity, Edison ushered in the leading edge of our HELL'S KITCHEN! How many wood ashes would be left over when we heated our homes and cooked our food with electricity - none! None! Our historical stream of nutritional minerals, used by humans since before recorded history was about to be abruptly shut off.

Unfortunately, there was no one in the forest to hear the tree fall - this was the moment of transition for human health and no one saw the danger coming - no one offered a replacement for the nutritional minerals we were about to lose.

Edison's "secret" was a mental creation rather than a reality - a thermal regulator to cut off or dampen the electrical current to the filament before it melted or burned out. Edison scholars Robert Friedel and Paul Israel underline his audacity: "For Edison, the search for a practical incandescent light was a bold, even foolhardy, plunge into the unknown guided at first more by overconfidence and a few half-baked ideas than by science. To suggest otherwise is to rob the inventive act of its human dimension and thus to miss an understanding of the act itself."

Other electricity hopefuls in arc and incandescent lighting projects had transmitted a large amount of current along a heavy wire to a low resistance filament. The important discovery that Edison had

made, after arguing it out with Charles Batchelor, was to raise the voltage to transmit a small level of current through a thin wire to a high resistance filament. His "secret" was the practical employment of the law proposed in 1827 by a German physicist by the name of George Ohm.

Edison later said, "At the time I experimented I did not understand Ohm's law. Moreover, I do not want to understand Ohm's law. It would stop me experimenting." Understanding the relationship connecting voltage, current and resistance was all important to the creation of the incandescent lamp, and he understood it "intuitively" even if he could not illustrate it in a mathematical formula."

Academics and scientists from America and England who were still caught up in the concept of low resistance and heavier wires, dismissed Edison's project to light New York out of hand as both "scientifically stupid and economically hopeless."

Edison's immediate need was to find a filament of high resistance that he could heat up to incandescence in a clear glass bulb, evacuated of oxygen to an "airless" state to inhibit oxidation. In 1878 Edison was not even close to solving these problems. His friend and lawyer, Grosvenor Lowrey quickly raised \$300,000 to form Edison Electric Light Co. The necessary filament was not as forthcoming as Edison had hoped; he had rejected carbon because it burned up very easily. Platinum wire produced low resistance but did not oxidize and therefore seemed to fit the need. He started with a design that featured long spirals of thin platinum wire, in an effort to increase the filament's resistance, but it was a tedious and extremely hazardous enterprise.

In mid-April, Lowrey brought a cluster of investors into the dark Edison laboratory, where he had set up twelve lamps with platinum filaments connected in a series. Edison instructed John Kruesi to turn on the current gradually. Francis Jehl, an assistant, noted: "I can see those lamps rising to a cherry red and hear Mr. Edison saying, 'A little more juice,' and the lamps began to glow. A little more.... And then one emits a light like a star after which there is an eruption and a puff, and the machine shop is in total darkness."

Batchelor replaced the faulty lamp; the same thing happened a few minutes later. Only Lowrey's eloquence and the steadfastness of

42-year-old John Pierpont Morgan kept the group together.

A second challenge was the ability to produce and keep a vacuum - no one yet had been able to pull enough air out of the bulb. Edison fell back on one of his unique skills, he looked for the right person for the job. He placed a classified add in the New York Herald for a glass blower, which produced an eighteen year old in a little red German student cap.

The burly mechanics had a hardy laugh at the appearance of the relatively small almost feminine Ludwig Boehm; however, his skills produced a better bulb more aligned with Edison's design and he helped design and perfect an original technique to evacuate a bulb by infusing mercury. The technique was tedious and often times frustrating; however, in September, after almost a month of trial, Edison and his team achieved a vacuum that was one hundredth of an atmosphere. Edison was so overjoyed to learn that at this level of reduced oxygen in the bulb that even a carbon stick did not burn up quickly, and the carbon gave a brighter light than the platinum had.

This was the good news. The negative was that resistance to this carbon filament was only about two ohms (which would mean more current, more copper). Resistance could be raised by creating a fine filament in a tight spiral, however, the filament would have to be no thicker than fifteen thousandths of an inch. Edison pushed all his help to chaos, attempting to convert carbon into "reeds" no thicker than thread. Day after Day, night after night, over and over the spiral "reeds" would shatter.

Two hectic, sleepless and fruitless weeks went by before Edison decided to change directions; his new thought was to bake the powdered carbon into a plain cotton thread. On the eighth attempt, on October 21, Batchelor carried the small hair-like thread to Boehm's house to insert into a bulb. "Just as we reached the glass blower's house, the wretched carbon broke," Edison recalled. "We turned back to the main laboratory and set to work again. It was late in the afternoon before we produced another carbon, which was broken by a jeweler's screwdriver falling against it. But we turned back again and before nightfall the carbon was completed and inserted in the lamp. The bulb was exhausted of air and sealed, the current turned on, and

the sight we had so long desired to see met our eyes.”

Thread No. 9, lit at 1:30 a.m., and lasted until 3:00 p.m. - 13 full hours, after which Edison added a stronger battery to boost the light to 30 candles, or three times gas light. They watched the thread-like filament react to the severe heat. The light continued for another hour. It was a crack in the glass that let air in that resulted in a burst of flame and then the lab went dark. Although their carbon-cotton filament was not perfect, the excited team had demonstrated to themselves that a carbon filament in a vacuum could work!

After inspecting the charred filament under the lens of a microscope, Edison set out to find another organic fibrous material, some form of cellulose that would create even more resistance than cotton. By November 16th, the team settled on a piece of everyday cardboard. Edison noted, “None of us could go to bed, and there was no sleep for any of us for forty hours. We sat and watched it with anxiety growing into elation. The lamp lasted about 45 hours, and I realized that the practical incandescent lamp had been born.”

Even at this early stage, Edison was setting up the commercial electricity business in New York, Paris and London. The laboratory team worked day and night, making filaments and assembling bulbs by hand, one by one, so that on New Year's Eve, when Edison held his open house in Menlo Park to a public exhibit he had about 300 bulbs. Over 3,000 people showed up to gawk and ask questions; the academics and experts still refused to be impressed.

Henry Morton of the Stevens Institute, who had been on the Rockies expedition to view the eclipse of the sun, claimed that Edison was selling “a fraud upon the public,” which spurred Edison to further promise that, “He would erect a statue of Morton at Menlo Park and shine an eternal electric light on his gloomy countenance.”

“There is a wide difference,” Edison noted, “between completing an invention and putting the manufactured article on the market,” - however, the marketing of the electric light bulb was the least of his problems. Edison had to invent the entire electrical industry! He had to create an entire system which would include every possible detail - and then commercially produce every part necessary.

Edison had to assemble a central power station; design and manu-

facture his own dynamos (generators) to convert steam energy into electrical energy; ensure a steady electrical current; connect a 14-mile network of underground wiring; insulate the wiring against moisture and the accidental discharge of electrical discharges or "short circuits;" invent and install safety devices against fire; design commercially efficient motors to use electricity in the daylight hours to run elevators, printing presses, lathes, fans, etc.; design and place meters to monitor individual home and business use of the electricity for billing purposes; invent, manufacture and market a catalogue full of switches, breakers, sockets, fuses, distribution systems and lamps.

In 1880 and 1881 Edison established Con Edison and General Electric and put their offices in a double brownstone mansion on 65 Fifth Avenue. For his central power production station, Edison purchased several deserted and broken down warehouses at 25 - 257 Pearl Street in full view of the Brooklyn Bridge towers which were still under construction.

In December 1881, Edison began digging up paving stone to lay conduit to carry the wires outward from the Pearl Street power generating station. Edison frequently was on his hands and knees in the trenches to check the connections himself.

On Sunday, September 3, 1882 Edison spent the day and night checking and double checking, rehearsing and double rehearsing the completeness and every function of the system for its unveiling and grand opening on Monday afternoon. "The gas companies were our bitter enemies, ready to pounce upon us at the slightest failure," Edison said.

When John W. Lieb, the chief electrician pulled the power switch on Monday September 4th at 3:00 p.m. only one of the six dynamo sets worked and the steam engine coughed and ran roughly for failure to be warmed up early; but Edison, sitting in the distant offices of Drexel, Morgan & Co. was ready for the historic moment when he would ceremoniously turn on 106 lamps, was not disappointed - they all snapped on and brightened the surrounding office. They all came on; 100 percent came on! At the offices of the *New York Times*, "In fairy tale style," fifty two filaments came to life and lit the after-

noon and well on into the night.

Only a dozen or so people had gathered in the warehouse to observe the dynamo open up a new chapter in human enterprise. The lights went on in the first district Monday September 4, 1882 at 3:00 p.m. in the full sunlight of a late summer afternoon, which resulted in a "rather feeble effect."

The event was anticlimactic - despite the optimistic talk of a central station that would light 2,500 to 10,000 structures by means of a single circuit, the Edison Electric Illuminating Company, opened for business with only eighty five customer's fully wired and bragging only 400 lamps.

The Pearl Street facility costs to that date came to a total of \$600,000. The first power station staggered forward with only one generator, which worked rather poorly - the advanced engines were not to arrive for two more months.

"Look, the Edison lamps have been turned on," observers simply said, when the first city power station turned on - "and that was that."

It was only later, after many months had passed that people began to realize what Edison had created at the dusty Pearl Street warehouse. "It was in fact, nothing less than the marriage of steam and electric force," said T. Comerford Martin, a long time Edison friend. The world was now rapidly moving from the steam age of James Watt into the electrical age of Thomas Edison.

It had taken more than 1,500,000 years to move from the first taming of fire to steam and then only 100 years to move from steam power to electrical power. The simple unassuming wood ashes that had carried the health and development of humans forward for so long were now becoming "an unnecessary drudge."

More than fifty years had come and gone since Michael Faraday had discovered the "mechanical production of induced electricity and had been possessed by a vision of the future (of) electric power." The future needed Edison, however, as well as Ohm, Faraday, Ampere, Arago, Franklin, Newcomen, Watt and others to bring the world into the Electrical Age. A London newspaper said it all - "There is but

one Edison.”

In Edison's day, the most revered fertilizer for home gardens was an equal mix of wood ash produced in fire places and stoves, bone meal and animal blood meal obtained at slaughter houses. The wood ashes in this mix would be removed from the stove's ash pan every morning and placed directly on the garden or added to animal manure or compost fertilizers.

Wood ashes were also used as “culinary” ashes and used to cut the more valuable salt 10:1 or 100:1 since before written history - in effect providing a constant stream of essential minerals for the growth, health and longevity of humanity

The fork in the road to Hell that we had been looking for had been found; it was at 3:00 p.m. on Monday September 4, 1882 that we took the off ramp and exited from the main road and took the fork in the road to obesity - the wrong path, the fork in the road to Hell. Edison's technician had loosed the seeds of obesity from Pandora's Box when he loosed commercially generated electricity!

The road to obesity wouldn't be free, it would be an expensive toll road to Hell - it wouldn't be a freeway, we would pay a grave price to stay on the road that would bring us to obesity, diabetes and the Metabolic Syndrome - \$117 billion per year. By 2004, sixty five percent of Americans would be overweight, thirty one percent of Americans would be obese and 1,000,000 new American diabetics each year would become the new metabolic pandemics that plagued the industrialized world (of course Americans led the way).



Chapter 8

"As one of the greatest of technical geniuses, (Edison) became a symbol of our Machine Age, one of the founders of our mechanical civilization, for better or for worse."

-Dr. Le Compte du Nouy

HELL'S KITCHEN

In 1761, the Colonial kitchen, with it's huge stone fireplace, was the center of family and community life rather than the church or parlor. In the other rooms, the rafters could be bare and the floors made of earth or raw hewn boards; however, in the kitchen there was room enough near the hearth for finishing chores, and for children to play and watch their mother cook and prepare meals in large iron pots.

On the hearth's back bar were hooks of varying lengths called trammels, hakes, pot-claws, potclips and pot crooks. There were also lug-poles, job-crookes, recons, gallows-balke and gallows-crooks. On these devices the housewife hung pots and kettles.

In the kettles and pots were hasty puddings and corn meal porridge, while corn dumplings were baked on the side - "if the colonialist travels he takes with him no-cake, or Indian corn, which is (an ear of) corn parched in ashes and carried in a leather bag."

Since the advent of commercial electrical generation in 1882 by Edison, there was a persistent, incessant and accelerating search for more uses of electrical power. Some of the more interesting inventions of progress included electrical, coal, gasoline, oil, natural gas

and propane kitchen appliances. They were marketed as instruments of progress “that would free the American house wife from the old drudgery and the dirt associated with the starting of the kitchen and heating wood or coal fires,” these new stoves were cleaner and more efficient than the old wood stoves - “no more soot to clean from your pots and pans.”

Early Stoves That Replaced Wood Stoves

Date	Style
1862	Ship Camboose or Cabin Stove F.D. Chase, Boston, MA (wood, coal)
1872	Air Tight Stove C.H. Tyler, Greenfield, MA (Wood, coal)
1878	Masonry Stove (attached to chimney) Menanite Kitchen, American Agriculturist (wood, corn cobs)
1882	Troy Charter Oak (Since 1854) The Metal Workers (wood)
1890	Oil Stove Perry Stove Co., Albany, NY
1891	All Cast Iron Ranges (wood, coal) Buckwalter Stove Co.
1895	Wood Stove Sears Catalog
1914	Electryl Electric Cooker Stove (Purcell & Nobbs) London, England
1914	Maud Lancaster's Electric Cooking Gilbert, England

The new electrical technology was proof that things could only get better; however, no one even considered the possible side effects on the world when we made the conversion from wood and coal fuels to electric, gas and propane.

Where would our nutritional minerals now come from? We still needed our essential minerals, yet no one seemed to worry about the looming deficit. We had been using the wood ash kitchen byproduct for thousands of years as a supplement combined with table salt, "culinary ashes" and a fertilizer for our gardens - it would be gone in but a few short years and we would be put metabolically naked into the ovens of HELL'S KITCHEN!

When one looks at the time line following the 1882 quickening of the commercial electric industry, the more frightened we become. As we achieved the next plateaus of scientific achievement, including "medical miracles," nuclear power, space technology, etc., we sped up technological progress, but we left our primal human needs for mineral nutrients in the empty kitchen ash bin.

The result of our failure to appreciate our loss, our failure to replace the ash minerals we had depended on for thousands of years has become a whirlwind of obesity, diabetes and the Metabolic Syndrome.

In 1866 Nikolaus Otto invented the first gasoline engine; at this time horses had been supplying power for carrying loads, plowing and pulling wagons for 6,000 years. As Otto's first engine was weak and seven feet tall it didn't initially look like it could replace the horse.

In 1874 Eli Whitney invented the cotton gin to end the labor intensive job of cleaning the seeds and hulls from the cotton bolls.

In 1874, an Austrian chemistry student created a new chemical compound, dichloro-diphenyl-trichloroethane, that he called DDT.

On May 10, 1876, the United States opened it's first centennial celebration at the grand Centennial Exhibition in Philadelphia. Over all, 9,000,000 people would visit the fairgrounds. One of the most popular attractions was a single gigantic coal fired steam engine that powered literally thousands of mechanical inventions set in the exhibition's Machinery Hall. At the grand opening, shortly after a

choir sang Handel's "Hallelujah Chorus" to an enthusiastic crowd of over 100,000 people, President Ulysses S. Grant stepped up to the giant engine and pulled its control lever.

With a hiss and cloud of steam, the engine's cross beam began to rock, its two huge pistons began to work, and eight miles of connecting shafts running through the hall began to turn. A huge cheer rose from the crowd as fourteen acres of shiny new machinery sprang loudly to life at the same moment spinning, sawing, sewing, pumping and printing without the need for the control and the muscle power of human hands - America became an industrial power, America joined the industrial revolution. America was headed faster and faster into HELL'S KITCHEN.

In 1885, the gasoline engine increased in efficiency to the point where Gottfried Daimler installed an engine on a bicycle to produce the first motorcycle; he waited until 1896 to build the first truck.

In 1877, fingerprints were first invented and used by William Herschel; he worked for the Indian Civil Service, he used "thumb print signatures" of army pensioners in Bengal to ensure they only collected their money once.

In 1888 the first electric transformer was invented, thus extending the distance, that electrical services could be provided.

In 1895 the first electric locomotive starts commercial operation.

In 1893 - 1895 recorded the appearance of the first gasoline powered automobile. Developed by Charles E. and J. Frank Duryea, and incorporated by almost all other automobile manufacturers, they first exhibited their automobile to an amazed public in 1900 at the first National Auto Show at Madison Square Garden, New York.

In 1897, British physician, Dr. Ronald Ross, proved that malaria was spread by a mosquito of the genus *Anopheles*.

In 1897, the Sears Roebuck Catalogue listed iron kitchen stoves that burned wood, soft coal or hard coal, gasoline stoves and oil stoves - the relentless move away from wood as a kitchen and heating fuel had begun in earnest:

"The Merit Sunshine is a substantial, well constructed stove of improved design, intended to meet the demand for a well made and attrac-

tive stove at a low price. The design is new and neat, and workmanship is equal to the best. The special features of this stove are: Cut top plates with heavy, deep edges; heavy rim covers and centers; heavy grate and fire box lining, dumping and shaking grates; nickel knobs and hinge pins; handsome nickel panel on oven door and on front doors; outside oven shelf; tin-lined oven door; improved, outside oven, with broad rack." Sears and Roebuck Catalogue 1897: 355 lbs. \$17.40.

"The Giant Acme Gasoline Stoves, with lay-down tank, has latest, best and most perfect burner yet made: lights and burns without smoke or odor, giving blue process flame at great reduction in price. It is operated by one valve, very simple; any child can operate it; has loose grate; safety lay-down tank; when tank is filled it must be lowered and when lowered all fires are extinguished immediately. Beautifully ornamented with full nickel name plate. Fully guaranteed, and can be returned at our expense if it does not do the work represented." Sears and Roebuck Catalogue 1897. 30 lbs. \$7.13.

"The Acme Central-draft, (three burner), round wick, oil cook stove is an embodiment of art and utility. They are modern in design, honest in construction, perfect in combustion. Elaborately finished in nickel and sure to give the best of satisfaction. We have spared neither labor nor money to make the the Acme Cook Stoves the finest finished, most durable and best operating oil cook stoves in the world. Modern ideas and improvements make them the most convenient and economical stove ever placed on the market. Our prices are always the lowest, quality considered" Sears and Roebuck Catalogue 1897. 35 lbs. \$5.50.

In 1900, Karl Landsteiner, an Austrian physician discovered that each human has a slightly different blood "type." He identified at least three "types" or "groups" of blood and called them A, B, and O. His discovery explained why people react in different ways when given a blood transfusion; if the donor blood does not match the recipient blood the red cells can clump together causing blood clots and death.

Yellow fever was proven to be spread by a mosquito of the genus

Aedes in 1900. Studying a yellow fever epidemic in Cuba under the watchful eye of U.S. Army surgeon Dr. Walter Reed, British physician Dr. James Carroll allowed an infected mosquito that had fed on yellow fever victims to bite him - his theory was confirmed when he developed yellow fever (he later died of the cardiac complications of yellow fever).

The 1900 Sears and Roebuck Catalogue no longer carried the gasoline or oil stoves - the petroleum stoves made food taste like gasoline or kerosene and they blew up. The Acme Stoves that used wood and coal hung on in the Sears Catalogue, however, the price went up to \$31.05.

In 1901 the police of Scotland Yard in London opened their official fingerprint department. Using a simple system of fingerprinting techniques, the police collected comprehensive records of convicted criminals. The technique was later improved by Sir Henry Galton, and developed by Edward Henry, chief commissioner of the London Police, who used the technology to offer a sure way of identifying criminals.

In 1901, Dr. Robert Koch, a German bacteriologist proposed a theory that it was rats that carried the bubonic plague.

In 1901, Dr. Gerrit Grijns, a Dutch doctor, has shown that Beriberi, a disease that affects the nervous system and is characterized by weakness, numbness, and paralysis is caused by the deficiency of a nutrient (thiamine - B1) removed when white rice is polished (removal of the bran).

The formal opening of the Texas oil fields in 1901, meant that gasoline was plentiful and cheap, making cars the new "king of the road."

In 1901, Wilhelm Conrad Rontgen received the first Nobel Prize for physics for a chance observation that lead to the discovery of the X-ray in 1895. His most famous experiment involved placing his wife's hand between the X-ray source and a photographic plate, resulting in the production of the first permanent X-ray of the human body. An estimated 3 million X-rays were taken in 1901.

Dr. George Perthes observed that X-rays can be used to reduce the size of cancerous tumors.

In 1901, the McCormick Harvesting Machine Co. Chicago, Illinois mailed out calendars that still featured a farm family warming themselves in their living room with a wood burning open fireplace.

In 1902, Landsteiner discovers a fourth blood group - AB.

In 1902, Ivan Pavlov, a Russian physiologist uncovered a fundamental type of nervous system response that he believes is the root of many aspects of human behavior - he called it the "conditioned reflex." Pavlov's theories are based on experiments in which biologically important events, such as feeding time for a dog is accompanied by a signal, such as the chiming of a bell. After a few days, the dog begins to salivate at the sound of the bell, whether or not any food is present.

William Maddock Bayliss and Ernest Henry Starling identified the first known hormone - secretin, produced by glands in the small intestine (by 1905 they soon called these chemical messengers hormones from a Greek word meaning to "rouse to activity").

In 1903 the American Ford Motor Company sells its first "Model A" automobile to a doctor in Detroit.

The German pharmaceutical giant Bayer, marketed a new pain-relief drug in 1903 under the name "Aspirin." The drug was synthesized by a Bayer chemist, Felix Hoffman, six years earlier, when he was searching for a remedy to relieve his father's arthritis pain. It was also suggested that Aspirin (acetylsalicylic acid) could also relieve headaches. A related compound, sodium salicylate (extracted from willow bark), was used for over twenty five years as a pain reliever for arthritis, however, it frequently produced gastric bleeding.

In 1905, Dr. Albert Einstein, a twenty six year old working in the Swiss patent office in Berne, developed the "special theory of relativity." The theory is based on the concept that the velocity of light remains constant regardless of the relative motion of its source or an observer.

In 1905, German zoologist Fritz Schaudinn and a colleague, dermatologist Erich Hoffmann, discovered the bacteria that caused syphilis; the organism has been named *Spirocheta pallida*. It had just been confirmed that syphilis can be transmitted from mothers to their babies and by sexual intercourse.

In 1906, Will Keith Kellogg founded a company to market a "healthy" breakfast called "corn flakes." The rolled and toasted wheat and corn "flake" was developed by Will Keith's brother John Harvey Kellogg, an "alternative healer" at the Battle Creek Sanitarium in Michigan. The Kellogg brothers believed that their revolutionary breakfast food would replace the traditional American fried breakfast.

In 1906, Frederick Hopkins, an English chemist, was the first to suggest that many foods contain "accessory factors," substances apart from fats and proteins that are essential for health. Vitamins are chemical substances, co-factors that are needed for growth and good health.

In 1907, the tungsten filament incandescent bulbs went on sale and began to replace Edison's original carbon filament. Tungsten filaments have an extremely high melting point of 6,170 degrees F.

In 1908, headlines state, "Ford's production line paves the way for autos for all." Ford, determined to capture the market for automobiles that more people can afford, Henry Ford has adapted and improved the mass production systems pioneered by Ransom Eli Olds.

Parts are manufactured in separate locations and delivered into the main factory so that technicians could assemble the complete auto. It takes just slightly more than twelve hours to assemble each car. Ford produces two models of his Model T, selling for \$850 and \$825 respectively. It still wasn't cheap in terms of 1908 dollars, however, Ford continues to develop ways to lower costs.

French physician, Dr. Claude Mantoux developed a skin testing technique for determining if an individual was infected with tuberculosis (consumption) or exposed to it. The technique features an injection of a sterile extract of the TB organism, called tuberculin, intradermally (between the layers of skin). If redness and swelling develop at the site of the injection, it indicates that the individual either has an active infection or has been exposed.

The first filtered coffee maker was manufactured by Melitta Benz, a German housewife. She lined a can with absorbent paper, punched holes in the bottom of a can, placed fresh ground coffee beans in it and then poured boiling water through it, which resulted in perfectly brewed coffee time after time. In 1909, she exhibited her filtered coffee pot at the Leipzig trade fair.

In 1909 the General Electric Company of Schenectady, New York began to market their first electric toaster. The toaster has a heating element made of wires that glow when plugged into the wall jack. The bread slice was laid on the wire elements, when one side is browned, the bread is manually turned over to brown the second side. The downside of this original electric toaster was that it had to be turned over and removed from the heat manually - left unattended the toast would burn.

In the October 1909 issue of the *American Family Journal*, classified ads appeared for dealing with obesity:

Fat is Dangerous

It is unsightly, uncomfortable, spoils the figure, causing wrinkles, flabbiness and loss of vigor. Excess fat weakens the heart, liver and lungs. I want to prove that my treatment will positively reduce you to normal and no matter where the excess fat is located, stomach, bust, hips, cheeks, neck, it will quickly and safely be reduced without exercise or dieting.

Rheumatism, asthma, shortness of breath, kidney and heart troubles will disappear as the fat goes away. - H. C. Bradford, M.D., New York.

Don't Stay Fat

No longer any excuse to be too fat. Try Kellogg's safe fat reducer.

E. J. Kellogg, Battle creek, Michigan

By 1909, ads began to appear in all magazines and newspapers for weight loss programs. The lack of wood ashes in the human diet began to have an effect on the modern urbanite. The effect was obvious enough that it created the opportunity for a commercial enterprise - the embryonic beginnings of the weight loss industry quickened.

Sren Peter Srenson, a Danish chemist, devised a new scale to measure acidity and alkalinity of solutions. The scale is called the

“pH” scale and ranges from 0 to 14. Neutral solutions have a pH of seven; strong acids such as sulfuric acid have a pH of 0 to 2. At the other end of the scale, strong alkalinity such as caustic soda will have a pH of 14. The pH of a solution is a measure of its concentration of hydrogen (H⁺) ions. The term pH stands for “potential for hydrogen.”

By 1910, millions of U.S. homes had telephones. Since Alexander Graham Bell had invented the telephone in 1876, many advances had been made in the telephone technology. Telephone lines and manual exchanges connect distant towns and cities all over North America. Some larger cities have automatic exchanges which allow the caller to dial the number directly without having to engage an operator first.

In 1912, Sydney Russell invented the first electrical heating pad, which a short time later became the electric blanket.

In 1912, Casimir Funk, a Polish American biochemist, called the small essential substances in food vital amines, later called vitamins - he isolated thiamine (vitamin B1) from unpolished rice.

In March 1912, *The Gentlewoman*, displayed classified ads for Weight loss:

Fat Vanishes

One pound a day

New Drugless Method

No more fat folks!

200,000 women made happy with this new knowledge!

My friends were charitable, and they called it “obesity”: others referred to me as being stout, but I knew it was just plain bulky fat! To reduce your weight, you must find the cause, you must get at the very reason.!

Marjorie Hamilton, Denver, Colorado

In 1913, American chemist Irving Langmuir, of General Electric, has developed a way of extending the life of light bulbs and increasing their brightness. He invented a gas filled, coiled tungsten filament lamp bulb. Vacuum tungsten filament bulbs had been manu-

factured for several years, however, at high temperature the filament gradually evaporates and then condenses on the cooler walls of the bulb resulting in the bulb blackening and dimming. Evaporation also thins and weakens the tungsten filament, thus shortening the life of the bulb.

Langmuir discovered that when he filled the bulb with nitrogen, the process of evaporation slowed down. His tungsten bulb also used a coiled filament, which reduces heat loss and evaporation, which increase the efficiency and life of the bulb.

Henry Ford continued to work towards his goal of having every American drive a car. He modernized his manufacturing facility in Highland Park, Michigan, which included a moving assembly line. The automobile bodies slowly move along a conveyer belt and workers on each side progressively added the appropriate parts. This modern process reduced the manufacturing time from twelve hours per car to 98 minutes to complete each car, resulting in considerable cost savings. Ford was able to reduce his Model T's (nicknamed the "Tin Lizzies") price to \$400 per car! It immediately became the cheapest and fastest selling car on the market.

In June 1914, the "*Modern Kitchen*" featured an ad for a six-burner natural gas kitchen stove.

In 1915, Knox Sparkling Gelatin of Johnstown, New York, mailed out free issues of Mrs. Knox's, *Dainty Dish and Dessert Recipe Book*, with receipt of 4 cents postage and address.

In 1916, August Fruehof, a Detroit blacksmith built the first tractor trailer, a truck with a detachable cab powered by a gasoline internal combustion engine. The cab can be attached to a variety of trailers or other farm equipment. The invention eliminated the need for multiple vehicles to operate a farm.

The first electric washing machines were invented by John Fisher, an engineer, who marketed the machine with a promise to "take the drudgery out of wash day." One filled the machine's tub with the laundry to be washed, warm water, soap and turned on the motor and the machine did the work. When the wash was done, the water was wrung from the clothes using the hand cranked rollers.

In 1917, Clarence Birdseye invents the process of freezing food

and begins marketing Birdseye frozen food products.

In January 1918, the *Hearth and Home* displayed ads touting weight loss programs:

Reduce Your Fat Now

We extend this free offer to you, reader, because we want you to learn from your own actual experience what Adipo, the new health giving fat reducer will accomplish in safely and easily taking off excess fat without starving, sweating, ridiculous exercising or other inconveniences to the user. You can eat and grow thin! - We send thousands of these free trial books around the world. - The Adipo Co. New York.

In 1918, the Luminous Unit Co. of St. Louis, Missouri, placed an ad in the *Saturday Evening Post* to market the Brascolite. The Brascolite saturates an atmosphere with light approaching the quality of daylight, it is the ideal work-light for every exacting service - in the laboratory, the drafting room, the library, office, or wherever particular or long eye-use is required. "It is better to let electricity wear the eye - glasses than to put them on yourself."

In 1918, the Cribben & Sexton Company of Chicago, Illinois placed an ad with the *Saturday Evening Post* to market the Universal Combination Range that burned natural or artificial gas, coal or wood:

"You won't have the troubles you now have, if you get a Universal. That's because with the Universal Combination Range you can use the fuel best suited to cooking, baking or roasting you have to do. For quick baking, use gas. For big roasts use coal, or use both coal and gas at the same time, if you want to.

To change fuels, you have only to turn the key little Mary is holding in her hand. Turn it one way for baking with gas. Turn it back again for coal. There are no parts to change as in ordinary combination ranges. And you can't make a mistake, because you have only to turn the key. That's all!" Sold for cash or easy payments.

In 1919, Charles Strite, an American inventor and mechanic from Minnesota patented the pop-up electric toaster. In Strite's toaster, you

push a slice of bread into a slot, where it sits between two heating elements. Both sides of the bread are toasted at the same time, and at a set time a spring drives the toast popping back out of the slot. The toaster has a clockwork timer to ensure the proper toasting time. The toaster was sold under the brand name "toastmaster."

In 1919, the Kellogg's Company of Battle Creek, Michigan placed an ad in the *Saturday Evening Post* to market Kellogg's Toasted Corn Flakes, that was sold only in flavor-holding 'waxtite' packages that "won its favor through its flavor."

In 1920, the National Lamp Works of the General Electric Company of Cleveland, Ohio placed an ad in the *Saturday Evening Post* to market National Mazda Lamps:



A 1919 photograph of Dr. Wallach's grandfather in his farmer's market. Note the wood stovepipe on the left and the Edison bulb hanging from the ceiling (center).

“Now it is possible to buy store lighting on known standards. The Foot-Candle Meter measures exactly the illumination you are actually getting - on the counter where the goods are examined and the sales are made!”

In 1921, Canadian physicians Frederick Banting and Charles Best have successfully isolated the pancreatic hormone insulin. Insulin was used to treat people with adult onset type II diabetes. They created diabetes in laboratory dogs by tying off their pancreatic ducts. They isolated insulin that was produced by the Beta cells of the pancreas, then injecting the hormone into the diabetic dogs, which resulted in a “cure.”

In 1923, Albert Calmette, a microbiologist from the Pasteur Institute in Lille, France, and veterinarian Jean Guerin created and tested a vaccine for tuberculosis. The new vaccine, BCG (Bacillus Calmette-Guerin) is prepared from a weakened strain of the bovine (cattle) tuberculosis bacteria. The vaccine produces a mild infection that increases the resistance to the human tuberculosis bacteria.

In 1924, Willem Einthoven, a Dutch physiologist received the Nobel Prize for medicine for inventing the electrocardiograph, a device designed to measure the electrical impulses of the heart. It has been used to diagnose many different forms of heart disease, irregular heartbeat, high blood pressure and thyroid disease.

The International Harvester Company developed the Farmall tractor. It is designed to be, as its name implies, an all purpose farm tractor that can plow, plant, cultivate and harvest

During the 1920s, a plethora of electric kitchen appliances appeared. On December 10, 1927, Hotpoint advertised electric kitchen appliances featuring an electric range in a full page Christmas ad in *The Saturday Evening Post* for prices ranging from \$2.50 to \$65.00.

In May 1925, the Postum Cereal Company, Ltd. of Toronto, Canada placed an ad in the *Delineator* magazine to market Grape-Nuts baked wheat and oat cereal:

“Can we add years to our lives - increase our strength and vitality - overcome our most dreaded diseases by what we eat? Yes, say the foremost authorities in the world today.

Old age diseases - breaking down of the heart, hardening of the arteries, disease of the kidneys, nervous and digestive disorders - are developing more frequently in younger people with each succeeding decade.

Death from these diseases has increased 100 percent in 30 years. There can be no longer any doubt that our food is one of the most important factors contributing to this condition.

In your choice of food lies the difference between real health - and that dangerous condition of low vitality that leads to trouble."

In 1925, Dr. George Hoyt Whipple, an American pathologist took large amounts of blood from dogs to produce anemia. Whipple fed the dogs varying diets to learn how different diets affect the production of red blood cells. He recorded that dogs that ate liver, which is rich in iron (and copper, selenium, vitamin B12, vitamin A, vitamin C, vitamin D, folic acid and essential fatty acids), formed red blood cells more quickly than those that did not.

In April 1925, the Morton Salt Company of Chicago, Illinois, placed an ad in the *Woman's Home Companion* to market Iodized salt:

"Now - a new table salt that prevents goiter. Save children! Nature's iodine restored - yet tastes no different.

Goiter, which threatens almost every child, can now be prevented by this new salt.

Now physicians know its cause - an insufficient amount of iodine in the thyroid gland of the throat. Give the thyroid the necessary iodine, then, disease cannot grip it.

Watch your children carefully for the signs of goiter - nervousness, depression, sluggishness. Then salt their food this new way."

In April 1925, the American Stove Company of St. Louis, Missouri, placed an ad in the *Woman's Home Companion* to market the Lorain Oven Heat Regulator, which maintains a controlled and timed heat, even while you're miles away:

"The bride of a week can now achieve perfect cooking results with her first try, if she uses a time and temperature recipe and a gas range equipped

with the famous Loraine Oven Heat Regulator.

Don't forget to ask the dealer about the unique Lorain Oven Method of Canning, and about cooking whole meals in the oven while you're miles away."

In 1928, Joseph Schick invents the electric razor.

In 1928, Alexander Fleming, a Scottish bacteriologist working in St. Mary's Hospital Medical School in London, discovers a penicillin mold that kills bacteria. One morning, he noticed that a mold had contaminated one of his bacterial cultures and was preventing the bacteria from growing. He identified the mold as *Penicillium notatum*, and called the antibacterial substance it produced "penicillin."

Fleming isolated the mold, he grew a pure culture of it in a broth - after a few days he was able to prove that the growth medium had acquired the ability to kill a large range of Gram-positive disease causing bacteria.

In 1928, Albert Szent-Gyorgyi, an American biochemist found that capsicum or cayenne peppers are rich in vitamin C.

In 1928, Hugh Hammond Bennett published a U.S. Department of Agriculture Circular:

"Thief of the Fertility of His Soil"

"Not less than 126,000,000,000 pounds of plant food material removed from the fields and pastures of the United States every year. Most of this loss is from cultivated and abandoned fields and overgrazed pastures and ranges. The value of the plant food elements (considering only phosphorus, potash, and nitrogen) in this waste, as estimated on the basis of the chemical analysis of 389 samples of surface soil, collected throughout the United States, and recent selling prices of the cheapest forms of fertilizer materials containing these plant nutrients, exceeds \$2,000,000,000 annually. Of this amount there is evidence to indicate that at least \$200,000,000 can be charged up as a tangible yearly loss to the farmers of the nation. These calculations do not take into account the losses of lime, magnesia, and sulfur."

In 1930, Otto Frederick Rohwedder, an American jeweler perfected a bread slicing and wrapping machine. He had actually invented the bread slicing machine in 1912, however, the inability to wrap the bread mechanically delayed marketing the machine.

In 1931, Paul Kerrer, a Swiss chemist, establishes the structure and function of vitamin A.

On October 18, 1931 Thomas Edison died at the age of 84 as a result of gastric ulcers, Bright's disease (kidney failure with protein in the urine), uremia and diabetes. It is ironic that the man whose electrical technology started the downward path of the health of Americans and people in the industrialized world, should die of diabetes himself.

We learned that adult onset type II diabetes could be "prevented and cured" with simple mineral supplement programs in 1957. The minerals that had sustained us and had prevented diabetes, were originally obtained from the wood ash product of the kitchen and heating fires - Edison's electrical world had eliminated the traditional source of dietary minerals universally used by humans.

In 1932, Charles Glen King, an American chemist isolated vitamin C from lemon juice; it had long been known (since James Lind) that citrus fruit could prevent and cure scurvy; a vitamin C deficiency disease.

In 1933, Tadeus Reichstein has become the first to produce a vitamin synthetically in the laboratory, and his "artificial form of vitamin C is identical to the natural one."

The first all-electric, automated tea-maker, the Teasmade, is introduced in Great Britain.

A carbonated drink, 7 up, formally marketed as Lithiated Lemon is introduced in America.

In 1936, the U.S. Senate entered into the record (as U.S. Senate Document No. 264) a reprint of an article from *Cosmopolitan* magazine, which stated, "Our physical well-being is more directly dependent upon the minerals we take into our systems than upon calories or vitamins, or upon the precise proportions of starch, protein or carbohydrates we consume."

The document's authors went on to state:

"Do you know that most of us today are suffering from certain dangerous diet deficiencies which cannot be remedied until depleted soils from which our food comes are brought into proper mineral balance?

The alarming fact is that foods (fruits, vegetables and grains) now being raised on millions of acres of land that no longer contains enough of certain minerals are starving us - no matter how much of them we eat!

This talk about minerals is novel and quite startling. In fact, a realization of the importance of minerals in food is so new that the text books on nutritional dietetics contain very little about it. Nevertheless, it is something that concerns all of us, and the further we delve into it the more startling it becomes.

You'd think, wouldn't you, that a carrot is a carrot - that one is about as good as another as far as nourishment is concerned? But it isn't; one carrot may look and taste like another and yet be lacking in the particular mineral element which our system requires and which carrots are supposed to contain.

Laboratory tests proved that the fruits, the vegetables, the grains, the eggs, and even the milk and the meats of today are not what they were a few generations ago (which doubtless explains why our forefathers thrived on a selection of foods that would starve us!).

No man today can eat enough fruits and vegetables to supply his stomach with the mineral salts he requires for perfect health, because his stomach isn't big enough to hold them! And we are starting to turn into big stomachs!

No longer does a balanced and fully nourishing diet consist merely of so many calories or certain vitamins or fixed proportion of starches, protein and carbohydrates. We know that our diets must contain in addition something like a score of mineral salts.

It is bad news to learn from our leading authorities that 99% of the American people are deficient in these minerals, and that a marked deficiency in any one of the more important minerals actually results in disease. Any upset of the balance, any considerable lack of one or

another element, however microscopic the body requirement may be, and we sicken, suffer and shorten our lives.

We know that vitamins are complex chemical substances which are indispensable to nutrition, and that each of them is of importance for normal function of some special structure in the body. Disorder and disease result from any vitamin deficiency.

It is not commonly realized, however, that vitamins control the body's appropriation of minerals, and in the absence of minerals they have no function to perform.

Lacking vitamins, the system can make some use of minerals, but lacking minerals, vitamins are useless."

Several pivotal things were happening concurrently at this point in time regarding the health of Americans and the entire industrialized world as well:

- 1) Great numbers of people moved off the land and family subsistence farms and moved into towns and cities to work in factories for cash wages.
- 2) Urban families had eagerly shifted their kitchens and homes from wood (and coal) as a fuel for cooking and heating to electrical, oil, gas and propane appliances - there were no longer any ash byproducts to put in our gardens and mix with our food and table salt.
- 3) Large specialized farms and orchards growing wheat, corn, rice, soybeans, citrus and fruit used simple NPK (nitrogen, phosphorus and potassium) fertilizers for maximum yields - few people put wood ashes in their gardens anymore. Gardens had become a toy of the serious hobbyist, rather than an important source of food and minerals for the average American.
- 4) World War II was on the horizon, technology's star was rising, infectious diseases were being wiped out by

sewage systems, clean water and improved personal hygiene and antibiotics.

- 5) Travel was easy, just about every family had a car; commercial bus, train and even air travel was cheap and widely available; gasoline was 12 cents per gallon.

In December 1937, the Edison General Electric Appliance Company of Chicago, Illinois, placed an ad in the *Ladie's Home Journal* to market the Hotpoint Electric Range:

"This year, in homes all over the land, women are going to receive the priceless gift of freedom! Freedom from long hours of kitchen toil. Freedom from pot-watching, from scouring soot-blackened pans and kettles, and from cooking failures.

Day after day, for years to come, this new Hotpoint Range can be a source of joy and contentment in your home! It's modern - never smudges kitchen walls and utensils. And with the help of the amazing Chef's Brain" (automatic timer clock) it actually cooks whole meals without any supervision!"

In 1938 fluorescent lamps were made available in the United States, delivering a low voltage to gas-discharge lights - they produce approximately four times as much light. The fluorescent lamp is made from a glass tube about two feet long and does not contain a classical filament typical of an incandescent bulb - instead the inside of the tube is coated with phosphor and filled with mercury vapor. This new tube produces invisible ultraviolet light in an electrical discharge - as a result the phosphor fluoresces, or turns the ultraviolet light into visible white light. Thomas Edison's General Electric Company had worked on this new development since 1934.

A new home use steam iron was marketed by Edmund Schreyer of Ridgefield, CT. Schreyer's iron used a jet of steam to dampen the clothes as the clothes were pressed. The new iron was fitted with an adjustable thermostat - one need only to set the required heat for the type of fabric and the iron maintained the temperature.

The Swiss company Nestle begins to market a successful instant coffee.

The first hip replacement operation was performed in humans in 1938.

In 1939, DDT was rediscovered by Paul Herman Muller, a Swiss chemist who discovered DDT's pesticide potential. During World War II, DDT was used to kill fleas and head and body lice. DDT killed insects and bugs in two ways - by simple contact on any part of the insect and secondly when the insect consumes the chemical. After World War II DDT was employed world wide as a crop spray.

In November 1940, *The American Home* displayed a General Electric ad for the electric sink:

General Electric Electric Sink

1. Washes all the dishes!
2. Disposes of all garbage!

“It's easy to stay young electrically!”

By 1940, the electric garbage disposal had all but eliminated household kitchen generated compost.

In 1940, Howard Florey and Ernst Chain, two Oxford University scientists have isolated and purified, from a species of mold, an anti-bacterial substance that they hope will become a useful drug. The drug, “penicillin,” is able to kill infectious bacterium inside the living body. Their discovery of penicillin followed an investigation to identify naturally occurring antibacterial substances. They quickly came across Alexander Fleming's 1928 work on penicillin and the small scale production of the drug.

Vincent Du Vigneaud, an American biochemist, identifies the sulfur-containing vitamin - biotin.

In 1941, a revolutionary type of farm was to cover the countryside - windmills that generate electricity. For centuries windmills had been used to generate power, however, the energy of the wind was transformed directly into mechanical energy to grind grain, pump

water and pull loads uphill. In the 1890s, P. LaCour, a Danish engineer, patents a system of sails that were rigged to a steel tower to power an electrical generator. Smith and Putnam, two American engineers, replaced the sail design with propeller blades similar to those that power an airplane - the windmill was first erected at "Grandpa's Knob" in Vermont, ushering in an environmentally friendly, renewable source of electricity.

On December 7, 1941 the Japanese attacked Pearl Harbor, destroying battleships of the American 7th Fleet and killing almost three thousand sailors, marines and soldiers. We were at war and no one was thinking of vitamins and minerals.

In 1942, the U.S. government set up the Manhattan Project which was commissioned to invent the technology required to build an atomic bomb before Nazi Germany did. The goal was reached in three years, at a cost of \$2 billion dollars (today's dollars would be \$200 billion).

In 1943, Selman Waksman, an American biochemist, discovered, Streptomycin, a new antibiotic that is highly effective in killing Gram-negative bacteria and the tuberculosis bacillus.

Streptomycin is a natural substance extracted from the mold, *Streptomyces griseus*. A mixture of penicillin and streptomycin will be commonly used as a "broad spectrum" antibiotic to kill both Gram-positive and Gram-negative bacteria.

In 1945, the citizens of Grand Rapids, Michigan were subjected to an experiment sponsored by the U.S. Surgeon General: fluoride was added to the drinking water supply to determine whether or not it reduces dental caries or cavities. Earlier, in 1936, Dr. H. Treadly Dean, head of the dental hygiene unit at the NIH, speculated that by adding fluoride to the drinking water in "safe levels," one might fight tooth decay.

The most potent weapon known to man, the nuclear bomb, was used to end the war in the Pacific. Allied scientists, led by Edward Teller, Enrico Fermi, and J. R. Oppenheimer, had been working since 1941 to develop a nuclear device ahead of the Germans. A small plutonium device was exploded on July 17, in the New Mexico desert. Following calls for the Japanese to surrender, on August 6, a single

uranium bomb "Little Boy," was air burst 900 feet above the city of Hiroshima. Analysts estimate that half of the cities 350,000 population would be dead by the end of the year as a result of radiation exposure. Three days later, an American B-52 bomber air burst a plutonium nuclear bomb, nicknamed "Fat Man," 1,650 feet above the city of Nagasaki, causing enormous infrastructure damage and killing an estimated 70,000 people.

Pillsbury introduced it's first "ready to cook" boxed convenience food - a ready mixed pie crust.

In February 1945, Ovaltine placed an ad in the *Ladie's Home Journal* to market Ovaltine:

"Authorities agree - vitamins do you more good in combination with certain other food elements.

If taking vitamins doesn't seem to be doing you as much good as you hoped - this may be the reason. Authorities now know vitamins do not work alone! They work as a team with certain other food elements (minerals). Therefore scientists, doctors, nutritionists are urging "Take your vitamins in - food for best results!

For this reason, thousands are switching to Ovaltine. Unlike mere "vitamin carriers" - Ovaltine contains not only extra vitamins, but nearly all the precious food elements necessary for peak vitality. Especially those food elements needed for vitamin teamwork.

For example, vitamin D, calcium and phosphorus must work together - and you get them all in a glass of Ovaltine made with milk. Vitamin A and protein act together in cell-building, and they're both in Ovaltine. Vitamin B1 and fuel-food are both necessary for sparkling vitality - and they're combined in Ovaltine."

In 1946, Linus Pauling, the renowned American chemist came up with a theory on how enzymes work - enzymes are proteins that do work by causing chemical reactions within living organisms. According to Pauling, enzyme activity is related to the shape and these three dimensional shapes are related to the "activated complexes," that form briefly during chemical reactions during the course that the reacting substances were turning into products.

In 1947, the first microwave ovens were marketed in the United States, however, sales were initially far less than had been hoped for.

In 1949, General Mills and Pillsbury began to market packaged ready made cake mixes ("Ready-made cake mixes provide home-made cakes without the work") with all of the dry ingredients in a box and in various flavors including chocolate, yellow, and white fudge. One only needed to add water, eggs, and sometimes butter.

By the 1950s just about 100 percent of urban kitchens in the industrialized nations had an electric, natural gas or propane stove and home heating furnace; just about seventy five percent of rural homes in industrialized nations had an electric, gas or propane stove and home heating furnaces.

Wood ashes were pretty much an inconvenience of the past, no more wood ash dust, no more spilled ash buckets, no more getting up early to rekindle the fire. Everyone marveled how wonderful life was without ashes.

The average family was overjoyed to give up the drudgery of dealing with the ash byproduct of the "old fashioned" kitchen and heating furnace or fireplace. However, they had unknowingly and even eagerly given up something so basic to human health and life itself - it had been hidden in plain sight, the lowly kitchen ashes!

We were now without the constant stream of nutritional minerals in our diets that wood, peat and kelp ashes had provided for tens of thousands of years - we were now rushing headlong, metabolically naked toward the pandemics of obesity, diabetes, the Metabolic Syndrome, osteoporosis, arthritis and Alzheimer's disease of the 20th and 21st centuries - and no one recognized the insidious loss! No One!



Chapter 9

"I die by the help of too many physicians; it was the crowd of physicians that killed me."

Alexander the Great, on his death bed.

"What sort of doctor is he, Asked the first (woman)? The second (woman) responded, Oh, well, I don't know very much about his ability, but he's got a very good bedside manner."

A Punch cartoon, 1884

"There are more things in heaven and earth...than are dreamt of in your philosophy."

- Hamlet, Shakespeare

The Vitamin Wars

Before written history, health care started out as a "family practice" - it was the mom, the dad, the grand mother, the grandfather, the aunt, the cousin, maybe a neighbor who was a self appointed healer, the nun or the pastor that held the child's hand, bandaged their wounds, soothed their fevered brow, mixed and dosed the herbs and applied the poultices.

The natural progression from the family dispensing health care to it's own members to the "angels of mercy" from religious sects going out and doing "good works" for the aged, the poor and religious pilgrims was a predictable series of events as hands became tribes, villages became towns and collections of towns became kingdoms and nations.

In the beginning of human development, these concerns about health were driven by the individual's desire to survive and true altruistic religious and scholarly motivations. However, as religions became entwined with the infrastructure and personalities of governments, kings and empires, public medicine previously driven almost exclusively by scholarly, religious and altruistic motivations, became part of the power structure and filled with intrigue, greed and driven by ego.

Chemistry became a powerful science in the development of clinical medicine during the 1800s. The pursuit of chemistry and biochemistry as related to health and disease led to the theory that certain diseases could be viewed as deficiency diseases, caused by the lack of vital elements in the diet.

The biochemical theory, now finding its way into the medical school, that disease could be caused by a "deficiency" was a significant shift in paradigm. "Disease is so generally associated with positive (causative) agents - the parasite, the toxin, the *materies morbi*," noted a report in 1919, "that the thought of the pathologist turns naturally to such positive (causative) associations and seems to believe with difficulty in causation prefixed with a minus sign."

Among the diseases to which this new and novel concept could be applied was marasmus or starvation in babies in poor areas (muscle wasting, fat loss, and significant loss of body weight). Another was Kwashiorkor, found in tropical Africa, characterized by muscle wasting and edema, enlarged liver, diarrhea and lassitude.

In 1755, Gaspar Casal (1680-1759), published an account of pellagra (a vitamin B3 deficiency), then "a new disease" in Spain. Pellagra is characterized by the "Three D's" including dermatitis ("a horrible crust, dry, scabby...crossed with cracks"), diarrhea, dementia and premature death - this disease flared up each Spring.

Casal noticed that pellagra's victims were poor and that corn meal was their staple food and they had little or no access to meat or milk. The same problem of pellagra existed in the poor people of Italy.

Various theories were offered for the cause of pellagra, the main one was the limited diet of corn. "Maize" was a New World plant that had been brought back to the Mediterranean area of Europe by the

Spanish Conquistadores. A French physician writing in 1845 suggested that the “predisposing” cause of pellagra was a diet with “too few animal products” and that the “efficient” cause was bad maize.

In 1865, a Mexican doctor pointed out that pellagra was not a problem in Mexico even though maize was the staple food of the peasant. He believed the reason pellagra was avoided in Mexico was “in the habit of soaking the maize in wood ashes before cooking, which had a disinfectant effect.” By contrast the European method of baking corn mush to polenta, “left it contaminated with moulds and toxins.”

Some theorized that pellagra was an insect borne disease like malaria and yellow fever, carried by flies from the feces of pellegrans to the meals of future victims.

Hungarian-born Joseph Goldfinger (1874-1929), of the U.S. Public Health Service, observed that in orphanages and mental asylums, where the “inmates” developed pellagra, the staff (naturally eating a wider variety of food) escaped the disease. When eggs and milk were added to the diet of the affected individuals, pellagra disappeared.

Goldfinger started a prisoner volunteer test in which he fed the participants only corn meal, when dermatitis developed, he theorized that pellagra was due to “a lack of dietary factors.”

In the early 20th century, pellagra was so common in the southeastern part of the U.S., 20,000 cases were reported in Georgia alone during its peak - the affected patient's diet's were all limited to corn and pork fat. A system of federal “retardation centers” was established for isolating pellagra patients, who were thought to be infectious in a similar fashion to tuberculosis.

In 1849, Thomas Addison reported the existence of a hitherto unknown fatal anemia, that was eventually called pernicious anemia - it was characterized by a low red blood cell count, but the cells that were observed were enlarged red blood cells. Vitamin B12 was eventually identified as the missing essential nutrient, however, there was another feature unique to pernicious anemia - the stomach had to produce two important substances: 1) stomach acid and 2) an acid dependent cofactor known as “intrinsic factor” - without either or both vitamin B12 could not be absorbed.

It was learned that vitamin B12 injections could, by bypassing the stomach be utilized to prevent and cure pernicious anemia. Stomach acid substitutes (i.e. - betaine HCl) could be used as a supplement to activate intrinsic factor and encourage the absorption of vitamin B12.

Beriberi (Sinhalese for "I cannot"), was a common disease in the Asian rice cultures. Beriberi was characterized by a weakness in the legs, hands, arms and congestive heart failure - death resulted from pulmonary edema and abdominal "dropsy."

In 1882, a Japanese naval vessel was sent on a long training tour with a starting crew of 275 sailors. Sixty percent of the crew developed beriberi and twenty five died. Dr. Takaki (1849-1915), a German trained naval surgeon noted that the protein content of their diet was far below the German standard, he quickly proposed that the protein content be increased.

On a second voyage the next year, Takaki, modified the sailor's diet by exchanging half of the rice with barley and he added, milk, beef and tofu (soy curd) - beriberi did not appear and no deaths occurred. Takaki, reported that he had prevented beriberi by increasing the protein in the diet.

In the 1890s, Christiaan Eijkman, a Dutch physician and the director of the research laboratory in Batavia, Jakarta, was also overwhelmed with beriberi cases. It was theorized that beriberi was due to a bacterial infection, so an army of bacteriologists were sent from the Netherlands to identify the germ. Eijkman noticed the hospital chickens developed polyneuritis (beriberi) when they were fed the same white rice that was being eaten by the people with beriberi, instead of their standard brown rice chicken feed.

When Eijkman fed whole brown rice to the affected chickens they quickly recovered - he theorized there was a toxin in the polished white rice and that the antidote was in the polishings or rice bran.

Eijkman's colleague, Gerrit Grijns (1865-1944), had a different theory - he believed that there was "some essential substance in the rice that was being destroyed by heat," he proposed that, "both diseases (fowl polyneuritis and human beriberi) were due to the ab-

sence from the diet of some factor present in the rice polishings."

Eijkman mailed a questionnaire to all of Java's prisons, inquiring on the incidence of beriberi, the physical environment and the diet. The questionnaire revealed no relationship between the rate of beriberi and the environment, however, the disease was a significant problem in seventy one percent of the prisons using white rice and only three percent in those using rough rice.

In 1906, Axel Holst (1860-1931), an Oslo bacteriologist was sent to study beriberi, which had become a problem in Norwegian sailors. In a simple experiment he fed guinea pigs on different kinds of flower - the entire population of guinea pigs died in thirty days. He further found that by simply adding cabbage or lemon juice supplements to their diet the guinea pigs all flourished and avoided the disease.

The same year, Frederick Gowland Hopkins (1861-1947), of Cambridge, fed young rats on casein, lard, starch, sugar and salts (minerals) -supposedly everything essential to health. He added milk to the diet of one subgroup and found that "only those receiving the milk thrived." "Astonishingly small amounts of certain substances in food, accessory food factors, were essential" he concluded, "for the body to utilize protein and energy for growth." The key to nutrition, he concluded, "lay in these missing links!"

In 1912, Casimir Funk (1884-1967), a Polish born biochemist, advanced the deficiency theory, by isolating the active substances in rice bran that would prevent beriberi. He named these dietary factors or missing link "vital amines" (later shortened to vitamins in 1920 when it was learned that not all vitamins were amines).

Funk went on to link vitamin deficiencies to beriberi, scurvy, pellagra and rickets. The theoretical basis for nutritional supplementation had finally been scientifically established.

The "deficiency" theory was beginning to collect a weight of evidence and take on a shape that all could understand. All facts were pointing to the "likelihood that scurvy, beriberi, pellagra and rickets could be caused by dietary deficiencies" of essential factors in the diet!

Rickets was a disease described in the 17th century by Francis

Glisson - it had become wide spread and a significant problem in the industrializing cities using coal as a fuel. In 1890, Theobald A. Palm reported on rickets - "It is in the narrow alleys, the haunts and playgrounds of the children of the poor, that the victims of rickets are to be found in abundance" - he recommended sunlight to prevent and reverse rickets.

In 1900, it was estimated that in Boston eighty percent of the children "under two, and twenty five percent of the children from poor families demonstrated signs of rickets." Dr. David Paul von Hansemann (1858-1920) of Berlin, believed rickets was due to a lack of fresh air, sunlight and exercise.

Edward Mellanby (1884-1955) induced rickets in puppies by feeding them diets lacking a factor found in certain animal fats. Supplementing the puppies with cod liver oil prevented and reversed the rickets. Mellanby attributed rickets to a deficiency of a fat soluble factor he called vitamin A. By then various researchers demonstrated that rickets could be prevented and cured by exposing the sufferer to sunlight or by supplementing with cod liver oil.

In April 1912, Professor A. von Wasserman of Berlin, reported in *Popular Mechanics* magazine that he had identified "a chemical substance (selenium) that will cure cancer in mice." He combined selenium with eosin, a diffusible stain, in order to ensure dispersal of the trace mineral and then injected the mixture into the mice - the tumors were destroyed.

Wasserman believed the "tumors in mice were so similar to the tumors in humans, that it is believed that an important advance had been made toward the cure of that veritable scourge (in humans)."

In 1922, it was established that the original vitamin A consisted of two separate factors. The original growth factor retained the name vitamin A, while the anti-rachitic factor was designated vitamin D. The fortification of milk with vitamin D soon followed in the United States.

In 1922, it was shown that when weanling rats were fed on a milk diet, they grew to maturity in a normal way, but they were usually sterile - the deficiency of a fertility factor was immediately theorized. Herbert McLean Evans (1905-1983), of the University of Cali-

fornia, and Katherine Scott Bishop (1889-1976), showed that the fertility factor was abundant in green leafy vegetables and wheat germ. The fertility factor was fat soluble but not related to any other vitamin so it was called vitamin E.

It was well documented since the 18th century work of Dr. James Lind, that several food items could ward off and cure scurvy. The 19th century pathologist, August Hirsch (1817-1894) identified potatoes as anti-scorbutic, and then there were the doctors in the London hospitals who had identified scurvy in children, treated them with diets of meat, milk and citrus juices. Scurvy was accidentally induced in guinea pigs in 1906 by Axel Holst, Theodor Christian and Brun Frolich while attempting to produce beriberi by giving an all flour diet.

Funk theorized that scurvy was a vitamin deficiency and a series of experiments were set up to isolate and identify it. During the investigative process it was found that many foods could prevent and reverse scurvy, however, some in particular (apples, lemon juice and cabbage) lost their anti-scorbutic effects when they were heated.

During World War I, scientist, including Harriet Chick (1875-1977) at the Lister Institute in London, started a series of studies to determine the anti-scorbutic properties of various foods. Arthur Harden (1865-1940) was able to isolate citric and other acids from lemon juice, including a residue that had an exceptionally high anti-scorbutic property that was not found in the other acids isolated from lemon juice.

In 1919, the anti-scorbutic factor had been universally accepted as an essential nutrient. Albert Szent-Gyorgyi (1893-1986), isolated "hexuronic acid" from the adrenal glands. In 1932, W. A. Waugh and Charles Glen King (1896-1988) isolated the "lemon vitamin" and demonstrated that it was identical to hexuronic acid.

In 1932, this vitamin (vitamin C) was synthesized by Szent Gyorgyi - in 1933 the vitamin's name was officially changed to ascorbic acid.

The concept that certain diseases (i.e. - scurvy, pellagra, beriberi, rickets, pernicious anemia, anemia, goiter, etc,) were caused by deficiencies was initially a novel thought, a really earth shaking thought,

however, it was initially a toy for but a few scholars, military physicians and philosophers - these beliefs of "deficiency diseases" started after Lind in the 1700s, who believed that the healing power of lemons for scurvy was as an herbal medicine, the same concept as using quinine to treat malaria.

Yet, the concept of "deficiency" diseases, hung on and produced a flurry of academic activity and discovery over the next 200 years. These 200 years were but a minor blip in the time line of human history - 30,000 years had passed since humans had developed the basic skills for the survival of modern man.

These deficiency disease studies set off a new war between the various schools of medical thought and theory - a 200 year war that continues on into the 21st century, because medicine as a profession, has lost it's altruism for seeking cures and there are no laws requiring doctors to cure people even when cures are available - so they don't.

The "sick man" it is said, "was to have dropped below the radar of the medical eye with the advent of the private clinic about 1800 - when the individual person was first given the generic label of patient," - a pathological sample or classroom exhibit riddled with disease. This Houdini-like "disappearing act" of the sick person accelerated over the next 200 years, relegating the individual to an algebraic "X-factor" in equations of economics, sociology, physicians' life styles, diagnostic technology, system's analysis and advertising programs.

Twentieth century medicine became identified with mind twisting political and social infrastructures and brick and mortar hospital structures and campuses that rivaled the most grand sky scrapers, medical universities, professional trade organizations that took on the look and functions of trade unions, international insurance conglomerates, pharmaceutical company's sales that surpassed most individual nation's gross national product, hospitals acting as teaching laboratories, research money generators that siphon funds from estates and large corporations, government departments (Medicare and Medicaid), international banking and corporate largess.

Medicine developing parallel with organized trades and governments was predictably connected by an umbilicus to political power,

economics, central and local governments, the law, social services, advertising and the mass media (Dr. Marcus Wellby, MD; Dr. Kildare: Dr. Ben Casey; E.R., and advertisements for Vioxx, etc.).

Once medical schools became part of universities, degrees were required to get licenses, medical societies, trade associations and guilds were formed, the practice of medicine became a toy almost exclusively for the wealthy. At this point, the industrial privateers and robber barons became interested in the "customer base" that patients provided.

During the transition into the 20th century, "scientific medicine" rapidly gained political and financial clout in the United States. The Federal bacteriological laboratory attached to the Marine Hospital on Staten Island, New York (1866), developed prominence and in 1930 became the National Institute of Health (NIH) and was moved to Bethesda, Maryland.

Medical research was mainly a function of private financing by the "robber barons," industrialists such as Andrew Carnegie (1835-1918), John D. Rockefeller (1839-1937) and his son John Jr. (1874-1960), who "laundered" their financial empires through "good works."

Scottish-born Carnegie grew his fortune through an iron and steel empire. Carnegie was a firm believer in the "Survival of the Fittest" and ruthless capitalism concepts, however, he also held that, "a man who dies rich, dies disgraced" - therefore, he devoted a great deal of his time and energy to philanthropy.

Carnegie was particularly fascinated with medicine and pharmaceuticals, so he committed himself to medical education and sponsored an investigation by Abraham Flexner (1866-1959) - a retired university president - into the status of medical education in America.

American medical schools had developed in a financially competitive market, initially depending on the student fees for survival. From the early 1800s, hundreds of these profit motivated medical schools were opened. These proprietary medical schools were particularly attractive to students because they were practical and did not require a "pre-medical" education prior to application.

Sixty two of these myriads of schools survived between 1802 and 1876, as well as eleven homeopathic and four eclectic schools.

Flexner's initial report, *Medical Education in the United States and Canada* (1910), claimed that quality had been sacrificed for quantity.

Flexner believed that a good medical education could only be provided by reducing the numbers of schools and improving their standards for selecting students and staff. Educated at Johns Hopkins and the University of Berlin, he was committed to making medical schools university departments. He demanded that these new medical schools be attached to big city universities to "attract better educated students and more clinical material for teaching."

Prior to the Flexner report only five medical schools were involved with research - Harvard, Johns Hopkins, and the Universities of Pennsylvania, Michigan and Chicago.

Flexner was a firm believer that the main requirement for true medical progress and education was basic science. He was critical of the French system, which he viewed as being too clinical, the English system he believed was too decentralized and too clinical, and he was particularly disturbed by the American system which he viewed as being "grossly commercial."

Flexner wanted the medical school to be part of a larger university and have a requirement that medical students should first have a degree in the natural sciences. He believed that the "medical school of the future should have it's own biomedical departments to encourage research; it had to have a teaching hospital; stiff entrance requirements for all students; and end with a doctoral degree."

Finding the existing laboratory facilities "wretched," Flexner insisted that the new medical schools he envisioned have scientific facilities unequaled by none.

Flexner, committed to raising medical school standards, suggested that the numbers of American schools be reduced to thirty one, and he recommended a system in which students were required to have a minimum of two years of college education prior to entry; and that the staff must be full time and not practicing physicians.

The majority of medical schools failed to meet Flexner's gold standard litmus test, which resulted in the closure of forty six colleges in ten years, including the homeopathic colleges and medical colleges for black and female students.

In 1910, 4,400 doctors graduated from medical school and obtained an M.D. degree, by 1920 these numbers of graduating doctors dropped to 3,047, resulting in amongst other things, an increase in the social status and incomes of the new graduates.

The medical schools that did survive got the message, they discarded their proprietary ways and they reinforced their relationships with hospitals and universities. In return, this new American breed of "scientific" medical schools was rewarded with enormous amounts of money from the philanthropic coffers of Carnegie and Rockefeller.

Rockefeller, a devout Baptist who made his multi-millions through his oil business, at first channeled most of his giving to what he felt were "worthy churches," and their altruistic endeavors, at the urging of others he later gave vast fortunes to endow medical schools.

One of Rockefeller's closest advisors was Frederick T. Gates, a Baptist minister, who had read William Osler's textbook, *The Principles and Practice of Medicine* (1892). Gates was mortified by "how few (human) diseases medicine could cure." Gates convinced Rockefeller to invest heavily in scientific medicine as a worthy cause for mankind.

The Rockefeller Institute for Medical Research was established in 1901- its mission was medical and pharmaceutical research, clinical practice was strictly forbidden. Abraham Flexner's brother, Simon was appointed director of the institute's laboratories. Simon Flexner went on to isolate the *Shigella* bacteria as a cause of dysentery.

Not all schools of thought were supportive of the move to "scientific medicine." Many mourned the loss of the clinical focus of the old medical schools - in 1933 the well respected pathologist Sir Andrew MacPhail (1864-1938) complained:

"I am well aware that in these days, when a student must be converted into a physiologist, a physicist, a chemist, a biologist, a pharmacologist and an electrician, there is no time to make a physician of him. This consummation can only come after he has gone out into the world of sickness and suffering, unless indeed his mind is so bemused by the long process of education in those sciences, that he is forever excluded from the art of medicine... In that case he is destined for the laboratory, the professor's

chair, or the consultant's office. What would have happened to Sydenham had he been put through this machinery is a problem in infinity which no human intelligence is competent to solve."

Once third party health insurance entered the scene, the patient was no longer the primary focus of the physician. The physician now seemingly had a bottomless pit of money available to them and all they had to do was learn the insurance codes and start sending in the bills - it worked! Physicians now had the perfect world - they were respected members of the power elite, even without royal blood lines; they had an endless stream of money collected from their customers in their behalf by the government (Medicare, Medicaid) and by private insurers; and the law treated doctors differently than they treated the general public.

By the 1920s, doctors had a "license to kill," they could "bury their mistakes," they were to be sued for malpractice rather than be charged with manslaughter or murder when they killed a patient; and even more importantly, doctors could crush their competitors socially, politically and financially without fear of reprisals! They were politically vaccinated against commercial equality normally provided by the Sherman Antitrust Act - in effect and in their own minds doctors had become omnipotent gods.

Standard Oil and "Ma" Bell were broken up, with the justification of the Sherman Antitrust Act to protect free flow of trade and promote competition - after all, competition amongst providers that stimulates better products, better pricing and better service, is good for the American public and the individual customer.

Everyone "knows" that chiropractic doctors, naturopathic physicians, traditional Chinese medical doctors, herbalists, Ayurvedic medical practitioners, etc. are "quacks" because medical doctors say it is so.

Patient's have "character roles" and "cameo appearances" relegated to them within the storyline of twenty first century medicine's grand scheme. Depending on who or which department is performing the audit or the budget, "patients are listed as demand, costs and ben-

efits, input or output, voters, clients and/or consumers of goods and services, owners of rights or litigants, the fracture in room number 110, frozen semen, an embryo or stem cells, a disease or clinical case example for students, bars on a budget or cash flow graph, or simple income prediction numbers." According to Roy Porter, author of *The Greatest Benefit to Mankind*:

"For it's fans, modern medicine, with it's microbe-hunters and micro-chips, has enabled westerners to escape from the valley of the shadow of death, living longer and healthier lives; for critics this is the era of the Holocaust and the Gulags, in whose unspeakable outrages doctors and psychiatrists were hardly reluctant participants. Scientific medicine may be the knights in shining armour or a new body-snatcher."

In 1951, Ruben Rausing invented the "Tetra Pak" milk carton; soon after the glass milk bottle disappeared.

Linus Pauling, an American chemist advanced the knowledge of proteins. He proposed that proteins consist of "polypeptide" chains (formed by smaller units called peptides). These chains contain a backbone of carbon and nitrogen atoms, to which other chemical units, including atoms of hydrogen and oxygen, are attached. Pauling showed how a polypeptide chain can coil up into a helix (3-D spiral) - weak bonds between the hydrogen and oxygen atoms in "successive" turns of the spiral provide stability.

In 1952, the first commercial jet service was introduced when BOAC flew passengers between London, England and Johannesburg, Republic of South Africa.

In 1953, American James Watson and Englishman Francis Crick, found that DNA (deoxyribonucleic acid) the "super molecule" that transmits biological information, has a "double-helical" structure, made of two intertwined spiral strands of linked chemical units called nucleotide bases.

The 3-D structure was quickly accepted; Pauling had already proposed the double-helix structure for proteins two years earlier; it was sufficiently complex to store a great deal of information; and it was simple enough to copy when cells divide.

In 1954, an electric pressure cooker (the original pressure cooker was the precursor of the steam engine in the 1700s) was widely marketed in the United States.

Dr. Jonas Salk, head of the virus research laboratory at the University of Pittsburgh, supervised the mass vaccination of American children with a chemically inactivated polio virus vaccine. More than 200 children developed polio as a result of a production error (all cases of polio diagnosed in the U.S. after 1982 were caused by the oral polio vaccine).

In 1955 a team of American surgeons successfully performed the first kidney transplant.

The United States launches the USS Nautilus, the first nuclear powered submarine, producing a tactical advantage over the Soviets in the Cold War.

In 1956, Dr. Werner Forssman, a German physician and American Doctor Andre Cournand and Dickson Richards received the Nobel Prize in Medicine for developing a practical cardiac catheter method for diagnosing cardiovascular disease. Forssman performed the first cardiac catheter experiments on himself (1929) using a fluoroscope to monitor the path of the catheter.

Birth control pills are tested on a massive scale in Puerto Rico.

Dr. Willem Kolff, a Dutch physician, developed an "artificial kidney" or dialysis machine, which was first employed in the United States. Blood from a patient's artery is diverted into the dialysis machine, which looks like a large rotating cylinder or drum. The blood enters a cellulose membrane, which is wrapped around the drum. As the drum rotates, the cellulose is submerged in a saline solution. The membrane allows metabolic wastes to pass out of the blood. The blood cells are too large to pass through the membrane. The cleansed blood is then pumped back into the patient.

In 1956, Dr. Carl Binger (1889 - 1976), an American physician wrote:

"Time was, and not so long ago, when the family doctor delivered babies and supervised their nursing, their weaning and their teething, when he vaccinated them and saw them through their measles and chicken pox.

and whooping cough. He told the boy about the facts of life and treated the girl for her menstrual cramps. He advised about diet and rest, gave spring tonics, clipped tonsils, set a broken arm, reassured father who couldn't sleep because of business worries, pulled mother through a case of typhoid or double pneumonia, reprimanded the cook who was found, on her day out, to have a dozen empty whiskey bottles in her clothes closet, gave advice about the young man's choice of college and profession, comforted grandma, who was losing her memory and becoming more and more irritable, and closed grandpa's eyes in his final sleep. He went on his endless, mysterious and incessant rounds leaving in his wake a faint odor of carbolic acid with which he disinfected his beard."

"But all that changed," Binger concluded, "This heroic figure is gone from our midst....Who killed Cock Robin?"

In 1958, BOAC initiated the first transatlantic jet service with eighty four passengers traveling between London and New York.

Dr. Ian Donald, a professor of gynecology at the University of Glasgow, developed the ultrasound diagnostic machine using sonar technology originally created to detect submarines.

In 1959, only twenty years after the production and widespread use of antibiotics, some bacteria have begun to show resistance to antibiotics. It was discovered that bacteria possess small pieces of DNA called plasmids; the plasmids are credited with the resistant genes - they replicate themselves and the plasmids pass from one bacteria to another when the bacterial cells come into contact.

In 1959, a Japanese fishing village on the shores of Minamata Bay was devastated by 2,500 people being hospitalized and more than 700 babies being born with catastrophic central nervous system birth defects.

The Minamata Bay disaster was attributed to industrial pollution with methyl mercury accumulation in the mud, algae and small fish in the bay; the medium sized fish ate the little fish; the large fish ate the medium sized fish; and humans ate the large carnivorous fish now heavily contaminated with mercury.

In 1960, U.S. regulatory agencies (FDA, etc.) sought control over

the supplement (vitamin/mineral) manufacturers. Their aim was to get supplements classed as drugs - fortunately they failed at the congressional level, however, the vitamin war is ongoing into the 21st century:

Stage 1 of the war - the FDA identified and targeted the "enemy." In 1961, the FDA stated, "The most widespread and expensive type of quackery in the United States today is in the formulation of vitamin products, special dietary foods and food supplements."

Stage 2- Debunking Four Nutritional Myths as defined by the FDA and the AMA in 1961:

1. All diseases are due to faulty diet.
2. Soil depletion and the use of chemical fertilizers cause malnutrition.
3. Modern processing methods and storage removes nutritive value from the food.
4. All people now suffer or will suffer from sub-clinical deficiencies.

In 1962, Rachel Carson, an American scientist and author, published, *Silent Spring*, a book that revealed that DDT was dangerous to the environment. She demonstrated that DDT not only kills insects, but also enters the food chain and had killed millions of birds that had either eaten poisoned insects or small food fish.

Stage 3- Declaring vitamin supplements unnecessary (1963) - According to D. G. Cooley, in an article, *What is a vitamin?* In Today's Health (an AMA publication, April, 1963) "So-called sub-clinical deficiencies, by definition do not present clinically detectable symptoms." "One can eat a bad diet for a long time before frank symptoms occur." "Self-chosen vitamin supplements cannot make up for a thoroughly bad diet that is deficient in many other nutrients." "Routine vitamin supplements cannot be expected to cure a disorder that the patient has diagnosed for himself."

Stage 4 - The Government continues it's anti-vitamin campaign using "hired guns." A book specifically written to debunk health claims for vitamins and discourage their use was published in 1968. Author, D. Takyon, participated in the 1967 senate hearings in Washington, D. C. to control the vitamin industry and "made substantial additions to the government's case." Takton died in 1988 at the age of fifty six.

Stage 5 - Forcing the issues - The FDA was the direct cause of catastrophic folic acid regulations (1963 - 1996).

In 1964, vitamin deficiency was, "declared not to pose any major public health problems in the U.S.A.

According to Victor Herbert, M.D. from Harvard Medical School, "When all cases of vitamin deficiency reported in the U.S. in a single year, are added up, they do not reach the figure of 20,000 (only) approximately one tenth of one percent of the population suffered from clinical vitamin deficiencies."

In 1966, Dr. James L. Goddard, FDA Commissioner, stated "Vitamin supplements or vitamin pills have little or no practical value to the average American. He can easily get the nutrients he needs in common foods with only a couple of exceptions, if he gets more than he needs, he excretes the excess." Goddard continued, "Take a look at today's American - bigger, stronger, and healthier than his forebears of fifty years ago - and you know that these vitamin pill promoters are talking nonsense."

The first published reports linking folic acid deficiency with birth defects:

Franklin, A. L., et al.: J. Biol. Chem. 169:427. 1947.

Nelson, M. M. And Evans, J.: Nutrition. 8:111. 1949.

Money, I. W., et al.: Anat. Rev. 120:119. 1949.

Baird, C. D. C., et al.: Circ. Res. 2:544. 1954.

Hubbard, E. D. and Smithells, R. W.: Folic Acid Metabolism and Human Embryopathy. The Lancet. 1:1254. 1965.

The FDA should have moved aggressively to institute folic acid fortification of foods and make revised recommendations for folic acid supplementation no later than 1965.

The 1965 Rate of Spina Bifida and Anencephaly

USA - 1/1,000

New Zealand 1/1,000

Australia 1/500

The 1963 restrictions on folic acid remained in place for thirty one years - by 1998, and the institution of folic acid supplementation, the spina bifida rate had dropped by twenty percent.

In 1996, the FDA (being dragged, kicking and screaming to the end), finally orders folic acid fortification of flour.

In 1966, Dr. Joel Wallach, an American veterinary comparative pathologist and conservationist, began performing autopsies for The Center for the Biology of Natural Systems. The center was established with a multimillion dollar grant from the National Institutes of Health and employed the facilities of the St. Louis Zoological Gardens, Shaw's Botanical Gardens and the biology department at Washington University, St. Louis.

The project was brought into being as a direct result of the sudden awareness of a possible world wide environmental meltdown as a result of industrial and agricultural pollution (i.e. - Minamata mercury disaster, DDT, etc.). Wallach was recruited as a comparative pathologist to look for captive wild species that might be ultra sensitive to pollution, species that could be used as "a canary in the mine," that would provide an early warning system for a possible industrial or agricultural pollution crises.

In 1967, Goddard abruptly resigned from the FDA - he was not to become the head of the "consumer protection and environmental Health Service," - a super agency he was planning to create.

In 1967, Dr. Rene Favaloro, an American Surgeon from Cleveland, Ohio performed the first coronary artery bypass graft surgery.

On December 3, 1967, at the Groote Schuur Hospital in Capetown, South Africa, a team of surgeons led by Dr. Christian

Barnard, transplanted the first human heart. Louis Washkansky, a fifty nine year old South African man, became the first recipient of a healthy heart removed from a dead donor.

The first home microwave ovens go on sale. The ovens work by bombarding food with ultra-short microwaves. When the microwaves strike the food, they cause the water molecules inside to vibrate millions of times per second which generates friction, which produces heat that cooks the food "from the inside out."

In 1968, pollution is blamed for the "biological death" of Lake Erie. Pollution, produced by industrial wastes, lead from leaded fuels, litter and garbage are becoming more important as environmental pollutants. Lake Erie, the fourth largest of the North American Great Lakes is listed as a "dead lake." Industrial and municipal wastes had been dumped into the lake for decades, resulting in it's waters being completely polluted. According to the U.S. Department of the Interior, 15,000,000 fish died each year in Lake Erie.

In 1971, Dr. Linus Pauling, double Nobel Prize winner, has supported the benefits of supplementing with large doses of vitamin C in his book, *Vitamin C and the Common Cold*. Pauling supported the large doses of vitamin C, both for fighting the common cold, but also for preventing cancer and heart disease. Many of the benefits of vitamin C arises from it's role in the manufacture of collagen, which is involved in the formation of connective tissue, bone matrix, skin and blood vessels. Pauling recommends that individuals take 10 - 12 grams of ascorbic acid or vitamin C each day.

In 1977, Dr. Joel Wallach, veterinarian and comparative pathologist, working at the Yerkes Primate Research Center, Atlanta, Georgia, reports the discovery of the first animal (rhesus monkey) to be diagnosed with Cystic Fibrosis (thought by "medical experts" to be a simple Mendelian, genetically transmitted disease found only in humans).

At first the scientific world was quick to confirm Wallach's diagnosis of CF, however, when they learned that the disease was caused by a congenital deficiency of the trace mineral selenium, Wallach was fired with twenty four hours notice.

Wallach shortly was appointed director of the Laboratory Ani-

mal Colony at St. Louis University - when the director of the Yerkes Primate Center learned of Wallach's new appointment, they contacted St. Louis University and he was fired again as he was moving into the new office.

In 1980, Dornier Systems, a West German aerospace company has produced the "Lithotripter," a machine that destroys kidney stones by means of focused high frequency sound shock waves. The "Lithotripter (from the Greek words meaning "stone" and "crusher"), causes the kidney stones to be broken up into sand-sized particles, that are easily passed through the urinary system.

In 1983, Dr. Joel Wallach finishes a 1,000 page reference book, *The Diseases of Exotic Animals*, published by W. B. Saunders, the book described therapies and the results of 13,000 autopsies of zoo animals - he concluded "every animal and every human being that dies of natural causes dies of a nutritional deficiency disease."

Wallach continued his pathology investigations, and by 1991 he had performed more than 17,500 autopsies in more than 454 species of animals and 3,000 humans - again confirming that "every animal and every human being that dies of natural causes dies of a nutritional deficiency disease."

In 1992, an Earth Summit, the United Nations Conference on Environment and Development, was held in Rio de Janeiro, Brazil. It's aim was to examine the negative impact of world wide economic development. With 117 heads of state and representatives of 178 nations attending the meeting, it was the largest gathering of world leaders in history. Issues discussed included global warming and mass destruction of the Amazonian rainforest, habitat for nearly half the world's species of plants. Countries attending this meeting agreed to protect endangered species and to reduce carbon dioxide emissions.

In 1993, the National Cancer Institute reported the results of a five year study on the protective value of antioxidants against cancer. The study was done in Henan Province of north-central China.

After selecting 29,000 people between forty and sixty nine years of age, the participants were placed in one of several nutrient investigation groups. Retinol, zinc, riboflavin, niacin, vitamin C, molybdenum, selenium beta carotene and vitamin E were compared to a

placebo.

The investigators reported that, "vitamin C, retinol, zinc, riboflavin, molybdenum and niacin had no statistically significant (protective) effect on cancer deaths after five years."

The group that took the combination of beta carotene, vitamin E and selenium for five years had dramatically positive results: 13 percent fewer deaths from all cancers, 21 percent reduction in deaths from stomach cancer and a 9 percent decrease in deaths from all causes.

In 1994, the September issue of the British medical journal, *Lancet*, reported in two featured articles that, "Doctors should take acting classes so they can at least pretend they're concerned about their patients' health." The articles go on to say, "that acting classes should be required in medical school so doctors can learn just when to provide a perfectly timed compassionate look, or a touch on the hand."

In 1996, a Scottish veterinary technician cloned a sheep; he "cloned" or copied a genetically identical sheep from a single cell taken from it. He transferred the nuclei from a sheep cell into unfertilized eggs from which the original nuclei had been removed. The eggs were replaced into the uterus of the original sheep, resulting in the birth of a lamb, named "Dolly." This proved that adult mammalian cells contain functional versions of all the genes required to produce an entire animal.

In December 1996, Dr. Larry Clark, of the University of Arizona reported on a randomized double-blind study (the "gold standard" in medical research) that followed 1,312 people for seven years. Half took 200 mcg of selenium daily, the remainder took a placebo. An analysis of the data showed that taking selenium slashed the occurrence of prostate cancer by sixty nine percent; colorectal cancer, sixty four percent; and lung cancer, thirty nine percent.

There are no drugs that will reduce the risk of cancer by one percent, yet doctors fail to recommend selenium as a cancer risk lowering approach for people with a high cancer risk - there is just too little money involved, so recommending selenium is just not worth their time.

In 1997, Wyeth Pharmaceuticals pulled Phen-Phen, a weight loss

cocktail, from the market because studies proved that the drugs damaged heart valves. The drug combination was used as a weight loss system by 6 million Americans, and it cost Wyeth \$16.6 billion in settlements.

Bayer Pharmaceuticals withdrew its cholesterol lowering drug Baycol because the drug produced a potentially fatal muscle breakdown disease called rhabdomyolysis. Bayer settled 2,861 cases out of court for \$1.1 billion and has an additional 7,577 cases pending as of 2004.

On May 29, 1997, an article in the *New England Journal of Medicine* reported that the twenty six year, \$30 billion "War on Cancer" was an abysmal failure. Despite an injection of an enormous amount of money and energy, the death rate from cancer continuously and consistently went up.

In 1997, the Coalition Against Insurance Fraud, reported that "the total doctor perpetrated health care industry fraud annually totaled \$54 billion nationwide, including \$34 billion from Medicare and Medicaid."

In September 28, 1998, Dr. Lester Mitscher, a chemist from the University of Kansas, reported that catechins or EGCGs "are the most powerful antioxidants there is." "EGCG is 100 times more powerful than vitamin C in halting oxygen (free radical) damage to DNA and twenty five times more effective than vitamin E," Mitscher reported last year.

On October 1, 1998, an article was published in the *Annals of Internal Medicine*, that pointed out that, in a study of 90,000 nurses (The Harvard Nurses Health Study), over a period of fifteen years, those that took a multiple vitamin/mineral supplement daily reduced their risk of colon cancer by seventy five percent - not a single medical association took up the information as a proactive recommendation to its members, the American Cancer Society failed to proactively pass on the information to the American public and the FDA failed to pass on the information to the general public.

On November 5, 1998, Dr. Raymond Worrell, Jr., a leukemia specialist at Sloan-Kettering and senior author of a study published

in *The New England Journal of Medicine*, said, "we now know that arsenic can safely bring patients with acute promyelocytic leukemia into remission."

In China, survivors have gone as long as ten years with no return of APL after being treated with arsenic. "Based on these highly sensitive molecular results, treatment with arsenic trioxide appears to exceed the effectiveness of any single drug to treat APL," said Dr. Steven Soignet. The arsenic works by selectively killing the cancerous cells that cause APL.

On November 18, 1998, Dr. Wei Zheng of the University of South Carolina reported that in a study of 930 women, those who ate their beef and pork cooked very well done increased their risk of breast cancer by 462 percent over those women who ate their meat cooked medium rare - it is not the amount of fat in the diet that counts, but rather, how you cook the fat that counts. Burnt animal fat is converted to heterocyclic amines which are extremely powerful carcinogens.

On January 21, 1999, an article was published in the *New England Journal of Medicine* that showed high fiber diets do not reduce the risk of colon cancer. In the Harvard Nurses Health Study that reviewed the health data of 90,000 nurses and 50,000 male doctors, showed that high fiber diets did not reduce the rate of colon cancer by any measurable amount - however, the general increase in the rates of dietary fiber consumption by humans increased the reported rates of mineral deficiency diseases (i.e. - obesity, diabetes, The Metabolic Syndrome, The Feedlot Syndrome, high blood pressure, arthritis, osteoporosis, Alzheimer's disease, hypothyroidism).

On March 10, 1999, an article published in the *Journal of the American Medical Association*, that the long held dogma, that high dietary fat intake increased the risk of cancer, was false. In a comparison of 90,000 nurses, the Harvard Nurses Health Study compared the rate of breast cancer in the women consuming less than twenty percent of their calories as fat against those that consumed more than fifty percent of their calories as fat - the data revealed that there was no measurable difference in the rate of breast cancer regardless of the fat intake.

On September 30, 2004, Merck (the \$22 billion per year "Pharma Giant") "voluntarily" withdrew Vioxx, their much vaunted arthritis and PMS pain killer, from the marketplace. Between 1999 and 2004, as many as 139,000 fatal heart attacks in the United States have been attributed to Vioxx prescriptions.

Merck and the FDA should have withdrawn Vioxx from the market early in 2000 (better yet it should have never been approved in the first place), as the earliest of studies clearly demonstrated that Vioxx doubled the risk of heart attacks. Twenty million Americans used Vioxx, which racked up \$2.5 billion in annual sales.

Merck is looking at \$27 billion in settlements and awards for damages. "For with Vioxx, Merck and the FDA acted out of short-sighted and irresponsible self-interest," wrote Dr. Richard Horton, Editor of *the British medical journal Lancet*.

David Graham, FDA drug safety reviewer of twenty years, says that the American public is "virtually defenseless" with regards to post approval safety issues. "What I'm painting is a picture of an FDA that is completely insensitive to drug safety," Graham said. Graham went on to say, "the FDA has an inherent conflict of interest that triggers denial, rejection and heat when safety questions emerge about products it has approved."

In Graham's view, drugs he believes require speedy removal from the marketplace include:

1. *Meridia*, a weight loss drug, that he believes should be taken off the market because he believes it doesn't pass the test, "whether it's benefits outweigh the risks of higher blood pressure and stroke among people taking it" - "I don't believe Meridia passes that test," says Graham.

2. *Crestor*, an anti-cholesterol drug. Graham says, "the government should reevaluate the occurrence of renal failure and other serious side effects among people taking Crestor." "there are other methods of lowering cholesterol that do not carry the risk of kidney failure," says Graham.

3. *Accutane*, an acne drug that is linked to birth defects, "represents a 20 year failure by the FDA and that sales should be immediately restricted," says Graham.

4. *Bextra*, a pain killer, "poses the same heart attack and stroke risk as *Vioxx*," said Graham.

5. *Serevent*, an asthma drug "that has shown with 90 percent certainty in a long term trial in England, to cause death from asthma," says Graham. "We've got case reports of people dying, clutching their *Serevent* inhaler."

Despite one pharmaceutical disaster after another, as of 2004, the FDA's position on nutritional health claims for vitamins and minerals had not changed since their inception:

1. "The general public was not able to understand the subject of vitamins and minerals and would make dangerous choices if given information, and they might take vitamins and minerals instead of prescription drugs."
2. "The First Amendment does not apply to the FDA."
3. "Pharmaceuticals approved by the FDA are safe and effective."

To protect the American public from the repressive anti vitamin and anti mineral supplement policies of the medical profession and the FDA, we decided to change our approach. For decades we had tried to win approval for a vitamin and mineral friendly political environment by holding up the gold standard of randomized double blind studies - we failed like everyone else who had tried because, the FDA constantly moved and raised the approval bar.

We consciously decided to approach the problem from a different perspective - the people's right to know what information was available at the present time. We decided to test the scope of the First

Amendment of the United States Constitution. This unique approach was the brainchild of our son Steve Wallach. As a result, we retrained our sights on the people's right to know and hired the best man for the job, First Amendment attorney, Johnathan Emord; the following is a small part of our First Amendment efforts:

Legal Resume Before the FDA as of 2004

1. Wallach v. FDA, Case No. 04 CV 0216BTM, U.S. District Court for the Southern District of California (Dr. Wallach and American Longevity, Inc. are co-plaintiffs). The Complaint was filed on February 3, 2004. Plaintiffs challenge 21 U.S.C. 343-2 as an unconstitutional restraint on speech because it drastically limits the circumstances in which dietary supplement companies can distribute peer-reviewed literature.
2. CSPI v. Thompson, Case No. 03-1962 (RBW), U.S. District Court for D.C.. (Co-intervenor/Defendant). Special interest groups challenge qualified claims process and application to conventional foods. (Dr. Wallach and American Longevity, Inc. asked the court to allow us to be defendants because we believed that the FDA would not vigorously defend our hard won selenium claim - the court agreed and we prevailed).
3. Whitaker et al v. Thompson, 248 FSupp.2nd 1 (D.D.C. 2002). (Wallach and American Longevity, Inc. Co-Plaintiff) Court granted Plaintiffs' Motion for Preliminary Injunction banning FDA Suppression of Antioxidants Health Claim.
4. Thompson v. Western States Medical Center, 535 U.S. 357 (2002). (Co-Amicus) U.S. Supreme Court upheld lower court's decision that Congress' ban on the advertising of compounded drugs violated the First Amendment.

Comments Filed to the FDA

1. Comments on FDA's Advance Notice of Proposed Rulemaking's (ANPR's) alternatives for regulating qualified health claims in the labeling of conventional human foods and dietary supplements. (filed February 20, 2004) (Co-Commenter)(Docket No. 2003N-0496. ANPR is tied to the CSPI case listed in previous section. No. FDA activity on ANPR.
2. Comments on Task Force for Consumer Health information for Better Nutrition (Co-Commenter), Docket No. 03N-0069; filed on 7/9/2003; no FDA activity on matter.
3. Reply Comments on First Amendment issues (Co-Commenter); Docket No. 02N-0209; filed on 10/25/2002. No FDA activity on matter.
4. Comments on First Amendment Issues (Co-Commenter). Docket No. 02N-0209; filed 9/12/2002; No FDA activity on the matter.
5. Comments on Guidance on Applying the Structure/Function Rule (Co-Commenter); No FDA activity on the matter.

Petitions Filed with the FDA

1. Petition for the Modification of Disease Connotations 21 C.F.R. 101.93 to include Structure Function Claims with Disclaimers that Eliminate Perceived Disease Connotations (submitted March 16, 2004)(Sole Petitioner).
2. Petition for Health Claims. Lycopene and Tomato-based foods containing Lycopene and reduction in the Risk of Certain Cancers (sole Petitioner)(filed 1/21/2004); No Docket Assigned yet; FDA response on preliminary review due on April 30, 2004.
3. Petition for Amended Health Claim. Omega-3 Fatty Acids and

Coronary Heart Disease (Co-Petitioner); Docket No. 03Q-0401; filed on 6/23/2003. qualified claim allowed by FDA on 9/8/2004.

4. Petition for Health Claims; Selenium and Reduction in the Risk of Certain Cancers and Selenium and Anticarcinogenic Effects (Sole Petitioner); Docket No. 02P-045; filed on 7/10/2002; qualified claims approved by FDA on 4/28/2003.

With these bold “shots across their bow,” we let the FDA know that we were serious about getting well documented nutritional health information to the American people and the entire world for that matter; we took action that they believed that no one would take.

We lost every battle in the Vitamin War with the FDA before the establishment of the DSHEA Act of 1994 and we have won every battle since.

On November 19, 2004 *USA TODAY* headlines read, “Scientist says FDA system ‘broken.’” A Food and Drug Administration scientist told a Senate committee that the U.S. public is “virtually defenseless” against another “terrible tragedy and a profound regulatory failure” like Vioxx.

“It is important that this committee and the American people understand that what happened with Vioxx is really a symptom of something far more dangerous to the safety of the American people,” David Graham, associate director for science and medicine in the FDA’s Office of Drug Safety, told the Finance Committee. “Simply put, FDA and its Center for Drug Evaluation and Research are broken.”

In 2004, the USDA added several more foods to the recommended antioxidant list including, “red beans, kidney beans, pecans, walnuts, ground cloves and cinnamon.”

Scientists praised a new cholesterol regulating compound in blueberries called pterostilbene. In a comparative study with “statin” drugs, pterostilbene equaled or exceeded the cholesterol lowering properties of the drugs; additionally pterostilbene does not have any negative side effects.

Mexican investigators determined that women who ate high car-

bohydrate diets had a significantly higher risk of breast cancer.

A University of Washington study showed that women who filled twenty five or more prescriptions for antibiotics over a seventeen year period had double the rate of breast cancer of those who didn't use antibiotics.

Scientists at the University of California at Santa Barbara and the Sansum Medical Research Institute found that some varieties of cinnamon have blood glucose lowering properties in adult onset type II diabetics.

The November 29, 2004, featured cover article of *Forbes* trumpeted that:

Pharma's New Enemy: Clean Living!

"There's a backlash building against the cost, risk and side effects of medication, and it's bad news for the pharmaceutical industry."

In December 20, 2004, the FDA forces Celebrex, a similar pain relieving drug as Vioxx, ads off the market because a double blind study showed a 200 percent increase in cardiovascular disease risk.

In December 21, 2004, the NIH pulled the plug on a study, "testing whether Celebrex or naproxen would reduce the risk of Alzheimer's disease was halted after researchers noted an increase in heart attack and stroke among participants who were taking naproxen (Aleve), an over the counter pain reliever on the market for nearly thirty years."

The study was stopped by the NIH after three years, when it was noticed that patients taking naproxen, sold under the brand name Aleve, had fifty percent greater incidence of cardiovascular events - heart attack or stroke - than patients taking a placebo.

In December 22, 2004, the *Los Angeles Times* reported under the headline, The National Institutes of Health: Public Servant or Private Marketer? - that:

"Doctors have long relied on the NIH to set medical standards. But

with its researchers accepting fees and stock from drug companies, will that change? A continuing investigation by the Times shows an unabashed mingling of science and commerce.

For 15 million Americans, it is a daily ritual: gulping down a pill to reduce cholesterol.

They do it because their doctors tell them to do it. Their doctors, in turn, rely on recommendations from the NIH and its scientists, such as Dr. H. Bryan Brewer Jr.

Brewer as a leader at the NIH, was part of a team that gave the nation new cholesterol guidelines that were expected to prompt millions more people to take the daily cholesterol lowering pill. He has also written favorably of a specific brand of cholesterol medication, Crestor. What doctors were not told for years is this: While making recommendations in the name of the NIH, Brewer was working for the companies that (manufacture and) sell the drugs. Government and company records show that from 2001 to 2003, he accepted \$114,000 in consulting fees from four companies making or developing cholesterol medications, including \$31,000 from the maker of Crestor.

Brewer was far from alone in taking industry's money: at least 530 government scientists at the NIH, the nation's preeminent agency for medical research, have taken fees, stock or stock options from biomedical companies in the last five years, records show.

Dr. P. Trey Sunderland III, a senior psychiatric researcher, took \$508,050 in fees and related income from Pfizer Inc. at the same time that he collaborated with Pfizer - in his government capacity - in studying patients with Alzheimer's disease. Without declaring his affiliation with the company, Sunderland endorsed the use of, Aricept, an Alzheimer's drug marketed by Pfizer during a nationally televised presentation at the NIH in 2003."

Dr. Abraham Flexner, the author of the Flexner Report in 1910, would turn in his grave if he knew what was happening in late 20th and 21st century American medicine. Funded by Rockefeller and Carnegie, Flexner rid America, if only briefly, of money driven medicine, connected medical schools with universities to encourage science based medicine and medical research, and set up a licensing

system for the protection of the American public.

Because government doctors are not audited yearly by the GAO (Government Accounting Office), they are free to accept enormous sums of money from pharmaceutical companies without fear of detection and punishment. Government doctors accepting fees from the industry that they are supposed to police are worse than bad cops because their felonious behavior affects everyone.

When it comes to anti-obesity programs the NIH and the CDC have degenerated to collectors of data and money, they are still holding onto the simple minded belief that obesity is simply eating too much and that exercising and getting on a CR diet will win the day.

If one throws out the 500 antibiotics from the total of 250,000 pharmaceuticals that are available to doctors to treat patients, this leaves 249,500 prescription drugs still available to doctors. Of the 249,500 pharmaceuticals available to doctors to treat patients for thousands of diseases including obesity, type II diabetes and the Metabolic Syndrome, however, not a single one is designed to cure a disease - they are all designed to milk the individual, private insurance and government insurance programs - and all have terrible side effects, many of which are disabling or life threatening.

We have eliminated almost 900 diseases in animals including obesity, adult onset type II diabetes and the Metabolic Syndrome with simple little nutritional formulas, we have doubled, tripled and quadrupled the animal's life spans using simple nutritional formulas rather than pharmaceuticals - these extraordinary improvements in animal health were not accomplished with wonder drugs, stem cells, genetically engineered proteins or heart transplants - they were accomplished with simple nutritional supplementation - what a concept!!



Chapter 10

"The low fat message backfired."
- Meir Stampfer Harvard Epidemiologist

The Feedlot Syndrome

The glut of grain produced by world wide peasant agriculture and large scale commercial farming has resulted in a series of mixed blessings and the opening of a modern "Pandora's Box" of "new diseases." More food has allowed greater numbers of humans to proliferate, survive and flourish; however, without the necessary systematic programs of essential nutrient supplementation, hundreds of nutritional diseases have become epidemic, many are pandemic, immune systems are compromised; the appearance of unfamiliar diseases has produced mass misery; and extended draught has resulted in catastrophic famine.

Even though the industrial revolution and the electrification of the world and their subsequent financial and scientific prosperity lifted and advanced some human cultures to dizzy heights, there was a painful and costly world-wide price. Industrialization has spread poverty, unsanitary conditions, pollution, degenerative diseases (i.e. - obesity, diabetes, Metabolic Syndrome, osteoporosis, cardiovascular diseases, Alzheimer's disease, etc.) and the appearance of "new" infectious diseases (i.e. - HIV, Ebola, Lassa, Marburg Fever, etc.) that were historically isolated in the jungles of Africa and Asia, are now efficiently spread by modern jet transportation far beyond the speed and distance capabilities of a mosquito's wings.

One of the unexpected observations that visitors to the United States report is that rich people are skinny and poor people are obese. The lower your income, the fatter you are! One out of every four of the people with below poverty incomes is obese, by contrast in households earning more than \$67,000 per year show an obesity rate of one out of six - minorities show an obesity rate of one out of three.

Wealthy people tend to be more educated and educated people tend to eat more protein and take more supplements when compared with the economically stressed.

If one disregards all other considerations, the “bang for the buck” when it comes to calories out of the grocery store, is carbohydrates. High carbohydrate, high sugar foods are cheap, tasty, they fill you up and they are commonly available.

In 2004, the University of California, Berkeley, reviewed the diet of 4,700 adults and found that soft drinks provide more than seven percent of the of the average daily adult intake of calories - the largest single source of calories in the American diet.

Together, soft drinks, alcohol, and sweets (candy), including pastries, account for more than twenty five percent of adult's calories. Add fruit drinks and salty, high carb snacks (chips, cake and cookies) and the numbers rise to thirty percent.

Twenty eight percent of Americans live in what is referred to as a “food desert,” - urban zones where grocery stores are a twenty to thirty minute drive away. People simply go to the local convenience store, even though prices are higher and the availability of fresh foods is lower.

The city of Philadelphia initiated a \$100 million project to build ten supermarkets in the “hoods,” because “you have people who are literally distanced out of healthy diets,” says Troy Blanchard, a sociology professor from Mississippi State University.

The preached and promoted remedies to obesity in America “are very middle class,” says Adam Drewnowski, director of the Center for Public Health Nutrition at the University of Washington. “They tell you, seek a healthy diet and exercise. Well, if you’re working two jobs and living in a trailer, you’re in no mood to get home and make a salad.”

With all of the statistics accumulated, with all of the current business, political and health interest in obesity, how can it be that we (Dr. Ma Lan and I) are the only ones who have figured it out!

On the world stage, we have traded the reputation for being “ugly Americans” to being “fat Americans,” people from the other industrialized nations have gone from being “westerners,” to “fat westerners.” This process of becoming fat gradually occurred over a fifty to sixty year period as we shifted from wood stoves to electrical, gas and propane kitchens.

This fattening process was initially slow and insidious, no one paid much attention, after all - we were just well fed Americans. However, something much more sinister than simply overeating and having made poor food choices had happened.

Something happened that had guaranteed we were all going to get fat - we had been placed in a virtual feedlot. American adults weighed on average, twenty five pounds more in 2004 than they did in 1960. American children weigh nine to ten pounds more today than they did in the 1960s. American teens weigh on the average twelve to sixteen pounds more today than they did in the 1960s.

According to the National Health and Nutritional Examination Survey, which is considered the gold standard of assessing American's weight, sixty five percent of Americans are overweight and “just going on a diet is not going to work.”

Survey statistics show:

Women, ages 20 - 74: weighed an average of 164.3 pounds in 2002, compared with 140.2 pounds in 1960.

Men, ages 20 - 74: weighed an average of 191 pounds in 2002, compared with 166.3 pounds in 1960.

Kids, ages 6 - 11: weighed an average of 74 pounds in 2002, compared with 65 pounds in 1963.

Boys, ages 12 - 17: weighed an average of 141 pounds in 2002, compared with 125 pounds in 1966.

Girls, ages 12 - 17: weighed an average of 130 pounds in 2002, compared with 118 in 1966.

Americans were placed in a classical "feedlot" environment, rich with a constant stream of dietary carbohydrate calories, low energy outputs and a diet almost devoid of minerals (which ensured a ravenous appetite) - the result is what we call "The Feedlot Syndrome."

A feedlot is a farm or ranch factory operation designed solely for the purpose of intensive growing and fattening of animals for human food. Raising prime beef successfully, requires a steady stream of carbohydrate, protein and fat calories in the proper ratios specifically designed for fattening.

The intense weight gain, high carbohydrate diet must be accompanied by optimal mixes of nutrients designed to encourage weight gain and drive the animal to eat with an endless and fervent hunger.

Feed, based on corn or other carbohydrates, is made available in a steady stream. Beef cattle will grow from a forty pound calf to 900 - 1,100 pounds in about a year under optimal feedlot conditions.

Growth rates slow at about one year of age and attempts to add more weight or fat become energy and economically inefficient as animals reach maturity. As a result, animals are usually killed for meat at young ages; "heavy" beef animals that can reach over 2,000 pounds at slaughter, produce large prime expensive steaks and are inefficient in terms of calories required to add the extra weight.

Meat produced in a feedlot is specifically designed for "prime" or "deluxe" upscale dining. Meat produced in a feedlot is heavily "marbled" and thickly covered with fat. Feedlot meat is tender and very tasty - it is the form of meat that is demanded by discerning dinners.

In 1920, Americans ate 165 pounds of red meat, poultry and fish per person per year; in 2002 we ate 230 pounds of animal protein per person per year.

The calories in the "modern" western diet, especially following the Korean War (1950s), primarily came from grain and carbohydrate based meals (i.e. - pasta, bread, dry cereals, potatoes, rice, soft drinks, fruits and juices, etc.). Doctors demanded that people eat

more carbohydrates and give up animal fat and switch to plant oils and margarines instead of using butter and lard - Americans dutifully complied with their doctor's instructions and joyfully climbed into the feedlot.

Despite the "healthful shift" in eating habits dictated by the medical profession, to a low fat diet by eighty five percent of Americans who revered their doctors, human weight began to go up at an ever increasing rate. As usual Americans led the way - we became fatter more rapidly than anyone else because we had devised the most efficient feedlot diet - The Seven Food Group Diet Pyramid!

The Seven Food Group Pyramid, which has been in vogue for the last twenty five years, could not have been designed more perfectly to drive human weight up and our percentage of body fat up, than if we would have tried.

The Seven Food Group Diet (the Feedlot diet) encouraged the consumption of six to eleven servings of carbohydrates each day as breads, pasta, potatoes and grains; limited the consumption of fats and didn't even talk about the need for supplemental minerals to eliminate deficiencies and to eliminate the munchies!

Tragically, Americans immediately began to gain weight, after all, we had be placed in the feedlot by the USDA, FDA and the medical profession.

We American's and individuals of other industrialized nations are fat because we have followed the misguided and ill-fated advice from the medical profession. Simply, we're fat because we eat at lot - considerably more than humans have ever eaten before - and almost all of the added food is some form of carbohydrate, a perfect feedlot diet. And our hunger was driven to psychotic levels by mineral deficiencies - after all, we had willingly given up our traditional source of dietary minerals to avoid "kitchen drudgery"- wood ashes were gone and fat bellies had arrived!

In 2004, adult women ate more than 335 more calories per day than they did in 1971, while adult men have added to their daily intake of calories over the same time period by 168 calories. The greater female increase in calorie consumption is related to their

monthly cycle, blood loss and iron deficiency resulting in greater degrees of pica and the munchies than in males!

On average, Americans ate 1,497 pounds of food per person in 1970. Personal consumption rose to 1,775 pounds per person per year by the year 2,000. A superficial glance would indicate that it is a good thing because we are eating more "vegetables" (encouraged by the 1992 USDA food pyramid), however, about one third of these vegetables are iceberg lettuce, French fries, ketchup and potato chips.

We humans were stampeded to eat more carbohydrates and less fat - we gladly followed the medical advice and we ate more servings of grain, however, the grains we ate were primarily processed white flour such as pasta, tortillas and hamburger buns which essentially have the same food value as white sugar - the whiter the bread, the sooner you're dead!

The reason that the percentage of dietary fat is down in the human diet is not that we eat less fat, but rather that we eat greater amounts of carbohydrate:

The United States produces enormous amounts of corn so economically that Americans have become very creative at figuring out new uses for it, including fuel to drive cars, fuel to produce electricity, plastics and animal and human food. Our pet animals eat corn, our livestock eats corn as high fructose corn syrup and we eat and eat corn.

Corn syrup is more economical than sugar - we Americans harvest 286 million tons of corn per year driven by American industrial and agricultural policies.

About fifty seven percent of the American corn that is produced is fed to livestock, five percent is converted to high fructose corn syrup, which is more economical, sweeter and easier to handle than granulated sugar.

Soft drink and juice manufacturers are encouraged to use the low priced corn syrup in just about every canned, bottled, frozen, baked and bagged "food," in which it replaced the far more nutritious materials.

From 1972 to 2002, the levels of sugar and syrup produced each year per American increased twenty one percent from 104 pounds to 126 pounds. During that same time period, the percentage of syrup sweetener in that total grew from less than one percent to almost fifty percent.

North America - U.S. states with the greatest obesity rates (Mississippi, Alabama, etc.) are the southern states. The more active, the supplement popping and outdoor oriented affluent American west (Colorado, Utah, etc.) have the lowest rates of obesity.

South America - As Latin America transitions from Third World economies to industrialization (including electricity, natural gas and propane), markets are filled with processed foods - they rose from only twenty percent of food sales in the 1980s to sixty percent in 2000.

Europe - Candy, fast food and dry sweetened cereals make up more than half of the food ads in ten European Union countries. The U.K. alone increased their snack food consumption twenty five percent in only five years.

Africa - In many parts of Africa, obesity is diagnosed in more children than classical "malnutrition." The country of Tunisia's urban population has shifted to processed flour breads.

Asia - In Shanghai, roads are today clogged with automobiles instead of the traditional bicycles. KFC opened a drive through restaurant in Beijing in 2002. In all Chinese cities, obesity went from four percent in the 1970s to the twenty-eight percent level in the year 2000.

Oceania - The shift to the western diet and urban cooking fuels has produced the fattest people on earth.

In all situations where a countries or regions populations shift from wood burning subsistence rural communities to urban communities the people become obese. As rural subsistence farmers move into the city, they give up their historical practice of using their wood cooking and heating ashes as "culinary" ashes and putting them into their gardens. More food is made available to the city dwellers, however, no one instructs them to take mineral supplements to make up

for their loss of “culinary” and garden minerals - pica, cribbing, cravings, binge eating and the munchies take over and they are unwittingly sucked into the Feedlot Syndrome.

Body Mass Index*

Body Mass Index (BMI) is an accurate measurement of the humans “body fat status.” BMI takes into account that short humans normally weigh less than taller people:

$$\frac{\text{Weight (pounds)}}{\text{Height (inches) squared}} \times 703 = \text{BMI}$$

Example

$$\frac{\text{Weight } 160 \text{ lbs}}{\text{Height (69 in) squared}} \times 703 = 23.6 \text{ BMI}$$

BMI	Class
< 18.5	Underweight
18.5 - 24.9	“Normal”
25.0 - 29.9	Overweight
30.0 - 39.9	Obese
40 and above	Morbidly obese

* Lemonick, M. D.: How We Grew So Big. Time. June 7, 2004.

The common denominator for all of the world’s movement toward a pandemic of obesity is in fact the migration of rural peasant populations, who had historically put the wood ashes from their cooking and heating fires into their gardens, to the urban centers that

were driven by coal, steam, electricity, natural gas and propane fuels for cooking and heating.

This pattern of modernization and movement to obesity will be repeated over and over until we commit ourselves to replacing the lost raw materials given up by giving up wood as fuel. Deficiency of wood ash (minerals) equals pica and the resulting obesity - it is that simple.



Chapter 11

"You just can't eat one." - Lays Potato Chips

*"Some seek not gold, but there lives not a man who does not need salt."
- Cassiodorus
Roman statesman*

*"Psyche and body react sympathetically to each other; it seems to me. A change in the state of the psyche produces a change in the structure of the body....conversely, a change in the structure of the body produces a change in the state of the psyche."
- Aristotle, Physiognomonica*

Pica, Cribbing and the Munchies

At least 30,000 years ago humans, driven by a salt hunger, began to mix wood ashes (wood ashes were used to cut the more valuable salt 3:1 to as much as 10:1 to extend its volume) with their dietary salt, add them directly to food as "culinary" ashes and additionally mixed the balance of each days production of ashes into their gardens as a mineral rich fertilizer, which resulted in greater fertility, survival rates of children, longer lives and stronger skeletons that survived the millennia.

The use of wood ashes by humans appeared at about the same time amongst all human cultures world wide in a parallel development. Salt and mineral hunger is not a newly acquired drive - it is as old as man himself.

Over the millennia, little changed in the human habit of con-

suming wood ash minerals as “culinary” ash added to their food and additionally using them as fertilizer until 3:00 p.m. on Monday September 4, 1882, when the first commercial electric generating plant was brought on line on Pearl Street in New York by a Thomas Edison technician.

By 1950, almost all American urban homes were heated with electricity, gas, oil, coal or propane; almost every American homemaker cooked on an electric, gas or propane stove; and ninety percent of rural American homes were heated with or cooked with electricity, gas or propane.

The new kitchen conveniences were successfully marketed by trumpeting, “the end of kitchen drudgery,” “no more messy stove or furnace ashes to take outdoors.”

About the same time wood fuels transitioned into electrical, gas and propane fuel, the home vegetable and corn garden became a labor of love for a few hobbyists as opposed to supplying a significant part of the family food supply as it had for more than 30,000 years.

The results of these technological advances and cultural shifts were catastrophic:

- 1) People moved off the land, left their traditional back yard gardens behind and purchased food from large grocery chains that obtained their products from commercially grown sources that used commercial NPK (nitrogen, phosphorus, potassium) fertilizers.
- 2) The traditional source of minerals (“culinary” wood ashes) necessary to meet our daily need for essential minerals was suddenly and even proactively replaced with “clean” fuels that left no “waste,” sooty pots or messy byproduct (ashes).
- 3) The availability of and rate of consumption of empty sugar sweetened calories (i.e. - breads, pasta, potatoes, snacks, soft drinks, etc.) rose dramatically.

- 4) Physical activity and personal expenditure of work energy was significantly reduced.

Being overweight and being obese is synonymous with being an American, however, other industrialized nations are not far behind in their rates of obesity. The "pica," "cribbing," and the "munchies" classic behaviors are always associated with being overweight and obese. These cravings are solely a reflection of mineral deficiencies - neither vitamin deficiencies, protein deficiencies, fat deficiencies, or calorie deficiencies will produce the munchies behavior - only mineral deficiencies will produce the munchies!

The snack food and fast food industries are aware of the relationship between mineral deficiencies and cravings (i.e. - sugar, salt, chocolate, carbohydrates, etc.), mineral deficiencies and bingeing, mineral deficiencies and the munchies and they use these behaviors to their financial advantage.

The snack food industry knew that mineral deficiencies produce a psychotic eating drive just as the giant tobacco companies knew that cigarettes were addictive.

The munchies behaviors are often extreme in small children and teen-agers because these age groups have the highest need for energy and minerals to support growth and development. As a result, snack food and soft drink companies have paid millions of dollars to be the sole suppliers of snacks and soft drinks in large county and municipal school districts - the kids pay for the advertising by buying the sweetened or salted treats and they become addicted to the brand while they are frantically looking for minerals.

Historically, the consumption of salt to satisfy the pica behavior was of value because salt contained some traces of other minerals and was mixed with wood ashes to extend the more valuable salt. Today, salt consumption, although contrary to popular belief, is not in of itself harmful; it does, however, present the problem of allowing our bodies to think we are also getting a wider supply of essential minerals than we really are.

"Cribbing" is a name given to a particular form of pica in domes-

tic animals. A good husbandry man knows when a horse or cow “cribs” (chews or gnaws on a wooden feed box, wagon rails, fence, hitching post or barn door, or eats dirt); the animal really has a craving for minerals that are deficient in their diets. Mineral starved animals will at first consume large quantities of supplemental minerals until their cravings are satisfied, they will automatically reduce their level of consumption to a maintenance level.

“Salt appetite” is very striking in both pregnant animals and humans. Bizarre cravings are legendary in the pregnant human (i.e. - corn starch, pickles, dirt, wood ashes, plaster, clay, etc.). From antiquity, the descriptions of cravings or “pica” are associated with pregnant women. The Hawaiian King Kamehameha’s mother Queen Kekuiapoiwa had cravings for eyeballs, although she specifically wanted chief’s eyes, she was given the eyes of sharks to eat.

“Geophagia,” or earth eating is very common in mineral deficient pregnant humans.

Pica, or perverted appetite or irresistible cravings is characterized by the eating of bizarre or unsuitable nonfood substances (or fattening non foods such as high sugar and high carbohydrate snacks and soft drinks) has been a curiosity for philosophers, doctors and priests for thousands of years. Pica as a symptom of mineral deficiencies is universal in the world and has been reported in animals and humans of all age groups.

In modern times the easy availability of candy, sugared-high calorie snacks, high sugar soft drinks, juices and salted junk foods of all types has led to obesity when people feel the obsessive, almost psychotic deep cravings of pica - “I have to eat and I have to eat it all!”

People wouldn’t crave or consume the empty calories found in snack foods if they were properly nourished with the essential minerals. No amount of advertising or give away coupons and gimmicks would drive humans to constantly consume snacks and junk food - when one is well nourished with minerals, junk foods and snacks are just not interesting.

When people are deficient in minerals “the call to eat” is an endless incessant obsession that can only be subdued by an individual’s will for but a short time - eventually the mineral

deficient individual will wake up in the middle of the night and eat an entire twenty pound ham or an entire German chocolate cake - in the end pica will always win the battle of the binge!

A doctoral thesis on pica by Augustus Fridericus Mergiletus (1701) begins with the definition of pica:

"The malady pica is called 'kitta' in Greek. Its derivation comes from the name of the common magpie because the bird ("pica" is the Latin name for "magpie") itself is believed to suffer from some malady (mineral deficiencies, yes, even wild animals suffer from mineral deficiencies) because it flits from tree to tree constantly seeking food or because it enjoys all sorts of foods."

A catalogue of bizarre examples of pica is found in Mergiletus' thesis. In men, he recorded one individual who ate leather, wood, nestlings and live mice; a second consumed woolen garments, leather, a live cat, and some mice; a third ate cattails and decomposed human bodies infested with maggots.

In Mergiletus' female subjects he recorded women who ate human flesh including one particularly horrible lady - "she lured children to her home with sweets, killed them and pickled them for her daily fare."

The most common description of pica by Mergiletus were of women's desire for mud and mortar scrapped from walls (just like modern children who eat caulking and lead paint) - we have often said that children who eat lead paint are screaming for minerals for their mineral starved bodies - give them minerals and they won't eat lead paint. Girls ate their own hair, cotton, thread, raw grain, starch and lizards.

Mergiletus also described pica in animals including cats that ate wood ash and pregnant hunting dogs that ate unusual objects.

Cooper (1957) in her classic report on pica refers to several ancient and medieval writers who emphasized the occurrences of pica in pregnant women: Aetios noted that pregnant women crave various and odd foods, some salty and some acid; some, he said, "crave for sand, oyster shells and wood ashes." Aetios recommended a diet

including "fruits, green vegetables, pig's feet, fresh fish and old tawny fragrant wine."

Boezo (1638) noted that pica occurred most often in pregnant women. Boezo viewed pica as a physiological problem, and is the first to mention iron preparations as a treatment for pica - he suggested "one and one half scruples of iron dross taken for many days as wonderfully beneficial for men and women." He also noted "the case of a virgin who was accustomed to devour salt in great quantities from which chronic behavior she developed diarrhea and wasting.

Christiani of Frankfurt (1691) reported a woman who ate 1,400 salt herring during her pregnancy. LeConte (1846) suggested, "that animals eating earth do so because of a want of inorganic elements."

In modern times, the substances frequently reported as eaten as a result of pica behavior in humans includes paper, metallic gum wrappers, ice, dirt, coal, clay, chalk, starch, baking powder, pebbles, wood, plaster, paint, chimney soot, road tar, hair, human and animal feces, cloth and carbohydrates!

Because of social constraints on our public behavior, most people under public scrutiny who display pica will eat sugar, snack foods, larger servings of food, soft drinks and smoke tobacco when they are really craving minerals.

Commonly, mineral deficient people will stand with the refrigerator door open, blankly staring into the refrigerator after consuming a 10,000 calorie, eighteen course holiday or celebration meal. When asked what they are doing, they will respond, "I don't know, I'm just looking for something!" It is not more calories they are looking for - it's the minerals that their deficient body is screaming for that drives them to the refrigerator!

Many individuals who smoke, drink excess alcohol, abuse prescription drugs or use street drugs and become addicted to sugar laden soft drinks are attempting to satisfy a deep, nagging and underlying craving that is caused by a mineral deficiency.

Ice eating (pagophagia) is also commonly observed in iron deficient children and adults.

Parasites have been reported as a cause of pica - *Ankylostomia-*

sis, ascariasis, oxyuriasis and others have been blamed. Parasites of course can cause anemia and malnutrition by direct blood sucking from the host or by competing with the host for food in the gut. The resulting anemia is in fact perceived by the body as an iron deficiency resulting in pica.

Pica is commonly seen in children, teens, pregnant women and the obese; however, David Livingston observed that "both slaves and the gentry developed pica" in Africa. In Africa, pica is viewed by tribal peoples as a magical superstitious behavior.

Henrock (1831) attributed pica to a "paucity of good blood and a lack of proper nutrition." Waller (1874) reported that David Livingston observed many cases of clay and earth eating (geophagia), a frequent form of pica in pregnant women.

Orr and Gilka (1931) and de Castro (1952) recognized that "edible earths might be rich in sodium, iron and calcium." Gelford (1945) reported that pica was common in Kenya amongst African tribes (Kikuyu) living mainly on a vegetarian diet; but absent in those people eating diets rich in animal flesh, blood and bones (Massai).

The reduced pica behavior in high animal protein diets, that naturally contain iron and phosphorus can partially explain why high protein, modest fat and low carbohydrate diets work to reduce weight. We have found that by adding the complete plant derived mineral supplement to the weight loss and weight management programs, the weight loss will be more dramatic for all and will be maintained.

Dickens and Ford reported that twenty five percent of children eat dirt. We believe that mineral deficiency pica occurs in American children at a much higher rate than twenty five percent; we believe it occurs at eighty to 100 percent, however, because of the social constraints against eating dirt, kids will instead consume more soft drinks, more juice, more French fries, more cookies, more cake, more pasta and generally more calories as a result of their mineral deficiencies.

This mineral deficiency driven pica, not playing outdoors, watching TV and playing video games, is the cause of obesity in kids. Give children and teens more minerals to replace the wood ashes that they gave up when their grandparents and parents installed electric,

gas, and propane stoves, and we will eliminate the obesity, type II diabetes and the Metabolic Syndrome epidemic in America and the pandemic in the industrialized world.

Richter (1955) published Kirkam and Birch's analysis of twenty samples of edible clay from Africa:

Clay Minerals	Percent
Silica	47.76
Aluminum oxide	22.03
Sodium oxide	4.87
Potassium oxide	2.88
Iron oxide	4.98
Manganese oxide	0.32
Calcium oxide	2.97
Magnesium oxide	1.08
Phosphorous pentoxide	0.13

Constant searches for calcium rich eatable clays and soils and territorial disputes over limited supplies often led to endless tribal wars.

Cooper (1957) reported a twenty one percent rate of pica in American children referred to the Mother's Advisory Service in Baltimore.

Lanzkowsky (1959) reported that twelve children with pica had hemoglobin that ranged from 3 gm% to 10.9 gm% with a mean of 7.89 gm% $\pm$ 2.64. The institution of iron (iron dextran i.m.) "resulted in a cure for pica in one to two weeks." Again, we have observed that when children are sufficiently supplemented with minerals they will not be obese.

McDonald and Marshall (1964) reported on twenty five children who ate sand. They divided the group in half - they gave one group iron and the other group saline (salt solution). After three to four months 11 of the 13 children given iron had lost their pica behavior compared with only three of the 12 given saline.

Reynolds et al (1968) reported that thirty eight people with anemia exhibited pagophagia (ice eating) as the most common form of pica; twenty two of twenty five had their symptoms of pica disappear after correcting the iron depletion.

Woods and Weisinger (1970) reproduced pagophagia experimentally in rats by withdrawing blood. The pagophagia in the anemic rats was cured when the anemia was cured. It continues to be of interest to note that pica and cribbing behaviors are not produced by vitamin deficiencies.

Two thirds of the 153 pregnant women studied by Taggart (1961) developed cravings. The most common craving was for fruit, pickles, blood pudding, licorice, potato chips, cheese and kippers. A craving for sweets, vegetables, nuts and sweet pickles came in second place.

Phosphate appetite was described by LeVaillant (1796) as the anxious search by cattle in phosphate deficient South African pastures for discarded dog bones (osteophagia); they also chewed on wood (pica) and each others horns.

Bone chewing (osteophagia, a form of pica) has been reported in many wild species of herbivores including reindeer, caribou, red deer, camels, giraffe, elephant, and wildebeest. We have seen elephant eating limestone road-beds and large termite heaps as ready and available sources of minerals.

It was demonstrated that calcium deficient weanling rats will consume large amounts of lead acetate solution (even though they may taste bad) when compared with calcium fed controls. The Oriental diet tends to be low in calcium, leading them to habitually and preferentially eat bones (when Ma Lan and I were courting she ate the bones and gave me the fillets) partially leached of calcium with vinegar.

Magnesium deficient animals will crave and eat excess salt. Magnesium tastes bad enough that animals will die of magnesium deficiency initiated seizures rather than consume a pure magnesium supplement. It has been suggested that recovery from a magnesium deficiency can be a very unpleasant process, enough so that the bad taste of magnesium supplements normally required for recovery ex-

ceeds the unpleasant features of the deficiency.

A natural potassium deficiency is highly unlikely in nature, as all grains, fruits and vegetables tend to be rich in potassium (NPK fertilizer) - a potassium deficiency can be found in individuals given diuretics and individuals whose only vegetable intake is French fries and ketchup.

Potassium deficient rats are hyper alert, hyper active and tend to lick everything including their metal cage, lab equipment, each other and urine puddles (beer and urine have a similar taste). It is also interesting to note that even with a severe potassium deficiency laboratory rats have a strong craving for sodium chloride (salt).

Chromium and vanadium deficiencies are manifested by thirst for liquids (soft drinks, juices, alcohol, etc.), with water being the last choice and an obsessive hunger for carbohydrates of all types. The "munchies," cravings for alcohol and candy cravings (Regan's cravings for jelly beans), especially chocolate, are sure signs of a chromium and vanadium deficiency.

Salt (NaCl) deficiency occurs commonly, and is oftentimes caused by ignorant and over zealous doctors who demand that "their" patients reduce or eliminate salt intake. The medical profession must think that humans are dumber than cows! The first thing that a farmer or rancher puts in the pasture for their livestock is a salt block (and they have been doing it for tens of thousands of years) - the animals have free choice access to salt and other minerals in uncontrolled and unlimited amounts. Aristotle noted in his *Historia Animalicem VIII*, that "sheep are healthier when they are given salt."

Where it was rare, salt was traded ounce for ounce for gold, brides or slaves. Salt and salt rich clays mixed with wood or kelp ashes were the first mineral food supplements consciously employed by man since the beginning of time and before recorded history - pica did not begin with the first appearance of snacks - pica began with the first man.

The Roman statesman Cassiodorus was quite observant when he said, "Some seek not gold, but there lives not a man who does not need salt."

The production of salt as a food supplement for humans and

beasts is as old as civilization itself. Salt was produced by evaporating sea water from shallow man made ponds and by mining rock salt from large land locked deposits.

Early villages sprang up around salt springs, humans were attracted by wild herbivores that were obligated to come daily to the springs to satisfy their driving need and hunger for salt and minerals. To obtain salt from these dilute springs, hunters kept wood fires continuously burning to boil away the water; these salt-boiling peoples activities were recorded as far back as the Neolithic Ages.

Whole forests were burned from Tyrol in the Alps, Moselle, and Franche-Count, France; the Soale and Lueburg, Germany, and Droitwich, England to support this salt production industry.

Before the Norman invasion of England, salt pans of Cheshire and Worcestershires (famous for its sauce) supplied England and Gaul, going south to the Thames River with the subsequent establishment of London.

From the 9th to the 14th centuries peat was soaked in sea water, dried, burnt and the minerals of the resultant ash were then extracted with sea water; many millions of tons of peat were harvested annually for this mineral/salt dietary supplement production process for the commercial trade.

Mesopotamian towns specialized in the salt production industry and transported salt up the Tigris and Euphrates Rivers. Jericho (8000 B.C.), near the Dead Sea and the salt mountain of Mo are the oldest known agricultural communities that participated in the salt trade.

The human salt hunger (pica) was also an important force in the African slave trade as captured children from warring tribes were sold into slavery in exchange for salt. In other parts of salt poor Africa, humans developed the practice of drinking cattle blood or urine to obtain salt.

The residents of the Sierra Leone coast gave all they possessed, including their wives and children in exchange for salt - because salt is an absolute requirement for life and because salt is not equally distributed on earth and coveted by the have-nots - it is said by African tribesmen that, "He who has salt has war."

The modern day human salt hunger is used by the American and

international snack industry to drive their sales. Heavily salted potato chips, corn puffs, popcorn, salsa, bacon, French fries and pizza dominate snack and fast food purchases; heavily sugared snacks, soft drinks, cakes, cookies and candy come in a strong second.

The child, preteen and teen age infatuation with snack foods, juices and soft drinks, drugs smoking and alcohol appear to be the result of pica, because kids are so mineral deficient they are driven to seek out something; anything to put in their mouth that might satisfy the irresistible craving. If the available choices taste better (salt or sugar), more will automatically be consumed and obesity is guaranteed.

When people stop smoking they will almost always gain weight because their mineral deficiencies were not dealt with, their cravings that had been “managed” by the use of tobacco are then no longer under control - the resulting cravings and munchies that appear when smokers give up tobacco will result in weight gain and obesity.

We humans happily replaced our historical source of “culinary” minerals derived from wood ashes with soft drinks and French fries and we are reaping the whirlwind of obesity, diabetes, the Metabolic Syndrome, and other degenerative diseases that have become a world wide pandemic since the 1950s.

God help us, through massive advertising by the Hollywood film industry, television, newspaper and magazine industries and the overwhelming availability of high carbohydrate liquid and solid snacks, and universal self inflicted mineral deficiencies our culture drives our children into the feedlot and to inevitable obesity.



Chapter 12

“...the Hunza soil is almost a manufactured soil. Every solitary thing that can serve as food for vegetables, field crops and fruit trees is diligently collected, stored and distributed in rationed equality over every square foot of the hundreds of terraces. Sunken compost pits are conveniently located, and into them go ashes from cooking and heating fires, inedible parts of vegetables, pulverized animal bones, dead leaves, rotten wood and the collected manure of animals.”

- Dr. Allen E. Banik
Hunza Land

Mineral Premix: Culinary Minerals

Since before written history, wood ashes (plant derived minerals) were the most universally available fertilizer. Wood ashes were predictable; they were a fine particle sized mineral dust and an immediately available source of plant food that was produced everyday in the human's daily life as a byproduct of cooking and warming our homes.

The systematic cultivation and fertilization of land to commercially produce food was first written about in 4000 B.C. in China. Chinese farmers used “organic material, wood ashes and silt reclaimed from streams and rivers that had been eroded from the hills and mountains to fertilize gardens and field crops.”

Wood ashes were universally available and contained a wide variety of plant derived minerals that were easily available to cultivated garden plants and directly available to humans as a source of nutritional minerals when mixed with table salt and applied to our food as “culinary” minerals.

“Culinary” ashes were of great and absolute importance to human survival. Wood, peat and kelp ashes were added to human food since before written history. The “munchies,” pica and cravings didn’t begin with the first potato chip or Hershey’s bar - the munchies began with the first man. From the first man to modern man we were and still are dependant on plant source minerals.

Food preparation, in it’s earliest and simplest form, was raw meat or crudely “cooked” in fire. Food was immediately eaten or preserved briefly by cooking, drying in the sun or by being salted.

Salt was a valuable and cherished substance and it was traded ounce for ounce for gold and slaves. As a result, salt was not squandered by simply eating it, it was mixed, “cut’ and diluted with wood, peat or kelp ashes. And so with the simple act of “extending” salt with “culinary” ashes, humans, from the beginning of time, unknowingly acquired essential supplemental minerals.

Fat Ladies of Malta

Maltese figurines of “fat ladies” date from the Upper Paleolithic era (25,000 years ago) to the earliest evidence of metal-using cultures in the Neolithic era. Some of these “fat lady” figurines have been found in western Europe, however, the greatest numbers of these figurines have been found in Egypt, the Levant, Turkey, Greece, Cyprus and the Balkans. The most detailed of the “fat lady” figurines were found on the islands of Malta dating back to 3000 B.C and 8000 B.C. in Catalhoyuk, Turkey.

The most dramatic of the “fat lady” figures is the “sleeping lady” of the Hypogeum. This relatively small statue depicts an obese lady lying on her side on a highly detailed woven bed.

These “fat lady” figurines have been interpreted by archaeologists to be the subject of religious cults who worshipped “mother goddesses” to ask for agricultural and human fertility. In Malta, the worship of obese goddesses grew into an all consuming pursuit.

Geological evidence shows that between 5000 and 7000 years

ago, Malta was a heavily wooded island. The first immigrants quickly and ruthlessly cleared and deforested the fragile island ecosystem of its original plant life. The inhabitants used the wood as fuel to boil sea water to produce salt, cook their food and to heat their dwellings.

Soil erosion soon plagued Malta, stripping the top soil and accessible nutritional minerals - crop yields dropped precipitously.

Prehistoric Maltese agriculture was carried out in an environment that had already been deforested. Studies of land snails as early as 4000 B.C. and fossilized pollen as early as 3600 B.C. confirm that Malta was already deforested and treeless - no trees - no wood; no wood ashes - no "culinary ashes - no ashes to cut salt and no ashes to put in the gardens and fields as fertilizer.

By the year 3000 B.C., Malta was affected by catastrophic ecological changes - soil erosion and environmental degradation effected the mineral values of their food grains and vegetables.

Malta became an island in trouble (i.e. - high population and an ecological and agricultural collapse) - village communities struggled to maintain earlier standards of living and produce their historical cultural foods and diet.

Wood ashes that had been plentiful in the early days of immigration, were no longer available for "culinary ashes," cutting salt or fertilizing the gardens.

The loss of pasture to wind and water erosion resulted in a significant reduction in the consumption of animal protein and an increase in the consumption of grain as a large portion of their diet. Low mineral intakes combined with high carbohydrate intakes threw the Maltese into the Feedlot Syndrome and the production of the "fat ladies."

By 2000 B.C. the entire agricultural industry in Malta vanished. The lack of the wood ash byproduct of heating and cooking produced mineral deficient grains, vegetables, fruit and a loss of culinary ashes. Wood ashes were the minerals that were left over when the carbon of the wood was burned.

The lack of the nutritional minerals produced pica, cribbing, cravings, binge eating and the munchies. The munchies, along with the high carbohydrate diet, produces the Feedlot Syndrome - ergo:

obesity!

Prehistoric Malta was a perfect model for us to understand obesity, diabetes and the Metabolic Syndrome epidemic in America and the pandemic in the industrialized nations of the world.

Unfortunately the early interpretations of the Maltese "fat ladies" were made by archaeologists whose expertise was limited to ancient cultures and religions. They failed to recognize the catastrophic results (obesity, diabetes and the Metabolic Syndrome) of a wood ash deficient culture!

The interest in substances and materials that would increase crop yields dates back to antiquity. The early Greek philosophers proposed the amazingly modern concept that plants are derived from a combination of chemicals.

Democritus of Abdera (460 - 360 B.C.) proposed, what would come to be considered, "the first writings on the atomic theory."

"Mother earth, when fructified by rain, gives birth to crops for the nourishment of men and beast. But that which came from earth must return to earth (from ashes to ashes) and that which came from air (returned) to air. Death, however, does not destroy matter, but only breaks up the union of it's elements, which are then recombined into other forms."

By contrast, Aristotle (384 - 322 B.C.), the tutor and mentor of Alexander the Great, believed that all material constituents were derived from the mutual convertibility of the four elements: earth, water, fire and air - a concept that was believed into the Medieval period and beyond. Aristotle assumed that, "plants assimilated organic matter through the roots," the progenitor of what has been known as the "humus theory of plant nutrition."

The incorrect belief that plants take up organic matter through the roots can be understood in light of the ancient observation that organic matter, particularly manure, compost and plant residue (ash) benefit plant growth.

Pliny, the Roman wrote, "It is universally agreed by all writers that there is nothing more beneficial than to turn up a crop of lupines, before they have podded, either with the plow or the fork, or

else to cut them and bury them in heaps at the roots of trees and (grape) vines.”

At 343 B.C. deserters from Alexander the Great's army established the Hunzikut culture in the Hunza Valley (19,500 foot elevation) on Mount Rakaposhi (25,550 feet elevation - the third highest mountain in the world). From the beginning of their written history 2,500 years ago, the Hunzas employed “culinary” ashes to extend table salt and used the remainder as fertilizer in their manufactured soil at an elevation of almost three times the timberline level.

In St. Matthew 13:3, Jesus, spoke in parables to the multitudes using the simple language of farmers and shepherds:

*“Behold, a sower went forth to sow;
And when he sowed, some seeds fell by the wayside, and the fowls came and devoured them up:
Some fell upon stony places, where they had not much earth: and forthwith they sprung up, because they had no deepness of earth:
And when the sun was up, they were scorched; and because they had no root, they withered away.
And some fell among thorns; and the thorns sprung up, and choked them:
But others fell into good ground, and brought forth fruit, some a hundred-fold, some sixtyfold, some thirtyfold.”*

Jesus knew that healthy well fertilized “good ground” produced good crops and that by giving his message to farmers and shepherds as a parable in their simple language, that they would hear and understand his deeper message.

In 10 A.D., the Roman, Lucius Junius Moderatur Columella, stressed the need to apply fertilizer so that, “the soil will not be tired and doesn't age when it receives fertilizer, and the farmer can have larger yields when the soil is supported by frequent and appropriate fertilization.”

But in Europe, it would not be until the 18th and 19th centuries that the idea of using animal and human feces as a source of humus was discussed systematically and promoted by German, Austrian and French scholars. However, in the early years of the 19th century it

was discovered first by Spiengel and then by Liebig, that it was not the humus as such, but rather the minerals in the humus that nourished plants.

Liebig demonstrated that potassium is one of the major essential nutrients required by plants - potassium had long been known as potash, because it was extracted from the ash left in iron pots after the burning of wood, straw and other plant materials.

"Culinary" ashes were used by all humans before the Hunzas, and have continually been used by all humans after the Hunzas. So it was with the pre-Columbian Indians. In the 13th century, the use of "culinary" ashes was recorded in North America by anthropologists.

In 1588, Thomas Harriot (1560-1621), tutor to Sir Walter Raleigh, was appointed "geographer" for his second Virginia expedition - he soon published in Hakluyt's *Voyages on North Carolina Indians* and how they raised corn. He described mountain spinach five foot high and how the Indians made a "fine broth from the seeds, and from the stalks of the herb, they produced a kind of salt by burning it to ashes, they then season their broth with it. This is the only salt they know, and they enrich their fields by burning weeds after drying in the sun. This they do to save themselves the labor of carrying it away, rather to enrich better the ground."

While the conscious use of manure continued in the Medieval times, there were no recorded theoretical contributions to plant nutrition until the Renaissance. Bernard Palissy (1510 - 1589), considered by most to be the founder of the study of agricultural chemistry, stated:

"The modern concept that the purpose of manuring was to replace substances lost (from the soil) by crop removal. Manure is carried to the field for the purpose of restoring to the latter a part of what had been removed proceeding, thus you will restore to the soil the same substances that have been removed by previous crops and which following crops will regain to their advantage."

Until the 1700s, scientists believed "that the life of animals was dependent on the consumption of food and that plants derived their

nutrition totally from the soil.”

Jan Baptista van Helmont (1577 - 1644), a Belgian physician offered the first experimental evidence that soil is not the sole source of plant nourishment. Van Helmont grew a small willow tree in an earthenware pot for five years, adding only water to the pot. At the end of five years, the willow had increased in weight by 164 pounds, while the earth in the pot had decreased in weight by only two ounces. On the basis of these results he reasoned (incorrectly) that all the substance of the plant was produced from the water and none from the soil:

“I took an earthen vessel in which I put 200 pounds of earth that had been dried in a furnace, which I moistened with rain water, and I implanted therein the trunk or stem of a willow tree, weighing five pounds. And at length, five years being finished, the tree sprung from thence did weigh 169 pounds and about three ounces. When there was need, I always moistened the earthen vessel with rain water or distilled water, and the vessel was large and implanted in the earth. Lest the dust that flew about should be co-mingled with the earth, I covered the lip or mouth of the vessel with an iron plate covered with tin and easily passable with many holes. I computed not the weight of the leaves that fell off in the four autumns. At length, I again dried the earth of the vessel, and there was found the same 200 pounds wanting about two ounces. Therefore, 164 pounds of wood, bark and roots arose out of water only.” (Jan Baptista van Helmont).

Eighty years later, John Woodward (1605 - 1728) was able to show that spearmint sprigs grew better in water containing small amounts of soil when compared to those grown in pure rainwater.

The beginnings of the clear understanding of the relationship of chemistry to plant nutrition goes back to two 17th century chemists, Johan Glauber (1604 - 1655) and Gabriel Plattes (1600 - 1655) through their analysis of “salts” such as wood ash, limestone and salt peter (potassium nitrate) and their affect on plant growth.

To compensate for the lack of manure that resulted from the Thirty Year War, Glauber invented a chemical fertilizer he called “philosophic dung” or “fattening salt.”

Despite these observations on the value of "salts" (the minerals found in the ash of burned food plants), the belief that humus (organic material) was the basic "food" of plants was upheld well into the 19th century and was supported by such renowned chemists as Theodore de Saussure (1767 - 1845) and Sir Humphrey Davy (1778 - 1829).

The source and function of the inorganic elements of plant ash was unknown. A prize was offered in Germany, "to resolve the question of whether the inorganic elements found in the ashes of plants were constituents produced by the plants or must be absorbed and what was their role in the plant."

The prize was awarded to A. F. Wiegmann and L. Palstroff based on experiments comparing the growth of plants using synthetic soil compared with plant growth in sand alone.

Fire, one of the five "elements" of the ancient Greeks, was of great interest to the alchemists and the Renaissance chemists who followed. What happens when a substance is consumed by fire was a favorite subject of study and argument between the learned chemists of the 1700s - this one question alone resulted in a revolution in the study and understanding of chemistry. The study of plant physiology played an enormous role in the chemistry revolution.

Joseph Priestley, the English clergyman and chemist, and member of England's Royal Society was extremely interested in the changes in matter that resulted from being burned in fire - ashes!

Priestley, a friend and contemporary of Franklin, was deeply involved with the problem's of coal mines in the early days of the industrial revolution, including air quality and flooding.

On August 17, 1771, Priestley "put a (living) sprig of mint into air in which a wax candle had burned out and found that, on the 27th of the same month (ten days later), another candle could be burned in the same air."

Priestley believed that, "he had accidentally hit upon a method of restoring air that had been injured by the burning of candles."

The "restorative which nature employs for this purpose," he stated, "is vegetation." Priestley pushed his observations further and demonstrated that "air restored by vegetation was not at all inconvenient

Elements Required by Plants for Optimal Health

Macronutrients	Form in which absorbed	Concentration in whole plant (dry weight)
Nitrogen (N)	NO ₃ /NH ₄	1 - 3%
Phosphorus (P)	H ₂ PO ₄ /HPO ₄	0.05 - 1.0%
Potassium (K)	K ⁺	0.3 - 6.0%
Calcium	Ca ⁺	0.1 - 3.5%
Magnesium	Mg ⁺	0.05 - 0.7%
Sulfur	SO ₄	0.05 - 1.3%
Micronutrients		
Iron	Fe ⁺	10 - 1,500 ppm
Chlorine	Cl ⁻	100 - 300 ppm
Copper	Cu ⁺	2 - 75 ppm
Zinc	Zn	3 - 150 ppm
Boron	BO ⁻	2 - 75 ppm
Manganese	Mn ⁺	5 - 1,500 ppm
Molybdenum	MoO ₄ ⁻	Trace
Needed by only a few plants		
Sodium	Na ⁺	Trace
Silica	Si	Trace

An Average Mineral Analysis of Wood Ash (1926)

Mineral	Percentage
Potassium	16 - 40
Phosphoric acid	2
Calcium carbonate	32
Magnesium oxide	2 - 5

to a mouse.”

When he was presented with a medal for his discovery (of oxygen), the citation included the statement, “For these discoveries we are assured that no vegetable grows in vain... but cleanses and purifies our atmosphere.”

It was Jan Ingenhouz (1730 - 1799), a Dutch physician who discovered that, “plants required light to produce oxygen, and that the sun by itself without plants could not produce oxygen.”

In the 19th century, Justus von Liebig (1803 - 1873), a dominant figure in plant nutrition, demonstrated that carbon was supplied by the air and not by humus (although he incorrectly believed that the carbon was absorbed by the roots).

Liebig also assumed incorrectly that most nitrogen was absorbed from the air (he was unaware of the nitrogen fixation process by soil bacteria) and not from the humus. He did conclude correctly that the common supplies of wood ash from cooking and heating fires were insufficient in volume to support the growing fertilizer needs of 19th century commercial agriculture.

At the same time, in 1856, large deposits of potassium were discovered in Germany. By 1860, the first results from potash field trials were published in Germany and throughout Europe. In 1891, the German potash industry founded the first agricultural research and advisory institute in Stassfurt. By 1910 some 32,000 field trials on potash as a fertilizer had been conducted. By 1952, the International Potash Institute was established in Berne, Switzerland.

In addition to carbon, hydrogen, and oxygen, which are obtained from water or from air, plants require a number of other elements. These elements, including minerals, are obtained from the soil.

The needs of plants for essential elements such as nitrogen or magnesium (for the production of chlorophyll) can be determined by an analysis of the molecules of which plant cells are made. Other requirements were determined by studying the ability of plants to grow in distilled water to which small amounts of various minerals are added.

Comparisons of the mineral content of plant cells with the mineral concentration of the water medium in which it grows, demon-

strate that there can be marked differences in the concentrations of various nutrients. Thus, we must conclude that substances from the soil do not diffuse passively into the root cells of the plant, but rather, they are carried across the root cell membranes, against the diffusion gradient by an energy requiring event called "active transport."

Plants only "need" sixteen essential elements, of which nine are minerals, for optimal health, they only "need" three elements (N, P, K) for optimal production in terms of tons and bushels per acre - humans need a minimum of sixty minerals out of a total of ninety essential elements for optimal health - two thirds of the essential nutrients necessary for optimal human health and longevity are minerals.

Pure wood ash is not a uniform substance or product. Different species of trees are concentrators of select minerals and minerals do not occur in a uniform blanket over the crust of the earth - minerals occur in veins like chocolate in chocolate ripple ice cream, minerals occur in veins like gold and silver making the composition of wood ash mineral content highly variable.

As a general rule, the softer woods contain less potassium and the harder woods contain more potassium, ranging from 16 to 40 percent respectively of the total mineral mix in the ash.

Wood ashes also have a very favorable physical effect on the soil, allowing for more aeration and greater capacity for retention of water.

Unfortunately, wood ashes are no longer a viable source of human mineral nutrition. Wood ashes are no longer a common byproduct of the everyday tasks of cooking and heating our homes, so we must find an alternative source of plant minerals if we are to get obesity, diabetes, the Metabolic Syndrome and other degenerative diseases under control and yes - even eliminate them.

Animal Diets Advanced Quicker than Human Diets

In 1914, there was enough knowledge regarding the need for vitamins and minerals for optimal health, maximal production and longevity, that nutritional companies sprang up that produced nutri-

ent enriched feeds for livestock. One such company was Purina Mills of St. Louis, Missouri.

The use of vitamins and minerals in animal feeds became commonplace to increase survival, fertility, production and to prevent and reverse diseases. Vitamins and minerals were added as a complete package or "pre-mix" to pellets comprised of alfalfa, corn, soybeans, wheat and oats.

To improve the efficiency of the enriched feeds and to prevent the animals from sorting through the feed and eating only what they wanted, the ration was compressed into homogenous pellets. Today, there are pellet rations for all laboratory animals, pet animals and farm animals.

Over the years this pellet ration system was so successful, that their use resulted in the elimination of as many as 900 different diseases including obesity (that still afflict humans in 2005) and doubled and quadrupled life spans of animals. This miraculous health benefit was not the result of the use of wonder drugs, genetically engineered proteins or stem cells, or heart transplants - the health benefits were simply due to the application of nutritional science. Agricultural scientists were not going to let the cow's food choices affect the farmers income.

In 1925, the W. T. Rawleigh Company produced a booklet that promoted, "Ideal Poultry Raising." The booklet informed the reader on the housing, care, feeding, culling the flock, diseases and their symptoms, nutritional prevention and treatment.

In 1934, Purina, a pioneer animal feed company, advertised in *The Country Gentleman*, "What a difference 90 days make!"

"Today, a toddling chick...in 90 days, a proud pullet ready to lay (eggs). There's something to think about! Tiny bones and little muscles have grown several times in size...a delicate fuzz has sprouted into hundreds of feathers...a chick weighing grams has grown into a pullet weighing pounds...all in 90 short days.

A wonderful change...and only one thing can do it...good feed! Pullets are made of feed. What they are...what they do for you...depends on what you feed from the time they're a few hours

old. This year consider either, Purina Startena Chow (mash) and Purina Chick Chow (scratch)....or Purina Startena Chow (all-mash) for the first six weeks...and then Purina Growena Chow (mash) and Purina Intermediate Hen Chow (scratch) until your pullets are laying in sixteen weeks.

Put these Chows before your chicks. In Startena Chow and in Growena Chow are all the needed ingredients mixed in just the right proportions to tell you a wonderful story in 90 days!"

"Feed Startena and see the difference! The difference in cost between Purina Startena and ordinary starting feeds is little. It only takes two pounds of Startena to grow a big sturdy chick ready to switch to Purina Growena. When a chick eats so little, you can't afford to take chances.

The difference in results between Startena and ordinary feeds is BIG. Because Startena is perfectly balanced and blended to furnish what a chick needs, it does a far better job of raising chicks than ordinary feeds. Greater livability, extra growth, better feathering and more yellow color in shanks and beaks - these are things you get from feeding Purina Startena, the feed that contains Pur-a-tene."

"What a Difference at 6 weeks! There's no question about seeing the difference Purina Startena makes in chicks at six weeks. They weigh more, they are better feathered. They have stretch and frame. They are sturdy and lively. It's this difference that Startena makes in chicks at six weeks that has brought the reputation to Startena as being America's best for starting chicks.

Two pounds Startena for the first six weeks - that's all it takes to build a big, sturdy chick. When so much depends on so little, the best is none too good.

Actual records last year, on 925,427 chicks fed Startena, show that 93 percent of them lived. At six weeks of age, the light breeds averaged 17.4 ounces per chick; the heavy breeds 19.2 ounces per chick. No wonder the big swing is to Purina Startena."

A Multiple Sample Analysis of Wood Ash (mg/G)*

Analyte	Amount
Calcium	271 - 307
Iron	2.2 - 3.7
Magnesium	16.5 - 46.5
Potassium	15 - 83
Copper	0.05 - 0.15
Zinc	0.10 - 0.92
Sodium	1.0 - 13.9
Chromium (ug/G)	0.004 - 0.005
Phosphorus	7.3 - 19.4
Silica	1.1 - 17.6
Aluminum	2.4 - 7.4

March 1998

The Navajo Indians

The diet's of the Navajo Indians have changed significantly since the pre-Columbian times. Archeological data suggests that their early diet, was the standard fare of hunter gatherers and consisted only of indigenous plants and wild game until 1200, when a northern Athabaskan tribe of Navajos migrated to the southwest and learned farming from the Pueblo Indians. Corn, laced with "culinary" ashes, became a staple in the Navajo diet. Two centuries later, the Spanish brought sheep and cattle, and the Navajo became shepherds.

Today the Navajo Nation numbers more than 151,000 and is the largest Native American Tribe. In recent decades, many processed foods have been introduced into the Navajo diet as a result of government rationing, food distribution programs and the growth of trading posts and food stores, yet the tradition of "culinary" ashes being added to their food remains in the more rural areas.

A Trace Mineral Analysis of a Multiple Sample of Juniper Ash (ppm)

Analyte	Amount	Analyte	Amount
Aluminum	9.41	Mercury	0.006
Antimony	0.062	Molybdenum	0.041
Arsenic	<0.01	Neodymium	0.048
Barium	1.42	Nickel	0.078
Beryllium	<0.005	Niobium	<0.01
Bismuth	0.011	Osmium	<0.005
Boron	0.759	Palladium	<0.005
Bromine	0.22	Phosphorus	8.69
Cadmium	<0.005	Platinum	<0.005
Calcium	698.0	Potassium	536.0
Carbon	107.0	Praseodymium	0.048
Cerium	0.064	Rhenium	<0.01
Cesium	0.055	Rhodium	<0.051
Chloride	620.0	Rubidium	0.121
Chromium	0.011	Ruthenium	0.067
Cobalt	0.041	Samarium	0.055
Copper	0.258	Scandium	0.12
Dysprosium	0.029	Selenium	0.060
Erbium	2.11	Silica	67.9
Europium	0.010	Silver	0.045
Fluoride	0.29	Sodium	26.1
Gadolinium	0.036	Strontium	2.44
Gallium	0.017	Sulfur	5.96
Germanium	0.099	Tantalum	0.018
Gold	0.021	Tellurium	<0.01
Hafnium	0.028	Terbium	<0.01
Holmium	0.01	Thallium	0.011
Indium	0.02	Thorium	<0.005
Iodine	2.15	Thulium	0.023
Iridium	<0.005	Tin	0.009
Lanthanum	<0.005	Titanium	0.486
Lead	0.016	Tungsten	<0.01
Lithium	0.036	Vanadium	0.085
Lutetium	0.013	Ytterbium	0.021
Magnesium	1760.0	Yttrium	0.011
Manganese	3.12	Zinc	1.87
		Zirconium	0.028

Juniper ash proves to be a good source of dietary calcium, iron and a moderate source of magnesium. One teaspoon of juniper ash provides the equivalent calcium levels found in one cup of milk; it supplies twenty eight percent of the RDA for iron and nine percent of the RDA for magnesium. The woody branches contain more calcium and silica, whereas the twigs and needles (the growth part of the tree) contained the highest levels of trace minerals.

Typical recipe that includes “culinary” ashes:

Shaped Corn Griddle Cakes

Mix one cup juniper ashes with water and strain through a grass brush into basin of corn flour, using about three and one half cups of the liquid. Add one half cup clear water to this greenish dough and knead until thick. Shape with moist hands into round cakes less than an inch thick and four inches in diameter. Bake on a fry griddle, rubbing it (the griddle) with a cloth between bakings, and moisten cakes on both sides when done.

Even today, the corn based foods of the Navajo contain “culinary” ash, which is mixed with boiling water and added after straining, to the blue corn flour “to make it blue.” Blue corn contains an anthocyanin pigment that turns a distinct greenish-blue color in the alkaline medium created by the addition of wood ash. The wood ash was also used daily to make a “tea” and was used medicinally for stomach and intestinal ailments.

Of the three major types of plants used to make the culinary ashes listed in the Navajo culture, juniper (*Juniperus monosperma*), commonly called cedar is used most frequently. The small branches and needles are most favorably used as they produce the finest particle ash (the minerals that are left when the wood and needles are burned).

Green juniper branches are gathered, placed on a grill or screen and set alight. The ash is collected in a pan placed underneath the grill. Juniper being absent, ash was also made from dried tumbleweed or greasewood.

The three sources of wood “culinary” ashes were all good sources of essential minerals. Iron concentration was higher in the grease-

wood; wood ash, although variable in analysis, tends to be rich in potassium, calcium, iron, copper and zinc, trace minerals and rare earths.

In the absence of wood ash, special clay was mixed with the salt and/or the Navajo food. Unfortunately in the 1940s, urban Navajo cooks began replacing "culinary" ashes with the more available baking soda - thus began the transition from the slim wood ash using Navajo shepherds to the obese baking soda using urban Navajo.

The Okinawa People

In 2003, a small group of our associates, including Blake Graham, and Dr. Wallach went to Okinawa to determine where these exceedingly long lived people obtained their nutritional minerals. We went first to the Department of Agriculture and learned that the soil of the entire Okinawa archipelago of 161 islands was made up of red, yellow or black clay - no coral calcium.

Many large studies, including the twenty five year long Okinawa Project that was conducted by two Harvard medical doctors, were produced in an attempt to determine the Okinawa secrete of longevity - so we felt we didn't need to ask all of the detailed questions of diet, religion, exercise and professions that had already asked over and over again. We decided that we were going to ask only one question - "what did you do with the wood ashes from your cooking and heating fires?"

We visited many of the Okinawa "old ones," and shared traditional meals of boiled pork, steamed sea vegetables, bitter melon and squash with them. We asked all of the old men over 100 years, "what did you do with the wood ashes from your cooking and heating fires each day?" The old men replied, "we don't know."

We asked the Okinawa women of ninety and 100 years or more, "what did you do with the wood ashes from your heating and cooking fires each morning?" They replied, "Everybody knows that, we put the ashes into our gardens and we added them to our rice to produce a rich yellow color!"

The Japanese and the people from Okinawa used "culinary" wood

ash! They preferred the hardwood trees for fuel because they burned hotter and burned longer - unknown to them, the hardwood trees also contained considerably more minerals than the soft wood trees.

A recent comparison of the mineral content of the bones from the modern Japanese and the 3,000 year old bones from ancient Japanese monks showed that the ancient bones contained a wider variety of minerals and a greater amount of each mineral - a testimony to the nutritional value of "culinary" wood ashes.

The mystery of the Okinawa longevity had been solved by asking the question that no one had asked before - "what did you do with the wood ashes from the heating and cooking fires each morning?" The simple answer was that they "used them in their gardens as fertilizer and in their rice as 'culinary' ashes."

It is often said that, "familiarity breeds contempt." This human failing has cost us dearly. Over time, because wood ashes were dirty, inconvenient and so "old time," we lost our respect for, our interest in and our understanding of wood ashes - wood ashes that were cherished by humans 30,000 years earlier. As a result, we made a catastrophic error as we willingly and proactively gave up the "drudgery" of the wood stove, scorched pots and wood ashes and moved almost overnight to electricity, natural gas and propane household fuels.

We have to understand, why we originally used wood ashes before written human history. We have to understand what we gave up before we can know how to replace it (just like alcoholics have to know they are in trouble before they can begin to solve their problem). We can't replace wood ashes with pharmaceuticals!

We Americans have to get busy, because our window of time for a healthful survival is rapidly closing. If we keep doing what we're doing our health care system will bankrupt us even as it fails to give us the "bang" for our health buck. Simply said, Obesity is here to stay unless we supplement our diets with the plant minerals we have so abruptly given up!



Chapter 13

“Artificially sweetened drinks might not help the (weight loss) diet, because they might make it harder for the body to know when to stop scarfing. Given a choice between (identically) flavored high and low calorie liquids, rats (preferentially) guzzle the high calorie stuff.”

*- Susan Swithers, Purdue University
International Journal of Obesity
July 2004*

Size Does Matter: the Key to Cellular Fitness

We know that if you do nothing else, consuming 1,200 mg of calcium each day will reduce your risk of being overweight by sixty five percent! If you are overweight, give up 500 calories each day (one burger), and consume the 1,200 mg of calcium per day and you will double your rate of weight loss - even without the need for exercise!

Supplemental minerals are truly the prevention and cure of being overweight and obese. However, all minerals are not created equal - it doesn't matter how many minerals or how much of each mineral you consume - what counts is how many minerals and how much of each mineral you absorb!

Remember, we require a minimum of sixty minerals to be optimally healthy, physically fit and to maximize our life spans. Unfortunately, with the advent of electric, gas and propane cooking stoves

and heating furnaces, we have given up the traditional and ideal source of nutritional minerals - plant minerals derived from the wood ashes that supported the human need for minerals since before written history.

Because the daily supply of wood, peat and kelp ash generated in the home could not fulfill the growth requirements of commercial agriculture, agricultural chemistry went another direction. Commercial fertilizer, NPK, appeared in 1840 to meet the need of a growth industry necessary to meet the food demands of the now urbanized population of the industrial revolution; shortly thereafter, the first commercial electrical generating plant came on line in 1882.

The urbanized citizen of the industrialized nations demanded food and lots of it - they traded the hours they had historically spent in their gardens and fields for hours in the factories and cash money. They now depended almost totally on commercial agriculture for their "daily bread."

Unfortunately their bread was not like their fathers bread - yes, there was lots of it, however, the new bread was critically short of minerals. In 1910, eighty five percent of Americans were still subsistence farmers or gardeners - and they were still putting wood ashes into their gardens and their food. By 1950, almost 100 percent of urbanites from all industrialized nations were solely dependent upon commercial agriculture for their food.

We must consciously supplement to make up the difference between the high levels of minerals that were in our father's bread and the significantly reduced mineral levels in our commercially prepared bread.

USDA Nutrient Data Base Comparing 1975 Values With 2004

Only when one looks at a USDA data base that compares the level of vitamins and minerals in certain common foods between 1975 and 2004, do we truly understand how depleted our foods have become since we gave up back yard gardens and wood ashes:

USDA Nutrient Data Base

Food	1975	2004	% Change
Iron (mg per 100 grams)			
Apples	0.3	0.12	- 60%
Blueberries	1.0	0.28	- 72%
Onions	0.5	0.19	- 62%
Tomatoes	0.5	0.45	- 10%
Carrots	0.7	0.30	- 57%
Spinach	3.1	2.71	- 13%
Corn	0.7	0.52	- 26%
Peas	1.9	1.47	- 23%
Broccoli	1.1	0.73	- 33%
Kale	2.7	1.70	- 37%
Strawberries	1.0	0.42	- 58%

Calcium (mg per 100 grams)			
Apples	7	6	- 14%
Blueberries	15	6	- 60%
Onions	27	22	- 18%
Tomatoes	37	33	- 11%
Green beans	56	37	- 33%
Corn	3	2	- 33%
Peas	26	25	- 4%
Broccoli	103	47	- 54%
Kale	249	135	- 46%
Strawberries	21	16	- 24%

Vitamin A (Int. units per 100 grams)			
Apples	90	54	- 40%
Blueberries	100	54	- 46%
Onions	40	2	- 95%
Tomatoes	900	623	- 31%
Corn	400	1	- 400%
Broccoli	2,500	660	- 74%
Strawberries	60	12	- 80%

Vitamin C (mg per 100 grams)			
Blueberries	14	9.7	- 31%
Onions	10	6.4	- 36%
Carrots	8	5.9	- 26%
Potatoes	20	11.4	- 43%
Green beans	19	16.3	- 14%
Spinach	51	28.1	- 45%
Corn	12	6.8	- 44%
Broccoli	113	89.2	- 21%
Kale	186	120.0	- 35%
Strawberries	59	59.0	0%

Immigrant Statistic

A simple barometer of the nutritional value of the American diet is the health of immigrants. We can look at what happens to people who leave their third world birth place and come to the U.S. for a new life. In a study published in the *Journal of the American Medical Association*, December 15, 2004, it was reported that, "The longer you live in America, the more likely you are to be obese."

The study examined the obesity rates among immigrants and native born Americans. It found that immigrants, regardless of race or country of origin, living in the USA for more than fifteen years resulted in an average of a ten pound weight gain - the product of being placed in the mineral deficient "Feedlot Syndrome" fostered by the American "four food group" mentality, and the loss of their historical "culinary" and garden ashes without any concerted effort to replace them.

Minerals Have Personalities

The most significant development in the trace element field of study in the last twenty five years of the 20th century was the recognition that trace mineral elements have a fundamental role in health maintenance for humans and that optimal intakes from food sources can no longer be taken for granted.

The essential role of trace elements in the support and maintenance of the human immune system against bacteria, viruses, fungi and cancer, degenerative diseases and birth defects are well documented. The mode of action and the tissue concentrations of trace elements varies considerably. The only common denominator of trace minerals is that they are all found in exceedingly low amounts in the healthy tissue. They are usually found in parts per million (ppm); micrograms/gram (u/g); or with some as low as parts per billion (ppb) or nanograms per gram (ng/g).

The concentration of trace elements in tissue or requirement levels does not represent their relative importance as an essential nutrient. The concentrations of trace elements and their functional con-

centration must be continuously maintained within narrow physiological limits if the functional and structural fabric of the tissues and organs of the animal and human body are to be maintained for optimal fertility, development, growth, health and longevity.

Higher species such as birds, mammals and humans have a "homeostatic" mechanism that maintains tissue concentrations of trace minerals at optimal levels at their physiological sites.

Almost all trace elements function as catalysts for a cellular or sub-cellular enzyme system. Enzymes are simply proteins that do work and in their role as catalysts their function ranges from that of a weak ionic (ph affects) modulator to highly specific metalloenzyme ("metallic finger") functions for DNA and RNA.

In metalloenzymes, the metal atom cannot be substituted by any other trace element. Although Vallee has demonstrated that Co and Cd can replace Zinc in certain enzymes with some changes in substrate specificity but the enzyme is not deactivated.

Because of the frequently wide disparity between the absolute and dietary requirements (results from different intestinal absorption efficiencies for individual elements) the most efficient form of trace mineral supplement, such as plant derived colloidal minerals should be consumed.

For the metallic form of iodine the ratio of absorption is almost 1:1; for iron it's 1:10 and for chromium it's 1:100.

The RDA (Recommended Daily Allowance) and RDI (Recommended Daily Intake) and other dietary recommendations for trace elements are conceived by the National Science Foundation to "meet the needs of practically all healthy people" - they are not applicable for people with "health challenges" or full blown disease states.

Initial Depletion Phase

Initial depletion of trace elements in animals and humans is characterized by changes only in the metabolism of the element itself in response to a suboptimal intake, the individual target tissue compensating so that no disturbance of biological structure or function are detectable by means of a test. Increased intestinal absorption effi-

ciency; and/or reduction in excretion; and gradual diminishment of body reserves occurs. Pica occurs in a mild form at this early stage of deficiency. This phase of trace mineral deficiency returns to normal (including the elimination of the pica behavior) quickly with a return of optimal intake.

Compensated Metabolic Phase

The compensated metabolic phase is characterized by a reduction of certain specific biochemical functions, such as trace mineral dependent enzyme reactions or receptor sites. The "homeostatic" mechanism becomes ineffective in maintaining the normal trace mineral concentrations at receptor sites causing a reduction in biochemical function (i.e. - low blood sugar, irregular heart beat, loss of hair color, weight gain, mild obesity, etc.). Pica becomes evident in the individuals everyday life - they become "chocoholics" or "cola addicts." Supplementation with minerals will quickly reverse the symptoms and behavior changes including pica and the more extreme "munchies."

Decompensated Metabolic Phase

The decompensated phase of trace element deficiency is characterized by the appearance of serious clinical symptoms or defects in the metabolic (i.e. - pre-diabetes, significant weight gain, obesity), cognitive (i.e. - learning disabilities, ADD, ADHD, etc.), emotional (Dr. Jykell/Mr. Hyde, "Bad Seeds," depression, manic depression, bipolar, anger, rage, anxiety attacks, panic attacks, etc.), developmental (dyslexia, gay behavior, spina bifida), structural (muscular dystrophy, obesity), loss of work-stamina and reduction of the longevity (average M.D. lifespan is 56 years of age) of a species. Obsessive pica is obvious to even the casual observer - the affected individual is aware that they can not help themselves in curbing their cravings and appetites by means of conscious will alone.

Clinical Phase

The clinical phase of trace mineral deficiency is characterized by the onset of full blown disease states and even death (i.e. - cardiomyopathy, diabetes, cancer, aneurysms, arthritis, goiter, morbid obesity, etc.). For most trace mineral deficiencies in the clinical phase, diagnosis by the "orthodox" physician is almost nonexistent; the cure however, is simply a matter of replacement of the trace element (a correct diagnosis can lead to a remarkably rapid positive clinical response ranging from forty eight hours to thirty days for recognizable results; including the loss of up to one pound of excess weight per day on a proper mineral supplement program); others do not reach full return to normal after permanent biochemical, chromosomal or gross physical damage has occurred.

There are three basic types of minerals; they are not of equal size or biological value and size does matter!

Metallic Minerals

Metallic minerals include egg shell, oyster shell, calcium carbonate, limestone, coral calcium, dolomite, inorganic mineral salts, sea water, Great Salt Lake water, mineral oxides (i.e. - iron oxide, copper oxide, etc.), vortex waters, sea bed minerals, "soils" (which are usually clay), sea bed clays, clays, "rock flours," and various antacids such as Rollaids and Tums.

Typically, metallic minerals are found in tablets and powders as gluconate (calcium gluconate, zinc gluconate, etc.), lactate, sulphates, carbonates and oxides (iron oxide, etc.).

Metallic minerals, despite wild claims to the contrary are only eight to twelve percent biologically available to younger animals and humans; after attaining the age of thirty five to forty years, the availability of metallic minerals plummets to three to five percent.

A typical metallic mineral supplement, either alone or as part of a multiple is calcium carbonate. Calcium carbonate can be purchased in 1,000 mg tablets or capsules (i.e. - coral calcium, limestone, egg

shell, oyster shell, etc.), which breaks down to 400 mg of metallic calcium and 600 mg of carbonate. Two 1,000 mg calcium carbonate capsules or tablets do not give you 2,000 mg of calcium - they only give you 800 mg of elemental calcium and at a ten percent availability rate you net eighty (80) milligrams of bioavailable calcium. Therefore, to get 2,000 mg of usable calcium from a 1,000 mg capsule or tablet of calcium carbonate, you would have to take twenty five (25) capsules each day! A 100 capsule bottle of calcium carbonate will therefore only last four days.

Amounts of Metallic Calcium in a 1,000 mg Tablet

Calcium Source	mg available
Calcium gluconate	90
Calcium carbonate	400
Calcium acetate	230
Calcium citrate	210
Calcium lactate	140
Cows milk (1000 mg fluid)	10

Chelated Minerals

Chelated (Key - lay - ted) minerals were an invention of the livestock industry. In the 1940s, the livestock industry, to ensure maximum availability of dietary and supplemental minerals fed to livestock being fattened for market, created a new form of minerals.

The original chelating agent used was calcium EDTA, a man-made amino acid that was created by the Germans just prior to WWII to antidote arsenic and lead exposure in chemical warfare attacks (calcium EDTA is used today for intravenous chelation therapy to open up obstructed arteries).

The term chelated, from the Latin, literally means "claw," but in more modern times it is used to describe the process by which an amino acid, protein or enzyme (enzymes are proteins that do work) is wrapped around the mineral molecule which enhances the

bioavailability of the metallic mineral.

Chelated minerals are approximately forty to sixty five percent bio-available, a 400 percent increase over the availability of metallic minerals - certainly a significant improvement over the amounts of elemental minerals absorbed.

Chelated minerals are easily recognized on the label of a multiple vitamin/mineral supplement or free standing calcium supplement such as selenomethionine, chromium picolinate, or sometimes as an "amino acid chelate."

Colloidal Minerals

Colloidal (Koal-oid-ul) chemistry while not a new concept, is not widely known about or understood by the general public. Simply said, a colloid refers to a substance that exists as ultra fine particles that are suspended in a medium of different matter. The colloidal state is the state of a solute (mineral or other substance such as a paint pigment) in a solution when its molecules do not separate into atoms as with a true solution (sodium chloride or salt separates into separate sodium and chloride atoms while in solution), but rather the colloidal particle remains grouped together to form solute particles.

The presence of these inorganic colloidal particles, which are approximately one hundred-thousandth to one ten-millionth of a centimeter in diameter (about 400 thousandths to four millionths of an inch), can often be detected by means of an electron microscope. As a result of the grouping of the molecules, a solute in the colloidal state cannot pass through a suitable semi-permeable membrane and gives rise to negligible osmotic pressure (they will pass through filter paper), depression of freezing point and elevation of boiling point effects.

These ultra-fine particles of the colloid are just barely larger than most molecules and are so small that they can't be seen with the naked eye - about one billion of these colloid particles would fit into 0.01 cubic inch.

The "solution" part of a colloid provides a solid, gas or liquid

medium in which the colloid particles are suspended. The suspended particles in a colloid can also be a solid, a gas or a liquid.

In 1925, H. Freundlich classified solutions into three categories:

1. True solutions
2. Colloidal solutions
3. Emulsions and suspensions

The four part method of classifying solutions is as follows:

1. Identify particle size
2. Determine presence of Brownian movement (random movement of particles suspended in liquids or gasses resulting from the impact of molecules of the fluid surrounding the particles).
3. Ability to pass through filter paper
4. Level of solubility

In 1975, S.S. Voyutsky (a Russian) wrote the classic text on colloidal chemistry. Voyutsky referred to solutions as “molecular dispersion systems” and “heterogeneous highly dispersed colloidal systems.”

The exact point between the molecular and colloidal degrees of dispersion cannot be established because the transition from molecularly dispersed systems to coarsely dispersed systems is a continuum.

A colloidal system must have three basic characteristics:

- 1) It must be heterogeneous (consists of dissimilar ingredients or constituents).
- 2) The system must be multi-phasic (i.e. - solid/liquid; gas/liquid, etc.)
- 3) The particles must be insoluble (do not dissolve in the solution).

Each one of these classifications interacts with the others to give colloids their unique qualities. The interesting thing about colloids is that they remain heterogeneous, multi-phasic and insoluble at differ-

ent concentrations as long as a larger number if not all of the particles are within the range of sizes of colloids (1n to 100 n).

The molecular groups or particles of the colloid solute carry a resultant electrical charge, generally the same sign (negative) for all of the particles. A small percentage of these inorganic colloids will pass through the intestine of a living animal or human because a natural chelating process takes place in the gut in the presence of protein-containing food.

Inorganic colloidal material which readily passes through filter paper may be separated from dissolved substances, such as starch, sugar or salt, by placing the mixture of mineral colloid and non-colloid in a parchment shell surrounded by distilled water. The inorganic colloids are "too large" to pass through the membrane, but the molecules of salt, starch and sugar or any other dissolved substance pass readily through the semi-permeable membrane (they separate into individual atoms or very small molecules). This kind of separation process is called dialysis.

In the process of digestion, the inorganic minerals in food or supplements soon become inorganic colloids and as an inorganic colloid they cannot penetrate the intestinal wall to enter the blood stream. In the presence of amino acids, a small percentage of the inorganic colloids form chelated minerals and organic colloids which are able to be dialyzed through the mucus membranes of the intestinal walls into the blood stream - this bio-available mineral state is known as a "crystalloid."

Crystalloids or organic colloids readily pass through cell walls, while inorganic colloids are "too large." Additionally, we must remember that in the living organism there are other physiological forces at work, notably "active transport," which interferes with or modifies the expected osmotic phenomenon.

Active transport, refers to the act of the intestinal mucosal cell reaching out with a pseudopodium, engulfing and transporting the particle into itself and then transported into the adjacent blood stream. In fact most absorbed food and nutrients are absorbed into the human body by active transport rather than by a passive trip through a semi-permeable membrane.

Colloidal mineral supplements and commercial colloids are found in four different forms:

1. Unprotected colloids are made of bare "rock flour," this is the form of inorganic metallic colloid found in the sea bed minerals, clays, "soils," and "glacial milk." This form of inorganic colloid is in fact a metallic mineral and is only available to plants when there is a healthy soil population of bacteria and fungi.

2. The second type of mineral colloid is found in living systems of bacteria, fungi, green plants (food crops), animals and humans and is coated by a water loving (hydrophilic) substance such as gelatin, albumin, albuminoids, or collagen. This coating protects the now "organic mineral colloid" and allows it to be a crystalloid for absorption, storage and physiological uses and thus maximizing its bioavailability up to 98 percent.

3. The third type of organic mineral colloids has a protective coating of carbon with a molecular chain length of 10 to 12 carbon atoms. This type of colloid is also found in bacteria, fungi, plants (including some forms of petrified wood), animals and humans and is thought to be the most stable form of natural organic mineral colloid.

4. The fourth type of mineral colloid is not found in nature, but rather is manufactured industrially by coating the metallic colloid with sulfated castor oil (lipophyllic or fat or oil loving) to form commercial detergents.

Humic Shale

Humic "shale" is a unique source of plant derived colloidal minerals. Humic shale originated from plants during the Carboniferous Era, those lush tropical plants took up some seventy seven plus metallic minerals available to them from a fertile soil that had as many as eighty four minerals.

The ancient soil was so rich with minerals that trees grew as much as twenty five feet per year; the great brontosaurus or "thunder liz-

ard” attained a body weight of 140,000 pounds (seventy tons) with a mouth no larger than that of a horse - the ability of the brontosaurus to attain such a bulk with so little a mouth, meant he was consuming plants rich with concentrated mineral nutrients.

A volcanic eruption covered these mineral rich forests with a thin layer of mud and ash, thick enough to create an air-tight “vault” and dried or desiccated the plants into a deep accumulation of “hay,” plant compost or a “pre-coal” form, but not deep enough or heavy enough to pressurize the dried plant material into coal or oil - in effect it became a prehistoric compost.

The entombed humic shale never fossilized or petrified, in other words they never became rocks - just a compressed, dried, prehistoric compost pile that contains a high level of plant derived colloidal minerals.

Humic shale can be used as an excellent soil conditioner for organic gardens, farms and ranches as it supplies a rich source of humus and no less than seventy five plant-derived colloidal minerals.

Humic shale can also be ground into a fine meal and soaked for three to four weeks in filtered spring water, it then provides a high quality plant derived colloidal liquid mineral supplement that contains not less than 19,000 mg of plant minerals per quart.

It takes seventy eight pounds of humic shale to produce the minerals in one quart of liquid plant minerals. The seventy eight pounds of humic shale represents approximately 1,034 pounds of lush prehistoric green mineral rich plants. The fluid extract of high grade humic shale contains no less than seventy seven plant derived colloidal minerals and is up to 98 percent available for animals and humans.

In the absence of wood ashes for use in cooking (“culinary” ashes), to cut table salt and to fertilize the garden, we must supplement with minerals to avoid mineral deficiency diseases including obesity (the Feedlot Syndrome), diabetes, the Metabolic Syndrome, arthritis, osteoporosis, periodontal disease, anemia, hypothyroidism, bone spurs, kidney stones, etc.

Our experience with literally millions of users, tells us that the plant derived colloidal minerals are an effective replacement for the

Essential Minerals Found in a High Quality Humic Shale *

Aluminum	Gold	Rhenium
Arsenic	Hafnium	Rubidium
Barium	Holmium	Samarium
Beryllium	Hydrogen	Scandium
Boron	Iodine	Selenium
Bromine	Iron	Silica
Calcium	Lanthanum	Silver
Carbon	Lithium	Sodium
Cerium	Lutecium	Strontium
Cesium	Magnesium	Sulfur
Chloride	Manganese	Tantalum
Chromium	Molybdenum	Terbium
Cobalt	Neodymium	Thulium
Copper	Nickel	Tin
Dysprosium	Niobium	Titanium
Erbium	Nitrogen	Vanadium
Europium	Oxygen	Ytterbium
Gadolinium	Phosphorus	Yttrium
Gallium	Potassium	Zinc
Germanium	Praseodymium	Zirconium

* Many of the minerals listed are not "classically" considered to be "essential," however, they all meet two or more of the criteria for an essential nutrient.

wood ashes that are no longer available to modern Americans.

Failure to consciously replace the lost plant minerals we historically harvested from our cooking and heating fires with colloidal plant minerals (plant derived colloidal minerals) will be the most catastrophic event in human history - the cost in human misery, death and increase in medical cost will destroy American society, a goal that eluded our cultural, financial and political enemies (both foreign and domestic) have worked towards since 1776.



Chapter 14

“The most exquisite folly is made of wisdom spun too fine.”

- Benjamin Franklin

Obesity, Diabetes & the Metabolic Syndrome

Sixty five percent of Americans are overweight and thirty one percent are obese (thirty pounds or more overweight); by 2005, obesity will surpass tobacco as the number one preventable killer of Americans; by 2010 seventy five percent of Americans are projected to be over weight and forty percent of Americans are projected to be obese.

Obesity, treated as a disease of gluttony, increased by ninety five percent between the 1990s and 2000; anti-obesity drugs, diet and surgery treatment programs have made 20th and 21st century doctors wealthy, however, obesity has cut into the bottom line of many other industries including the travel industry.

On December 8, 2004, the Northwestern University Medical School of Chicago, Illinois released a report that was published in the *Journal of the American Medical Association*, the report highlighted the American obesity epidemic's effect on the Medicare system - “with current trends of increasing overweight and obesity afflicting all age groups, urgent preventive measures are required not only to lessen the burden of disease and disability associated with excess weight but also to contain future health care costs incurred by the aging

population.”

The report from the Feinberg School of Medicine, “found that the annual average Medicare charges for severely obese men were \$6,192 more than for non-overweight men - 84 percent higher.”

For severely obese women, “the annual average Medicare charges were \$5,618 more, or 88 percent higher than for women not overweight.”

For men, “the total average annual Medicare charges for those not overweight were \$7,205, for the overweight \$8,390, for the obese \$10,128 and for those men who were severely obese \$13,674.”

The total average annual Medicare charges “for women in the same weight groups were \$6,224, \$7,653, \$9,612 and \$12,342.”

Approximately 12 percent of American adults over the age of sixty five years were considered obese in 2004, a figure that is projected to increase to 20 percent by the year 2050 - obesity usually occurs concurrently with hypertension, diabetes, coronary artery disease and the Metabolic Syndrome.

During the 1990s alone the average American's body weight went up ten pounds. Seemingly benign, these ten pounds have crushed the American airlines industry with their weight. These extra ten pounds caused the airlines industry to spend an extra \$275 million to consume an additional 350 million gallons of jet fuel to carry the additional blubber in the year 2000 alone (*American Journal of Preventive Medicine*).

“The obesity epidemic has unexpected consequences beyond direct health effects,” said Dr. Deron Burton of the CDC. “Our goal was to highlight one area that had not been looked before.” According to the study, the additional fuel burned to carry the additional weight also had a devastating environmental impact; an estimated 3.8 million extra tons of carbon dioxide were released into the atmosphere in the year 2000 which directly contributes to our earthly problems of global warming and desertification.

The average human has 40 billion fat cells, and they can easily and quickly reproduce and duplicate into a population of 100 billion fat cells in the obese; and they are tough, resilient and defend themselves vigorously almost as if they were directed by a brain that

had a defense plan.

The most simplistic view of obesity is that, “the fat person consumes more calories than they burn.” However, it is quite clear that obesity is not that simple. Fat people are driven to eat! Their conscious will can only override this command to eat for so long and eventually they will succumb to the command and eat, and eat, and eat.

This almost psychotic urgency to eat in spite of being obese, in spite of having high blood pressure, in spite of heart disease, in spite of having diabetes, in spite of social rejection is a powerful and irresistible force.

This force that commands a fat person to eat despite all human effort to resist can only be created by one possible condition - mineral deficiencies that result in pica a.k.a. the “munchies.”

Types and Locations of Fat

Fat cells are known to function differently in various parts and zones of the body. Fat accumulated on the hips and thighs, the “pear shape,” is thought to be less dangerous metabolically; this is a double edged sword because hip and thigh fat deposits are also the hardest to get rid of.

Visceral fat or abdominal fat inside the body cavity is blamed for a higher risk of diabetes, cancer, cardiovascular disease, high blood pressure and elevated triglycerides.

Visceral fat actually produces “inflammation” encouraging the production of free radical compounds that today are thought to be the cause of arteriosclerosis and stimulates the production of clot-promoting chemicals. Here is where obese and even overweight individuals can protect themselves from the free radical damage to their arteries by ramping up on their antioxidant intake - it is best to error on the side of high intake of antioxidants. The good news is, visceral fat is the first fat to be lost as a result of exercise.

While unsightly and cosmetically undesirable, subcutaneous fat tends to be metabolically benign. A 2004 study conducted by Washington University in St. Louis showed that removing an average of

twenty two pounds of subcutaneous fat with liposuction did not improve the health of the obese person; elevated blood pressure did not fall; elevated cholesterol and triglycerides did not drop; and elevated blood sugar did not improve following liposuction procedures.

There have been uncountable diet plans throughout the centuries: low calorie diets; low fat diets; low carbohydrate diets, etc. Yet, the American obesity epidemic and the world wide pandemic of overweight and obese kids and adults, diabetes and the Metabolic Syndrome (large deposit of visceral abdominal fat, high blood pressure, elevated cholesterol and triglycerides, insulin resistance, etc.) continue to spiral out of control.

If improper diet in terms of percentages of carbohydrates, fats and proteins was the root cause of obesity, we would have long ago eliminated the pandemic and epidemic weight problem. The purpose of the research that led to the creation of this book, was to identify the actual bottom line cause, prevention and cure (in the purist sense) of obesity. So we will begin with the successes and failures of various peoples.

The Amish

The Old Order Amish arrived in North America during the 1700s, and yet today in the 21st century, they maintain their old ways. The Amish have rejected almost all modern conveniences including electricity, natural gas, propane, internal combustion engines and telephones.

The Amish still use horses to pull their family carriages and buggies instead of automobiles, they still use horses to power and pull their farm implements, they use kerosene or fireplaces to light their homes at night, they use wood fires to cook and heat with, and in the 21st century they are still adding the wood ash byproduct of their kitchen stoves and hearth to their composts and gardens.

There are 30,000 Old Order Amish in the U.S. today, compared with almost 300 million Americans who have willingly and gleefully given up the old ways and transitioned their daily lives to electricity, automobiles, farm tractors and telephones.

The rates of obesity (4 %) and type II diabetes (6 %) in the Amish population are only ten percent and fifty percent of those reported by their “modern” neighbors living in the same county but on the other side of the cultural fence.

Most casual observers believe the lower rate of obesity and diabetes in the Amish is due to a better, more wholesome diet and the physical demands of an 18th century farm life.

We have spent a considerable amount of time living with the Amish in Lancaster County, Pennsylvania and we can assert with complete certainty and honesty that the Amish consume more sugar, more fat, more red meat, and more dairy products than anyone in North America. They are famous for Shoo-fly pie, mince meat pie, apple pie, sweet-potato pie, hot oatmeal with brown sugar, potatoes of every type, gravy, breads, pancakes, molasses and waffles with maple syrup.

The Amish are heavy consumers of whole milk, butter and cheese; they eat pork, beef, goat and mutton three times each day; and they are big consumers of poultry and eggs.

The carriage and draft horses do most of the physical work on an Amish farm, so what gives them a significantly lower rate of obesity and diabetes? It's the wood ashes! The wood ashes are the humble source of essential minerals that are dutifully put into their gardens every morning. Wood ashes are like oxygen, we can't do without them (or their replacement), yet they are hidden in plain sight! They don't get any respect!

Nez Pierce Indians

Obesity and diabetes are relatively recent disease problems for the Native American population of North America - prior to the 1940s, obesity and diabetes were rare diseases. An examination of the health and morbidity reports compiled by civilian and military physicians between the 1830s and the 1930s confirms that diabetes were rare diseases in Native Americans.

Since the 1940s the rate and incidence of obesity and diabetes in the Native American population has grown to pandemic and epi-

demic levels. A CDC report estimated 15.3 percent of the Nez pierce Indians were diagnosed with diabetes in 2002. Approximately thirty three percent of all Native Americans fifty five years of age and older were diagnosed with diabetes by 2003. In some tribes, diabetes and obesity affects up to fifty percent of the adult members over the age of thirty five years. These rates of diabetes and obesity in Native Americans are the highest incidence reported in any culture in the world.

Prior to the 1940s, Native Americans still lived on the reservation and lived a traditional lifestyle including a traditional diet. Studies of photographs of Native Americans, at the end of the 19th century, showed that most Indians were lean and physically fit. Their diets consisted of goat, mutton, beef and unrefined grains, including corn that was prepared with "culinary" wood ashes - wood ashes were still being put into their gardens as fertilizer.

During the 1960s and 1970s the Indian Health Services began to notice a dramatic rise in the rate of diagnosis of adult onset type II diabetes and obesity. Unfortunately, the IHS focused on screening, early traditional treatment and diabetic drug programs and surgery (amputations). The diabetes therapy program developed for Native Americans is so drug driven that pharmacists have been designated as the "primary" educators and monitors of the diabetic Native American patient.

Between 1998 and 2001, respondents in an Idaho, Oregon and Washington survey reported that obesity was 250 percent higher in Native Americans when compared to all races.

A 2004 pilot supplement study (Wallach, J.D., MacCoy, B., Henry, L. and Schrauzer, G.N.) with the Nez Pierce Indians of Idaho, has shown that almost all participants lost weight, significantly reduced blood pressure and reduced fasting blood sugars by thirty to fifty percent. All were able to significantly reduce or wean off of their medication.

The Nez Pierce diabetes and obesity program was based on a mineral supplement program (plant derived colloidal minerals) that replaced the "culinary" and fertilizer wood ashes given up by Native Americans as they transitioned into modern electric, natural gas and

propane kitchen and heating fuels.

Pima Indians

The Pima Indians are a band of Native Americans who live on the invisible border dividing Arizona and Mexico. The Pima band on both sides of the border are from the identical genetic pool; yet the Pima Indians on the Mexican side of the border have almost no type II diabetes and have only a ten percent obesity rate compared with the twenty two percent diabetes and seventy percent obesity in the Arizona Pimas!

With a small population of 12,000 individuals, the Arizona Pimas ran up a \$7 million annual Federal cost for pharmaceutical treatments; additionally \$3 million was spent annually for dialysis charges for diabetics. As a result of the epidemic and pandemic rate of diabetes and obesity, the Arizona Pima Indian's life span is shorter than the average American population by 15 - 19 years.

The Pima Native Americans from Arizona live in trailer homes that use natural gas or propane, electric stoves and microwave ovens for cooking and heating - they use automobiles, pickup trucks and tractors and they buy food and government Native American "commodities" from the grocery store like everybody else.

The Arizona Pima's transition from subsistence farming and wood stoves, and their traditional diet to the American lifestyle began in the 1940s and by 1960 the epidemic of obesity and type II diabetes, seemingly targeting the Arizona Pimas, had started in earnest.

By contrast the Pima Indians living south of the Arizona border in Mexico, are still subsistence farmers, they still cook and heat their homes with wood, and, as we go to press they are still putting their wood ashes into their gardens every morning.

What Then is Obesity?

Obesity, bottom line, is a deficiency disease - a deficiency of an entire category of essential nutrients - minerals. Certainly the symptoms and clinical presentation of the simple uncomplicated deficiencies are rarely seen anymore, as they are exacerbated or aggravated

by the increased intake of sugars and carbohydrates and reduction in exercise.

The deficiencies of just about any or all essential minerals will result in pica, cribbing, cravings, binge eating and the munchies in varying degrees. The deficiency of even one essential mineral results in a slight increase in food cravings and the “stand in front of the open refrigerator door” behavior.

It is the intersection of multiple mineral deficiencies and the excess intake of calories that produce obesity.

The deficiencies of essential minerals also explains why women gain excess weight during a pregnancy and breast feeding, they lose minerals to their babies, and if they fail to replace minerals faster than the babies steal them, the resulting pica and munchies cause the new mom to eat, and eat, and eat.

The deficiency of two or more essential minerals will produce a significant level of pica, cravings and binge eating. Iron and phosphorus deficiencies occurring at the same time will result in an almost psychotic binge eating and soft drink consumption rampage that results in morbid obesity.

The act of reducing calories from carbohydrate, fat and protein can result in weight loss, however, reducing calories alone does not solve the systemic effects of mineral deficiencies, which explains why individuals who drop off of the highly restricted diets of all types will regain the weight that they had lost; it also explains why many individuals maintain or gain weight even when they are truly cutting back on calories.

Glucose is an “all-purpose” fuel sugar that can be used as a source of energy by the body. However, the overt intake of sugars is not necessary, as adequate levels of blood glucose can be made from dietary or stored fat and protein. Sugars and carbohydrates are not essential nutrients, they are quick sources of energy, they can be fun to eat, however, they are not necessary or essential to consume.

A portion of whatever we eat, including bread, rice, or beans is converted to glucose. To keep the blood levels of glucose at the level each of us requires, the hormone insulin is produced by the Islet's of

Langerhans, and released to regulate the rate of glucose metabolism.

Insulin can't perform properly without several cofactors including the trace minerals chromium and vanadium. As we consume food, insulin is released into the blood from the pancreas to regulate the use and storage of sugar and carbohydrate.

Carbohydrates are fractured into the simple sugar glucose and used for an instant energy source, glycogen (animal starch) for storage and later use or converted to body fat for long term storage. The body's ability to control or manage the use of blood sugar is referred to as "glucose tolerance."

The absence of or insufficient levels of dietary or supplemental chromium and vanadium, results in blood glucose being converted into blood lipids and body fat, rather than efficiently being burned for energy. The concurrent deficiencies chromium, vanadium and other minerals results in ravenous hunger, cravings, binge eating, pica, cribbing and the munchies.

The increased intake of carbohydrate in the form of sugar (one half pound per person per day in 2004 compared with one half pound per person per year in the 1700s) and the deficiencies of minerals are the basic root causes of obesity. It is an interesting fact, that as the Asian and Latin American countries "catch up to western countries in terms of lifestyles and economics," their rates of obesity catch up as well!

The reason that the transition from third world cultures to industrialized cultures results in an almost immediate increase in obesity is that they give up using wood for fuel - they convert to electricity, natural gas and propane for heating and cooking fuels! There are no wood ashes left to put in the gardens or for use as "culinary minerals."

The resulting mineral deficiencies produce the munchies, cravings and binge eating; add to these deficiencies a significant increase in carbohydrates in the diet and only one thing can happen - obesity!

In 1971, 14.5 % of Americans were obese; in 2004 the rate rose to 30.9 %. The number of American deaths attributed to obesity is 400,000 per year and expected to rise by 2005 to be the highest cause of preventable deaths.

From 1996 to 2001, 2 million American teens and young adults swelled the ranks of the clinically obese. According to the *Journal of the American Medical Association*, March, 2004, the total cost for this level of obesity is a whopping \$117 billion per year.

The rate of obesity in American teens between the ages of 12 and 19 was five percent in 1980, the rate rose to fifteen percent in 2004.

Serving Size Changes

Food	Size Change		Calorie Change	
	1954	2004	1954	2004
Burger	2.8 oz	4.3 oz	202	310
Fries	1955 2.4 oz	2004 7 oz	1955 210	2004 610
Hershey's Bar	1900 2.0 oz	2004 7.0 oz	1900 297	2004 1,000
Coca Cola	1916 6.5 oz	2004 16.0 oz	1916 79	2004 194
Movie Popcorn	1950's 3 cups	2004 21 cups	1950's 174	2004 1,700

The average American women's dress size in 1950 was an 8, in 2002 it rose to a 14.

The obese individual suffers more frequently from a plethora of diseases and "complications" than does the slim individual.

The adipose or fat tissue is the primary storage organ for energy in the human. It is the main reservoir for stored calories ingested in excess of basal needs, and it gives up calories when caloric intake is delayed (missed meals) or in times of starvation.

Excess energy is stored in adipose tissue in the form of triglycerides, when this process becomes excessive obesity occurs.

What is Adult Onset Type II Diabetes?

Adult onset type II diabetes (insulin resistant diabetes) is a simple mineral deficiency disease. Insufficient dietary intakes, or reduced absorption efficiency of chromium, vanadium and other minerals produces increased blood levels of insulin. Increased blood levels of insulin produces hypoglycemia, hyperinsulinemia, narcolepsy, pre-diabetes and adult onset type II diabetes (insulin resistant diabetes).

Adult onset type II diabetics produce more insulin than non-diabetics, adult onset type II diabetes is not an insulin problem; rather, it is a simple mineral deficiency problem.

The relationship between chromium, vanadium, insulin and heart disease shows why the most common cause of death in diabetics is cardiovascular disease. This relationship is described by Walter Mertz, M.D., who reported that chromium was an essential nutrient in 1955 - he was soon appointed to the Chairmanship of the USDA Human Nutrition Research Center in Beltsville, Maryland.

"The most consistent effect of marginal chromium deficiency is elevated insulin response, and the first effect of chromium supplementation is the restoration of a normal response. It appears....that if we improve glucose tolerance, restore normal insulin levels and at the same time lower cholesterol levels - particularly (LDL cholesterol - the bad cholesterol) - we may reduce the risk of cardiovascular disease."

Richard J. Doisy, Ph.D., professor of biochemistry at the State University of New York, Upstate Medical Center, tested the effect of supplemental chromium on normal twenty to twenty five year olds. He found when healthy individuals were given chromium supplements, their pancreas did not have to produce as much insulin to manage their blood sugar levels. "This could be interpreted that even in 'normal' subjects the dietary intake of chromium is marginal,"

Dr. Doisy, concluded. "My contention is that probably one-fourth to one-half of the people in this nation are deficient in chromium... We have relatively high tissue chromium levels at birth and it's downhill from there on in. And the result for the health of Americans is that millions have maturity (type II) onset diabetes. And millions

suffer from vascular disease,” he reported.

“You tend to increase (the rate of) your chromium excretion when you’re eating excessive carbohydrates,” said Dr. Doisy. “And if the diet is not providing sufficient chromium (to make up for the increased rate of loss), then you are going to be in negative balance.”

We have known since the 1957 reports by Walter Mertz, that the deficiencies of chromium and vanadium are the primary causes of diabetes, however, the concurrent deficiencies of other essential minerals will “throw gasoline on the fire” of diabetes and produce all of the complications associated with the chronic form of the disease.

Chromium is required for circulating glucose tolerance factor to function properly. Chromium is an essential trace element necessary for the normal glucose metabolism, It’s function is directly related to the function of insulin.

Average American Serum Chromium

Mean Cr blood levels (u/L)	Year
28 - 1000	1948
180	1959
520	1962
170	1962
28	1960
23	1968
13	1971
10	1972
4.7 - 5.1	1973
0.73 - 1.6	1974
0.16	1978
0.43	1980
0.14	1983
0.13	1985

Only the trivalent state of chromium is biologically active and it works by forming complexes with organic compounds. Insulin-chromium relationships are not restricted to glucose metabolism. Animal studies have shown that chromium stimulates amino acid transport and protein synthesis.

Deficiency of chromium is characterized by glucose intolerance (hypoglycemia, insulin resistance, hyperinsulinemia and diabetes), poor growth, peripheral neuropathies, negative nitrogen balance (loss of muscle mass - sarcopenia) and a decreased respiratory quotient.

The dramatic drop in serum chromium started when the wood ash byproduct of cooking and heating was no longer available for "culinary" ash - and, when the ashes were no longer being placed in our gardens.

Signs of Chromium Deficiency in the Human

- Impaired glucose tolerance (hypoglycemia, hyperinsulinemia, diabetes)
 - Elevated circulating insulin
 - Glycosuria (sugar in the urine)
 - Fasting hyperglycemia (elevated blood sugar)
 - Impaired growth
 - Decreased longevity
 - Elevated serum cholesterol and triglycerides
 - Increased incidence of aortic plaque (arteriosclerosis)
 - Peripheral neuropathy
 - Brain disorders (dementia)
 - Decreased fertility and lowered sperm count
-

Anderson, 1981

By 1977, K. N. Jeejeebhoy reported that chromium could eliminate the need for medical insulin in human patients. His intravenous use of chromium in human diabetics eliminated peripheral neuropathies and well as eliminated the need for insulin treatments to stabilize blood sugar.

On August 31, 2004 Harvard investigators released the results of

a fifteen year study on toe nail chromium levels and their relationship to the rate of diabetes. The Harvard School of Public Health scientists analyzed the toenails of 33,000 men ranging from 40 to 75 years of age. They found that the men with the lower levels of chromium in their toenails had higher rates of heart disease and diabetes. The study was published in the September 2004 issue of *Diabetes Care*, the official journal of the American Diabetes Association.

In 1991, the University of British Columbia declared that, "the trace mineral vanadium could replace insulin for adult onset type II diabetics."

Vanadium appears to function like insulin by altering cell membrane function for the ion transport process; therefore, vanadium has a very beneficial effect for humans with glucose tolerance problems by making cell membrane receptors more sensitive to insulin.

Vanadium inhibits cholesterol synthesis in animals and humans; this is followed by a drop in plasma cholesterol and reduced aortic cholesterol deposits.

The number of Americans with diabetes in 1980 was 5.8 million, in 2004 the numbers rose to 13.3 million with an estimated 17 million American diabetics undiagnosed or at the pre-diabetic state.

In between 1988 and 1994, 45.7 % of Americans over the age of eighteen years with diabetes were obese and between 1999 and 2002, 54.8 % of Americans with diabetes were obese.

What is the Metabolic Syndrome?

The Metabolic Syndrome is characterized by insulin resistance (hyperinsulinemia, hypoglycemia, narcolepsy, pre-diabetes, etc.), hypertension, elevated cholesterol and triglycerides and abdominal obesity.

The cause of the Metabolic Syndrome is not genetic, but rather a collection of multiple essential nutrient deficiencies including deficiencies as illustrated on the table of next page.

**Nutritional deficiencies associated
with the Metabolic Syndrome**

Nutrient Deficiency	Clinical disease
Omega -3 essential fatty acids	elevated total cholesterol, elevated LDL, obesity, peripheral neuropathies, thrombotic disease, atherosclerosis
Calcium	hypertension, osteoporosis, peripheral neuropathies, insomnia, pica, obesity
Chromium, vanadium	diabetes, insulin resistance, hyperinsulinemia, hypoglycemia, pica, anxiety, depression, obesity
Selenium	cancer, cardiomyopathy heart disease, cataracts, macular degeneration, hypothyroidism, arteriosclerosis, atherosclerosis, pica
Niacin	pellagra, elevated cholesterol, dermatitis, dementia, diarrhea
Phytonutrients (antioxidants)	inflammation, arteriosclerosis, macular degeneration, cataracts, obesity, Metabolic Syndrome, Alzheimer's disease, cancer, cardiovascular disease.

The Nigerian Clue

On February 1999, the journal *Scientific American* ran a cover article that looked at hypertension in African Americans. The article featured a blood pressure study including third world Nigerian tribes still living in the remote bush country and African Americans of Ni-

gerian decent in the U.S.

The tribal Nigerians had an average of seven percent of the individuals with hypertension, while there was a thirty three percent rate of hypertension in the U.S. urban individuals - obviously, hypertension in black people is not genetic, if hypertension was genetic, then the blood pressure of both groups would be seven percent and seven percent!

What then causes the difference in the hypertension rates between the two groups? The primitive tribal individuals still add the wood ashes from their heating and cooking fires into their gardens, mix the ashes with their salt and use the wood ashes as "culinary" ashes mixed with their food - while their city cousins use electricity, natural gas and propane to cook and heat their homes with, and the rate of mineral supplementation in the African American community is very low.

The Metabolic Syndrome sets up a systemic "inflammation crises" that produces arteriosclerosis and atherosclerosis of small and medium caliber arteries (cerebral, coronary and carotid arteries) by causing free radical damage to the single layer of intimal cells that line arteries, and by damaging (free radical damage) a portion of the circulating cholesterol to produce a higher level of LDL or "bad cholesterol."

Total cholesterol is analogous to paint on a house. All of the paint on the house came from the same paint can, yet the paint on the western exposure often times blisters, powders, oxidizes and flakes before the rest of the paint because as the sun sets in the west it produces an intense heat damage. Like wise the total circulating cholesterol comes from the same bucket, however, unprotected by antioxidants, a portion of the total will be damaged and become LDL. LDL is not a separate species of cholesterol, it is just damaged cholesterol and damaged cholesterol produces inflammation.

The number of Americans with "high" cholesterol as defined by medical doctors was 37 million in 2004. The most popular drug prescribed to lower cholesterol was Lipitor (Pfizer) with 2003 sales listed as \$5.8 billion resulting in an annual cost per patient of \$1,100. Side effects of Lipitor include kidney damage, muscle weakness and

rhabdomyolysis, a muscle breakdown disease.

The non drug alternatives that can be used to bring cholesterol into the normal range (220 - 270), includes a complete supplement program that provides all of the known 90 essential nutrients. In particular, chromium, vanadium, essential fatty acids, niacin and antioxidants (selenium, green tea, vegetables, dark berries, etc.) and a low carbohydrate diet are all known to bring cholesterol into the normal range.

In 2004, the number of Americans with hypertension is listed as 50 million. The most popular drug prescribed to bring blood pressure under control is Norvasc (Pfizer). In 2003 sales reached \$1.9 billion for an average annual cost per patient of \$500. Side effects include dizziness, edema, flushing and palpitations.

Non drug alternatives for controlling hypertension include a complete supplement program that would promote healthy blood pressure levels, including a focus on calcium, weight loss (intense mineral supplement programs), the elimination of sugar, carbonated drinks, and caffeine - and the significant reduction of carbohydrates.

A Brief History of Snacks

Amelia Simmon's cookbook, originally published in Hartford and then Albany in 1796, is considered the first American cookbook, not only because it was published after the Revolution, but because it was written by an American for Americans.

Simmons was famous for huge dessert cakes, including her famous "Independence Cake" - the recipe called for twenty pounds of flour, fifteen pounds of sugar, ten pounds of butter and twenty four eggs. Many of her dessert recipes called for "pearl-ash" (white wood ash) or potash.

Simmons Honey Cake

6 pound flour	1 oz ginger
2 pound honey	Orange peel (orange zest)
1 pound sugar	2 tsp pearl ash
2 oz cinnamon	6 eggs

Dissolve the pearl ash in milk, put the whole together, moisten with milk, bake 20 minutes. Amelia Simmons, *American Cookery* 1796.

In 1853, Cornelius Vanderbilt, tired of thick American fries, ordered chef George Crum, of the Moon's Lake lodge in Saratoga, NY, to create the first thin "potato chip." Crum sliced potatoes paper thin, fried them, and the "Saratoga chip" was created and the snack food industry was born.

In 1876, Charles Hires sold his root beer at the great Centennial Exposition. It contained sixteen roots and herbs including birch bark, spikenard, sarsaparilla and hops. He had sold his drink for seven years at his Philadelphia pharmacy after creating it on his honey moon in 1870. In 1877, Hires sold his root beer by mail order - a 25 cent packet that made five gallons.

In 1880, Ira Remsen published an article describing his accidental discovery of saccharin, a chemical substance he says is, "300 times sweeter than an equal amount of sugar." Saccharin is a white crystalline, aromatic compound that had no nutritive value as it passed through the body unchanged.

Remsen felt that saccharin would have great commercial value because, "it could prove quite useful for diabetics, for fat people and others that could not use sugar."

In 1882, on Monday September 4, at 3:00 p.m. the switch was pulled and Edison's first commercial electric generator hummed into action, current flowed and a new era of technological progress was joyously welcomed in. By 1950 we were totally captives and victims of HELL'S KITCHEN and the epidemic of obesity, Metabolic Syndrome and diabetes started in America unopposed.

Between 1882 and 1950 there was not a single voice raised to expose the health dangers of a wood ash (mineral) free society. No matter how lofty our goals were for reaching out to space, no matter how many multiple organ transplants we performed and artificial hearts we installed and no matter how many politically correct dreams and laws we wrote for cultural progress, we

ultimately would be held back for the lack of the basic elements required for the maintenance and repair of our flesh and blood bodies.

In 1885, a new "tonic brain food and exhilarant" (Dr. Pepper) went on sale at Wade Morrison's Old Corner Drug Store. Morrison moved to Waco, Texas from Virginia after his courtship with Miss Pepper was ended by her father who also happened to be his boss. Morrison headed west and bought the pharmacy, where a clerk named Charles Alderton and Morrison concocted a new drink - Morrison named it after Miss Pepper's father, Dr. Pepper!

In 1886, John Styth Pemberton, a Civil War veteran and local Atlanta, Georgia pharmacist invented Coca Cola, which he sold for 5 cents per glass. Sufferers of all kinds found "consolation" in the new coca-extract brew. It was billed as, "the intellectual beverage and temperance drink."

Coca Cola was said, "to cure dyspepsia (indigestion) and headaches." Pemberton claimed, "that it lifted the spirits as well." The Atlanta pharmacist formed the Pemberton Chemical Company and registered a trade mark for French Wine of Coca, "an ideal nerve tonic and stimulant." To attract the teetotalers he took out the alcohol and added caffeine-rich kola nut extract.

Coca Cola competed with other drinks including Imperial Inca Cola, Coca-coffee and Cocafeine. The original coca drink, vin Mariani, had won prizes from organizations all over Europe, one of which had called it a "wine for athletes."

Experts of the day said that, "cocaine can cure everything from head colds, to the mental depression that accompanies the hysteria (PMS) in the female to lack of energy."

Pemberton also created Globe of flower cough syrup, Indian Queen Hair Dye and Triplex Liver Pills.

Pemberton joined forces with entrepreneur, Asa Chandler, who immediately saw the wider commercial value of the drink (Coca Cola) and turned a local curiosity into a business. The average sales of the original product per pharmacy reached nine bottles per day.

In 1889, Chandler attributed mythical properties to Coca Cola and began to advertise it as a "sovereign remedy." Many scientists of

the day disagreed - by 1890, 400 clinical cases of acute suffering from cocaine abuse had been reported in medical journals. Physician Albrecht Erlenmeyer calls the drug "the third scourge on mankind," after morphine and alcohol. Parke Davis sells it in cigarettes, ointments, tablets and injections.

Angelo Mariani, whose vin Mariani was the drug's first commercial application, collected a list of well known users including Thomas Edison, Sigmund Freud and Pope Leo XIII who was said to keep a flask of vin Mariani under his belt.

In 1892, The Edison General Electric Company merged with Thomson-Houston Electric to form General Electric. Edison was said to have been bitter because the new company did not bear his name.

In 1894, Joseph Biedenharn, was the first to market Coca Cola in a bottle.

In 1894, Milton Hershey, a Lancaster, Pennsylvania candy maker, declared that, "caramels are only a fad, chocolate is a permanent thing." He developed a chocolate product in the shape of a rectangular slab. The new chocolate "bar" was priced cheaply so it could become an everyday treat. The Hershey Milk Chocolate Bar, which also comes with almonds, is "prepared with skimmed milk from Pennsylvania cows."

In 1895, the USDA reported that Americans consumed one half pound of sugar per person per year.

In 1898, Caleb Bradham, a New Bera, North Carolina Pharmacist, prompted by the success of Coca Cola, created a similar carbonated drink. Originally the new fountain beverage was called "Brad's Drink," but it was later changed to Pepsi Cola and bottled to widen sales.

In 1899, two lawyers, Benjamin E. Thomas and Joseph Whitehead secured exclusive rights to bottle and sell Coca Cola for one dollar.

In 1916, the Root Glass Company of Terre Haute, Indiana created the famous Coca Cola female body shaped bottle. The motto was, "You can identify Coca Cola, even in the dark."

In 1900 Coca Cola served two bottles per day and by 1920 Coca Cola served 1,000 bottles per day.

In 1920, brand named and packaged foods hit the market place because masses of Americans and peoples in all industrialized nations had moved off the land into the cities. People no longer had their subsistence gardens, people were no longer independent and people had begun to switch from wood as the primary fuel for cooking and heating to coal, oil, kerosene, propane and electricity - the wood ashes we had grown up with as humans were becoming "as scarce as hen's teeth."

Underwood sardines, La Choy Chinese food, Maxwell House coffee (Teddy Roosevelt coined the phrase "its good to the last drop"), Sunkist oranges, Campfire marshmallows, Kellogg's All-Bran cereal, Good Humor Ice Cream Suckers, and Baby Ruth bars (named for the daughter of former President Cleveland) and Brillo to scour pots and pans were the products of the day.

The "ice box" that one had to fill with block ice to keep food cool and keep milk sweet was replaced with the electric Frigidaire. Silk stockings are out and the new rayon hosiery from Du Pont Fibersilk Company was the rage of the day. The variety of automobiles was staggering - Cadillac, Chalmers-Franklin, Chevrolet, Maxwell, Mercedes, Milburn Electric, Packard, Peerless, pierce-Arrow, Salient, Stephens, Stutz and Ford Model T (the "fliver" is good for the liver).

In 1923, Robert Woodruff, five years after his father purchased the Coca Cola Company, began to distribute the product world wide.

By 1925, Coca Cola sales were generating \$6 million in sales per day; four years after Albert Einstein won the Nobel Prize in physics for his theory of relativity.

In 1928, Coca Cola ads followed the American Olympic Team to the Amsterdam Olympics; the Coca Cola ads were emblazoned on racing dog sleds in Canada and Bull fighting arenas in Spain.

Coca Cola introduces the "six-pack" and the open topped cooler concepts.

In October 18, 1931 Thomas Edison died at the age of 84 from Bright's disease (kidney disease with protein in the urine), uremia, gastric ulcers and diabetes (he was no longer putting wood ashes from heating and cooking fires into his garden and his daily fair always included sugar in the form of apple dumplings and Coca Cola).

In 1933, the Tennessee Valley Authority (T.V.A.) was created in the first 100 days of the FDR administration. It came under constant attack by the private power generating sources, including a power corporation headed by Wendell Willke, a powerful Wall Street attorney, who argued, "To take away our market is to take our prosperity."

The Supreme Court ruled that the constitution did not give anti-competition protection to private utility companies. The T.V.A. was formed to bring electrical power and industry to the poorest part of America. The states that benefited were seven Southern and border states ranging from the Smokey Mountains, coal towns, bleak mountains and drought ridden cotton farms. T.V.A. promoted flood control, produced clean hydroelectric power and developed land and forest management programs.

In the beginning of T.V.A. there were only two out of 100 people in the Tennessee Valley who had access to electric power and many families lived on less than \$100 per year.

In 1941, Coca Cola's Woodruff ordered, "Every man in uniform gets a bottle of Coke for 5 cents, wherever he is, and whatever it costs the company."

In 1943, General Dwight D. Eisenhower, sent an urgent cablegram to Coca Cola, requesting the shipment of enough materials to build ten bottling plants. Between 1941 and 1960, the number of countries bottling Coca Cola doubled.

In 1957, Walter Mertz, an American chemist discovered that he could prevent and cure adult onset type II diabetes in laboratory animals, pets and farm animals with a simple supplement program that contained chromium and numerous cofactors necessary for the chromium to support insulin's job of getting sugar from the blood through the cell membrane into the cell.

Adult onset type II diabetes is not an insulin problem or insulin deficiency, patients with adult onset type II diabetics often times make considerably more insulin than non diabetics. Adult onset type II diabetes, like obesity, is simply a mineral deficiency problem.

The obvious question is, if diabetes is a simple mineral deficiency problem, how come it still plagues modern man? Unfortunately there are no laws requiring doctors to cure patients when there are cures.

available - so they don't!

We would all like to believe in our heart of hearts that medical doctors are altruistic and would turn Heaven and earth to find a cure for their patients. Unfortunately, most doctors went to medical school to learn a trade, to have the skills necessary to generate a great lifestyle and climb the social ladder, not to eliminate obesity, high blood pressure or diabetes.

According to the GAO (General Accounting Office), the congressional budget watchdog, over the twenty five to forty years that a diabetic lives after diagnosis, they are worth \$750,000 to their doctor if they follow the standard medical approach to diabetes.

Also, urban populations no longer put wood ash minerals into their gardens resulting in widespread mineral deficiencies.

After seventy five years, Coca Cola began to introduce additional flavors to the original product line. By 1966 the international sales grew in many countries including Cambodia, Montserrat, Paraguay, Macau and Turkey.

By 1977, Walter Mertz's 1957 animal work with chromium and diabetes was repeated in humans by Dr. K. N. Jeejeebhoy of the University of British Columbia.

In 1978 the Coca Cola Company secured the sole rights to market packaged cold drinks in the Peoples Republic of China.

In 1981, Robert E. Goizueta became Chairman of the Board and CEO of the Coca Cola Company and installed a new policy he called, "Intelligent risk taking."

In 1985, Coca Cola became the first soft drink in space, Diet Coke was introduced, and Goizueta tried to introduce a sugar substitute sweetened drink to replace the original sugared Coca Cola, in a move that critics called, "The biggest marketing blunder ever." Overwhelmed by enormous customer pressure, including class action law suites, the original sugar sweetened formula was returned to the marketplace as Coca Cola Classic.

In 1991, researchers at the University of British Columbia, Vancouver, Canada, stated that the trace mineral vanadium could replace prescription insulin for treating and managing type II diabetics.

In 1995, the USDA reported that Americans consume 157 pounds of sugar per person per year (up from one half pound per person per year in 1895) - approximately one half pound of sugar per person per day! This unbelievable rate of sugar consumption is driven by a hoard of sugared carbohydrate foods, drinks and snacks.

In 2004, world wide Coca Cola Sales reached \$1.3 billion per day.

On June 8, 2004, the eight year Harvard Nurses Health study of data from 91,000 nurses over the age of thirty five showed that "nurses drinking one sugar sweetened soft drink per day had an eighty five (85) percent increased risk of developing adult onset type II diabetes."

The intersection of mineral deficiencies that cause pica a.k.a. the munchies, critical deficiencies of chromium and vanadium and the universal availability and high intake of sugar and carbohydrates is the basic root cause of obesity, diabetes and the Metabolic Syndrome. It is that simple. If everyone were to supplement properly, with the correct mineral mix, we could eliminate obesity, high blood pressure, adult onset type II diabetes and the Metabolic Syndrome in America within a year - and the world would soon follow our example.



Chapter 15

“Vice knows she’s ugly, so puts on her mask.”

- Benjamin Franklin

Safe Cooking

It is quite clear that what one eats and what one doesn't eat will determine one's health and length of life. It was determined after having performed 17,500 autopsies on more than 454 species of animals and 3,000 humans - every animal and every human that dies of natural causes dies of a nutritional deficiency disease.

To do everything we can to maximize health and longevity we have to guarantee ourselves an optimal stream and proper amounts of all essential nutrients each day - we cannot depend on good luck; we must be proactive!

Once we assure ourselves that we are consuming optimal amounts of the essential nutrients, we must also protect ourselves from dangerous choices of foods, food ingredients, storage practices and cooking methods. So we are going to look at and review some pretty basic stuff we all need to be aware of every time we buy, store, cook and consume food.

Free Radicals

The term free radical is a generic term that covers a broad classification of dangerous food chemicals that are produced either directly or indirectly by man. Free radicals increase our risk of many of

the dangerous side effects of obesity and weight loss programs, they increase our risks of cancer, cardiovascular disease, diabetes, Alzheimer's disease, multiple sclerosis, ALS (Lou Gehrig's disease), Parkinson's disease, macular degeneration, cataracts, fibromyalgia and they accelerate the aging process.

By eliminating the intake of free radicals one can significantly reduce their risks of many diseases. A good insurance policy for protecting yourself against the accidental or unknowing consumption of free radicals is to flood yourself with multiple antioxidants at every meal. The old adage, "An ounce of prevention is worth a pound of cure" surely fits here - if taking antioxidants everyday can significantly reduce one's risk of many terrible diseases, why would a lucid person not take antioxidants? Why wait until a disease strikes you before you have a motivational concern?

Additionally, as a person who is overweight or obese loses weight, the byproducts of "burning" fat can turn into free radicals that can increase the risk of many diseases, including inflammation of the intima (the lining of the arteries) which then produces arteriosclerosis, cardiovascular disease, cancer, Alzheimer's disease, etc. To reduce the risk of the complications of weight loss, people need to flood themselves with antioxidants in order to protect themselves.

Antioxidants

Selenium	Catachins (EGCG)	Leutines
Lycopines	Phenols	Xanthones
Xanthines	Zeaxanthines	Procyanins
Vitamin C	Vitamin E	Beta carotene

ORAC Value

A good way to measure the antioxidant activity of foods including juices is by means of the ORAC point system. ORAC stands for Oxygen Radical Absorption Capacity, a measure of the scavenging

capacity of antioxidants against the peroxy radical.

ORAC Rating	Food	Units (per 100 g)
1	Prunes	5,770
2	Raisins	2,830
3	Blueberries	2,400
4	Blackberries	2,036
5	Kale	1,770
6	Strawberries	1,540
7	Spinach	1,260
8	Raspberries	1,220
9	Brussels sprouts	980
10	Plums	949

Trans Fatty Acids

Trans fatty acids are a form of free radicals that are universally found in margarines, cooking oils (yes, even olive oil), salad dressings and processed foods that contain these ingredients. Trans fatty acids are extremely dangerous food chemicals that are formed by oxidized (chemically rancid) fat. These types of fats are found in just about all processed foods recommended by doctors, and the FDA estimates that there are more than 42,000 products containing trans fatty acids on grocery store shelves in America.

Harvard estimates that trans fatty acids directly kill 30,000 Americans each year and many more indirectly. The American romance with trans fatty acids began in the 1950s driven by the medical trades unrealistic paranoia of saturated animal fats and cholesterol (which humans had been eating for thousands of years) and a false belief that they were the "boogey man" that caused cardiovascular disease.

Following the lead of the medical hysteria, food manufacturers rushed to find something that could replace animal fat - they grabbed at vegetable oils (a product that had been abandoned by the animal feed industry because plant oils had produced many new diseases including muscular dystrophy, pancreatitis, cardiomyopathy, and

fibromyalgia).

Oils and margarines were cheaper than animal fat (lard, suet, etc) and butter and had a longer shelf life. Chefs could buy vegetable oils in large containers that “didn’t need refrigeration,” thus allowing them to keep it on the self, however, as the oil sat on the shelf it slowly but steadily oxidized and metamorphosed into trans fatty acids.

Cooks liked the new oils because they could be kept hot all day without smoking up the kitchen- they could use the same oil over and over, however, the constant heat quickly converted them into trans fatty acids - a free radical that few in human medicine knew anything about.

Trans fats were a dream come true for the fast food and junk food industries. Trans fatty acids made potato chips, crackers and cookies crispier and they allowed manufacturers to add tasty fat in foods that had never had fat in them before (Oreo cookie filling, etc.). Trans fatty acids are solid at room temperature, oils on the other hand are liquid and would have turned the foods into a slimy greasy mess. Manufacturers could now “lock” fat into their products.

“Besides increasing the number of LDL (bad) cholesterol and triglycerides in the bloodstream, trans fats also lower your HDL (good) cholesterol,” says Bruce Holub, Ph.D., a professor of nutrition at the University of Guelph, Canada. “Trans fats also increase blood levels of a compound called lipoprotein a. The more of this lipoprotein you have in your system, the greater your risk of developing heart disease,” he said.

In a current review of the Harvard Nurses Health Study, for 80,000 nurses over the age of thirty five, found that getting only three (3) percent of your total daily calories from trans fatty acids increases your risk of heart disease by fifty (50) percent! Three percent of your calories from trans fats (7 grams of trans fat) are provided by a single order of fries.

Trans fatty acids increases your risk of diabetes, sarcopenia (muscle wasting associated with age) and cancer. When Harvard researchers reviewed all the previously published data on trans fats, they found that men and women with the highest daily intakes were also the most at risk for developing diabetes.

Common Foods that Contain Trans Fatty Acids*

Food	Trans Fat (Grams)
1 Chicken pot pie	8
2 Biscuits	8
1 Large fries	7
1 Cinnamon roll	6
6 Fish sticks	5
1 Order nachos with cheese	5
1 Tblsp Margarine	5
6 Oreos	4
1 Waffle	4
9 Tater Tots	4
4 cups Movie popcorn	3
1 Slice apple pie	3
1 pak Cheese and crackers	3
2 Home made chocolate chip cookies	3
5 Piece chicken tenders	2.5
1 Brownie	2.5
1 Jelly doughnut	2
1 cup Chex party mix	2
1 Hamburger bun	<1
1 slice Cheese pizza	<1

*Samantha Heller: The Hidden Killer. Men's Health. October 2004.

There are no "safe limits" for trans fatty acids - do not eat them!

Heterocyclic Amines

Heterocyclic amines are free radicals that are found in well done meat and burnt animal fat. They are extremely dangerous and can increase one's risk of breast, prostate and colon cancer by as much as 462 percent!

In a Harvard Nurses Health Study of 90,000 nurses over a twenty year period, they compared the rate of cancer between those who consumed less than twenty percent of their calories as fat versus those

who consumed more than fifty percent of their calories as fat and found there was no difference in the rate of cancer.

In a 1998 study by the University of South Carolina, they found that consuming burnt animal fat increased the risk of cancer by a staggering 462 percent!

It is not the amount of fat in your diet that increases your risk of obesity, diabetes and cancer it's the type of fat and how you cook the fat!

Acrylamides

Acrylamides are free radicals that are generated by "super heating" carbohydrates and sugars. Croutons, fried potatoes, crackers, cookies, cold cereals, hard crusted breads, browned sugars and pie crusts are loaded with acrylamides. These free radicals are known to increase the risk of cancer significantly.

Red Meat

Red meat is an unequaled source of animal protein and essential lipids - the secret is to not eat meat cooked well done or eat the fat burnt as they become heterocyclic amines. Medium rare and medium are the safe way to eat beef. Steak Tartar is a half pound of raw, freshly ground sirloin, served with chopped onions, pepper corns and topped with a raw egg - it doesn't get any better than this!

Salt

Salt your food to taste. Salt does not cause high blood pressure and restricting salt will not reduce your risk of high blood pressure or add ten seconds to your life - in fact restricting salt will shorten your life and make your very short life miserable.

Salt is the raw material needed to manufacture stomach acid. Stomach acid is hydrochloric acid that is necessary to digest food, ensure absorption of minerals, activate pepsin (protein digestive enzyme produced by the stomach) and activate the "intrinsic factor"

necessary to absorb vitamin B12.

In April 1997, The Sodium Task Force reported that those individuals who restricted their salt intake to one gram per day had a 600 percent higher rate of heart attacks than those who consumed more than two grams of salt per day.

Restricted salt intakes will make you heat intolerant and over the long term will result in digestive problems including belching, burping, intestinal gas and G.E.R.D. (gastro esophageal reflux disease).

Restricted salt diets with the resultant hypochlorhydria (low stomach acid) will increase risks of nutritional deficiency diseases including obesity (reduced absorption of minerals increases the munchies), diabetes, high blood pressure, arthritis, osteoporosis, kidney stones, cataracts, fibromyalgia, etc.

Sugar

To avoid obesity, diabetes, cardiovascular disease, the Metabolic Syndrome, dementia and reduce your risk of cancer, it is important to eliminate sugar (natural and processed) from your diet. Your body makes all of the blood sugar, blood glucose and glycogen (animal starch) it needs by reconfiguring fats and proteins. Sugars and carbohydrates are not essential nutrients.

According to the USDA, Americans were eating one half pound of sugar per person per year in 1895, and in the year 2002 Americans ate 157 pounds of sugar per person per year - that's almost one half pound of sugar per person per day.

Not only will you make it easier to lose those unwanted pounds and keep them off by giving up sugar, but you will think clearer, you will be less irritable; you will have boundless energy and if you have hypoglycemia, the Metabolic Syndrome or hyperinsulinemia you will be heading in the right direction necessary to solve your problems:

- No apple juice
- No grape juice
- No honey
- No molasses (cane juice, turbanado sugar, brown sugar, etc.)

- No table sugar
- No corn syrup
- No fructose
- No granola or grain based energy bars
- No processed meats
- No BBQ sauce, no pasta sauce, no chili
- No candy
- No colas, non colas or other carbonated drinks
- No ice cream (exceptions include no sugar added ice cream), no sherbert
- No pastries
- No boxed or bagged cold or instant hot cereals
- No low fat yogurt or 98% fat free yogurt (they are loaded with sugar)

If during the process of giving up sugar you get the “munchies” or feel the need to go on the classic “chocoholic” binge, take one ounce of plant derived colloidal minerals per 100 pounds of body weight - even if it’s ten times per day at the start (remember - cravings or pica are the sure sign of mineral deficiencies!!).

Sugar Substitutes

Sugar substitutes can satisfy your sweet tooth without adding significant calories to your diet. There is a considerable amount of public controversy surrounding the use of sugar substitutes, however, used in moderation they are “considered” safe and have been used for years. Typically, Splenda can be used teaspoon for teaspoon of sugar for baking.

1. Aspartamine - (also known as Nutrasweet and Equal)
A man made sweetener created by joining two amino acids.
2. Splenda - (also known as Sucralose) A natural substance extracted from sugar cane.

3. Saccharin - (also known as Sweet and Low) A man made chemical discovered in the 1800s.
4. Stevia - An herb (*Stevia rebaudiana*).

During 2004, Splenda became the number one blended sweetener in the U.S., selling more than all the other brands of “faux” and real sugar. Sugar substitutes have failed miserably for totally replacing sugar for baking. Sugar does more than just sweeten food, it helps the cake rise, giving it volume and texture; it caramelizes, producing the golden brown color associated with quality baked goods; it provides greater shelf life.

In 2004, Splenda and Equal began to market “baking blends: Equal Sugar Lite - a mix of sugar, maltodextran, aspartamine, acesulfame potassium (a common no-cal sweetener) and artificial flavor - can substitute for sugar at one cup for one cup in recipes; Splenda Sugar Blend for baking is part granulated sugar and part no-calorie sweetener Sucralose - you only need half as much of the product as you would need sugar.

Reading Labels

In January 1, 2006, food manufacturers were required to list trans fats on the nutrition facts and some supplement facts panels, right below saturated fats. Trans fats are formed when manufacturers “hydrogenate” vegetable oils to make them solid at room temperatures. Trans fats are found in margarines, fried foods, cooking oils, baked cookies and cakes, chips, seasoned popcorn and croutons.

Label Dictionary

Fat-Free - Less than 0.5 gm of fat per serving, with no added fat or oil.

Low-Fat - 3 gm of fat or less per serving

Cholesterol-Free - Less than 2 mg cholesterol per serving and 2 gm or less saturated fat per serving.

Reduced-Calorie - At least 25% fewer calories per serving than the comparison food.

Low-Calorie - 40 calories or less per serving

Light (Fat) - 50% or less of the fat than in the comparison food (i.e. - 50% less fat than our regular cheese)

Light (calories) - 1/3 fewer calories than the comparison food

High-Fiber - 5 gm or more fiber per serving

Sugar-Free - Less than 0.5 gm of sugar per serving

Sodium-Free (Salt-Free) - Less than 5 mg of sodium per serving

Healthy - A food low in fat, saturated fat, cholesterol and sodium, and contains at least 10% of the Daily Values for one or more of vitamin A, vitamin C, iron, calcium, protein or fiber.

“High,” “Rich in” or “Excellent Source” - 20% or more of the Daily Value for a given nutrient per serving

“Good Source of,” “contains” or “Provides” - Meal or dish provides 10% to 90% more of the Daily Value for a given nutrient than the comparison meal or dish.

The “devil is in the details” and this concept certainly holds true when it comes to obesity, diabetes and the Metabolic Syndrome. One could perform ninety nine percent of a health program correctly and become obese, develop type II diabetes or the Metabolic Syndrome simply by missing the mineral supplement program.



Chapter 16

“Eat to live, and not live to eat.”

- Benjamin Franklin

Dr. Wallach's See Food Diet

While there are literally thousands of weight loss programs and diets available to everyone, they all have many common denominators. All weight loss and diet programs are not guaranteed, they are not successful 100 percent of the time, most starters drop off of the program over time and most gain all or more of their starting weight back again after dropping off the diet.

The basic reasons for the universal failure of “weight loss diets” of any ilk, is that they don't deal with the basic root cause of obesity - the mineral deficiencies. Therefore, the primary difference of the Dr. Wallach's See Food Diet, the principal that sets it apart from the others is that the pica, cribbing, cravings, binge eating, the munchies and the “stand in front of the open refrigerator” syndrome is dealt with and eliminated - these basic failure factors having been removed will guaranty success for everyone.

The diets generally fall into three basic types:

1. Low calorie - The old standby is the low calorie diet. The thinking is simply that a calorie is a calorie and the fewer you eat the less you will weigh and the more weight you can lose. The low calorie diet proponents believe if you eat fewer calories than your basal needs,

the more weight you will lose.

Promoters of the low calorie diets falsely claim that dietary supplements are not required. You have to count calories, points, or portion sizes. (Jenny Craig, Weight Watchers).

In 1963 there were 400 clients registered with weight watchers; by 2003 there were 34,600,000 registered clients.

2. Low fat - The proponents believe that fat is the enemy and tend to eat lots of plant based foods rather than animals foods. Unfortunately, the devotees don't differentiate between good fats and bad fats, which results in many health problems, including an increased risk of stroke, Alzheimer's disease, seizure disorders and asthma.

The low fat approach to weight loss suggests less than ten percent of daily calories come from dietary fat. The concept is, by cutting out fat you can eat one third more total food each day without taking in more calories. The claim is that the body only needs four to six percent of the calories from fat to maintain health (in actuality, thirty five percent of daily calories provided by fat is a more correct amount). This diet is very difficult to adhere to and is potentially dangerous. (Ornish, Pritikin)

3. Low Carb - The total amount and the type of carbohydrates are limited to reduce the body's glucose and insulin levels in an effort to make weight loss and weight management easier.

In the diet's initial phase, one can eat unlimited meat and fat, however, carbohydrates must be limited to twenty grams of carbohydrates or less per day (approximately one slice of bread).

For the first fourteen days, the dieter consumes no fruit, bread, pasta, grains, milk or even ketchup. The stated purpose of the low carb diet, "is to create a state of ketosis, encouraging your body to burn fat for energy rather than carbohydrates."

Eugene Fine and Richard Feinman, of the SUNY Downstate Medical Center of New York believe that "critics of low-carbohydrate diets always invoke the laws of thermodynamics, but they're not understanding them properly." Feinman claims there are two laws, not one,

that are involved.

Opponents of the low carb diets often emphasize the first law of thermodynamics - that the total energy in a system is always conserved. Feinman considers the second law - that energy tends to dissipate over time - just as important, it determines what happens to energy in food when it is broken down in the body.

"We know on a molecular level that the body's pathways are less efficient at turning protein calories into glucose," says Feinman, "That means that more energy is lost as heat." So more of each calorie will convert into heat if it comes from protein; more of the calories will remain in the body if they come from carbohydrates.

The idea is to satisfy the taste buds with large amounts of rich, sumptuous foods to reduce the cravings. They certainly have forgotten the minerals and an unsupplemented high protein diet can increase the risk of kidney stones. (Atkins, Southbeach, The Zone, Dr. Wallach's See Food Diet)

In May of 2004, The Valen Group, a consulting firm for the Fortune 500 companies, stated, "59 million U.S. adults are controlling their intake of carbohydrates and an additional 40 million said they were considering a low carb diet in the next twelve months. Collectively, those figures are fast approaching half of all U.S. adults."

Dr. Wallach's See Food Diet is perhaps the best of the best when it comes to low carbohydrate, low glycemic index diets. Our low carb diet differs from all others in that our concerns for the mineral intake is actually more heavily weighted and more important than parsing words on the percentages of carbohydrates, fats and proteins.

There has been an endless parade of diets for various purposes including weight loss, health and longevity going back to the days of the Chinese alchemist thousands of years ago. Yet the problem of obesity, diabetes, the Metabolic Syndrome and other degenerative diseases has reached epidemic and pandemic proportions.

If low carbohydrate diets were the answer to our obesity problem we wouldn't have the projection of forty percent of Americans being obese by 2010. To be fair, we must say that low carbohydrate diets and low glycemic loads can speed up the weight loss process, how-

ever, if low carbs are one's sole consideration - you are in for a rude awakening.

If you live for a low carb diet that will work, you will be on an endless crusade and treadmill. To lose weight safely and prevent the common pitfalls of weight loss (i.e. - heart attacks, liver damage, high blood pressure, Beriberi, arthritis, osteoporosis, periodontal disease, cancer, sudden death, stroke, kidney stones, etc.), one has to protect themselves from the ravages of mineral nutritional deficiencies that result in psychotic eating and free radical damage generated by burning body fat.

The burning of body fat is not a danger free process. The combustion of all fuels, including wood, coal, oil, gasoline, sugar and fat, produces wastes, pollution and dangerous combustion byproducts including free radicals. These dangerous byproducts cannot be treated lightly or dismissed out of hand, they must be prevented and neutralized with an aggressive proactive antioxidant plan.

The Principals of a Low-Carb Diet

In the shortest possible description and purpose of a low carb diet, one can say that it is conceived to keep blood sugar levels from having wild fluctuations. The purpose is to "steady" blood sugar levels, which will help reduce episodes of ravenous hunger.

Foods that produce a slower rise in blood sugar after being consumed are said to have a low glycemic index. Fat has a glycemic index of zero because it does not affect blood sugar directly or immediately. Whole grains, vegetables and proteins are considered low glycemic foods because they produce a very slow, gradual, time release increase in blood sugar.

The reality is, low or zero glycemic foods do help you feel full longer, however, this theory presumes the reason that a person is overweight or obese is just because they can't control their eating habits for lack of education, personal will or self control.

The original Atkins Diet critically slashed carbohydrates (i.e. - bread, pasta, rice, sugared soft drinks, potato chips, cookies and fruit.) which was driven by the Seven Food Group Pyramid and promoted by the USDA, and medical doctors had caused sixty five percent of

Americans to be overweight and forty percent to be obese and turn adult onset type II diabetes into an epidemic disease.

Atkins' followers were allowed large portions of high fat red meat, cheese, butter, and oils that traditional low fat diets had prohibited. The dieter could eat large bowls of greens and low carb vegetables.

The South Beach Diet restricts carbs, fruits, sugars, fat laden meat and cheese.

According to many experts, the reason that low carb diets worked was that, "they were boring, at some point we become so tired of any given thing that we eat, that it has no continuing hedonic effect on us," says Brian Wansink, Ph.D., University of Illinois. Additionally he found that, "the more variety in your diet, the more food you'll eat, as everyone who has dined at a lavish buffet can attest."

Therefore, they theorize, "taking the boredom out of a low carb diet with low carb comfort foods, such as ice cream, chips, and beer could reduce it's effectiveness." "Originally, the Atkins diet was black and white," said Alice H. Lichtenstein, D.Sc., professor of nutrition science and policy at Tufts University. "I can have the steak but not the baked potato. Now I can have the steak, but I can also have the low carb pasta on the side."

In summary, experts believe the reason that the Atkins approach worked was that blood sugar was more stable and that a low carb diet was boring and didn't encourage over eating of total calories.

Net Carbs and Sugar Alcohols

Sugar alcohols and fermentable carbohydrates are safe, although their consumption may produce diarrhea and gas in susceptible people or following over consumption. There are some studies that show that they can aid in blood sugar control, but there is no evidence that by consuming these foods weight loss will be easy.

In low carb labeling, "net carbs" refers to the total grams of carbohydrates per serving minus grams of fiber and sugar alcohols. The term "net carbs" has no legal recognition by the FDA as of 2004.

Sugar alcohols, or polyols include erythritol, hydrogenated starch hydrolysates [HSH], isomalt, lactitol, maltitol, mannitol, sorbitol, and xylitol.

Polyols are not digested in the small intestine the way that regular sugars are. Instead, they pass through to the large intestine, where they are digested by bacterial fermentation. Thus they do not elevate blood sugar and insulin levels as much or as quickly as table sugar does, giving them a low glycemic index. But like sugars, which have 4 calories per gram, they do contribute calories, ranging from a low of 0.2 per gram for erythritol to 3 calories per gram for HSH.

Fermentable carbohydrates include inulin, resistant starch, and polydextrose - this "catchall" refers to smooth, bland-tasting, highly processed ingredients (derived from corn) that can be used in place of high carb white flour and counted as fiber. Fermentable carbohydrates are digested the way that sugar alcohols are. They have between 1.6 and 2.5 calories per gram (compared with 4 calories per gram for table sugar and processed starch).

Dr. Wallach's Plan

The Dr. Wallach's See Food Diet is based on principles rather than a carbohydrate, fat or calorie counting regime. I like an approximation of forty percent protein, forty percent fat and twenty percent carbohydrates. This may seem like a lot of calories to the average person, however, when one is well saturated with all of the essential minerals, reasonable portions become automatic - we are no longer driven by a command to eat.

If an individual is physically active or they need to lose twenty five pounds or more, the best approach is to eat four to six small meals per day, or have a hand full of nuts as a between meal snack. It is important to use protein and fat to stabilize blood sugar rather than sugar or low glycemic index carbohydrates.

It is important to eat minimal amounts of fruit and fruit juices, if you are more than twenty five pounds over your ideal weight totally avoid fruit and juices altogether - eat more vegetables.

Drink eight glasses of water each day (if you are active and physically working or exercising you will want to add a complete nutrient sports-energy drink concentrate to your daily water supply).

Do not eat after 7:00 p.m. (except for celebrations or special events).

No fried foods, margarines, cooking oils, meat cooked well done or burnt fats.

No carbonated drinks, caffeine or sugar (there can be celebration exceptions, however, these exceptions should be rare).

Switch to gluten-free diet, which is primarily to treat gluten-intolerance; gluten causes inflammation in the small intestines.

People with gluten - intolerance may also benefit from it.

Breakfast should always be eaten, preferably including eggs, unprocessed meat and vegetables (tomatoes are to be preferred over fruit - they have far more nutrients than fruit and a lot less sugar).

Lunches can be large salads with grilled meat, fish or poultry added. Remember, croutons are bread. Put the dressing on the side.

At home you will want to use a high ORAC value juice (noni, mangosteen) or juice blend for a salad dressing. When dining out you will want to use "Dr. Wallach's Salad Fork Trick."

Dr. Wallach's Salad Fork Trick is simply performed by dipping a salad fork 1/4 inch into the dish of salad dressing and shaking off the excess. This simple procedure will allow you to taste the salad dressing, yet give up ninety five percent of the calories.

The most important part of Dr. Wallach's See Food Diet has been left until the last. The supplement program should include all of the ninety (90) essential nutrients. Specifically, the minerals should come from a plant source. Obviously, for most of us ("culinary" ashes from wood fires are no longer available) that means the consumption of a liquid plant derived colloidal mineral supplement with a minimum of seventy seven minerals at the concentration of 19,000 mg per quart.

These liquid minerals should be taken at the rate of one ounce per hundred pounds of body weight as a maintenance dose. For example, a 100 pound person would take one ounce per day in a split dose or one half ounce (one tablespoon) twice per day. A 200 pound person would take two ounces per day; a 300 pound person would take three ounces per day and so on.

For those individuals needing to lose twenty five pounds or more the supplement level can be increased to one and one quarter ounces (1 1/4 oz) per 100 pounds of body weight.

Chapter 17

“Meaningful weight loss is difficult to achieve, particularly when one considers that the goal of any reducing program is not merely to lose weight but to keep it off. The results of most treatment programs of obesity have been uniformly unsatisfactory. Clearly, the therapeutic approach currently employed by most physicians is inadequate. This situation is likely to prevail until a more fundamental understanding of the regulation of eating behavior and body weight is achieved.”

- Lester B. Salans

*Chief, Laboratory of Cellular Metabolism and
Obesity; Associate director, Diabetes, Endocrinology
and Metabolic Diseases. NIH*

Burn, Baby - Burn!

Antoine Lavoisier (1743 - 1794) and his coworker P. S. Laplace (1749 - 1827), a mathematician, confined a guinea pig in oxygen for ten hours and measured the carbon dioxide produced. He also measured the amount of oxygen used by a man active compared with him at rest. In these experiments, he was able to show that combustion of carbon compounds with oxygen to carbon dioxide and water is the true source of animal heat and that oxygen consumption increases during physical work.

“Respiration is merely a slow combustion of carbon and hydrogen, which is similar in every respect to that which occurs in a lighted lamp or candle, and, from this point of view, animals (and humans) that breathe are really combustible bodies which burn and are consumed.”



Lavoisier was guillotined on May 8, 1794 during the French Revolution. The judge presiding over Lavoisier's fate was reported to say, "The Republic has no need of savants (scholars)."

Ingenhousz adopted Lavoisier's thoughts on gases and theorized that the plant was not simply exchanging "good air" for "bad." In the sun, he suggested a plant "absorbs the carbon from carbon dioxide, throwing out at that time the oxygen alone, and keeping the carbon to itself as nourishment."

Nicholas Theodore de Saussure (1767 - 1845) used Lavoisier's theories of quantitative measurements and showed that, "equal volumes of CO_2 and O_2 are exchanged during photosynthesis, and that the plant does indeed retain the carbon (and store "solar income")." He also showed that, "during photosynthesis the plant gained more weight than could be accounted for by the carbon taken in as carbon dioxide." In other words, "the carbon in the dry matter of plants comes from carbon dioxide, but, equally important, the rest of the dry matter (with the exception of the minerals from the soil) comes from water."

Now, all of the steps of photosynthesis were identified - carbon dioxide, water, and light:



Photosynthesis is respiration or combustion reversed! In photosynthesis (which occurs in specific organelles called chloroplasts), light energy is converted to chemical energy, and carbon is fixed into organic compounds such as sugar, carbohydrates, oils, structural polysaccharides (including cellulose), amino acids, enzymes, DNA and RNA.

Humans take in the stored "solar income" chemical energy as food and convert it to another form of chemical energy (metabolism) or mechanical energy (movement). Excess intake of dietary energy above the basal needs for any reason will result in a net weight gain.

There are many approaches to accelerating weight loss which as

a group are referred to as "thermogenics." They will be discussed here because humans are impatient and they want immediate results from any activity, and it can certainly be appropriate to add thermogenics to a weight loss program.

We like to provide and feature the nutritional program that will eliminate pica, cravings, binge eating and the munchies that will result in a steady loss of weight and a return to the individual's ideal weight. With this said, we appreciate the emotional support that thermogenics can provide.

Exercise

Humans are designed for exercise; in an anatomical sense, the human's ability to run separates us from the tree dwelling apes. According to researchers from Harvard University, there are no less than twenty six anatomical and physiological traits that make humans capable of long distance running. Over long distances, humans can outrun monkeys, dogs and horses.

While humans cannot run as fast, for short distances, as monkeys, horses, greyhounds or antelopes, humans can keep on going and going.

Benjamin Franklin determined that the best way to measure exercise, was not by it's duration, but rather "by the degree of warmth it produces in the body." Franklin's theories on exercise were amongst the first linking exercise to calories of heat produced and the number of calories burnt. For example, Franklin explained, "walking a mile up and down stairs produced five times more body warmth than walking a mile on a level surface.

When swinging weights, Franklin calculated, "that this raised his pulse from 60 to 100 beats per minute." He correctly calculated that body "warmth generally increases with quickness of pulse."

Beside one's basal requirements for energy, physical exertion or exercise is the second greatest influence on the energy requirements of humans. Exercise, as a consumer of energy, may be as little as ten percent of the total energy requirements (bed ridden patient, computer technician, executive, etc.) of an individual, or up to fifty per-

cent in laborers or athletes.

A man performing heavy physical activity (farmer) may require 4,800 or more kcalories per day, while an individual of the same body build, weight, age and height in the same environment, however, performing light work (book keeper) would require only 2,500 kcalories per day.

The inactive or sedentary human usually requires thirty percent more kcalories in addition to the basal requirement; by comparison, a lightly active human may need fifty percent more kcalories beyond basal requirements; a moderately active human would require seventy five percent additional kcalories above basal needs; and a laborer or athlete might require an additional 100 percent more kcalories over basal needs.

Body weight also affects energy requirements. The greater the

Activities That Consume Calories (calories per hour)

Activity	Weight (pounds)	
	120	170
Aerobic dance	330	460
Basketball	330	460
Bicycling	220	310
Golfing	250	345
Hiking	330	460
Rowing	385	540
Soccer	385	540
Swimming	330	460
Tennis	385	540
Walking	250	345
Weight training	165	230

body weight, the greater the caloric requirement and therefore the kcalorie expenditure for an activity. It takes more energy for an obese human to walk for ten minutes than it does for a slim individual.

On December 23, 2004 the Harvard Nurses Health Study showed that obese women who exercise for 3 1/2 hours per week will still die sooner than women who are toned and trim. Women who were both sedentary and obese were almost 2 1/2 times more likely to die.

"There has been some suggestion that if you are particularly active, you don't have to worry about your body weight, or about your diet. That's very misleading," says Frank Hu of the Harvard School of Public Health, lead author of the study published in the *New England Journal of Medicine*.

The study of 116,500 women, showed that obese women who worked out for at least 3 1/2 hours weekly had a death rate that was 91 percent higher than lean women who exercised for 3 1/2 hours per week. Lean women who exercised less than that increased their risk of early death by 55 percent, compared with lean women who worked out more frequently - for inactive obese women the premature death rate was 142 percent greater.

Walking or light "express" exercise programs can produce the same weight loss result as a heavy weight or machine program in a gym after twenty weeks, when combined with a 1,200 to 1,500 calorie diet. At the end of twenty weeks the individual can lose seven to eight percent of their starting weight, after forty weeks their weight loss totaled eight to nine percent according to the University of Pennsylvania.

In 1987 there were 17,351,000 health and fitness club memberships in the USA; in 2003 the numbers climbed to 39,400,000.

In another study, investigators at the University of Pittsburgh monitored the weight loss of fifty nine women who were limited to 1,200 to 1,500 calories per day, found that behavior modification, walking, walking and light strength training, or walking and yoga lost twenty two to twenty eight pounds in six months.

Experts recommended that people try to take 10,000 steps per day (about five miles). Walking a mile requires about 2,000 to 2,500 steps.

Holiday food	Steps required to burn off calories
Stuffing/dressing (1 cup)	6,992
Candied yams (1/2 cup)	3,910
Roasted turkey (3 oz)	2,737
Red wine (8 oz)	3,266
Pumpkin pie (1 slice)	5,267
Pecan pie (1 slice)	1,569

The average American walks only 5,000 steps per day, or about two miles. Research shows that walking an extra 2,000 to 5,000 steps and reducing calories by 100 per day will stop weight gain.

Calories Burned per Activity*

Activity	Calories burned
E-mail (1 minute)	2
Ride elevator (2 minutes)	3
Walk to partner's office (1 minute)	4
Order take out meal (1 minute)	1
Take stairs (2 minutes)	19
Load dishwasher (10 minutes)	23
Watch TV	35
Go to car wash	35
Play cards	52
Play video game	53
Cook meal	70
Wash dishes	80
Mow lawn/riding mower	88
Wash car (at home)	104
Mow lawn/power mower	193
Play basketball	280

* Calories calculated for 30 minutes of activity (unless noted): 150 lb human

Pet Exercise

Walking one's dog helped individuals to lose weight and keep it off. A study reported in 2004 that when people and dogs were put on a diet and exercise weight loss program, they both lost weight and kept it off; although dogs did better than their humans and the program did not increase begging.

Sixty five percent of Americans and twenty five percent of dogs are overweight or obese. "If you're looking for motivation and social support to lose weight, you probably don't have to look any further than the pet in your own home," said Dr. Robert Kushner of Northwestern Medical School in Chicago, who led the study.

The study was funded by Hill's Pet Nutrition, which makes Science Diet and a prescription diet dog food. The study was made up of three groups: 56 people; 53 dogs; and 36 dogs and their owners. The participants attended weekly counseling sessions at Northwestern on diet and exercise, and were encouraged to walk at least twenty minutes and limit calories to 1,400 per day - dogs were fed a weight loss prescription diet.

The dog owners did slightly better than the dieters who walked and dieted without the dog. Overall, people lost eleven pounds or five percent of their starting body weight in the first four months and kept it off for the next eight months. The diet and exercise program was more enjoyable for the dog owners.

The weight loss for the dogs was even more effective. They lost an average of twelve pounds - fifteen percent of their initial weight - one dog actually lost 35 pounds. "Begging behavior did not go up," said Dennis Jewell, a Hill's animal nutritionists who did the dog-diet part of the study.

We would be remiss if we didn't point out that "exercise without supplementation is suicide." There has never been a professional athlete from a major professional sport who has lived to be 100. The reason for this surprise observation is that athletes sweat more than the average person.

Sweat is not just water, sweat is a soup that contains all of the known essential nutrients - if the lost nutrients are not replaced prop-

Steps Required by Activity

Activity	Number of Steps
One mile	2,000 - 2,500
Five miles	10,000
Golf (9 holes)	8,000
City block	200
Soccer (90 minutes)	8,000 - 10,000
Short walk (10 minutes)	1,200
Female 30 - 39 (ave. per day)	5,819
Male 30 - 39 (ave. per day)	5,162
Female Amish (39 hours/week)	14,000/day
Male Amish (43 hours/week)	18,000/day

erly the athlete will fall victim to a wide range of nutrient deficiency diseases, including a breakdown of joints, tendons, ligaments, cartilage, connective tissue and bones. The rate of sudden death in athletes is higher than in the average individual.

Examples include Brian Maxwell, the legendary marathon runner and founder of the Power Bar empire who died in March 2004 at age 51 from a cardiomyopathy heart attack.

Domenic Mobilio, the legendary Canadian soccer player died at age 35 from a sudden heart attack November 2004.

Sergei Zholtok, hockey player for the Nashville Predators, died suddenly while playing at age 31.

Jeff Halvorson, Canadian football running back, collapsed and died during a practice at age 21 September 2004.

Rachael Townsend, 29 years of age, died October 2003 while running a marathon.

Ronanda Pierce, age 19, a six foot five inch freshman basketball player for FSU died of a ruptured aneurysm June 2004.

Additionally there are thousands of athletes of all ages, partici-

pating in many different sports, who die suddenly each year.

The Knife Diet

While not classically thought of as a “thermogenic,” bariatric surgery should be considered as an option for those who are morbidly obese. As with all surgery, there are risks associated with bariatric surgery. Bariatric surgery has a complication rate of seven (7%) percent including death.

In a major government decision that attempted to turn obesity from the status of a personal failure to a medical problem, Medicare announced that it would remove barriers to covering anti-obesity treatments after forty years of stating, “fat was not an illness and not covered.”

The change was anticipated to open the door for government funding of various bariatric surgeries and intensive diet and exercise programs. The program will “put more pressure (on private insurers) in terms of medical and surgical procedures,” says health financing researcher John Newman of Georgia State University.

In 1992, 16,200 bariatric surgeries were performed; by 2002 the number had risen to 63,100; and by 2003 the annual numbers reached 144,000.

The medical community was overjoyed at this decision, because it will cause obese people to preferentially opt for expensive surgery without any effort on the part of the individual rather than choose the non-surgical programs. With sixty five percent of Americans described as being overweight and thirty one percent of them obese, the increase in the financial pressure on Medicare and private insurance will prove to be catastrophic.

1. **Roux-en-Y gastric bypass:** The most popular of the three most common “bariatric” surgeries - staples are used to create a small pouch in the stomach connected to the bowel by a piece of the small intestine, bypassing the majority of the stomach. It accounts for up to 90% of the “bariatric” procedures performed in the USA.

2. **Biliopancreatic diversion:** This procedure involves removing a

large portion of the stomach and bypassing much of the small intestine.

3. **Gastric banding:** This procedure uses a band to pinch off a small portion of the stomach. The band can be adjusted to allow more or less food to enter the stomach. About 9% of the bariatric procedures in the USA employ this technique.

4. **Jaw wiring:** Jaw wiring is a primitive surgical approach to weight control. Yes, by starving the patient they will lose weight, however, when the jaw is unwired the weight comes right back because the basic problem of mineral deficiencies has not been dealt with.

HGH

Human Growth Hormone: Human growth hormone or its precursors are known to conserve body protein and muscle mass while encouraging the burning of body fat.

The reduced production of HGH associated with aging is sometimes called "somatopause." Somatopause is characterized by symptoms of aging including wrinkles, gray hair, decreased energy, decreased sexual function, increased body fat (obesity), increased risk of cardiovascular disease, osteoporosis, etc.

Features of Growth Hormone Deficiency

Anabolic Tone

Reduced lean body mass (sarcopenia)

Reduced skeletal muscle strength

Reduced exercise performance

Increased total body fat

Increased abdominal and visceral fat

Lipid changes

Elevated LDL cholesterol

Decreased HDL cholesterol

Elevated apolipoprotein-B

Bone Changes

Osteopenia (early osteoporosis)

Metabolic effects

Insulin resistance (in obese individuals)

Hypoglycemia

Abnormal resting metabolic rate

Reduced T4 to T3 conversion

Protein Synthesis

Loss of skin fat (thinning skin)

Loss of collagen in skin

Organ atrophy and wasting

Decreased rate of skin, hair and nail growth

Dehydration

Reduced urine production

Reduced volume of sweat (difficulty in thermo regulating)

Decreased cardiovascular output

Increased vascular resistance (hypertension)

Emotional Function

Reduced energy levels

Volatile behavior

Dementia and reduced concentration capacity

Depression

Withdraws from social activities

Loss of goals and direction

Reduction or loss of sex drive

The amino acids L-arginine and ornithine can cause HGH to be released from the pituitary gland if supplemented in significant doses.

According to studies published in the *New England Journal of Medicine*, "HGH can reverse symptoms of human biological aging:

Restoring muscle mass

Increases thickness of skin and reduces wrinkles

Can to some degree restore lost hair.

Restores hair color

Increases energy

Increases sexual drive

Modulates cholesterol levels

Encourages increase in organ size

Improves eye sight

Can improve memory

Improves sleep patterns

Elevates mood

Improves blood pressure

Increases stamina

Increases cardiac output

Improves immune function

Increases wound healing capacity

Anorectic Drugs

Anorectic drugs: Anorectic drugs include pharmaceuticals and herbs that have a profound effect on the central nervous system to suppress the appetite. All anorectic drugs have serious health and life threatening side effects.

Thyroid Hormone

Thyroid hormone: It is clear that both the caloric content and the carbohydrate content of the diet affect thyroid metabolism. The thyroid gland functions as a thermostat for the body's basal temperature and metabolic rate. Increases in calories and carbohydrates increase the metabolically active hormones, while extreme restriction decreases the levels of active thyroid hormone.

Pyruvate

Pyruvate: Is a part of the Krebs Cycle, the cellular metabolic pathway that generates energy by burning carbohydrate. When supplemental

pyruvate is added to the metabolic system, more energy is burned - on a low carbohydrate diet, pyruvate will increase the rate that the body converts stored fat into carbohydrate or ATP which is then burned. This process takes some time, so there can be a delay of as much as six to eight weeks before significant weight loss occurs - once weight loss begins, however, it will be steady and predictable.

Chocolate

Chocolate is thought of as an indulgence, as something that overweight and obese people gorge themselves on - yet chocolate can be one of the great natural herbal weight control products ever known to man.

Studies published in the *American Journal of Clinical Nutrition* and the *Journal of the American College of Nutrition*, reported that chocolate does not raise cholesterol or increase LDL.

In a 2003 issue of the journal, *Hypertension*, it was reported that the flavonoids in cocoa could help increase blood flow to the heart, brain, hands and legs.

An article in the *Journal of Agriculture and Food Chemistry*, reported that cocoa has a higher antioxidant rating than tea or red wine.

When people crave chocolate, they are really mineral deficient (pica, munchies). In fact, bitter chocolate has a very low glycemic index because up to seventy percent of chocolate's calories come from fat, and fat does not raise blood sugar.

The fat in chocolate, cocoa butter, is comprised of stearic acid (34 percent), palmitic acid (20 percent) and oleic acid (30 percent). Both stearic and palmitic acids are saturated fats; oleic acid is a monounsaturated fat that is also found in olive oil.

Cocoa powder is made when the fat, or cocoa butter, is pressed out of the "liquor" - most cocoa powders have little or no fat.

Chocolate is a rich source of cocoagin, caffeine, theobromine, flavonol and flavonoids. These compounds have thermogenic and antioxidant properties that make bitter chocolate a great weight loss and free radical neutralizing herb.

Meal-Replacement Diets

Meal-Replacement Diets: Meal replacement diets are very useful in weight loss and weight management programs because calories are very accurately controlled and most contain some added nutrients.

We like to have meal replacements used for one or two meals per day; two days per week the meal replacements can be used for all meals. The remainder of the meals should contain varying types of proteins, green leafy vegetables, steamed vegetables and a serious anti-munchies mineral supplement.

Meal replacement diets should be high protein, moderate fat, low carbohydrate and minimal to zero sugar.

The basic root cause of obesity, it's symptoms and it's complications is a multi-mineral deficiency - this deficiency must be dealt with to eliminate obesity and maintain the individuals optimal weight. The benefit of any individual or combination of thermogenic programs is to support and accelerate the process of weight loss - used as the sole approach to solving a cosmetic or clinical weight problem, thermogenics, used alone will always fail over the long haul.

Thermogenics by themselves cannot safely or permanently produce a healthy weight loss program, as these weight loss approaches only deal with the symptom of obesity - the fat, and they do not deal with the general systemic cause of obesity - the mineral deficiency! And they don't deal with the need for antioxidants to protect the individual from the free radical damage that will be generated as a result of "burning" body fat.



Chapter 18

“Eating as an art is very old, but as a science, very young.”

*- Charles G. King, Prof. of Chemistry
Columbia University*

Recipes to Live For

Human beings need to consume “90 essential nutrients” on a daily basis to promote optimal health, fitness and longevity - fortunately our food plants (grains, vegetables fruits and nuts) can take carbon dioxide from the air, use the sun’s energy or “solar income”, and with a process known as photosynthesis manufacture long carbon chains and oxygen. The oxygen is released into the atmosphere and the carbon chains are then configured by the plants into starch (grains, potatoes, squash, etc.), sugar (sugar cane, fruit, beets, etc.), cellulose (plant structure, wood, plant fiber, etc.), vitamins, amino acids, fatty acids and phytonutrients (antioxidants, flavonoids, alkaloids, glycosides, etc.).

Unfortunately, plants cannot manufacture minerals and two thirds (60) of our ninety essential nutrients are minerals. Plants obtain minerals daily for their own use from the soil. Plants only need nine (9) minerals to be optimally healthy, humans need sixty (60) essential minerals - typically farmers only add three (3) elements (NPK - nitrogen, phosphorus, potassium) to their fields, because plants will produce maximum yields in terms of tons and bushels per acre when fed these three elements.

Unfortunately, mineral nutrients do not occur in a uniform blan-

ket throughout the crust of the earth - minerals occur in veins like chocolate in chocolate ripple ice cream and in veins like gold and silver. One is more likely to win \$1,000,000 a day in Las Vegas than they are to get all of their mineral nutrients just by "eating well."

The bottom line is, we as humans cannot get all the nutrients we need just by eating well - we could spend a million dollars per day on exotic food and we still could not get everything we need in optimal amounts - how many wealthy people live to be 100 years of age?

We should eat to live rather than live to eat - when one is in the mode of "I live to eat" they are mineral deficient (pica, munchies), however, with this said, there is no reason that food shouldn't be fun, tasty, nutritious and presented well. Food and dining are much more than just protein, carbohydrate, fat and fiber - dining is more than just putting stuff in our mouth - eating is a family event, a social event, a production, an educational experience and on many occasions, a business event.

Henry A. Wallace, former Secretary of Agriculture, stated, "Man, for 99.9 percent or more of his life here on earth lived without the 'benefit' of the starchy grains like wheat, oats, barely, rice and Indian corn." Man from the beginning was first a carnivore and only became an omnivore (eats both meat and grains) some 10,000 years ago when organized agriculture almost totally replaced hunting in the "Fertile Crescent" of Iraq.

Therefore it is not surprising that people today do better on low or moderately low carbohydrate diets. This chapter, "Recipes to Live For," reflects recipes with a relatively low carbohydrate intake consistent with man's early low carbohydrate nutritional history and expects that each of us will be personally responsible for our daily intake of the supplemental ninety essential nutrients with a special emphasis on the sixty essential minerals.

The application of wood ashes to early gardens, the early use of "culinary" ashes that were added to food, either directly or indirectly mixed with salt began in a parallel world wide fashion about 30,000 to 50,000 years ago. In urban communities worldwide today, available food no longer contains the wood ash minerals that had supported and fulfilled man's mineral needs for tens of thousands of

years.

From 30,000 B.C. to Monday September 4, 1882 at 3:00 p.m. man's mineral (wood ashes) enriched food provided a steady stream of essential minerals - this all ended when Edison's technician pulled the lever to start the first commercial electrical generating plant - by 1950, wood ashes as an essential part of urban mans life and culture had gone the way of the buggy whip following the appearance of the automobile.

We surely don't miss the buggy whip, that was the "accelerator" for the horse, because we have an accelerator or gas pedal in the car; however, we have developed a pandemic and epidemic of mineral deficiency diseases in humans including obesity, diabetes and the Metabolic Syndrome because we have given up our wood ash source of minerals without replacing them with another source of plant minerals.

The knowledge, recognition and acceptance of food's nutritional chemical limits and mineral shortcomings, still allows us to eat well and yet recognize that food alone can no longer provide all that humans require for optimal health and maximum longevity.

Nutritional supplements, especially minerals, should then be as comfortable on the family and commercial dining table as salt and pepper shakers, spices and sauces.

Breakfast

Poached, boiled and scrambled eggs

2 poached eggs

Breakfast meat 4 oz (i.e. - ground beef, steak, pork chop.

Salmon, trout, poultry)

Cheese

Tomatoes

Onions, peppers

2 soft scrambled eggs, in butter, low temperature

Cottage cheese

Broccoli

Onions (baby pearl)

1/2 poppy seed bagel with one pat butter or Philadelphia Cream Cheese

Whey protein/soy protein shake powder

8 oz water or milk

Add raw or lightly poached eggs

Add 1/4 cup dark berries

Add 1/2 cup chopped ice

Blend for 2 minutes

Omelet

2 egg soft vegetable omelet (tomatoes, slivered almonds, mushrooms, onions, bell peppers, broccoli, etc.)

Tomatoes, sliced

1/2 poppy seed bagel with pat of butter

Spinach-cheese omelet

Nonstick cooking spray

4 eggs, dash salt

Dash cayenne pepper

1/4 cup shredded sharp cheddar cheese (1 oz)

1 Tblsp snipped fresh chives, flat-leaf parsley or chervil

1 cup fresh spinach leaves well washed to remove sand

Red pepper relish

1. Lightly coat an unheated 8-inch nonstick skillet with flared sides with a non-stick cooking spray. Preheat skillet over medium-high heat.

2. In a medium bowl, combine eggs, salt and cayenne pepper. Beat with a wire whisk until frothy. Pour into prepared skillet; cook over medium heat. As egg mixture sets, run a spatula around edge of skillet, lifting egg mixture so that the uncooked portion flows underneath.

3. Continue cooking and lifting edges until egg mixture is set, but is still shiny and moist. Sprinkle with the cheddar cheese and

chives, parsley or chervil. Top with 3/4 cup of the spinach and 2 Tblsp of the red pepper relish.

4. Using the spatula, lift and fold an edge on the omelet partially over filling. Top with remaining spinach and 1 Tblsp of the remaining red pepper relish. Cut omelet in half for two servings.

Red pepper relish: in a small bowl, combine 2/3 cup chopped red sweet pepper, 2 Tblsp finely chopped onion; 1 Tblsp cider vinegar; and 1/4 tsp ground black pepper. Makes 2/3 cup of relish.

French Omelet

2 eggs

2 Tblsp water

1/8 tsp salt

Dash freshly ground black pepper

1 Tblsp butter

Filling (mushroom, cheese, fruit)

1. In a small bowl, combine eggs, the water, salt (omit salt if using cheese filling) and pepper. Beat lightly with whisk until combined but not frothy. Heat an 8-inch non-stick skillet with flared sides over medium-high heat until skillet is hot.

2. Add butter to skillet. When butter has melted, add egg mixture to skillet; lower heat to medium. Immediately begin stirring the eggs gently but continuously with a wooden spatula until mixture looks like very soft scrambled eggs. Stop stirring. Cook for 30 to 60 seconds more until eggs are set, but still glistening and soft.

3. Add a filling. With a spatula, lift and fold an edge of the omelet about a third of the way towards the center. Remove from the heat. Fold the opposite edge of the omelet toward the center; transfer to a warm plate.

Mushroom filling: In a small skillet, cook 1/3 cup sliced fresh mushrooms in 1 tsp hot butter until tender - spoon across center of omelet.

Cheese filling: Sprinkle 1/4 cup shredded cheddar, Swiss or Monterey

Jack cheese across center of omelet.

Fruit filling: Spread 2 Tblsp dairy sour cream or yogurt across center of omelet. Fold in two opposite sides. Top with 1/4 cup halved fresh or frozen strawberries, blueberries or raspberries.

Farmer's breakfast

- 1 Tblsp olive oil
- 1/2 Cup loose-pack frozen, diced hash brown potatoes
with onion and peppers (thawed).
- 1/2 cup chopped, cooked ham or hamburger meat
- 3 Jumbo eggs
- 3 Tblsp water
- 2 Tblsp shredded cheddar cheese

1. In a medium skillet, heat oil over medium heat. Add potatoes in a single layer. Cover and cook for 5 to 7 minutes or until potatoes are tender, stirring once. Stir in meat.

2. In a small bowl, combine eggs and the water; beat with a rotary beater. Pour over potato mixture. Cook over medium heat, without stirring, until egg mixture begins to set on the bottom and around the edge. Using a spatula or large spoon, lift and fold the partially cooked egg mixture so that the uncooked portion flows underneath.

3. Continue cooking over medium heat until egg mixture cooked through, but is still shiny and moist. Sprinkle with cheese. Remove from heat. Cover and let stand for 1 to 2 minutes or until cheese is melted. Makes two servings.

Salmon

- 4 oz grilled or poached salmon, trout or halibut
(cover each fillet with salted butter pats, chopped
chives, dill or sweet basil)

Red onion slices

Tomatoes, sliced

1 poached egg

1/2 onion bagel with 1 pat butter or cream cheese

Lunch

Large green leafy salads with carrots, celery, radishes, mushrooms tomatoes, grilled chicken breast, beef steak, tuna steak, mineral enriched noni juice dressing (tastes like vinaigrette).

6 - 8 oz burger (medium rare or rare)

1/2 bun (better yet - no bun!)

Sliced onion

Sliced tomatoes

Sliced "stackers" dill pickles

Swiss cheese

Spicy mustard

Mixed green salad

Grilled chicken breast sandwich (multi-grain bread)

Sliced tomatoes

Sliced onions

Spicy mustard

Cottage cheese

Sardines (packed in mustard sauce, tomato sauce or water)

Tomatoes, sliced

Artichoke hearts

Pimento stuffed green olives

Steamed carrot sticks

Sardine stuffed Peppers

2 large bell peppers, cored and seeded

3 1/2 oz sardines chopped

1 cup steamed rice

1/2 medium onion, chopped

1 cup tomato sauce

1 beaten egg

Salt and pepper to taste

Fill pepper cavities with mix and bake at 350 degrees F
for 20 minutes.

Serve with 1 glass California white wine

All beef bratwurst or hotdog (without bun)

Spicy mustard

Chopped onions

Chopped tomatoes

Catsup

Sour kraut

Beets

6 ounces of light beer

Zucchini and Squash

2 large yellow squash, sliced to 1/2-inch rounds (4 cups)

1 medium onion, coarsely chopped

3 ounces goat cheese

3 Tblsp butter

1/4 cup corn meal

2 large eggs, beaten with a whisk

1 cup whole milk

3 Tblsp fresh dill chopped, additional sprigs for garnish

1 Tblsp Sucralose

1 tsp Kosher salt

1/2 tsp freshly ground black pepper

Preheat oven to 400 degrees F. Place squash and onion in a large saucepan in enough water to cover them. Cook over medium-high heat, covered, until squash is tender (15 minutes). Strain well and mash slightly with a potato masher or large fork. Add the goat cheese and butter while still hot and stir until melted. Mix in the cornmeal, eggs, milk, dill, Sucralose, salt and pepper. Pour mix into an 8 x 8 inch baking dish or 6 individual ramekins that have been coated with a cooking spray. Bake the mix at 400 degrees F for 40 minutes (35 minutes for the ramekins), or until golden brown on top. Serve warm or room temperature.

Zucchini Pie

3 cups grated zucchini

1 small onion, chopped

1 cup all-purpose flour

- 1 cup grated provolone cheese
- 3 large eggs, beat with a whisk
- 1/4 cup olive oil
- 4 Tblsp grated Parmesan cheese
- 2 tsp chopped fresh basil
- 1 tsp baking powder
- 1 tsp Kosher salt
- 1/2 tsp freshly ground black pepper

Preheat oven to 350 degrees F. Combine all of the ingredients in a large bowl, reserving 1 Tblsp of the Parmesan cheese. Spoon the zucchini mixture into a ten inch round glass pie plate or metal pie pan that has been coated with a cooking spray. Bake until golden brown (45 to 50 minutes). Sprinkle with the reserved Parmesan. Cool ten to fifteen minutes before slicing.

Roasted tomatoes with shrimp and feta cheese

- 5 large tomatoes cut into eighths
- 3 Tblsp olive oil
- 2 Tblsp minced garlic
- 3/4 tsp Kosher salt
- 3/4 tsp freshly ground black pepper
- 1 1/2 lbs medium shrimp, peeled and deveined
- 1/2 cup chopped fresh parsley
- 2 Tblsp lemon juice
- 1 cup Feta, crumbled

Preheat oven to 450 degrees F. Place the tomatoes in a large baking dish. Spoon the olive oil and garlic over the tomatoes. Sprinkle with the salt and pepper and toss. Place on top rack of oven and roast for 20 minutes. Remove baking dish from oven and stir in the shrimp, parsley and lemon juice. Sprinkle the mix with the feta cheese. Return dish to the oven for an additional ten to fifteen minutes (until shrimp are cooked). Serve warm.

Dinner

Steak (6 - 10 ounces) 1 inch thick, medium rare.

Rub with garlic salt.

Large mixed, Spring green salad (include onions, tomatoes, mushrooms), mineral enriched noni-juice for salad dressing. Steamed vegetables.

One glass red wine (Merlot).

2 scoops no sugar added ice cream.

Grilled or poached fish - 4 oz. (Salmon, trout, halibut, mahi-mahi, snapper).

Slivered almonds

Sweet basil or thyme

Steamed broccoli or baby carrots

Vegetable soup or mushroom soup

One glass white wine

1 inch wedge of cheese cake (no topping)

1/2 roasted chicken, baste with melted 1/2 Tblsp salted butter, sprinkle with 1/2 tsp sweet basil.

Large green salad (include onions, tomatoes, steamed baby carrots or "coin cut," celery, green peppers).

Mixed steamed vegetables

1 sesame seed roll with salted butter

1 glass white wine

2 pork chops, 4 lamb chops or a lamb shank roasted medium or medium rare and juicy.

Mixed steamed vegetables or 1 bunch of asparagus

Large green salad, include wedges of lime

Herbal tea (peppermint)

1 glass of rose' or white wine

Peppered pork tenderloin with tomato-mint salsa

2 pork tenderloins (12 oz each)

1 Tblsp mineral enriched noni juice

6 Tblsp olive oil

2 tsp Kosher salt

3/4 teaspoon freshly ground black pepper

4 large tomatoes, cored and chopped (3 cups).

- 1/2 cup chopped sweet onion
- 1 large garlic clove, minced
- 1/2 cup chopped fresh mint
- 1/4 cup red wine vinegar

Place the tenderloins in a medium bowl. Add the mineral enriched noni juice, 1 Tblsp of the olive oil, 1 tsp of the salt, and 1/2 tsp of the pepper. Combine these ingredients to coat the pork well - cover and set aside. Combine the tomatoes, onion, garlic and mint in a medium bowl. Whisk together the red wine vinegar and the balance of the olive oil, salt and pepper. Pour over the tomato mixture and toss well; cover and set aside. Place a large nonstick skillet or a grill pan coated with cooking spray over medium heat. When hot, add the tenderloins and cook for three minutes on each side. Reduce heat to medium and cook 6 to 8 minutes on each side to brown evenly. Remove from skillet to a platter and let stand for five minutes. Slice each tenderloin into eight equal pieces, and serve with 1/3 cup of the tomato-mint salsa. Serve with sliced avocado and warm flour tortillas.

Appetizers

Smoked Salmon Cucumber Rolls

- 12 oz sliced smoked salmon
- 1/2 cup honey mustard
- 1 English cucumber, peeled, seeded, trimmed into 2-inch julienne strips

Trim the smoked salmon to 40 strips 2 x 4 inches each. Spread a thin layer of honey mustard on each strip, place a small clump of cucumber julienne at one end, and roll into a tight bundle. Arrange on tray, cover loosely with clear wrap and refrigerate until it is served.

Zesty Lemon Olives

- 1 pint oil or brine packed olives, green, black or mixed
- 1/4 tsp Kosher salt
- 1/2 tsp peppercorns
- 3 Bay leaves

Rosemary or thyme (several sprigs)
 1/2 tsp fennel seeds, lightly crushed
 Garlic cloves, 4-5, halved lengthwise
 Red chili flakes (large pinch)
 2 Lemons, medium
 2 Tblsp extra virgin olive oil

If using brine packed olives, drain them. In a medium bowl combine the olives, salt, peppercorns, bay leaves, herb sprigs, fennel seeds, garlic and chili flakes. Add the zest of one lemon and thin slices of the second. Add oil and mix well. Put into a covered jar and refrigerate for more than thirty minutes to blend flavors.

Baked Herbed Feta

3/4 inch thick wedge or slab of feta
 Red chili flakes, dried, large pinch
 Oregano, dried, large pinch
 2 Tblesp extra virgin olive oil
 Lemon wedges, one medium lemon

Preheat oven to 350 degrees F; put the feta on sheet of foil. Sprinkle the chili flakes and oregano evenly over cheese - drizzle with one Tblsp olive oil. Wrap the feta in the foil, put the package in a small baking dish, bake until slightly soft (not runny). Unwrap foil at the table and drizzle with more of the oil. Serve with toasted pita chips.

Goat Cheese, Pesto and Sun Dried Tomato Terrine

10 oz goat cheese
 1/2 cup heavy cream
 Kosher salt
 Black pepper, freshly ground
 3 Tblsp basil pesto
 5 oil packed sun dried tomatoes, drained, finely chopped
 1/4 cup pine nuts, toasted, coarsely chopped
 Extra virgin olive oil for drizzling
 Pita chips

Line the inside of a two cup steep sided bowl with plastic wrap

(let ends hang over sides). In a mixing bowl, mash the goat cheese and 1/4 cup of the cream with a fork and season with 1/4 tsp salt and a few grinds of pepper; add more cream. Spoon approximately one-third of the cheese into the lined bowl and pack into an even layer. Spread the pesto almost completely to the edges of the first layer of cheese. Top with another one-third of the cheese, the sun dried tomatoes, and all but 1/2 Tblsp of the pine nuts. Top with the remaining cheese. Pack evenly, fold plastic cover over the mix and refrigerate for a minimum of thirty minutes. Thirty minutes before serving, remove bowl from refrigerator, pull the edges of the plastic to free terrine from the bowl. Invert terrine onto serving platter, drizzle lightly with olive oil, let sit for thirty minutes at room temperature to warm up. Sprinkle with remaining pine nuts and season to taste with fresh ground pepper. Serve with toasted pita chips.

Steamed Artichokes with Garlic Butter

For the Artichokes

2 tsp pickling spices
4 large fresh whole artichokes
1/2 lemon, medium

For the Garlic Butter

3/4 cup unsalted butter
1 large clove garlic, minced
1/4 tsp Kosher salt

Put one inch of water and the pickling spices into a pot wide enough to hold the artichokes. Set a steamer rack into the pot, but not touching the water. Bring the water to a simmer over moderate heat. Trim artichokes, slice off stems for steaming later, rub exposed base with sliced lemon. Remove top one third of artichoke with serrated knife. Use scissors to trim off points of each artichoke leaf. Rub cut surface with sliced lemon. Place artichokes in steamer, stem side up. Cover the pot and adjust the heat to bring water to a simmer and steam artichokes until base is tender - about 30 - 40 minutes.

To make garlic butter, melt butter in small sauce pan over low heat. Add garlic and salt; warm slowly for one minute.

To serve, drain artichokes on paper towels, serve warm on plates, divide garlic butter into four small side bowls.

Peppered Shrimp

- 1 cup extra virgin olive oil
- 2 Tblsp toasted sesame oil
- 3 inch ginger root, sliced thin
- 1 bunch scallions, coarsely chopped
- 1 Tblsp cracked black pepper
- 1 Tblsp soy sauce
- 1/2 tsp cayenne
- Zest strips from 2 oranges
- 40 jumbo shrimp, peeled and deveined
- Kosher salt

Heat olive oil and sesame oil until smoking, add ginger and scallions and combine for one minute. Remove from heat, cool for one minute and add pepper, soy sauce and orange zest. Cool completely, then add raw shrimp - marinate for thirty minutes at room temperature. Remove shrimp from marinade, lightly pat dry with paper towel and season lightly with salt. Heat the shrimp in a heavy cast iron skillet, until hot and opaque throughout. Let cool and serve at room temperature on large garnished serving platter.

Snacks and Desserts

Nuts

Nuts are a wonderful low glycemic index snack - their high fat content does not raise blood sugar and they are a source of essential fatty acids. Nuts are energy dense and nutrient dense - 160 calories per ounce. A study done by the Center for Health Research at Loma Linda University, California showed that adding 250 calories of nuts (36 almonds) to an individual's diet for six months did not lead to a weight gain.

Eating a handful of nuts each day without cutting back on carbohydrates and other sources of fat will translate into a 10 to 20 pound weight gain over a year.

Giving up potato chips and candy will make nut consumption safe in terms of simple calorie exchange. If a person is still in the clutches of pica or the munchies, and will eat until the container is empty, portion control can be used - measure out the daily limit of nuts into small zip-lock bags.

Even the FDA allows, "Scientific evidence suggests, but does not prove, that eating 1.5 ounces per day of most nuts, as part of a diet low in saturated fat and cholesterol, may reduce the risk of heart disease."

Dr. Walter C. Willett, author of *The Harvard Medical School Guide to Healthy Eating*, writes, "Contrary to popular opinion, nuts are not junk food. They are actually a great source of protein and other nutritional goodies. One of the more surprising findings from nutrition research over the past decade is that people who regularly eat nuts are less likely to have heart attacks or die from heart disease than those who rarely eat them."

Walnuts: English walnuts are a good source of iron, phosphorus, zinc, potassium, and vitamin E; they are the only nut with significant levels of omega-3 fatty acids. One ounce (14 halves) contains 190 calories and 18 grams of fat.

Almonds: are rich in vitamin E and calcium (6 percent of the USDA recommended daily value). One ounce (24 almonds) contains 160 calories and 14 grams of fat.

Hazelnuts (Filberts): 24 nuts provide twenty percent of the USDA recommended daily value of vitamin E and ninety percent of the manganese. One ounce (20 nuts) contain 180 calories and 17 grams of fat.

Brazil nuts: can concentrate the trace mineral selenium if it is in the soil in which they are growing. One ounce (6 - 8 nuts) contains 190 calories and 19 grams of fat.

Pistachios: a handful of pistachios gives twenty five percent of the USDA recommended daily value for vitamin B6. One ounce (50 dry roasted nuts) contains 160 calories and 13 grams of fat.

Pecans: contain a relatively high percentage of fat, however, it is "good fat" (monounsaturated fat) - one ounce (20 halves) contains 200 calories and 20 grams of fat.

Macadamias: are "butter smooth" and contain the highest level of fat (including saturated fat) of all nuts. One ounce (10 to 20 roasted nuts) contains 200 calories and 22 grams of fat.

Cashews: a handful contains 20 percent of the USDA recommended daily allowance of magnesium, 30 percent of copper, and 10 percent of iron. One ounce (18 roasted nuts) contains 160 calories and 13 grams of fat.

Pan-Roasted Coriander and Chili Almonds

1/2 Tblsp almond oil

1 1/2 cups whole natural almonds

1 - 3 small dried red chilli peppers, seeded and finely Diced

1 tsp coriander seeds. Crushed

1/2 tsp Splenda

1/4 tsp Kosher salt

Gently heat almond oil in a large skillet. Add almonds; cook and stir until roasted and fragrant. Stir in chili, coriander, Splenda and salt. Serve warm or at room temperature.

To bring out the maximum flavor of nuts, toast in a preheated oven at 350 degrees F for ten to fifteen minutes.

Dark Berries

Dark berries are a great source of antioxidants, taste and fun. Freezing allows you to buy them during harvest season and have them in recipes throughout the year. Strawberries, raspberries, blackberries and blueberries can be used as great tasting and healthful additions to whey shakes, eaten as a berry salad before or after a

meal, they can be used as toppings for no sugar added ice cream or just served with heavy cream as a desert.

The texture of frozen berries is considerably different than that of fresh berries. The natural water in the frozen berries forms ice crystals that cut through the cell walls. When the frozen berries are thawed, juices flow out of the damaged cell walls and the character of the berry turns from firm to soft. Blueberries tend to maintain their shape and firmness better than other berries. Berries that are handled properly and frozen in rigid containers or freezer friendly zip-loc plastic bags can maintain their taste and eating quality for a year at 10 degrees F or below.

An important part of the freezing process for berries is the preparation:

Strawberries - put just 3 or 4 strawberries at a time in a quart container of water, swirl briefly to wash off sand and other debris and gently remove - do not allow them to soak. Using a sharp knife, remove stem and leaves and halve or keep whole. There are two good methods for freezing strawberries.

1. For whole, unsweetened berries, remove berries from the wash water, place on several layers of paper towels and gently pat dry. Line a cookie sheet with plastic wrap or wax paper, then place the berries in a single layer and cover with a paper towel. Place in the freezer. When the berries are frozen hard, repack them in freezer containers or plastic freezer bags - remove as much air as possible without scuffing the berries and return to the freezer.

2. Add a small quantity of pectin and sugar to enhance the retention of shape and taste. Wash (using above method) 1 gallon of whole berries. Remove stems and leaves and halve berries. Mix together 1/2 cup granulated sugar, 1/2 cup Sucralose and 2 Tblsp of powdered pectin. Stir gently into berries; let stand for ten minutes to allow the dry ingredients to dissolve and juices to be produced. Place in freezer containers or plastic freezer bags - leave 1 inch of space per container. This mix will make 6 pints of sweetened frozen berries.

Caneberries (red raspberries, black raspberries, boysenberries, lo-

ganberries and Marionberries) - place berries in a colander and lightly rinse. Drain well and place in a single layer on cookie sheets lined with plastic wrap or wax paper. Cover with paper towels. Place in the freezer until firm (1 hour). Remove berries and place in freezer containers or freezer bags. Caneberries freeze well without added sugar, however, if you wish you can add 1/8 to 1/4 cup of Sucralose per quart of berries.

Blueberries - choose blueberries that are a deep blue color and covered with a fine white powdery dust ("bloom") on the skin. Put blueberries in a colander and rinse carefully (washing after thawing is not recommended). Drain well and place in freezer containers or plastic freezer bags and place in the freezer.

Grilled Nectarines with Blackberries and Ice Cream

2 pints blackberries

1 tsp fresh lemon juice; more to taste

2 Tblsp unsalted butter, melted

2 Tblsp honey

1/2 tsp ground cinnamon

4 large nectarines

Vanilla ice cream (no sugar added)

Crisp butter cookies

Fresh mint sprigs

In a blender, mix 1 pint of blackberries with 1/3 cup of water and the lemon juice. Puree until smooth. Pass the puree through a fine sieve, pressing down with the back of a ladle. Adjust the flavor with more lemon juice or a little Splenda to taste. Reserve this coulis.

Heat the grill to medium high. In a shallow bowl or lipped plate, stir together the melted butter, honey and cinnamon. Cut each nectarine in half and remove the pit. Slice a bit off of the skin side so that the fruit will lay flat when served. Roll each half in the melted butter mix. Grill, pit side down, until the fruit is softened and the pit side begins to darken and caramelize (about 4 minutes).

Put a nectarine half (or two halves if they are small) pit side up in the center of each plate. Drizzle some of the blackberry coulis around

the fruit and sprinkle with the blackberries. Top each nectarine while still warm with a scoop of the no sugar added vanilla ice cream. Garnish with a crisp butter cookie and sprig of mint - serve before ice cream melts.

Creamy Mixed Berry Parfait

6 oz cream cheese, softened

1/3 cup Splenda Lite, sifted

1 cup heavy cream, chilled

Pinch table salt

1 1/2 Tblsp fresh lemon juice

1 tsp finely grated lemon zest

1 Tblsp finely chopped fresh mint; plus 4 sprigs for garnish

4 cups fresh or frozen berries (1 1/3 cups each of raspberries, strawberries and blueberries).

Combine the softened cream cheese, Splenda Lite and salt in a medium bowl. Using a hand-held electric mixer, beat on low speed until the cream cheese is smooth and the Splenda is incorporated. Add the heavy cream and beat on low speed until the mixture is smooth, stopping to scrape down the sides of the bowl as needed. Increase the speed to medium high and beat until the cream is billowy and holds medium firm peaks (30 to 60 seconds). Add the lemon juice, zest and the mint. Stir briefly to blend. Line up four parfait glasses or stemmed wineglasses. Beginning and ending with the berries, evenly layer all the berries and all but 2/3 cup of the cream into the glasses. Top with a dollop of the remaining cream and garnish with a mint sprig.

Blackberry Breeze

1/4 cup frozen unsweetened blackberries

1/4 cup fat free blackberry yogurt

1/2 tsp cinnamon

2 tsp Splenda

3/4 cup water

1 cup ice cubes

In a blender, combine the frozen blackberries, yogurt, cinnamon,

Splenda and water. Blend for thirty seconds, until smooth. Add the ice cubes and blend for thirty seconds more, until thick.

Mueslix and Berries

- 1 cup low-fat vanilla yogurt
- 1/4 cup old fashioned rolled oats
- 1 apple, peeled, cored and coarsely chopped
- 1/2 cup strawberries
- 1/2 cup blueberries
- 5 whole almonds, coarsely chopped

In a large bowl, combine the yogurt, oats, apple, strawberries, blueberries and almonds; mix well until blended. Serve immediately, or store in fridge for up to six hours.

Whey/soy sugar free protein shakes or pudding

Key lime pie - mix one serving of vanilla Nature's Whey shake according to directions.

Add 2 Tblsp frozen lime juice

1 Graham cracker

3 ice cubes

Blend on high speed for 45 seconds.

Egg nog - mix one serving of vanilla Nature's Whey shake according to directions.

Add 1/4 tsp ground allspice

1 lightly poached egg

3 ice cubes

Blend on high speed for 45 seconds.

Cinnamon Roll Supreme - mix one serving of vanilla Nature's Whey shake according to directions.

Add 1/4 tsp ground cinnamon

1 tsp softened butter

3 ice cubes

Blend on high speed for 45 seconds.

Strawberry cheese cake - mix one serving of vanilla

Nature's Whey shake according to directions.

Add 3 Tblsp of Jello's no-bake, reduced fat cheese cake mix

3 low-fat vanilla wafers

3 - 5 strawberries

3 ice cubes

Blend on high speed for 45 seconds

Chocolate peanut butter cup - mix one serving of chocolate

Nature's Whey shake according to directions.

1 heaping Tblsp of peanut butter

3 ice cubes

Blend on high speed for 45 seconds

Double deluxe chocolate fudge - mix one serving of

chocolate Nature's Whey shake according to directions.

1 packet of sugar-free, fat free hot cocoa mix

3 ice cubes

Blend on high speed for 45 seconds.

Cookies and cream - mix one serving of chocolate

Nature's Whey shake according to directions.

Add 4 drops of peppermint extract

3 ice cubes

Blend on high speed for 45 seconds.

Then add 4 low-fat chocolate wafers and blend at low speed for 10 seconds.

Chocolate mocha mint - mix one serving of chocolate

Nature's Whey shake according to directions.

Add 1 1/2 heaping Tblsp of General Foods International Coffees Swiss Mocha sugar free, fat free instant coffee.

4 drops of peppermint extract

3 ice cubes

Blend on high speed for 45 seconds

Banana split - mix one serving of chocolate Nature's Whey shake according to directions.

1 banana

1 oz Osteo-fx

1/4 cup fresh pineapple, chopped, drained

3 - 5 strawberries

3 ice cubes

Blend on high speed for 45 seconds.

Salads and Steamed Vegetables

Broccoli

Broccoli, carrots, cauliflower

Green beans, slivered almonds, pearl onions

Peas, carrots (1/4 inch cubes)

Peas, pearl onions

Mixed bell peppers (red, green, yellow, orange)

Beets, sliced onions

Okra, cooked tomato slices, pearl onions

Lima beans, pearl onions

Tomatoes, sliced and sliced onions

Romaine lettuce

Shredded provolone cheese

Mixed spring greens (miner's lettuce, bib lettuce, red cabbage, dandelion leaves and flowers, etc.).

1/8 to 1/4 head iceberg lettuce, tomato wedges, radishes, bell pepper rings.

Greek salad

8 Tblsp extra-virgin olive oil

1 cup loosely packed fresh basil leaves, plus 6 sprigs

1/4 tsp salt, plus more to taste

1/2 lb chunk Feta cheese

2 14 oz cans stuffed grape leaves

1 medium seedless cucumber, peeled and cut into bite-sized chunks.

- 1 8 oz package cherry or grape tomatoes, halved
- 3/4 cup black olives (Kalamata)
- 1 12 oz jar pepperoncini, drained
- 1 lemon, quartered and seeded
- Kosher salt
- Freshly ground black pepper

To make basil oil, combine the olive oil, 1 cup basil leaves and 1/4 tsp salt in blender. Blend until smooth. Set aside. Divide the Feta, stuffed grape leaves, cucumber, tomatoes, olives, pepperoncini and basil sprigs equally on six plates or bowls. Squeeze on lemon juice; drizzle with the basil oil. Add salt and pepper to taste.

Grilled everything salad

- 2 Tblsp prepared horseradish
- 3 Tblsp red wine vinegar
- 5 Tblsp extra virgin olive oil
- 1 tsp Kosher salt
- 1/2 tsp freshly ground pepper
- 1 10 oz package (2 bunches) fresh spinach, stems removed
- 1 1/2 lbs grilled flank steak medium rare (6 cups thinly sliced)
- 1 large onion, thickly sliced and grilled (3 cups)
- 3 carrots, thinly sliced and grilled (3 cups)
- 3 plum tomatoes, halved and grilled

Vinaigrette: whisk together the horseradish, vinegar, olive oil, salt and pepper in a small bowl. Set aside. Arrange the spinach on a serving platter. Slice the steak into 1/4 inch stripes, arrange them on top the spinach, along with the grilled vegetables. Drizzle with the vinaigrette and serve.

Gouda or Brie cheese wedges, sliced tomatoes

Crab and Avocado salad

- 1 Tblsp heavy cream
- 2 Tblsp buttermilk
- 1/2 tsp lemon zest
- 2 tsp fresh lemon juice; more for sprinkling

1/2 tsp Dijon mustard
Kosher salt and freshly ground black or white pepper
8 oz crab meat (lump) - remove all shell pieces
1/2 cup finely chopped celery (inner ribs and leaves)
3 Tblsp chopped fresh chives (1/2 in pieces)
1 ripe avocado, peeled, pitted - cut into 1/2 in pieces.
1 head Boston or Bibb lettuce (for lining plates)

In a small bowl, whisk the heavy cream, buttermilk, lemon zests, lemon juice and mustard. Season with salt and pepper to taste. In a medium bowl, mix the crab meat, celery and 2 Tblsp of chives. Fold in the dressing gently to avoid breaking up the crab meat chunks. Gently fold in avocado. Taste for salt and pepper. Line four salad plates with lettuce leaves, heap a scoop of the crab salad on each, garnish with the balance of the chives. Sprinkle each salad with 1/2 tsp lemon juice and serve.

Bistro salad with warm goat cheese Dressing

1/2 cup toasted hazelnuts (skinned)
4 oz log fresh goat cheese
1 Tblsp sherry vinegar
3 Tblsp hazelnut oil
Kosher salt
Freshly ground black pepper
4 hands full of small lettuce leaves (3 varieties) - 6 cups
Washed and thoroughly dried

Preheat oven to 400 degrees F and lightly coat a baking sheet with a cooking spray. Roughly chop half of the hazelnuts and finely chop the remaining half. Slice the goat cheese into 4 equal portions. Press the finely chopped hazelnuts into the cheese rounds to coat them on all sides. Set the coated rounds on the baking sheet and bake until heated (6 - 8 minutes). In a small bowl, whisk together the vinegar, oil, a pinch of salt and a sprinkling of freshly ground pepper. Taste and adjust seasonings. When the cheese is ready, toss the lettuce with the dressing until the leaves are evenly coated. Mound the lettuce on individual plates, top with a round of warm goat cheese,

scatter the balance of the hazelnuts over all dishes.

Soups

Vegetable beef or vegetable chicken soup

Dice broccoli, onions, carrots, peas, tomatoes, string beans, celery, okra, etc.). Four ounces of grilled chicken (diced), ground beef or roast beef cut into 1/2 inch cubes. Garlic salt and pepper to taste.

Seven-bean soup (great laxative!) simmer
for at least 2-4 hours

Continue to add water as needed.

2 quarts water

1/2 cup kidney beans

1/2 cup pinto beans

1/2 cup navy beans

1/2 cup lima beans

1/2 cup butter beans

1/2 cup soy beans or chick peas

1/2 green beans

1/2 cup pearl onions

Garlic salt to taste

Diced green tomatoes

Fresh Zucchini soup

For the soup:

12 sprigs fresh cilantro (1 packed cup); more leaves
for garnish

1 small head garlic, cross cut at middle; plus 1 Tblsp
minced garlic.

3 Tblsp unsalted butter

1 large onion, sliced (to yield 1 1/2 cups).

2 tsp Kosher salt; add more to taste

1 Tblsp minced fresh ginger

2 Tblsp minced jalapeno

1/2 tsp cumin seeds

1/2 tsp coriander seeds
 1/8 tsp cardamom seeds or one green pod
 1 lb zucchini (unpeeled), sliced 1/2 in thick
 1/4 tsp fresh lime juice; more to taste

For the raita:

1/2 cup whole-milk yogurt
 1/4 tsp Kosher salt; more to taste
 1 tsp olive oil
 1/2 tsp brown mustard seeds
 1/4 tsp cumin seeds

To make the soup: Put the cilantro, halved garlic head, and 5 cups of water in a large saucepan. Bring to a boil, reduce to a simmer, and cook uncovered for 15 minutes. Strain the broth. Melt the butter in a large heavy saucepan over a medium heat. Add the onion, 2 tsp salt, ginger, jalapeno, cumin seeds, coriander seeds and cardamom. Cover and cook, stirring occasionally, until the onion is translucent, about 10 minutes - don't let it brown. Stir in the Tblsp minced garlic and the zucchini, cover, and cook, stirring occasionally until the squash is softened (about 5 minutes). Add 4 cups of the broth; bring to a simmer; and cook until the zucchini is tender (about an additional 3 minutes). Puree the mixture in a blender and strain through a fine sieve. Add the lime juice; taste and add salt if needed. Serve warm, garnished with 1 - 2 Tblsp raita and the cilantro leaves.

Make the raita: In a small bowl combine the yogurt and salt. Heat the oil in a small skillet. When hot, add the mustard and cumin seeds. When they start to pop, add the oil and the seeds to the yogurt and stir.

Cucumber soup with avocado

1 large (8 oz) white sweet onion, peeled and quartered
 1 lb cucumbers, peeled, seeded, and cut into large chunks
 1 medium ripe avocado
 1 1/2 cups plain whole-milk yogurt
 3 Tblsp extra virgin olive oil
 1 - 2 cloves garlic, minced

3/4 tsp ground cumin (toasted and freshly ground)
 1 tsp Kosher salt; more to taste
 1/3 cup loosely packed fresh basil leaves
 2 Tblsp fresh mint leaves
 2 -3 Tblsp fresh lemon juice
 Freshly ground black pepper

Put the onion chunks in the food processor, pulse to chop finely. Scoop into a sieve, rinse under cold water, and set aside to drain. Put the cucumber chunks in the food processor, pulse to chop finely. Add the drained onion and pulse to combine. Set aside 3/4 cup of the cucumber onion mix to stir into the soup at the end. Cut the avocado in half and remove the pit. Scoop out the pulp and drop it into the food processor with the remaining cucumber onion mixture. Add the yogurt, olive oil, garlic, cumin and salt. Puree until smooth. Using a sharp knife, slice the basil and mint leaves into shreds and then stir the into the soup along with the lemon juice, 1 cup cold water and the reserved cucumber onion mixture. Season to taste with salt and pepper. Cover and chill for at least a half hour (several hours are preferred). Before serving, check for consistency, gradually adding another 1/2 to 1 cup cold water to thin the soup if necessary. Add more lemon juice and salt if needed and serve.

Portuguese Canja: Chicken soup

6 cups homemade or low salt chicken broth
 1/2 tsp Kosher salt; more to taste
 1/2 cup long-grain rice (basmati)
 1 boneless, skinless whole chicken breast, cut into
 1 x 1/2 -inch strips or 1/2 in dice.
 1/4 cup fresh lemon juice
 Fresh ground black pepper to taste
 6 Tblsp chopped fresh mint

Combine 1/2 cup of the broth, 1/2 cup water, and the salt in a small, heavy saucepan. Bring to a boil and stir in the rice. Reduce the heat to low, cover the pan and simmer until all the liquid is absorbed and the rice is just tender (20 minutes); set aside. In a large, heavy

saucepan, bring the remaining 5 1/2 cups broth to a simmer over a minimum heat. Add the chicken, reduce the heat to a gentle simmer, and poach until the chicken is firm and just cooked through (3 - 5 minutes). If the broth is cloudy, you can remove the chicken, strain the liquid through a cheesecloth - lined strainer, and return the chicken to the broth. Add the reserved rice and the lemon juice and season with salt and pepper to taste. Sprinkle on the mint and serve immediately.

Tomato bisque

- 2 Tblsp olive oil
- 1 small onion diced
- 1 clove garlic, minced
- 3 large sprigs fresh thyme
- 1 can (28 oz) crushed tomatoes in puree
- 1 1/2 cups homemade or low salt chicken broth
- 3 Tblsp honey
- 1 1/2 tsp Kosher salt; more to taste
- 1/4 tsp finely ground pepper; more to taste
- 1/3 cup heavy cream
- 2 Tblsp chopped fresh flat leaf parsley

In a medium pot, heat the oil. Add the onion and cook over medium heat, stirring frequently, until tender and lightly browned on the edges (approximately 7 minutes). Add the garlic and thyme; stir until fragrant (1 minute). Add the tomatoes, broth, honey, salt and pepper. Bring to a boil over high heat. Reduce the heat and simmer, stirring frequently until reduced by a quarter and thickened (approximately 15 minutes). Using a stand or immersion blender, puree about half the soup; it will be chunky and thick. Return all the soup to the pot and stir in the cream. Heat gently and adjust the seasoning. Ladle the soup into bowls, sprinkle with the parsley, and serve the soup immediately.

Chilled sugar snap pea soup

- 2 Tblsp butter
- 1 cup chopped onion (1 medium onion)

- 1 1/2 lbs sugar snap peas, washed and cut into 5 cups of 1/2 in pieces.
- 1 tsp Sucralose
- 1/2 tsp Kosher salt
- 1 cup half and half
- 2 Tblsp chopped fresh chives

Melt the butter in a large skillet over medium heat. Add the onion and cook 5 minutes. Add the snap peas and cook, stirring occasionally, until bright green and crisp-tender, about 5 minutes. Mix in the Sucralose and salt and cook 1 minute more. Stir the half and half with 3 cups of water in a large measuring cup. Place half the snap pea mixture in a blender or food processor. Pour in half of the water mixture and puree until smooth (30 - 60 seconds). Repeat with the remaining ingredients. Pour the soup through a strainer and chill for at least 4 hours. Sprinkle with the chives just before serving - makes 6 bowls.

Sauces

Tomato Marmalade

- 1 cup Splenda Lite sugar substitute
- 3 cups red wine vinegar
- 20 tomatoes, peeled, seeded, and finely diced
- 1 Tblsp cumin seed
- 4 - 8 jalapenos, seeded and finely diced
- 20 black peppercorns
- Zest and juice of 2 large oranges
- Zest and juice of 2 lemons

1. In a large saucepan, combine the sugar and vinegar and simmer over medium heat for 5 to 10 minutes, until the mixture reaches a syrup texture.

2. Stir in the tomatoes, jalapenos, cumin seed, peppercorns, orange and lemon zest and juices. Cook over low heat, stirring frequently, until marmalade reaches a thick consistency, 20 to 25 minutes, depending on the juiciness of the tomatoes.

3. Pour into a jar or other container and store in the refrigerator. Serve cold. Can be used on breakfast toast or bagel or on a hamburger.

Picasso Blue Salsa

- 2 cups fresh blueberries, divided
- 1 - 2 serrano peppers, chopped (with seeds)
- 2 Tblsp creamed hot pepper honey
- 1/2 tsp grated ginger root
- Juice of 1/2 lime
- 1/2 tsp salt
- 2 Tblsp minced purple onion
- 4 - 6 fresh mint leaves, coarsely chopped

1. Place 1 cup blueberries, serranos, honey, ginger, lime juice and salt in food processor.
2. Process about 20 seconds until pureed.
3. Add remaining blueberries, onion and mint leaves, then pulse several times. Mixture should be slightly chunky.
4. Refrigerate for at least one hour before serving.

This salsa is good on salmon, smoked turkey and pork tenderloin.

Rubs

As the name implies, rubs are rubbed into the food with your fingers prior to cooking to impart an "extra measure of flavor."

Cajun Rub

This rub is especially useful for pork, chicken and seafood. In a small bowl, combine 1 1/2 tsp ground white pepper, 1 1/2 tsp black pepper; 1 tsp cayenne pepper; 1 tsp dried thyme, crushed; 1 tsp onion powder; 1 tsp garlic powder; and 1 tsp salt.

To use, sprinkle rub evenly over pork, chicken or fish; rub in with your fingers - grill or broil until done. Can store in a tight container for up to three months. Mixture makes seven tsp, enough for

about 3 1/2 pounds of meat poultry or fish.

Lemon-Herb Rub

This zesty lemon rub will add a zippy accent to pork or beef. In a small bowl, combine 8 cloves garlic, minced; 1 1/2 tsp fine lemon zest; 1 tsp dried rosemary, crushed; 1/2 tsp salt; 1/2 tsp dried sage, crushed; and 1/2 tsp ground black pepper.

To use, sprinkle mix evenly over beef or pork; rub in deeply with fingers. Grill or broil until done (medium rare). Makes 2 Tblsp of mix - enough for two to three pounds of beef or pork.

Jamaican Jerk Rub

Jamaican jerk (barbecue) can be wet, dry, sauced or rubbed, and all of these styles have their fans. In a small bowl, combine 2 tsp Splenda; 1 1/2 tsp onion powder; 1 1/2 tsp dried thyme, crushed; 1 tsp ground allspice; 1 tsp ground black pepper; 1 tsp cayenne pepper; 1/2 tsp salt; 1/4 tsp ground nutmeg; and 1/8 tsp ground cloves.

To use, sprinkle mix evenly over meat or chicken; rub in with fingers. Grill or broil meat (medium rare) and poultry (medium) until done. Make 8 tsp or enough for four pounds of meat or chicken.

Chili Rub

Chili rub can be used effectively for beef, pork and poultry. In a small bowl, combine 1 Tblsp paprika; 1 tsp ground cumin; 1 tsp chili powder; 1/2 tsp salt; 1/2 tsp ground coriander; 1/2 tsp ground black pepper; and 1/2 tsp ground cayenne pepper.

To use, sprinkle mix evenly over beef, pork or chicken until done. Make six tsp, enough for three pounds of beef, pork or chicken.

Water

Water is essential for life, and we are made up of sixty to seventy percent water. Good water is essential to any successful weight loss and weight control program. Eight to ten glasses of water each day will help the absorption of mineral nutrients (which helps to eliminate pica), water helps to flush out the dangerous byproducts of "burn-

ing” fat and water gives the sensation of fullness.

Distilled water is 100 percent water with absolutely no contaminants but it tastes “flat” and over time will pull minerals from your body. A good compromise is to use distilled or double filtered water that has been “enriched” with liquid colloidal minerals at the rate of one ounce of the liquid colloidal minerals (19,000 mg of minerals per quart) per gallon of distilled or double filtered water.

We get lots of questions about water, so we looked for and found the opinions of professional wine tasters for different brands of bottled water. A panel comprised of Rod Smith, wine columnist for the *Los Angeles Times*; Peg Melnick, wine writer for the Santa Rosa Press Democrat; Jeff Cox, contributing editor for *The Wine News* (and *Men's Health*); Bill Traverso, wine buyer and judge for many prestigious national wine competitions; and Dan Kosta, wine maker at the Kosta Browne Winery in Sonoma County, California.

The waters were graded on a 1 - 20 scale (20 being the best of the best):

Fiji Natural Artesian Water - Bottled at the source on the main island of Viti Levu in Fiji, this slightly alkaline water contains dissolved calcium, magnesium and colloidal silica. **Score:** 19. **Comments:** “Tastes like spring water.” “Fat, round with a hint of flowers.” “Clean.” “Sweet and pleasant.”

Dasani Purified Water - Made by the Coca Cola Company from filtered municipal water, which is then “spiked” with minerals. **Score:** 17. **Comments:** “The cleanest tasting water in the bunch.” “Dry, clean - even crisp.” “It’s smooth and pure tasting.”

Ice Age Premium Glacier Water - Glacial meltwater collected at Alpine Creek in Toba Inlet, British Columbia. **Score:** 16. **Comments:** “Fresh and dry.” “Big in the mouth, but not dense.” “Dry.”

Crystal Geyser Natural Alpine Spring Water - Bottled at the (non-erupting) source springs in three mountain ranges - the Sierra Nevada, Cascades and Blue Ridge. **Score:** 16. **Comments:** “It has a pleasant, fat taste.” “Clean and dry with just a hint of minerals.”

Aqua Mountain - bottled under the authority of the 7Up Bottling of

San Francisco, is purified tap water. **Score:** 14. **Comments:** "It tastes like the plastic bottle that it comes in." "A slight creaminess and full body." "Plastic flavor."

Alhambra Drinking Water - The water from Mt. Shasta in northern California, is subjected to microfiltration and ultraviolet light "for a fresh, clean taste," according to the label. (exposure to ultraviolet light is for a germicidal effect). **Score:** 13. **Comments:** "Seems like tap water." "A neutral, mineral taste." "Plump water, round and soft."

Evian Natural Spring Water - Bottled in the French Alps at Cachat Spring in Evian, France. The label says, "filtered through the mountains," whatever that means. **Score:** 12. **Comments:** "Tastes like water." "Doesn't taste softened."

Glaceau Smartwater - The name "Glaceau" suggests it comes from a glacier, but no such luck. It's distilled water with calcium chloride, magnesium chloride and potassium bicarbonate added. **Score:** 11. **Comments:** "Chlorine on the nose and palate." "Kind of dull."

Aquafina Purified Drinking Water - The label promises "nothing but crisp, clean refreshment." The panel thought otherwise. **Score:** 10. **Comments:** "Bitter edge, tooth-gritting."

Tap Water - Only Dan Kosta liked it and thought it was "minerally."

Alcohol

A report released during the summer of 2004 by the National Institute on Alcohol Abuse and Alcoholism concluded that, in general, a single alcoholic drink each day can be beneficial. Wine may protect a sixty five year old man from cardiovascular disease, but may increase the risk of a car accident in a twenty two year old.

Two drinks per day can increase the risk of liver cirrhosis, pancreatitis and colon cancer.

One or two drinks per day can reduce the risk of a heart attack by thirty to forty percent about the same as statin drugs. Alcohol is thought to protect against gall stones.

According to a study discussed in the report, "If all current drinkers

were to quite drinking, the USA would suffer from an additional 80,000 cardiovascular deaths per years." There is a considerable body of evidence which shows that the "benefits of alcohol" are really the benefits of the phytonutrients in the alcoholic drink rather than the alcohol itself.

The USDA suggests that moderate alcohol consumption, one drink per day for women and up to two per day for men - even these suggested limits might be too high when one is planning on driving a car or operating machinery. Three drinks in an hour can be extremely dangerous while three drinks per day may be alright. The amount of alcohol in one twelve oz beer equals the alcohol in four ounces of wine and in one ounce of alcohol in spirits.

Extended or extensive drinking or binge drinking increases the risk of dementia and Alzheimer's disease. Low levels of alcohol consumption can decrease the risk of senile or vascular dementia caused by blood clots and thrombosis type strokes.

Approximately five percent of American drinkers become addicted to alcohol or abuse alcohol. Alcoholism affects 16 million Americans and only 1.5 million seek therapy or treatment.

A single drink can impair judgment, sexual function and sleep patterns.

Research suggests that increased levels of alcohol consumption increases the risk for mouth, pharynx, larynx and esophageal cancer. Some studies indicate that this risk of cancer goes up twenty six percent, while other studies indicate an increase of as much as eighty three percent.

Alcohol is known to interact in a negative way with many pharmaceuticals.

One large study indicates that the lifetime risk of breast cancer is 8.8 percent in nondrinkers and 10.1 percent in those women who consume two drinks per day - research also shows that folic acid supplementation can reduce the risk of breast cancer for nondrinkers and drinkers alike.

Alcohol drinking in women who are breast feeding can reduce milk production and impairs a baby's motor development. Pregnant women should not drink at all to prevent the fetal alcohol syndrome,

where emotional, behavioral and intellectual development are at risk.

Wine

The Chinese were consuming fermented beverages, including wine, as far back as 7,000 B.C. according to an anthropological discovery reported in the December 2004 issue of *Proceedings of the National Academy of Science*. "Fermented beverages are central to a lot of our religions, social relations, medicine, in many cultures around the world," said Patrick E. McGovern, of the University of Pennsylvania Museum of Archaeology and Anthropology.

These alcoholic drinks "have played key roles in the development of human culture and technology, contributing to the advance and intensification of agriculture, horticulture and food-processing technologies," he reported.

McGovern's team collected pieces of sixteen pottery vessels from Jiahu, a Stone Age village in China's Henan Province. It's the same site where archaeologists have found the earliest evidence of musical instruments, including a flute. The ceramic vessels were dated back to 7,000 B.C., and the chemical residue found in the vessels were analyzed - the results of the analysis revealed that, "they matched residues from modern rice and rice wine, grape wine, grape tannins and ancient and modern herbs including hawthorn berries."

"The most straightforward interpretation of these data is that the Jiahu vessels contained a consistently processed beverage made from rice, honey and fruit," the team concluded.

The team also reported on an analysis of a 3,000 year old liquid found in sealed bronze vessels from the Chinese city of Anyang. These vessels contained rice and millet wines, flavored with herbs and flower essential oils.

"It's similar to the rice wine that's produced in the area southwest of Shanghai" today, McGovern said. "I have never tasted the ancient samples, I've only smelled it, but it does remind you of these Shaoxing rice wines with the floral aromatic scents," he said.

Prior to these 2004 findings, the earliest known chemical evidence of wine, dating to about 3,500 B.C., came from a 1990 discov-

ery in Godin Tepe in the Zagros Mountains of western Iran. That was followed two years later with the earliest chemically confirmed barley beer, in another vessel from the same room at Godin.

In 1994, chemical testing confirmed resinated wine inside two jars excavated by a University of Pennsylvania archaeological team at the Neolithic site of Hajji Firuz Tepe, Iran, dating to about 5,400 B.C.

Prior to the discovery of wine in China dating back to 7,000 B.C., evidence for the growth of viticulture or organized commercial wine making is connected to the appearance of European villages, towns and cities - particularly in the Mediterranean region. It is thought that nomadic tribes from Europe were fermenting wild grapes and berries into wine and wine-like beverages as far back as 4,000 - 5,000 B.C.

The establishment of vineyards from the 4th millennium B.C. in Egypt and Mesopotamia and in the Aegean in 2500 B.C. followed the appearance of crude vineyards in India as far back as the 6th millennium B.C. The oldest tools and jugs known to be used for wine making are dated to the 5th and 6th millennium B.C. in Georgia and Armenia and in the 4th millennium in Iran.

The establishment of agriculture and the appearance of settlements were soon followed by the cultivation of grape vines, olives and figs.

It is said that, "wine is made up of a thousand substances." Many of these phytochemicals and phytonutrients in wine have not yet been analyzed. The greater number of these phytonutrients found in wine, including vitamins and minerals come from the grapes, the makings of the wine, however, the balance of the wine's content such as ethanol and glycerin are byproducts of the wine making fermentation. Some substances that begin in wine, such as sugar and vitamin C are either partially or completely removed by the fermentation.

Water makes up seventy five to ninety percent of the wine volume - the fifteen percent difference is made up of varying amounts of tannic acid, organic acids, mineral salts and pectins, which form the wine's extract.

The second largest constituent of wine is ethyl alcohol. Alcohol

levels will vary based on the type of wine and are typically listed on the label as a percentage of by volume (% volume). A wine with an alcohol level of 11% volume will be light and fun - wine with an alcohol content of 13% volume can produce rapid intoxication. The determining factor will be the structure of the wine and the ratios of the different constituents including plant acids, residual sugar, alcohol, tannins and color compounds.

The sugar levels of dry wine is typically less than 0.25 ounces per gallon (2 grams per liter), while in a botrytized sweet wine it can reach almost 67 ounces per gallon (500 grams per liter).

Over time, wine became known as a medicinal drink and not just a luxury. In many countries wine is a basic drink that is included in all meals, usually mixed with water to satisfy thirst. Old publications called wine the "most hygienic drink," and it was often safer to drink wine than it was to drink the water.

Serge Renaud, a French scientist observed that, "there was a 'French paradox' - while the French ate as much or more fat than Americans, the French had a 2.5 times lower rate of deaths from cardiovascular disease than Americans." The study demonstrated that the higher a country's wine consumption the lower was its mortality rate from cardiovascular disease.

For those who would like to have the health benefits of wine without the alcohol intake, there are juice and juice blends that contain the same phenolics and antioxidants as wine:

Phytonutrients found in Wine and Juice Blends (G/l)

Phytonutrient	Must	Wine
Polyphenols		
Flavonoids	Trace	0.0 - 0.05
Anthocyanins	0.004 - 0.90	0.0 - 0.50
Xanthones		
Proanthocyanins		
Tannins	0.1 - 1.50	0.1 - 5.0

Mineral Content of Wine (G/l)

Mineral	Must	Wine
Potassium	1.0 - 2.50	0.7 - 1.50
Calcium	0.04 - 0.25	0.01 - 0.20
Magnesium	0.05 - 0.20	0.05 - 0.20
Sodium	0.002 - 0.25	0.002 - 0.25
Iron	0.002 - 0.005	0.002 - 0.02
Phosphorus	0.08 - 0.50	0.03 - 0.90
Manganese	0.0 - 0.05	0.0 - 0.05

Vitamin Content of Wine (mg/l)

Vitamin	Must	Wine
Ascorbic Acid (vitamin C)	38.0 - 95.0	0.0
Thiamine (Vitamin B1)	0.10 - 0.50	0.04 - 0.05
Riboflavin (Vitamin B2)	0.003 - 0.08	0.008 - 0.30
Pantothenic Acid (Vitamin B5)	0.5 - 1.0	0.4 - 1.20
Pyridoxine (Vitamin B6)	0.3 - 0.50	0.2 - 0.50

Beer

Beer is often referred to as “liquid bread.” Beer contains many nutrients including significant levels of silica, a mineral required for the optimal health of bones, skin, hair and nails.

“Anti-ageing Ale”

A dark German beer, tasting like a malt beer contains 4.8 percent alcohol and is brewed with mineral spring water, algae, iron, vitamin A, vitamin D and flavonoids. Regular beer contains twice the antioxi-

dants found in white wine, and the anti-ageing beer has ten times the antioxidants as the regular beer.

Low Carb Beer

Brand	Serving	Calories	Total Carbs	Net Carbs	Fat
Michelob Ultra	12 oz	95	3	3	0
Miller Lite	12 oz	96	3	3	0
Rolling Rock green Light	12 oz	84	2	2	0

Eat well, enjoy the shopping, acquisition, preparation, cooking, the social aspects of food and the actual eating of food, however, always remember that you can't get everything your body needs from the food you consume. The only way you can guarantee that you have all 90 essential nutrients on a daily basis is to supplement with all 90 essential nutrients each day - yes, each and every day.

It is hard work to remember to supplement with all 90 essential nutrients each day, however, people do remember to make their coffee or tea every morning - the rewards for remembering to take all 90 essential nutrients each day and developing your supplement program as part of a "lifestyle" are worth more than 100 years of healthy living. Believe us - it's worth it!



Chapter 19

"There are more old drunkards than there are old doctors."

- Benjamin Franklin

"Longevity is only desirable if it increases the duration of youth, and not that of old age. The lengthening of the senescent period would be a calamity."

- Alexis Carrel

Man the Unknown (1935)

"Every man desires to live long, but no man would be old."

- Jonathan Swift

"It's not the years in your life, it's the life in your years."

- Abraham Lincoln

Longevity

The vast majority of humans that have ever been born died before they were ten years of age, primarily from nutritional deficiency diseases, exposure and trauma. Over the millennia, we humans have increased our life spans to seventy five years of age, give or take a few years. This apparent increase in longevity did not happen because of medical miracles, but rather, the increase in longevity happened because of public works, including sewers, clean water and depend-

able sources of energy and food. Contrary to claims by the medical profession, medicine has not added anything to human maximum life spans - no, not even a millisecond!

Experts, including geneticists and gerontologists, agree that all humans, regardless of race, religion, gender or creed, ~~humans~~ do not have a maximum life span and all have the genetic capacity and the genetic potential to live well beyond 100 years of age. The reason that we all don't live to be 100, is not because we can't, but rather, we simply do too many of the bad things and not enough of the good things.

Aubrey de Grey, a biogerontologist at the University of Cambridge in England, says, "that, under the right circumstances, humans born in the 22nd century could live up to 5,000 years." De Grey, who proposes that we use all the technology available to us to develop a "true cure for aging," is agreeably at the extreme optimistic end of the scientific age curve.

"I think people will someday live substantially longer than today," said Steven Austad, a biologist at the University of Texas Health Sciences Center in San Antonio. "Living into your 100s will be fairly routine, and up to 150 for the outlier (a longer lived person who is the exception to the rule). I think this (is possible) because we have been so successful at figuring out how to get animals to live longer."

People do not die from obesity as most "experts" would have you believe - people die from the things that cause obesity - mineral deficiencies. Specific mineral deficiencies are the cause of obesity, type II diabetes and the Metabolic Syndrome - these same mineral deficiencies either directly or indirectly are the cause of Alzheimer's disease, macular degeneration, cataracts, cardiovascular disease, cancer, osteoporosis, arthritis, lupus, fibromyalgia, etc.

Dr. Dilip V. Jeste, director of the Stein Institute for Research on Aging at UCSD says, "The prevention of hypertension, diabetes and obesity may help increase life span significantly."

The Honolulu Heart Program began in the mid-1960s, for three decades it evaluated 6,505 men between the ages of 45 and 68 years who were in "good health" when the program began. Of the men who survived to the ages of 71 to 95, forty percent remained free

Maximum Life Spans (Years)

Giant tortoise	177
Human	122
Macaw	64
Horse	62
Indian elephant	57
Alligator	56
Sturgeon	50
Chimpanzee	37
Pigeon	35
Dog	34
Buprestid beetle	30
Medicinal leech	27
Salamander	25
Cat	21
Bird spider	15
Earthworm	10
Mouse	3

from both physical and cognitive impairment.

The most reliable predictors of successful aging were low blood pressure, low blood sugar, being tobacco free and not being obese. All of the "predictors" are maintained in the normal values by proper supplementation of all of the known essential nutrients with a special focus on the 60 essential minerals - remember since we gave up wood as a cooking and heating fuel, the ash byproduct that had sustained humans for thousands of years as a source of dietary minerals is gone - we must, without exception, supplement with a new source of plant minerals to make up the difference.

The causes of obesity (deficiencies of essential minerals) are also the causes of a shortened life span - without a steady stream of dietary plant minerals, humans become overweight, obese and suffer

from numerous afflictions that make our lives shorter and more miserable:

The Leading Causes of Adult Death in America

Men	Women
1. Heart disease	Heart disease
2. Cancer	Cancer
3. Accidents	Stroke
4. Stroke	COPD (chronic obstructive Lung disease)
5. COPD	Diabetes
6. Diabetes	Alzheimer's disease
7. Pneumonia and influenza	Accidents
8. Suicide	Pneumonia and influenza
9. Kidney disease	Kidney disease
10. Liver disease	Septicemia (blood infections)

In all cultures and races of humans, women live longer than men, however, biology alone can't account for the longer lifespan of women. Females tend to eat better than males in terms of eating more vegetables (antioxidants) than men, females tend to nibble all day as they prepare food and cook, while men tend to gorge on one, two or three meals per day, women tend to smoke fewer cigarettes, consume less alcohol and their place in a culture or society tends to be less dangerous (weaving, gathering, cooking, child rearing, office work, etc.) than the pursuits of men (hunting, warfare, farming, truck drivers, shepherds, miners, mechanics, lumber jacks, etc.).

A 1997 survey by the USDA, reported that men consume an average of 96 grams of fat each day, which makes up forty four percent of their daily calorie intake. Women, on the other hand get thirty two percent of their calories from fat. By the year 2000, American men had dropped their dietary fat intake to thirty three percent, however, their vegetable intake (dietary antioxidant source) was still low.

During the 1990s women lived 0.1 years longer than men in Bangladesh, 0.6 years longer in India, 6.7 years longer in the US, 5.3 years longer in the U.K. and 7.8 years longer in France.

In general, obese people do not live as long as people who more closely adhere to their ideal weight. People who are overweight can be "fit," however, when it comes to longevity being overweight is still a big negative.

With the "party line" professed, there is current information that recommends that seniors forget dieting unless they are morbidly obese:

1. Don't go on a weight loss diet unless you are morbidly obese.
2. Mild exercise increases muscle mass and bone density well into old age.
3. Rapid weight loss (five percent over ninety days) can signal the onset of dental problems, or serious illness (diabetes, cancer, etc.).
4. High protein diets with a serious supplement program are essential for optimal health, prevention of sarcopenia (loss of muscle mass) and a long life. Seniors should beware of extreme heart healthy diets - do not eliminate fat or salt from your diet.
5. Avoid fried foods, margarines, meat cooked well done, burnt animal fats, sugar and carbonated drinks to eliminate direct and indirect sources of free radicals.
6. With diabetes, eat six to eight small meals per day with animal protein, green leafy vegetables, and complex whole grain carbohydrates - "the whiter the bread, the sooner you're dead."
7. Provide more finger foods for seniors, roasted chicken wings and drumsticks, sandwiches, wraps, etc.
8. Serve more of the nutrient dense foods to seniors such as milk, cheese, nuts, ground meat, squash, baked potatoes with butter, no sugar added ice cream several times each day to avoid sarcopenia (loss of muscle mass).

The Human Scorecard for Longevity

The September 2000 issue of *Scientific American* was almost en-

tirely devoted to human longevity and people who lived beyond 100 years of age. The various articles were written by a wide variety of physicians, gerontologists, geneticists, dieticians and nutritionists. A summary of the information reported in the September SA pointed out that the individuals who followed their doctor's advice to the letter, didn't make it to 100 years of age, while people who ate salt, eggs, red meat, butter and chicken skin did.

The special September 2000 human longevity issue of *Scientific American* had a summary, that was, in effect an editorial head scratching. The "experts" couldn't figure out why the people who were classed as "rebellious" patients lived to be 100 more frequently than ones who followed their doctor's advice perfectly.

Obviously, their doctor's advice wasn't very good. To give specific numbers to the lack of success by the American doctors when it comes to longevity we don't have to look very hard. In an April 1990 study, conducted by the World health Organization and the Center for Disease Control, of Atlanta, Georgia a comparison was made between the top thirty two industrialized nations for industrial production, energy consumption, education, per capita income, birth rate, death rate and longevity.

When the study was presented to the world, everyone was shocked to find that the United States ranked 17th in the world for longevity with an average life span of 75 years - there were sixteen other countries whose peoples lived longer than we did.

Even though we spent more money for health care than all the other countries in the world put together we still only ranked 17th - we certainly weren't getting the "bang for our buck" when it came to health care.

In 1990, the Japanese ranked number one in longevity with an average life span of 79.1 years, even though western doctors claimed the Japanese did everything wrong:

1. The Japanese were heavy smokers - sixty three percent smoked three to five packs of cigarettes per day, yet they had eighty five percent less cardiovascular disease and eighty five percent less cancer than we did.

2. The Japanese consumed an enormous amount of salt each day compared to Americans. The Americans were admonished by their doctors to eat less than one gram of salt per day. The Japanese by contrast consumed fifteen grams of salt per day, yet they had eighty five percent less cardiovascular disease and hypertension than did the Americans.

3. We dropped two atomic bombs on the Japanese to end World War II in the Pacific - they had more radiation per person than any other country in the world, yet they have eighty five percent less cancer than did the Americans.

The Study was rerun ten years later and the results released in June 2000. The Japanese remained number one in world ranking for longevity, although they dropped in average life span from 79.1 in 1990 to 74.5 in 2000.

In the 2000 study, the Americans dropped to an average lifespan of 70 and slipped in the world rankings for longevity down to 24th.

In 2000, the entire world spent \$2.7 trillion for health care, out of that princely sum, the USA alone spent \$1.6 trillion, more than half of the world's total expenditure, yet we only rank 24th in the world when it comes to health and longevity.

It is quite obvious that a lack of money, technology, drugs, doctors, nurses and hospitals is not the problem. Every country in the world sends their brightest young doctors to America to study our health and treatment methods. We are the health Mecca to the rest of the world. World leaders come to the USA for state of the art treatments.

A special 2004 longevity issue of *Scientific American* pointed out that the "oldest old" over ninety five years of age were usually healthier than individuals twenty years younger. Fifty percent of ninety year olds have Alzheimer's disease while only thirty three percent of 100 year olds have the disease.

Time magazine, August 2004, featured a special cover article, "How to Live to be 100 (and not regret it). The authors pointed out that by avoiding alcohol, tobacco and caffeine, the clean living (Mormon)

people of Utah live an average of eight years longer than the average American - in 2005 we dropped to 26th in the world for longevity!

The “Eat Less” Myth

The concept of “under nutrition without malnutrition” goes back to the work of Clive M. McKay in the 1920s. McKay, a Cornell University nutritionist, wrote his landmark papers in 1935 - he reported that by restricting carbohydrate calories by thirty percent (while keeping the fat, protein, vitamins and minerals the same) he could increase the life span of his test rats by 30 percent; when he cut the carbohydrate calories by sixty percent (while keeping the fat, protein, vitamins and minerals the same) he could increase life spans by sixty to 100 percent.

Effects of Calorie Restricted Diets Fed to Weanling Rodents and Monkeys

Effects Related to Altered Growth, Development and Metabolism

- A. Lower body temperature
- B. Delayed sexual maturity
- C. Delayed skeletal maturity

Effects Pointing to Improved Health

- A. Lower body weight
- B. Reduced abdominal fat loads

Reduced Risk for Chronic Age-Related Diseases

- A. Greater sensitivity to insulin
- B. Lower levels of fasting insulin
- C. Lower fasting glucose levels
- D. Lower total cholesterol and triglyceride levels
- E. Increased levels of HDL (“good cholesterol”)
- F. Reduced rate of decline for the hormone DHEA

Effects Found in Rodents and Still Being Investigated in Monkeys

- A. Delayed onset of cancer
 - B. Increased rate of cellular suicide
 - C. Longer average life span
 - D. Increased maximum life span
-

Another way of looking at CR diets is:

CR slows the rate of cell division. CR is thought to slow the rate of cellular metabolism enough so that division occurs less often, effectively slowing the aging process - this may help explain why laboratory animals on CR diets have a lower rate of late-life cancer.

CR boosts immunological response and increases levels of protective hormones.

CR reduces mitochondrial damage from free radicals produced by metabolizing energy fuel.

A human study by John O. Holloszy, a professor of medicine at Washington University in St. Louis, published in early 2004, noted that 18 people who had used a CR diet for three to fifteen years showed "dramatically reduced risks of developing diabetes or clogged arteries."

"It's very clear that calorie restriction has a powerful protective effect against diseases associated with aging," Holloszy said. "We don't know how long each individual actually will end up living, but they certainly have a much longer life expectancy than average because they're most likely not going to die from a heart attack, stroke or diabetes."

This concept of "under nutrition without malnutrition" has been touted as a calorie restricted diet (CR), with the emphasis on restricted calories, by many modern gerontologists; however, they fail to give equal weight to an essential part of the program - the "with-

out malnutrition” part.

A CR diet without a serious mineral intake will feel like a starvation diet because of the pica or munchies factor and therefore almost impossible to stick with.

Using Clive McKay’s original theory, modern scientists have repeated his work in fruit flies, worms, monkeys and many other laboratory animals with the same success. Dr. David Sinclair, of Harvard, put himself on the same 1,600 calorie diet eaten by members of the Calorie Restriction Society - Sinclair was able to maintain the CR diet for only a week. “It was too tough,” he said, “My hats off to the calorie restrictors. Now I’m hoping to find drugs that can give people the benefits of a CR (diet) without the (actual) diet.”

Dr. Roy L. Walford (1924 - 2004), a gerontologist at the University of California, Los Angeles, kept himself on a 1,600 calorie diet for three decades - in April 27, 2004, Walford died of amyotrophic lateral sclerosis (Lou Gehrig’s disease) at the age of seventy nine years. Walford continually preached the gospel of CR diets, however, he paid little attention to the “without malnutrition part” and as a result he died an unnecessarily early death.

CR Diet Benefits for Laboratory Animals

Protozoan (+85.7%)

Normal diet

Average: 7 days; maximum 14 days

Caloric restriction

Average: 13 days; maximum: 25

Bowl and Doily Spider (+80%)

Normal diet

Average: 50 days; maximum 100 days

Caloric restriction

Average: 90 days; maximum: 139 days

Guppy (+ 39.4%)

Normal diet

Average: 33 months: maximum: 54 months

Caloric restriction

Average: 46 months; maximum: 59 months

White Rat (+43.5%)

Normal diet

Average: 23 months; maximum: 33 months

Caloric restriction

Average: 33 months; maximum 47 months

Essential Nutrients and Antioxidants

Our dietary requirements for essential nutrients, goes up as we get older because our ability to assimilate them decreases with age. We like to error on the side of excess when it comes to daily intakes of our essential nutrients, especially when it comes to the needs of seniors - vitamins, amino acids, essential fatty acids and minerals - and don't wait until you're obese or old to take antioxidants, you should start before conception, during pregnancy and from the moment of birth.

While it is easy to produce deficiencies of essential nutrients and wind up with debilitating and life threatening diseases simply by not supplementing, conversely, it is difficult to overdose and damage yourself.

Supplement properly with plant minerals and you can avoid obesity, type II diabetes and the Metabolic Syndrome - if you have these chronic longevity stealing diseases, supplement properly and you can reverse these post September 4, 1882 diseases.

It is easy to forget that we humans supplemented with minerals for tens of thousands of years with the simple household byproduct of wood ashes (the minerals that are left when you burn the carbon in wood) from our heating and cooking fires - we put them in our gardens, we used them as "culinary" ashes and we used them as "salt" or we cut the more valuable table salt with them.

Most experts agree that Albert Einstein was one of the more intelligent human beings that have ever lived. He won the Nobel Prize in physics in 1921 "for his services to theoretical physics and especially for his discovery of the law of the photoelectric effect." Yet he died in 1955 at the age of 76 years of age of a ruptured aortic aneu-

rysm, something that even a turkey wouldn't die from (remember, we put everything that turkey's need nutritionally into their little alfalfa pellet) - ten years earlier we had eliminated ruptured aneurysms in turkey's (a cause of devastating losses) by simply increasing the copper levels in their food pellets.

We are sure that Einstein had the very best doctors that money could buy and that medical science could provide - yet Einstein died of a simple copper deficiency - a deficiency that he had warnings of (grey hair and deep skin wrinkles) for decades before he died.

Next time someone says, "Why should I supplement, they didn't supplement hundreds of years ago!" you now have the proper answer - yes they did, and here's how they did it - it was the wood ashes.

We believe that antioxidants will be added to the list of essential nutrients - without them we cannot protect ourselves from dietary and environmental free radicals and attain maximum life spans. Antioxidants are found in substantial quantities in:

Fruit - blueberries, cherries, kiwis, pink grapefruit, oranges, plums, prunes, raisins, raspberries, red grapes, strawberries, boysenberries, pomegranate, noni and mangostein.

Vegetables - alfalfa, beets, broccoli, Brussels sprouts, eggplant, kale, tomatoes, red bell peppers, spinach, tomatoes.

Green tea - EGCG

Antioxidants are essential to a long and healthy life and they are especially important to the obese human and the obese person who wants to loose weight. When body fat is burned, waste products (trans fats, oxidized fats and ceroid lipofucsin) are produced. These fat metabolism waste products are free radicals and they significantly increase the risk of cancer, heart disease, vascular disease (arteriosclerosis, atherosclerosis), thrombotic disease (stroke, deep vein thrombosis, pulmonary embolism, and coronary thrombosis), macular degeneration, cataracts and Alzheimer's disease.

Unfortunately, we have seen numerous people who were very good at losing weight, but they refused to take adequate levels of antioxidants and six months to a year later wound up with some terrible disabling or fatal disease that could have been prevented with a serious antioxidant program.

It is predicted that by 2025, 14 million Americans will die of Alzheimer's disease, a disease that is caused by oxidative damage to the myelin of the brain. Myelin makes up about seventy five percent of the human brain weight and myelin is almost 100 percent cholesterol.

The byproducts of oxidized cholesterol and fat includes, trans fatty acids and ceroid lipofuscin, which builds up and occupies space in all cells. Typically, thirty five year olds have as much as fifty percent of their neuron volume filled with ceroid, fifty year olds have sixty percent of their neuron volume filled with ceroid and 100 year olds have as much as eighty five percent of their neuron volume filled with ceroid.

Externally ceroid shows up as liver spots and age spots - if you have a dozen obvious deposits on the outside, you have millions of them on the inside. Fortunately, by eliminating the multiple sources of free radicals and significantly ramping up on your intake of antioxidants you can remove the ceroid from your body and slow the aging process.

In 1992, the University of California, San Diego and the Salk Institute jointly released a study that was published in *Biochemical/Biophysical Research Communications*, the study showed that the simple antioxidant vitamin E can ease memory loss in Alzheimer's patients.

In March of 1996, Johns Hopkins University released a 14-year study of 2,065 elderly people who were tracked to determine long term effects of Ibuprofen use for arthritis pain - they were surprised to learn that those individuals taking Ibuprofen had a 60 percent reduction in the risk of Alzheimer's disease (the effect was attributed to a unique antioxidant property of Ibuprofen).

On April 24, 1997, the University of California, San Diego released a study showing that vitamin E could "slow the devastating decline for some Alzheimer's patients, significantly delaying the need

for nursing home care.” “This is the first study showing there are (antioxidants) that can delay important milestones of deterioration in the life of a patient with Alzheimer’s,” said Dr. Leon Thal, chairman of the department of neurosciences at the UCSD School of Medicine.

On November 22, 2001, Dutch researchers at the University of the Netherlands released the two year study of 6,989 patients who took Ibuprofen for pain. The study showed that those taking Ibuprofen had a reduced Alzheimer’s risk of 80 percent.

On November 18, 2003, researchers at Tufts University reviewed the fatty acid levels of blood collected from 1,137 patients on the Framingham Heart Study, their diets were assessed and those free of dementia were followed for a mean of ten years. During the study they were assessed for the onset of new dementia including Alzheimer’s disease. Those with, “diets rich in Omega-3 fatty acids had a significant 48 percent reduction in the risk for the onset of dementia.”

On January 20, 2004, researchers from the Johns Hopkins University School of Public Health, released the results of a study on 4,740 men and women over the age of sixty five that showed, those taking vitamin C and vitamin E had a reduced risk of Alzheimer’s disease by 78 percent.

In summing it all up, we can confidently say that the simple intake of antioxidants and Omega-3 essential fatty acids can reduce the onset of Alzheimer’s disease by as much as forty eight to eighty percent.

Life Spans of the Rich and Famous

To our knowledge, with the exception of one professional bowler, there has never been a professional athlete live to be 100; to our knowledge there has never been a billionaire ever live to be 100; with the exception of Katherine Hepburn, Bob Hope and George Burns, there has never been a movie star ever live to be 100; with the exception of the Queen Mother of England who died at 101 and Madam Jiang Jie Shi of Taiwan who died at 104, there has never been any political figures live to be 100; and there never has been a Nobel

Prize winner live to be 100.

One would think, that if money, political power, connections, physical fitness, or intelligence had anything to do with healthful longevity that the notables in each category would have the longevity advantage - however, statistics point out that there are more 100 year olds and that there are more people over the age of 105, 110 and 114 from simple families and third world countries than from the industrialized nations.

The year 2004 was a typical year:

January:

Brian Gibson, 59. Director of acclaimed films, including "What's Love Got to Do With It?" Cancer.

James E. "Doc" Counsilman, 83. Innovative swimming coach ; led Indiana University to six NCAA championships and coached 48 Olympians, including Mark Spitz.

Michael Straight, 87. Former *New Republic*, publisher; one time spy recruit who helped unmask Anthony Blunt, a Soviet agent.

John Toland, 91. Won 1971 Pulitzer Prize for nonfiction for "The Rising Sun," on the Japanese empire during World War II.

Tug McGraw, 59. Relief pitcher with New York Mets and Philadelphia Phillies; known for the slogan "You Gotta Believe." Father of country music star Tim McGraw. Brain cancer.

Norman Heatley, 92. British scientist who worked on penicillin production.

Francesco Scavullo, 82. Fashion photographer who made beautiful women even more so; produced *Cosmopolitan* covers for decades. Heart failure.

Ingmar Thulin, 77. Swedish actress acclaimed for work with Ingrid Bergman ("Wild Strawberries," "Brink of Life.").

Alfred Pugh, 108. Last known combat-wounded U.S. veteran of World War I.

Thomas Kindness, 74. Six-term Ohio congressman.

Philip Geyelin, 80. Pulitzer Prize winning journalist, credited with turning *The Washington Post* editorial page against the Vietnam War.

Spalding Gray, 62. Actor-writer who exposed his personal life in acclaimed monologues such as "Swimming to Cambodia." Suicide.

Harold Shipman, 57. British medical doctor found guilty of killing at least 215 elderly patients. Hanged himself in prison.

David N. Henderson, 82. Eight-term North Carolina congressman; helped create Cape Lookout National Seashore.

Uta Hagen, 84. Actress who enchanted Broadway audiences for more than 50 years, notably as the brutal Martha in Edward Albee's "Who's Afraid of Virginia Wolf?"

Ray Stark, 88. Hollywood power broker; produced "Funny Girl," "The Goodbye Girl" and "The Way We Were." Died after a "long illness."

Harry "The Cat" Brecheen, 89. St. Louis Cardinals pitcher; won three games in the 1946 World Series.

Harry Claiborne, 86. Federal judge; was impeached and removed after conviction for filing false tax returns.

Jerry Nachman, 57. Colorful and award winning journalist; editor of the *New York Post*; later vice president and host on MSNBC. Gallbladder cancer.

Bernard Punsly, 80. The last surviving member of the movies' "Dead End Kids."

Ann Miller, 81. Long-legged tap dancer famous for movie stardom in musicals, including "On the Town," "Easter Parade" and "Kiss Me Kate"; she entertained Broadway in "Sugar Babies." Died of lung cancer.

Bob Keeshan, 76. Gently entertained generations of children as TV's

Captain Kangaroo; he became an outspoken opponent of violence in children's television.

Helmut Newton, 83. Acclaimed fashion photographer who was a "trailblazer" in exploring power, gender roles and an icy erotic, sexuality in his pictures. Died in a car accident.

Adella Wotherspoon, 100. Believed to be the last survivor of the 1904 sinking of the excursion ferry, General Slocum, that killed more than 1,000 passengers in New York.

Jack Paar, 85. Made the "The Tonight Show" the talk show everybody talked about, breaking the trail for Johnny Carson and others that followed. Died of a stroke.

Elroy "Crazy Legs" Hirsch, 80. NFL running back for the Los Angeles Rams and Hall of Famer; earned his nickname for an erratic running style.

Former Navy Cmdr. Lloyd "Pete" Bucher, 76. Helped his USS Pueblo crew survive 11 months of brutal captivity in North Korea; then returned home to face criticism for surrendering his ship without a fight.

Louie B. Nunn, 79. As Kentucky governor, he supervised the modernizing of the state's mental health system.

Mary-Ellis Bunim, 57. Co-creator of MTV's Real World and other reality TV shows. Breast cancer.

Julius Dixon, 90. Rock 'n' Roll songwriter who wrote the 50s hit "Lollipop."

Olivia Goldsmith, 54. Former management consultant who wrote "The First Wives Club." Heart attack during surgery.

Jake Hess, 76. Southern gospel pioneer and onetime member of the Statesmen Quartet. Died of pneumonia following a heart attack.

Billy May, 87. Trumpeter and bandleader who played with Glenn Miller, Frank Sinatra and others. Heart attack.

Ron O'Neal, 66. Star of the 1972 "blaxploitation" movie, "Superfly." Died of pancreatic cancer.

George Woodbridge, 73. *Mad* magazine caricature artist and illustrator. Emphysema.

Janet Frame, 79. Reclusive author of the book "An Angel at My Table." Leukemia.

Fanny Blankers-Koen, 85. Dutch athlete who in 1948 became the first woman to win four Olympic gold medals in track and field and was named the top female athlete of the 20th century by the International Association of Athletics Federations. Complications of Alzheimer's disease.

Eleanor Holm Whalen, 91. Swimming gold medalist, kicked off of the U.S. Olympic team after she was caught drinking and throwing dice. Kidney failure.

February:

Warren Zimmerman, 69. Last U.S. ambassador to Yugoslavia before it's breakup.

Francis Partridge, 103. British diarist; a member of the literary Bloomsbury Group.

Retired Adm. Thomas Moorer, 91. Former chairman of the Joint Chiefs of Staff; headed Navy during the Vietnam War.

Jose Lopez Portillo, 83. President of Mexico from 1976 to 1982 through oil-driven boom to debt-induced bust. Complications of pneumonia.

Daniel J. Boorstin, 89. Former Librarian of Congress; social critic, Pulitzer Prize winning author and best-selling historian; he wrote "The Americans: The Democratic experience." Pneumonia.

Jozef Lenart, 80. Former Czechoslovakian prime minister; cleared of treason over role in 1968 Soviet-led invasion.

Frank del Olmo, 55. Pulitzer Prize-winning *Los Angeles Times* journalist and a voice for Hispanics. Heart attack.

Jerome F. Lederer, 101. Aviation safety expert; aided Charles Lindbergh's 1927 solo trans-Atlantic flight and for NASA missions.

Fred Benninger, 86. Gambling executive; helped build some of Las Vegas' best-known properties.

Jan Miner, 86. Actress known for her character Madge in Palmolive commercials.

Marco Pantani, 34. Italian cycling champion. Drug overdose.

March:

Marge Schott, 75. Rough talking past owner of the Cincinnati Reds; repeatedly suspended for offensive remarks.

Paul Winfield, 62. Oscar-nominated actor ("Sounder") known for versatility on stage and screen. Heart attack.

Mike O'Callaghan, 74. Nevada governor in the 1970s.

Mercedes McCambridge, 87. Oscar-winning actress; provided demon-possessed girl's voice in "The Exorcist."

Abul Abbas, 56. Palestinian who planned the 1985 hijacking of the cruise ship *Achille Lauro*. He was captured by U.S. commandos died of a heart attack while in U.S. custody.

Robert D. Orr, 86. Championed educational reform as Indiana's governor in the 1980s.

Sidney L. James, 97. Founding editor and first managing editor of *Sports Illustrated*. Prostate cancer.

William H. Pickering, 93. Former director of the Jet Propulsion Laboratory; oversaw launch of the first U.S. satellite in 1958. Pneumonia.

John "J.J." Jackson, 62. Disc jockey who helped start the music

video era as one of the original video jockyes on MTV. Heart attack.

Former Queen Juliana, 94. Reigned 32 years in the Netherlands between 1948 and 1980. Pneumonia.

John C. West, 81. Former South Carolina governor; helped smooth racial tensions after patrolmen in 1968 killed three black protesters.

Sheik Ahmed Yassin, 70. Founder of Hamas, an Islamic militant group. Killed by Israeli Security Forces.

Joshua Eilberg, 83. Six-term Pennsylvania congressman who was on the House Judiciary Committee during the Nixon impeachment hearings in 1974.

J. Edward Roush, 83. Indiana congressman for 16 years who was ousted by Dan Quayle in 1976.

Sir Peter Ustinov, 82. Won 2 Oscars in an acting career that ranged from the evil emperor Nero in "Quo Vadis" to Agatha Christie detective Hercule poirot.

Jan Berry, 62. Half of surf music duo Jan & Dean ("Dead Man's Curve").

Alistair Cooke, 95. Urbane host of television's "Masterpiece Theatre" for 22 years; interpreter of U.S. culture for decades on the BBC's "Letter from America" for 58 years.

Frances Dee, 94. Actress who starred in the movie "An American Tragedy." Stroke.

Spalding Gray, 62. Actor and writer for the New York stage; famous for monologues that were converted to movies, including 1984's *Swimming to Cambodia*, 1996's *Gray's Anatomy*. Body found in New York's East River.

Genevieve, 83. Parisian singer and comedian known for speaking in her broken English to Jack Paar on "The Tonight Show" in the late 1950s and early 1960s. Died following a stroke.

Claus Josef Riedel, 79. Former president of his family's Austrian

glassware company, Riedel Crystal - he revolutionized wine drinking by realizing that the size and shape of the wine glass affects the wine's taste. Heart attack.

Edward Zubler, 79. Inventor; discovered the halogen lamp. Heart failure.

Brian Maxwell, 51. PowerBar inventor and a world-ranked marathon runner. Heart attack.

April:

Carrie Snodgrass, 57. Oscar-nominated actress who played a frustrated housewife ("Diary of a Mad Housewife"). Heart failure.

Fred Olivi, 82. Co-pilot of the plane that dropped the atomic bomb on Nagasaki.

Phil Sokolof, 82. Nebraska multimillionaire who used his wealth to promote better nutrition.

Jim Cantalupo, 60. McDonald's CEO who helped engineer the company's turnaround by focusing on service and healthier food. Heart attack.

Frank Morrison, 98. Governor of Nebraska during the turmoil of the 1960s.

John Maynard Smith, 84. Leading British evolutionary biologist.

Norris McWhirter, 78. Co-founder of Guinness Book of Records in 1954; the book was created to settle trivia disputes in pubs. Heart attack.

Mary McGrory, 85. Columnist for *The Washington Post*, who won a Pulitzer Prize for her Watergate writings.

Pat Tillman, 27. Safety for the NFL's Arizona Cardinals who traded a multimillion-dollar contract to serve as an Army Ranger in Afghanistan. Killed in action.

Estee Lauder, 97. Built multimillion dollar cosmetics empire.

Hubert Selby Jr., 75. Wrote acclaimed 1964 novel "Last Exit to Brooklyn." Lung disease.

Gaetano Badalamenti, 80. Once described as "boss of all bosses" of the Sicilian Mafia.

Pierre Koenig, 78. California architect whose glass and steel frame houses brought modern architecture to the suburbs following World War II. Leukemia.

Soundarya, 32. Indian film star who tried to get parts beyond the roles of "Bollywood" actresses. Plane crash.

Claude "Fiddler" Williams, 96. Jazz violinist who was part of the Kansas City jazz community in the 1930s and a guitarist in the Count Basie Orchestra. Pneumonia.

Henry Viscardi, 91. Champion for jobs for the disabled; he was born with malformed legs, he founded groups that created the National Center for Disability Services.

Roy Walford, 79. Gerontologist who believed in CR diets; he lived for 30 years on a 1,600 calorie diet. He died of Lou Gehrig's disease.

May:

Marvin Runyon, 79. Postmaster general in the 1990s who stressed customer service and tight budgeting.

Darrell Johnson, 75. Managed Boston Red Sox to the 1975 World Series.

Retired Army Gen. Robert F. Seedlock, 91. Led his troops in the construction of the Burma Road during World War II.

Elizabeth Ann Swift Cronin, 63. One of those held hostage in the U.S. Embassy in Iran.

Akhmad Kadyrov, 52. Kremlin-backed leader of Russia's troubled Chechnya province. Assassinated in a World War II Victory Day celebration bombing.

Paul F. Wehrie, 82. Infectious disease expert; helped battle smallpox and polio.

Arnold Beckman, 104. Prolific inventor of scientific instruments (including the pH meter) and a philanthropist.

Col. Robert Morgan, 85. Commander of famed Memphis Belle- B-17 bomber during World War II.

Alan King, 76. Witty comedian who went from standup in the Catskills to Broadway; known for "tirades against everyday suburban life." Lung cancer.

Tony Randall, 84. Comic actor; the fastidious Felix Unger in "The Odd Couple" and pal in a series of Rock Hudson-Doris Day movies.

Marjorie Courtenay-Latimer, 97. In 1938, she discovered a live coelacanth, a prehistoric fish long thought to be extinct.

Elvin Ray Jones, 76. Renowned jazz drummer in John Coltrane's quartet. Heart failure.

Jack Eckerd, 91. Turned three failing drugstores in Tampa Bay into a pharmacy franchise empire that bears his name; philanthropist. Pneumonia.

Samuel Johnson, 76. Built family's South Carolina Johnson Wax Company into a consumer products giant.

David Dellinger, 88. Peace activist; one of Chicago Seven tried for protests during 1968 Democratic National Convention.

Roger W. Straus Jr., 87. Co-founder of publishing house Farrar, Straus & Giroux.

Sam Dash, 79. Attorney whose questions during Senate Watergate hearings made him a household name.

Archibald Cox, 92. The special prosecutor fired in 1973 in the "Saturday Night Massacre" by President Nixon for refusing to curtail his Watergate investigations.

Ramona Trinidad Iglesias Jordan, 114. Puert Rican woman recognized as world's oldest living person.

Alberta Martin, 97. One of the last widows of a Civil War veteran; "belle" of Confederate history experts. Heart attack.

Clement Dodd, 72. Reggae producer who signed Bob Marley and the Wailers; he popularized Jamaican music world wide. Heart attack.

Brenda Fassie, 39. First black South African pop star to gain international fame. Complications of an asthma attack.

Joergen Nash, 84. Danish artist who claimed he cut the head off of Copenhagen's famed Little Mermaid statue in 1963.

June Taylor, 86. Emmy-winning choreographer who founded the June Taylor Dancers. Natural causes.

Syd Hoff, 91. *New Yorker* cartoonist and author of the children's book "Danny and the Dinosaur." Pneumonia.

June:

William Manchester, 82. Historian who brought a novelist's flair to biographies of such political notables as Sir Winston Churchill and John F. Kennedy; author of "The Death of a President." Two strokes.

Dr. Charles Kelman, 74. Developed outpatient cataract operation.

Frances Shand Kydd, 67. Princess Diana's mother.

Former President Ronald Reagan, 93. The "cheerful crusader" who devoted his presidency to winning the Cold War; scaling back government and making people believe it was "morning again in America."

James M. Roche, 97. As chief executive of General Motors, he helped promote equal opportunity.

Ray Charles, 73. Music giant that sang all forms of music; hits included "What'd I Say," "Georgia on my Mind" and "I Can't Stop Lov-

ing You.” Side effects of cholesterol lowering pharmaceuticals.

Robert Teeter, 65. Influential Republican pollster who worked in several presidential races. Cancer.

Al Lapin Jr., 76. Co-founded International House of Pancakes Chain with his brother in 1958.

Clayton Kirkpatrick, 89. Managed numerous changes as editor of the *Chicago Times*.

Mattie Stepanek, 13. Child poet; inspirational verses made him a best-selling writer (“Heartsongs”) and an activist for those diagnosed with muscular dystrophy.

Bob Bemer, 84. Computer pioneer who published early warnings of the Y2K risks.

Agnes “Sis” Cunningham, 95. Co-founder of the folk journal *Broadside*.

Bernard Grant, 83. Soap opera star on “Guiding Light” and “One Life to Live”; voiced English dubs for foreign films. Lymphoma and pneumonia.

Whitman Knapp, 95. Federal judge who led the Knapp Commission which exposed police corruption in New York City in the early 1970s. Pneumonia.

Jacek Kuron, 70. Polish academic widely viewed as the intellectual force behind Lech Walesa’s Solidarity Movement; he later became a labor minister.

Ralph Wiley, 52. *Sports Illustrated* writer and essayist on race in America. Heart failure.

Dorothy Brown, 90. Tennessee legislator and the first black female surgeon in the South, according to the National Library of Medicine. Heart Failure.

Dick Durrance, 89. Winner of 17 national ski championships who helped turn Aspen, Colorado into a top ski resort.

Karol Kennedy Kucher, 72. With her brother Peter, they became the first American ice skating pair to win a world championship in 1950. Pneumonia.

July:

Marlon Brando, 80. Revolutionized American stage and screen acting as Stanley Kowalski in "A Streetcar Named Desire"; created the character of Vito Corleone in "The Godfather."

Isabel Sanford, 86. Starred as Weezie on "The Jeffersons."

Laurance Rockefeller, 94. Conservationist, philanthropist; one of the six children of John D. Rockefeller Jr. Pulmonary fibrosis.

Carlos Kleiber, 74. Celebrated German-born conductor.

Charles W. Sweeney, 84. Piloted the plane that dropped an atomic bomb on Nagasaki.

George Busbee, 76. Georgia governor from 1975 to 1983; campaigned as "a workhorse, not a showhorse."

Robert Smylie, 89. Three-term Idaho governor; got sales tax passed.

David A. Wallace, 87. Urban planner who revived Baltimore's Inner Harbor.

Ann McGill Gorsuch Burford, 62. Reagan's Environmental Protection Agency Chief; quit under fire from environmentalists.

Richard Bloch, 78. Co-founder of H&R Block, world's largest tax preparer.

Ed Lewis, 86. Won the Nobel Prize in physiology /medicine in 1995 for research into how genes regulate body development.

Illinois Jacquet, 81. Saxophonist who played with almost every jazz icon of his time; best known for his solo in Lionel Hampton's 1942 song Flying Home.

Wilton Mkwayi, 81. Apartheid foe who was jailed alongside Nelson

Mandela.

Fred LaRue, 75. Special assistant to John Mitchell, Nixon's attorney general; served a prison term for Watergate.

Francis Crick, 88. Nobel Prize-winning scientist who, with James Watson, discovered the double-helix structure of DNA in 1953; long-time faculty member at the Salk Institute for Biological Studies in La Jolla, California. Colon cancer.

Carmine De Sapio, 95. Revived Tammany Hall after World War II and was its last chairman, a position that gave him power over New York City politics.

Jackson Beck, 92. Voice-over artist who introduced the Superman radio show and was the voice of Bluto in nearly 300 Popeye cartoons.

Antonio Gades, 67. Artistic director of the National Ballet of Spain, who popularized flamenco and is generally considered the greatest Spanish dancer of his generation. Cancer.

Jerry Goldsmith, 75. Composer who wrote the musical scores for almost 200 movies including "Patton," "Chinatown" and "The Omen," which won him an Academy Award in 1977; he also scored television shows such as "Star Trek: The Next Generation." Cancer.

Edward Killingsworth, 86. Modernist architect of the Case Study Houses, a Landmark Los Angeles-based project that promoted modern design and cost effective building in the postwar housing boom.

John Cullen Murphy, 85. Illustrator best known for the Sunday comic strip "Prince Valiant," which he created for 34 years.

Syreeta Wright, 58. Motown recording artist and songwriter with her ex-husband, Stevie Wonder. Breast Cancer.

Albert Casey, 84. Former chairman and CEO of AMR Corp., parent company of American Airlines.

William Mitchell, 92. Food scientist and creator of Pop Rocks candy;

he also owned the patents for inventions that led to Cool Whip and Tang. Heart failure.

Paula Danziger, 59. Author of more than 30 children's books including, "The Cat Ate My Gymsuit." Heart attack.

David Webb Chaney, 88. Chemist, who created AstroTurf. Cerebral hemorrhage.

Emile Peynaud, 92. Transformed the wine making industry into a modern industry that used science to standardize product.

Roger Ward, 83. Two-time winner of the Indianapolis 500; one of auto racing's biggest stars in the late 1950s and 1960s.

Thomas Klestil, 71. Former president of Austria; spoke out against Austria's involvement with the Nazis during World War II. Multiple organ failure.

Jeff Smith, 65. Minister; best known as PBS's Frugal Gourmet until a sex scandal ended his career. Heart disease.

Joe Gold, 82. Founder of Gold's Gym and an early promoter of body building in America. Heart failure.

August:

Alexandra Scott, 8. Cancer patient who started Alex's Lemonade Stand Charity. Childhood cancer.

Henri Cartier-Bresson, 95. Acclaimed French photographer, whose pictures defined the mid-20th century and inspired generations of photographers that came after him; co-founder of the Magnum photo agency.

Rick James, 56. Funk legend known for the 1981 hit "Super Freak." Heart attack.

"Red" Adair, 89. Legendary oil field firefighter ; inspiration for the John Wayne film "Hellfighters."

Fay Wray, 96. The damsel in distress, held atop the Empire State Building by the giant gorilla in "King Kong."

Julia Child, 91. Brought the secrets of French cooking to Americans through her ground breaking TV show and books.

Czeslaw Milosz, 93. Polish poet and Nobel laureate famous for his intellectual and emotional works about some of the worst cruelties of the 20th century.

William D. Ford, 77. Fifteen-term Michigan congressman helped to expand educational opportunities.

Paul Ngei, 81. Hero of Kenya's independence movement revolt against British colonial rule.

Sune Bergstroem, 88. Swedish scientist; shared 1982 Nobel Prize in medicine.

Hiram L. Fong, 97. First Asian-American elected to the U.S. Senate in 1959; represented Hawaii for three terms until 1977.

Elmer Bernstein, 82. Oscar-winning composer who scored such films as "To Kill a Mockingbird," "The Magnificent Seven," "The Ten Commandments" and "The Great Escape."

Elisabeth Kubler-Ross, 78. Psychiatrist who revolutionized the way the world looked at the terminally ill with "On Death and Dying."

Laura Branigan, 47. Grammy-nominated singer had a platinum hit in 1982 with "Gloria." Ruptured brain aneurysm.

Fred L. Whipple, 97. Harvard astronomer who proposed the "dirty snowball" theory that revolutionized the study of comets.

Leon Golub, 82. Artist known for paintings that featured monumental human figures and depict war, torture and suffering. Complications after surgery.

Isidro "El Indio" Lopez, 75. Saxophonist, singer and founder of the Isidro Lopez Orchestra; thought to be the father of Tejano music.

Complications from a stroke and brain aneurysm.

Carl Mydans, 97. Original photo-journalist for *Life* magazine; a pioneer in magazine photojournalism; he photographed General Douglas MacArthur wading ashore on his return to the Philippines. Heart failure.

Robert Linn, 95. Recognized by the Guinness World Records as the World's longest serving mayor, first taking office in Beaver, Pennsylvania in 1946 and being elected 14 additional times over the following 58 years.

J. Irwin Miller, 95. Former CEO of the Fortune 500 engine manufacturer Cummins Engine. Congestive heart failure.

Nuha al-Radi, 63. Iraqi ceramist and painter best known for her 1998 book "Bagdad Diaries." Complications of Leukemia.

Gloria Emerson, 75. Author and New York Times foreign correspondent. Suicide.

Sir Godfrey Hounsfield, 84. British electrical engineer who shared a 1979 Nobel Prize for his work on the invention of the CAT scan.

Larry Desmedt, 55. Builder of custom motorcycles; known to enthusiasts as "Indian Larry". Died from injuries suffered in a motorcycle crash.

Al Dvorin, 81. Concert announcer for Elvis Presley, whose signature line "Elvis has left the building," became a pop culture classic. Auto accident.

E. Fay Jones, 83. Architect and Frank Lloyd Wright student known for his design of Thorncrown Chapel in Eureka Springs, Arkansas. Heart and lung failure.

Frank Sanache, 86. Last of eight Meskwaki Indian "code talkers" who used their native language as a code during World War II.

September:

Herbert H. Haft, 84. Multimillionaire retailer offered goods from books to prescription drugs at discount prices.

Kirk Firdice, 70. Tough businessman, who, in 1992 became Mississippi's first Republican governor in more than 100 years.

Richard G. Butler, 86. Notorious white supremacist; given the name - "elder statesman of American hate."

Brock Adams, 77. Transportation secretary under President Carter; represented Washington state in the House and Senate. Parkinson's disease.

Fred Ebb, 76. Wrote lyrics for hit musicals "Chicago" and "Cabaret" as well as the song "New York, New York."

James David Barber, 74. Political scientist; wrote influential book "The Presidential Character: Predicting Performance in the White House."

Reynaldo Garza, 89. First Hispanic federal judge, appointed in 1961.

Johnny Ramone, 55. Born John Cummings, co-founded influential punk band The Ramones. Prostate cancer.

Marvin Mitchelson, 76. Hollywood divorce lawyer who pioneered the "palimony" concept - alimony for unmarried couples who separate. Cancer.

Russ Meyer, 82. Producer-director who helped develop the "skin flick," and later gained industry respect for his films such as "Beyond the Valley of the Dolls" and "Faster, Pussycat! Kill! Kill!" Complications of pneumonia.

Eddie Adams, 71. Photojournalist who took the Pulitzer Prize-winning Associated Press picture of a Viet Cong fighter being executed in a Saigon street during the Viet Nam War in 1968. Lou Gehrig's disease.

Francoise Sagan, 69. French author; Famous among teenagers for best-selling "Bonjour Tristesse."

Marvin Davis, 79. Billionaire oilman, philanthropist and, in the 1980s, owner of 20th Century Fox Studios.

Geoffrey Beene, 77. Designer of classic styles, that put him at the cutting edge of American fashion; his clients included first ladies Pat Nixon and Nancy Reagan.

Izora Armstead, 62. Half of the singing duo, the Weather Girls, known for their 1982 hit, "It's Raining Men." Heart failure.

Skeeter Davis, 72. Grand Ole Opry star, who had a cross over hit with "The End of the World." Breast cancer.

Frank Thomas, 92. Animator and one of Walt Disney's famed "Nine Old Men" responsible for the scene of the two star dogs sharing a strand of spaghetti on their first date in the 1955 classic, "Lady and the Tramp." Cerebral hemorrhage."

John Mack, 74. Harvard psychiatry professor who won a Pulitzer Prize for his biography of Lawrence of Arabia. Struck by a car.

Richard Butler, 86. Founder of the white supremacist group Aryan Nations. Heart attack.

October:

Rodney Dangerfield, 82. A "bug-eyed" comedian who had a self-deprecating act ("I don't get no respect") that brought him stardom in Vegas clubs, TV and the big screen.

Richard Avedon, 81. Redefined fashion photography as an art form while achieving acclaim for his stark portraits of the powerful; photographic subjects included Dwight Eisenhower.

Janet Leigh, 77. Wholesome "girl next door" beauty whose character's shocking murder in the Alfred Hitchcock thriller "Psycho" is a cult classic.

Gordon Cooper, 77. Youngest of the original seven Mercury astronauts; achieved many firsts, including first astronaut in space for 24 hours.

Maurice Wilkins, 88. British scientist who shared the 1962 Nobel Prize for early work on DNA.

Johnny Kelley, 97. The heart and soul of the Boston Marathon, running it a record 61 times - winning twice.

Jacques Derrida, 74. World-renowned thinker who founded the school of literary analysis known as "deconstructionism," arguing that the meaning of a collection of words is not fixed. Pancreatic cancer.

Maxime A. Faget, 83. NASA engineer who the designed original spacecraft for Project Mercury.

Christopher Reeve, 52. "Superman" actor, became America's spokesman for spinal cord injury research following an accidental fall while jumping horses.

Ken Caminiti, 41. Former San Diego Padres infielder and Major League Baseball MVP in 1996; admitted to steroid use. Died of a drug overdose.

Martin M. Kaplan, 89. Secretary general of the Nobel Peace Prize-winning Pugwash conferences on disarmament.

Pierre Salinger, 79. President Kennedy's press secretary; later a respected correspondent for ABC News. Heart failure.

Paul H. Nitze, 97. Arms control advisor, who served under eight presidents and helped design the Cold War policy of "containment." Pneumonia.

Robert Merrill, 87. Metropolitan Opera superstar who was equally comfortable singing the national anthem at Yankee Stadium.

Vaughn Meader, 68. Attained instant notoriety for satirizing John F. Kennedy in the multimillion-selling album "The First Family"; his

fame died following Kennedy's assassination.

Edward Oliver Leblanc, 81. Former Dominica premier who guided the island nation transition into independence.

Bruce Palmer, 58. Bass player for the folk-rock band Buffalo Springfield. Heart attack.

Lewis Urry, 77. Inventor considered the "father of alkaline" batteries.

James Chace, 72. Historian and author, biographer of Dean Acheson. Heart attack.

Ray Boone, 81. Patriarch of a three-generation family of Major League Baseball players. Complications following surgery.

Bill Bennett, 73. Hang-gliding pioneer who made the world's highest and longest unassisted free flight. Hang-gliding accident.

Samuel Lee Gravely Jr, 82. First black admiral and first black commander of a U.S. Navy fleet. Complications of a stroke.

Joyce Jillson, 58. Nationally syndicated astrologist who claimed to be astrology adviser to the Reagan White House. Kidney failure.

November:

Theo van Gogh, 47. Outspoken Dutch film director; great-grand-nephew of the painter, Vincent van Gogh; created a controversy and was threatened over his film "Submission," about violence against women in Islamic Societies. Shot and stabbed by Islamic radicals.

Sheik Zayed bin Sultan Al Nahyan, 86. Longtime king of the United Arab Emirates; he brought the UAE into the 20th century and built it into a high-tech world center.

Howard Keel, 85. Baritone in MGM musicals such as "Kiss Me Kate" and "Annie get Your Gun"; he later starred in the TV series "Dallas."

Yasser Arafat, 75. Palestinian guerrilla leader turned Nobel Peace Prize Winner; was hated as a sponsor of terrorism.

Ol' Dirty Bastard, 35. The rapper, born Russell Jones, and founding member of the Wu Tang Clan whose unique rhymes and wild lifestyle made him the biggest character in hip-hop. Drug overdose and heart attack.

Thomas M. Foglietta, 75. Six-term Philadelphia congressman and ambassador to Italy.

Elmer L. Andersen, 95. Liberal Republican who, as governor of Minnesota pressed for human rights.

Reed Irvine, 82. Founder of the media watchdog group Accuracy in Media.

Bobby Frank Cherry, 74. Ex-Klansman convicted in the 1963 Birmingham, Alabama, church bombing that killed four black girls.

Cy Coleman, 75. Composer of Broadway musicals ("Sweet Charity," "City of Angels") and pop songs ("The Best is Yet to Come").

Sir John Vane, 77. Shared 1982 Nobel Prize in medicine for discovery how aspirin works.

Fred Hale Sr., 113. Documented as the world's oldest living man.

Ancel Keys, 100. Developed "K rations" eaten by soldiers in World War II and theorized that heart attacks were caused by dietary fat.

Joseph John Sisco, 85. Diplomat whose State Department career spanned five presidential administrations; participated in Henry Kissinger's "shuttle diplomacy."

Arthur Hailey, 84. Best-selling author whose novel "Airport" generated a parade of disaster movies.

Philippe de Broca, 71. French director whose 1960s films "The Man from Rio" and "King of Hearts" brought him recognition and respect.

The Rev. Billy James Hargis, 79. Leading radio and TV evangelist and anti-communist crusader.

Ed Kemmer, 84. Actor who played Buzz Corry in the 1950s TV

show Space Patrol. Complications of a stroke.

Terry Melcher, 62. Producer and songwriter son of Doris Day who worked with The Byrds and the Beach Boys and helped shape the 1960s surf music sound. Cancer.

Zachary Solov, 81. Former dancer and chief choreographer with the Metropolitan Opera. Heart failure.

Larry Brown, 53. Oxford, Mississippi fireman who began writing novels in his spare time and won acclaim for his blunt raw portrayals of common people. Heart attack.

Iris Chang, 36. Best selling author of "The Rape of Nanking" and "The Chinese in America." Suicide.

December:

Prince Bernhard, 93. German-born father of the Netherlands' Queen Beatrix who earned the respect of the Dutch during the Nazi occupation by walking while wearing the yellow Star of David required of the Jewish people to show their support and solidarity for their Jewish citizens.

Dame Alicia Markova, 94. One of the 20th century's greatest ballerinas and a co-founder of the English Ballet.

Josef Schwammberger, 92. Sadistic Nazi labor-camp commander who hid for 40 years in Argentina before he was captured.

Jay Van Andel, 80. Billionaire co-founder of the Amway Corporation, which sells household products through network marketing and home meetings. Parkinson's disease.

Darrell Abbott, 38. Known as, "Dimebag," acclaimed guitarist with Grammy-nominated heavy metal band Pantera and more recently Damageplan. Shot to death while performing on stage.

Barry Corbet, 68. Member of a climbing team that forged a new route to the summit of Mount Everest in 1963. He and his team were the first Americans to conquer Everest; he gave up his chance to

reach the summit himself to his other team members.

Oscar M. Ruebhausen, 92. Former advisor to Governor Nelson Rockefeller advisor; he was an intimate but unpaid advisor; he was a Lend-Lease attorney during World War II; he was instrumental in forming the National Science Foundation.

Harold Benjamin, 80. Established the first Wellness Community Network for cancer patients and their families in 1982; he wrote "The Wellness Community Guide to Fighting for Recovery From Cancer." Pulmonary fibrosis.

Jack Newfield, 66. He was a trash and "muckraking" reporter who wrote books on Robert Kennedy and boxing promoter Don King. His books include "City for Sale: Ed Koch" and the "Betrayal of New York."

Sidonie Goossens, 105. Principal harpist of the BBC Symphony Orchestra from the 1930s until her retirement fifty years later; she was the last survivor of a musically distinguished family.

Agnes Martin, 92. Artist of the abstract Expressionism movement who was admired for the meditative quality of her work. Pneumonia.

Renata Tebaldi, 82. Italian soprano considered to have one of the most beautiful voices of the 20th century and a star at both the New York Metropolitan and Italy's La Scala.

Princess Kikuko, 92. Aunt of Japan's emperor and outspoken supporter of allowing women to assume the imperial throne. Blood poisoning.

Reggie White, 43. Renowned NFL defensive end who played for the Philadelphia Eagles and the Green Bay Packers, White was also an ordained minister; generally known as the "Minister of Defense." Heart attack.

Francess Lantz, 52. Wrote novels aimed at girls including the "Luna Bay" surfer girl series; her novel "Stepsister From Planet Weird," be-

came a Disney Channel movie. Ovarian cancer.

Isaura E. Fuentes, 103. She was a housewife who was born in Mexico and moved to San Diego.

Susan Sontag, 71. Writer, advocate of avant-garde and essayist; self-proclaimed "zealot of seriousness." Leukemia.

Jerry Orbach, 69. Popular Broadway star who became famous as the Law & Order TV cop, Lenny Brisco for twelve years; he was the voice of the candelabrum in "Beauty and the Beast"; he won a Tony Award in 1969 for playing Chuck Baxter in "Promises, Promises"; he played the first Billy Flynn, the lawyer in "Chicago." Prostate cancer.

John D. Deardourff, 71. Pioneer political consultant who served a string of prominent Republican clients. His clients included Governor Nelson A. Rockefeller of New York, Mayor John V. Lindsay of New York City, Senator Charles H. Percy of Illinois and John C. Danforth the former Missouri senator and U. S. delegate to the United Nations; he was considered one of the founding fathers of political consulting, a trade that was in it's infancy during the 1960s.

Maxwell McCrohon, 76. Former *Chicago Tribune* editor and UPI chief. Lung cancer.

Donald Robinson, 85. Had a 30 year career with the National Park Service; he turned the Cabrillo National Monument into the most visited national park in America as the best gray whale watching and most accurate gray whale counting site.

Artie Shaw, 94. Originally named Arthur Arshawsky, became a jazz clarinetist and big-band leader who successfully challenged Benny Goodman's reign as the King of Swing with his recordings of "Begin the Beguine," "Lady Be Good" and "Star Dust" in the late 1930s; Shaw was married and divorced eight times - his wives included Elizabeth Kern (daughter of song writer Jerome Kern), Lana Turner, Ava Gardner, Evelyn Keyes and Kathleen Winsor (author of "Forever Amber").

Julius Axelrod, 92. Nobel Prize winner; a NIH neuroscientist who won the prize in 1970 for his work on how nerve cells communicate

and affect behavior; his work formed the foundation for a host of new antidepressants in the class of Prozac and Zoloft.

Rod Kanehl, 70. He hit the first "grand slam" in the history of the New York Mets in 1962; he won the heart of manager Casey Stengel by crashing through fences to catch balls. Heart attack.

Hank Garland, 74. Legendary country, rock and jazz guitarist who performed with Elvis Presley, the Everly Brothers, Roy Orbison, Patsy Cline, Charlie Parker and many others in the 1950s and 1960s; he had his first million-selling hit at age 19 with "Sugar Foot Rag." Staph infection.

Jane Gray Muskie, 77. Wife of former presidential candidate, Edmund Muskie. Alzheimer's disease.

Alexander Marshack, 86. A self taught anthropological researcher who was the first to interpret Stone Age artifacts as primitive lunar calendars; Marshack analyzed small incisions in bone plaques found in southwest France that dated from the Paleolithic Period, about 30,000 years ago in the latter century's of the last Ice Age; he determined that humans were keeping monthly calendars 25,000 years before the appearance of writing and almost 20,000 years before the appearance of agriculture - his findings were published in *Science*, November 1964; because of his discovery he was hired as a research associate at the Peabody Museum of Archaeology and Ethnology at Harvard in 1963. Heart failure.

The Longevity Realities

We like to look at birthday announcements in the newspapers as well as obituaries so we know what the longevity possibilities are - in reality, what is our reward in terms of health and longevity if we eat right, supplement properly and maintain a proper weight. In other words, bottom line, is it all (the goal of healthy longevity) in the final analysis worth the conscious effort. The answer turned out to be a resounding - YES!

We will share with the reader a few of our favorite birthday and death announcements because they are a well traveled roadmap of how to do things right and how to get there from here.

On January 1995, Masoomeh Dousti, died at her home in the Iranian village of Kalatenbala in Khorasan province at the age of 161 years - she was survived by six living children ranging in age from 120 to 128 years of age. According to her son, she never visited a doctor or took any chemical medications - she only used herbs for medication.

On June 27, 1995, Dorah Ramothibe of South Africa turned 114 years of age - she was born in 1881 when Queen Victoria was on the throne of England. Her diet consisted of locusts (grasshoppers), pumpkin seeds, tortoise meat, wild herbs, dried fruit and coffee.

On April 1998, Bir Narayan Chaudhuri, a cattle farmer from a town outside of Katmandu, Nepal died at the age of 141 years.

On October 1999, Ralph Charles, of Somerset, Ohio turned 100 years of age - what made him interesting to us was that he was still piloting commercial charter flights, and his pilot's license had no restrictions.

On March of 2003, Maria Etelvina Dos Santos, a Brazilian, died at the age of 124 years. According to her birth certificate, she was the descendent of African slaves and was born on July 15, 1878 - she enjoyed a healthy life throughout her years.

On October of 2003, Sek Yi, a Cambodian tiger hunter and martial arts expert died at the age of 122 - his surviving widow, Long Ouk, was 108 years of age.

On September 1, 2004, a 93 year old woman who has six children, thirty five grandchildren, seventy five great grand children and ten great great grandchildren performed a sky dive from an airplane to celebrate her 93rd birthday- "Because it's a new adventure," explained Luree Kohtz, of Burley, Idaho.

A feisty Etna Powers rode in a horse drawn carriage during the 2003 Portland, Oregon Rose Parade on her 100th birthday as she had done six earlier times.

Sarah Clark Knauss, was born September 24, 1880, she turned 120 years of age in 2000 - older than the Eiffel Tower. She was born

in Hollywood, Pennsylvania.

On May 1, 2001, Harold Stilson, of Deerfield Beach, Florida, turned 101 years of age and went to play golf for his birthday - he hit his sixth hole in one - a 108 yard par three hole!

On January 2001, the *Guinness World Records*, listed Antonio Todd, a shepherd from Sardinia as the worlds oldest man at 112 years of age. His secret, "drink a good glass of red wine every day."

Jeanne Calment of France was born on February 21, 1875 - she died at the age of 122 years and 164 days August 4, 1997 which was documented by the *Guinness World Book of Records*.

The world's oldest man Shigechiyo Izunis of Japan lived to be an authenticated 120 years and 237 days. He was born June 29, 1865. He was six years old when Japan had its first census in 1871. He worked until he was 105 years of age, drank shochu (liquor distilled from barley) and took up smoking at the age of 70 - he died February 21, 1986.

The oldest dwarf was Hungarian born Susana Bokonyi, alias "Princess Susana," of New Jersey until she died at the age of 105 on August 24, 1984. She was three foot four inches in height.

The oldest person to fly as a passenger was Charlette Hughes of Redcar, England who was given a flight on Concorde from London to New York as a 110th birthday present in 1987. She flew again in 1992 at the age of 115.

The world's oldest athlete, Boba Joginder Singh was 105 years of age when he got the Gold Medal for his winning the discus throw at the 1998 Indian National Meet for Veterans, held in Thane, Bombay (Mumbai) - he was the only competitor over the age of 100 years. He represented India in a 1910 World Championship event held in Berlin, Germany.

The oldest person to ever have an operation was James Henry Brett, Jr., whose hip was operated on in Houston, Texas on November 7, 1960 at age 111 years and 105 days.

On August 6, 2004, Verona Johnson turned 114, and was documented as the oldest living American - there have been older Americans before her, and there will be older Americans after her. At 114, she "still remembers names pretty well." Her mind is still sharp enough

to solve crossword puzzles and remember joke punch lines - she can still rattle through the names of her thirty six grandchildren and great-grandchildren.

Johnson's daily snack consists of orange juice, one cracker, one cinnamon drop and one cashew - "That's enough," she claims.

Johnson gave up traveling by herself at age 100 and she gave up playing bridge at age 110. Johnson once exclaimed, "Electricity was the most important thing that happened to us." She is famous for saying, "You can get too old to enjoy life - (but) I never got that old!"

Johnson was quite correct, electricity was one of the more important things that happened to us, however, it was a two edged sword. Electricity provided a type of universal power that could be generated using a variety of fuels (i.e. - wood, coal, oil, nuclear, etc.) and could be used to drive everything from kitchen appliances to computers.

Unfortunately, when electricity was brought on line at 3:00 pm, Monday September 4, 1882 - few, if any knew that the genie had been let out of the bottle. The wood ash byproduct of cooking and heating was not replaced with a complete mineral supplement and the resultant consequences produced obesity, adult onset type II diabetes, the Metabolic Syndrome and hundreds of other mineral deficiency diseases.

Living healthfully beyond 100 years of age is in fact as easy as making a pie. There are many recipes for making pies and most cook books show a picture of what the pie will look like if one but follows the recipes.

In a similar fashion, we know what the typical seventy five year old human will look like if they follow the instructions of the average western doctor. We know what the average obese thirty five year old looks like and we know their reduced life span statistics.

To maximize ones health, fitness and longevity requires a significant level of personal responsibility of each individual. We must make sure we give our body all of the raw materials that it needs to maintain and repair itself - all 90 essential nutrients with a special emphasis on the sixty essential minerals.

We must eliminate the bad stuff that directly or indirectly gener-

ates free radicals - no fried foods, margarines, cooking oils, carbonated drinks, caffeine, excessive alcohol, sugar, excess carbohydrates, burnt animal fats and meat cooked well done. By eliminating these dangerous foods and substances we can add a healthful eight to 20 years to our lives; add in the 90 essential nutrients one can add another 10 to twenty years; add in the antioxidants you add more healthful years and significantly reduce the risks of cancer, cardiovascular disease and Alzheimer's disease.

Doing the correct things will allow us to avoid obesity, adult onset type II diabetes and the Metabolic Syndrome - these diseases are not inevitable, genetic or hopeless, they are in fact preventable and curable as they are caused by simple mineral deficiency diseases.

Remember, humans have the genetic capacity to live to be beyond 100 years of age - if we accept death at the age of sixty five as having had a "good life" when we could have lived to a healthful 100 years of age - we squandered and gave up thirty five years of a healthful and useful life that was given to us for free.

We fight hard to ensure that we get the full measure of our car, dishwasher, and television warranty - why not fight hard and intelligently to get the full measure of the healthful life that we are equipped to get.



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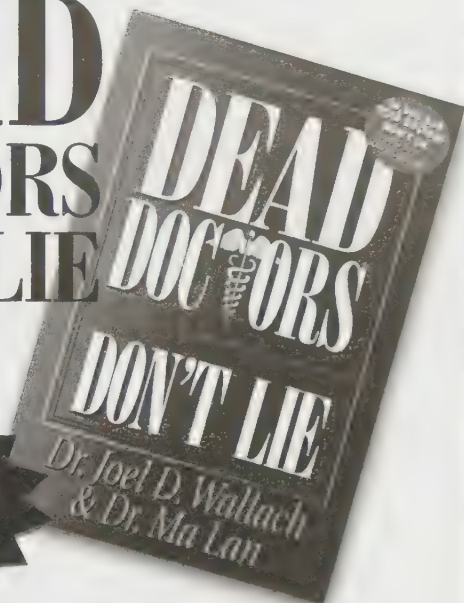
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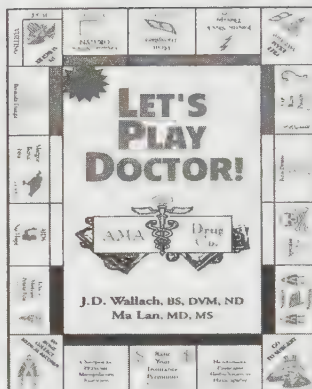
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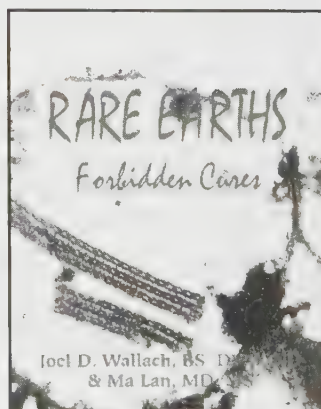
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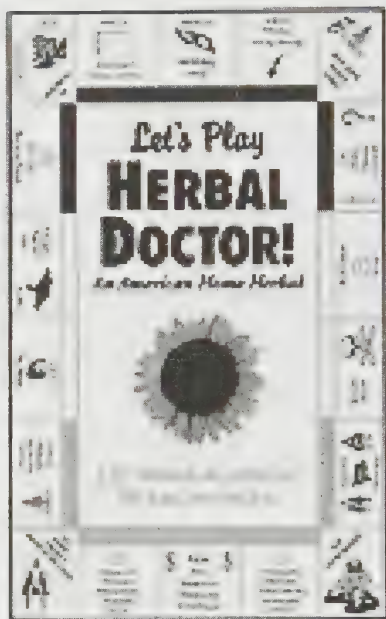


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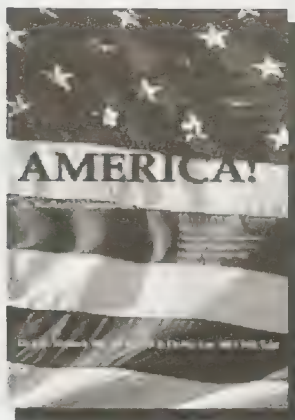


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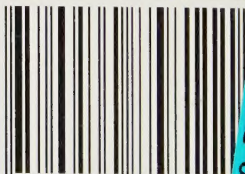
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